

*AWS Certified Generative AI
Developer – Professional*

Course Outline

1. *Architecture considerations for Building Production Grade AWS GenAI Apps – 15 Decisions*
2. *Retrieval Augmented Generation (RAG) -10 Key Architecture considerations to build RAG Apps*
3. *AI Agents and Agentic AI Architectures on AWS*
4. *Data Foundations for Generative AI on AWS*
5. *Responsible AI*
6. *Other AWS AI Services (Non GenAI)*
7. *AWS GenAI Security*



Course Introduction & Few Important Tips

Promise to complete in under 2 mins😊

Course Introduction & Few Important Tips



Welcome to my AWS Certified Generative AI Developer certification course!

I'm Rahul, and I'll guide you through every architecture decision you need to master for this exam.



Sections 2 & 3 of the course are **MOST CRITICAL** for the exam — expect multiple scenario-based questions from these topics.



Additional mini mock exams will be added over the next 1-2 weeks.



Adjust playback speed anytime to match your pace (0.75 or 1.25x or 1.5x).



Found an issue, missing topic, video, quiz discrepancies ? Message me on Udemy or email - I'll address it within 24-48 hours.



Need further guidance on AWS Certified Generative AI Developer exam, feel free to reach out via

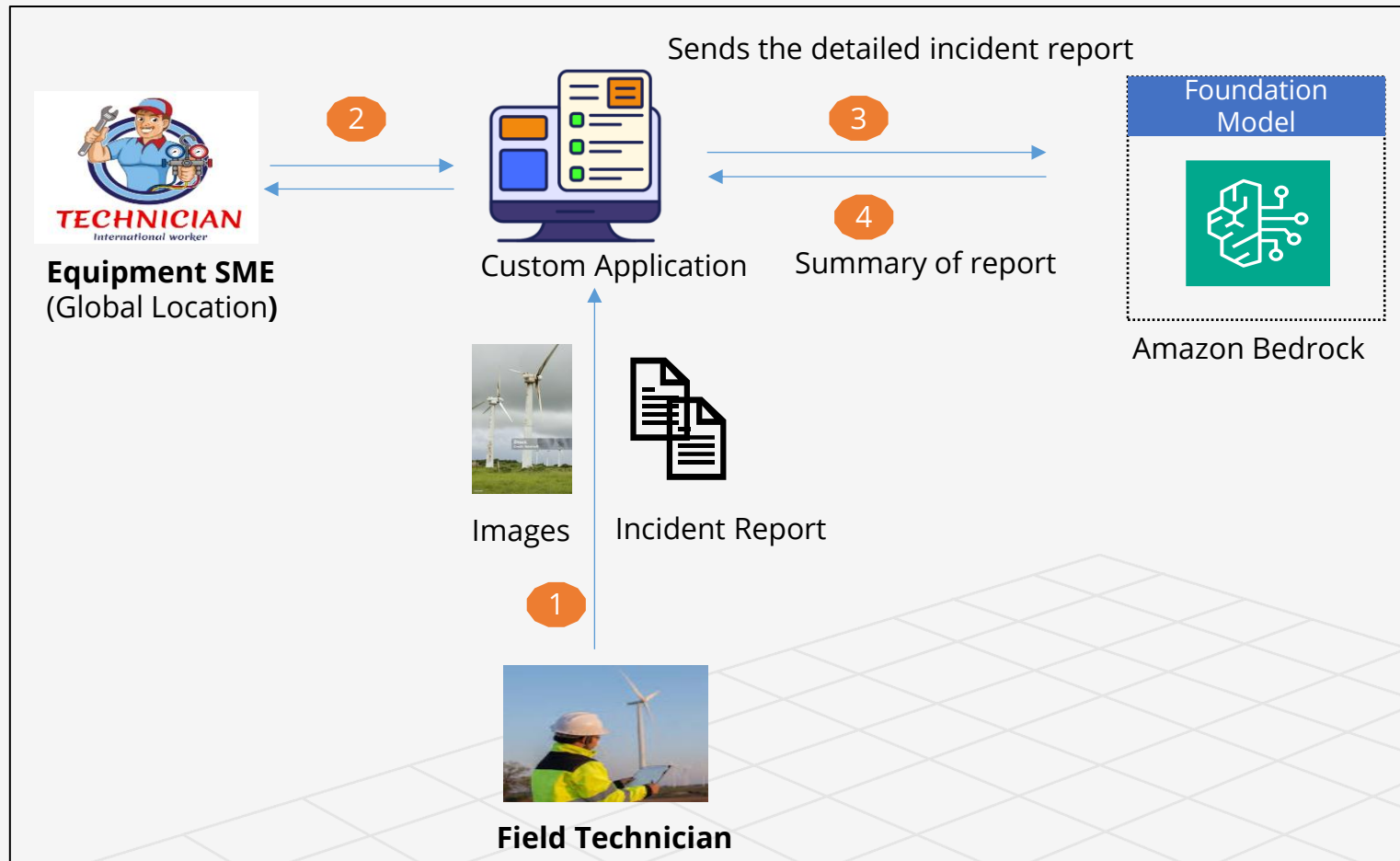
LinkedIn - <https://linkedin.com/in/rahul-trisal-7709628/> or trisalrahul@gmail.com

Section on : Use Case 1

***Architecture considerations for Building
Production Grade AWS GenAI Apps***

Use Case 1 : GenAI-Powered Equipment SME Assistant

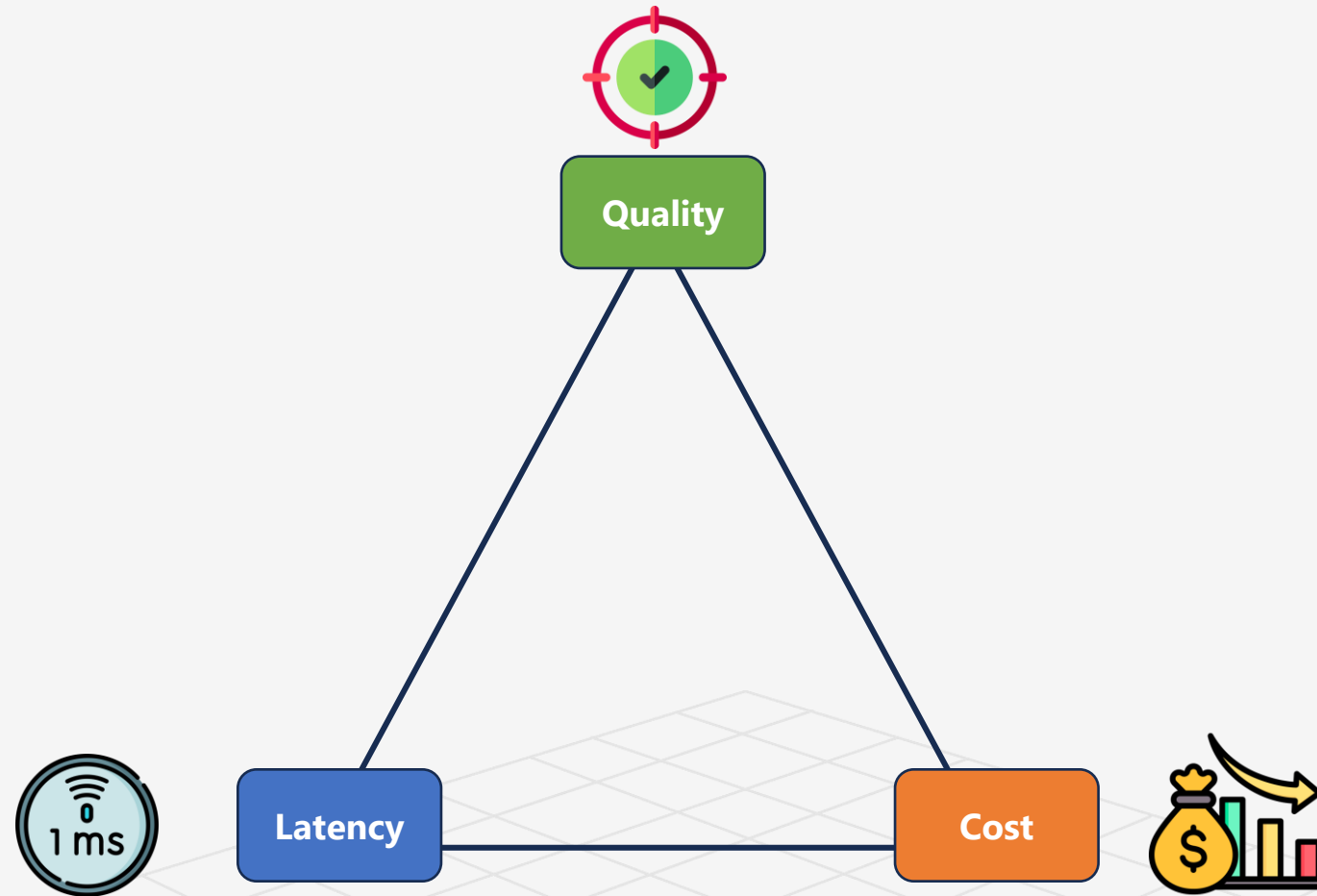
Business Use Case : Large Manufacturer wants to build a **GenAI-Powered Equipment SME Assistant** to reduce Wind Turbine Downtime.



Business Drivers

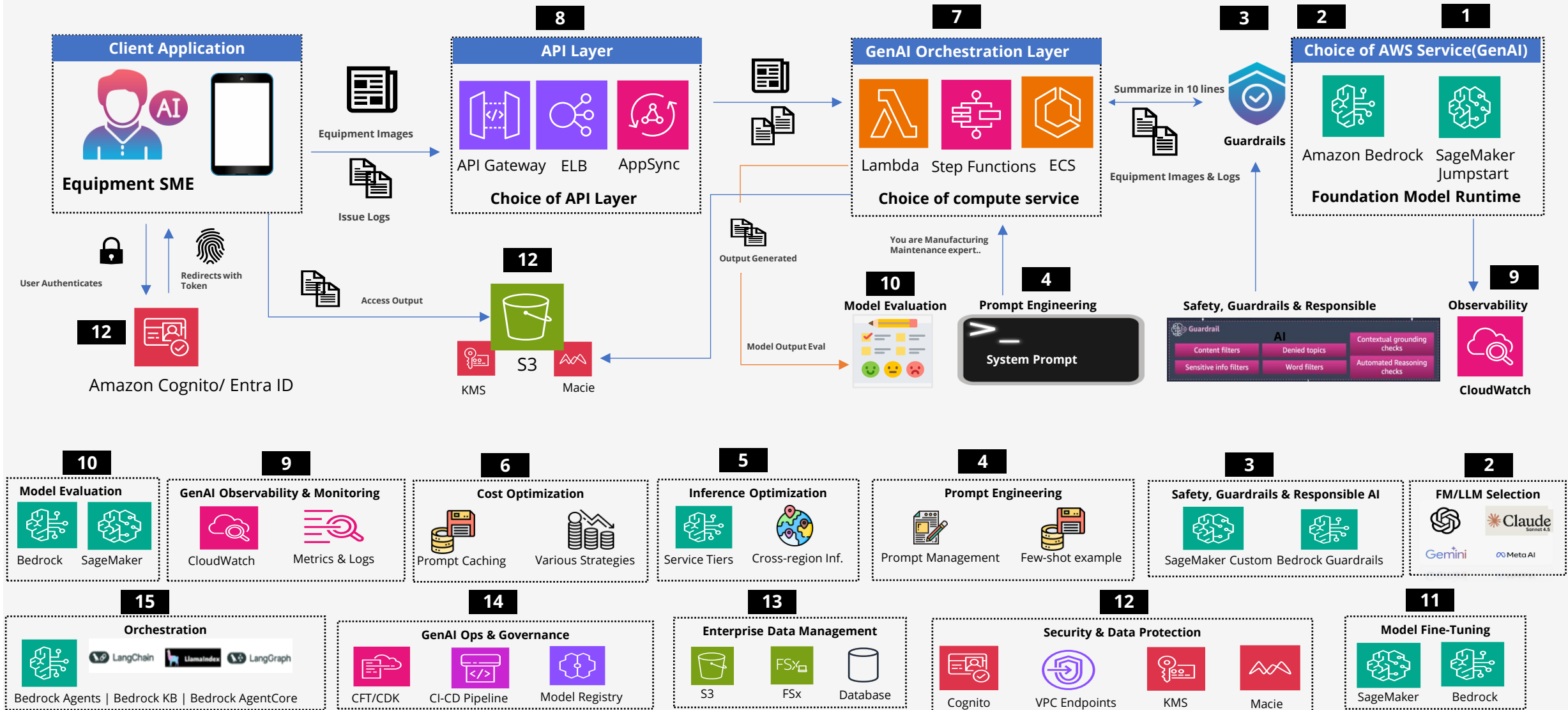
- Reduce wind turbine downtime
- Improve the productivity of Equipment SME

The GenAI Trilemma: Balancing Quality, Latency & Cost



Production GenAI architecture is defined by trade-offs between quality, latency and cost.

15 Key Architecture Considerations for building Production-Grade AWS Generative AI Applications



Decision Point 1:
Select right Generative AI Service

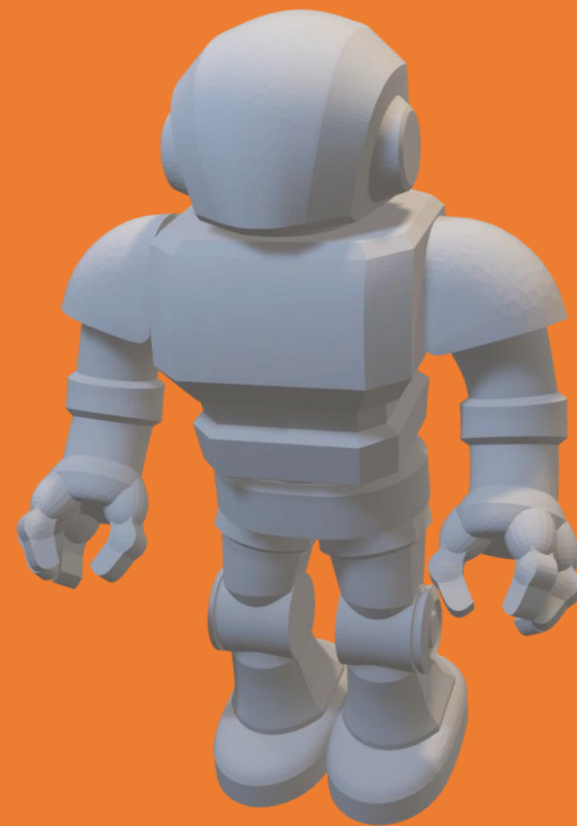


Select right AWS Generative AI Service

Option 1 - Amazon Bedrock

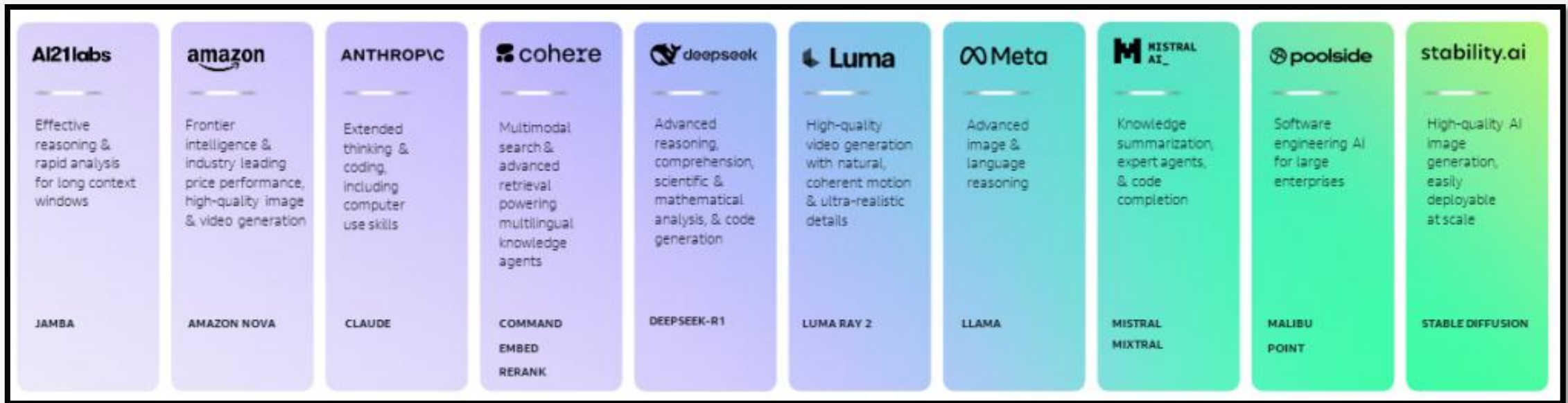


Overview of Amazon Bedrock

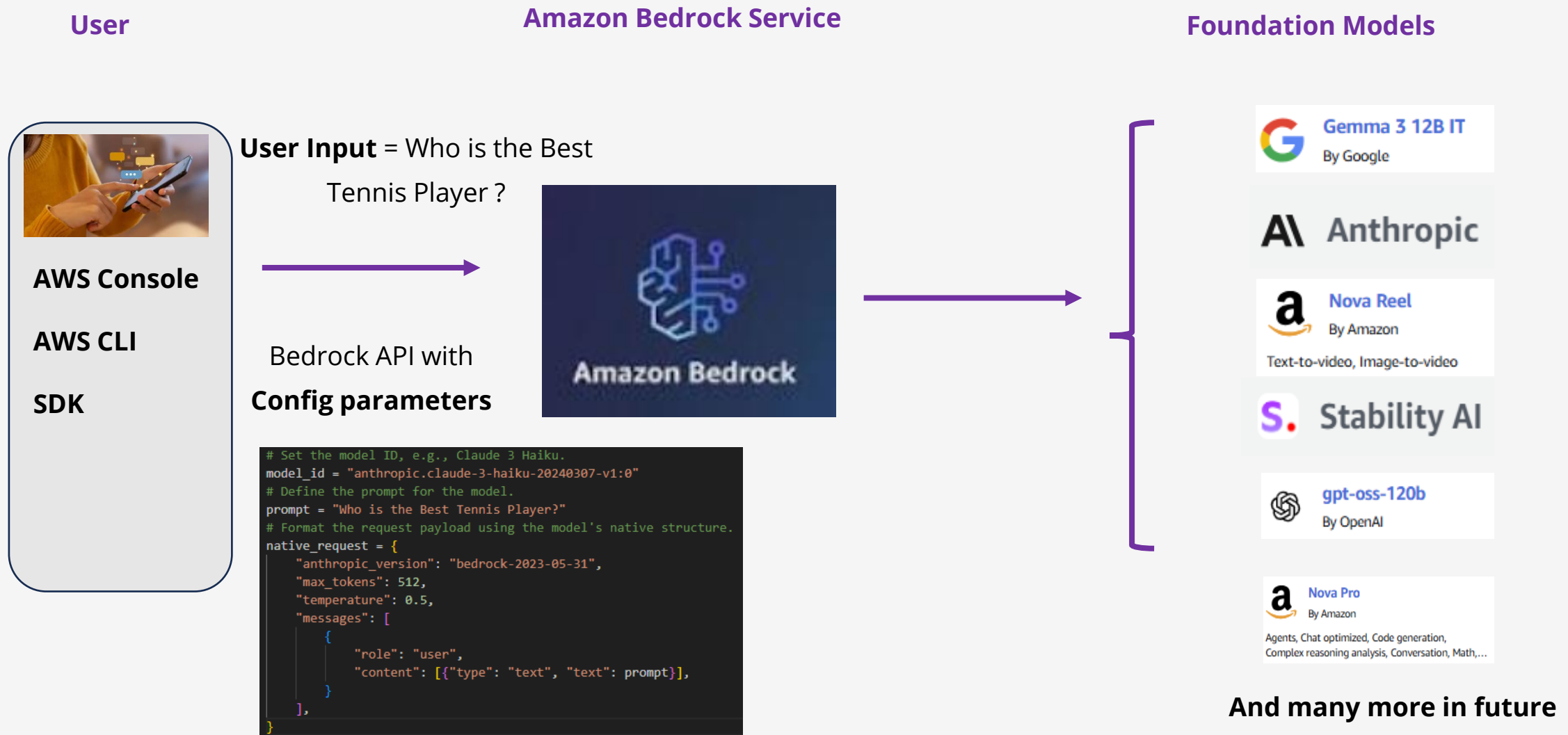


What is Amazon Bedrock ?

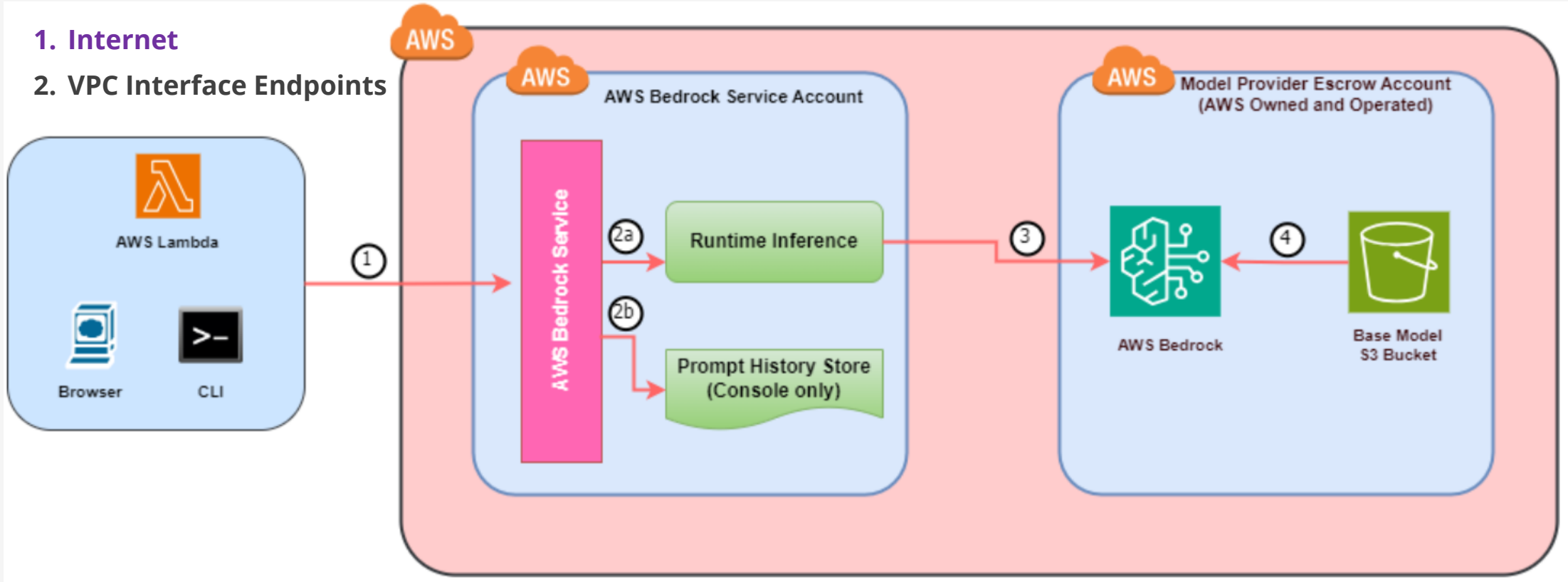
- Amazon Bedrock makes available **base Foundation Models from Amazon & third-party model providers** accessible through **an API**.
- It is a **fully managed, serverless service** from AWS.



How does Amazon Bedrock work – High Level ?



Amazon Bedrock – Architecture



- **Runtime Inference** : Used to **redirect to right model endpoint** based on the API request
- **Base Model : Baseline model** provided to every AWS account

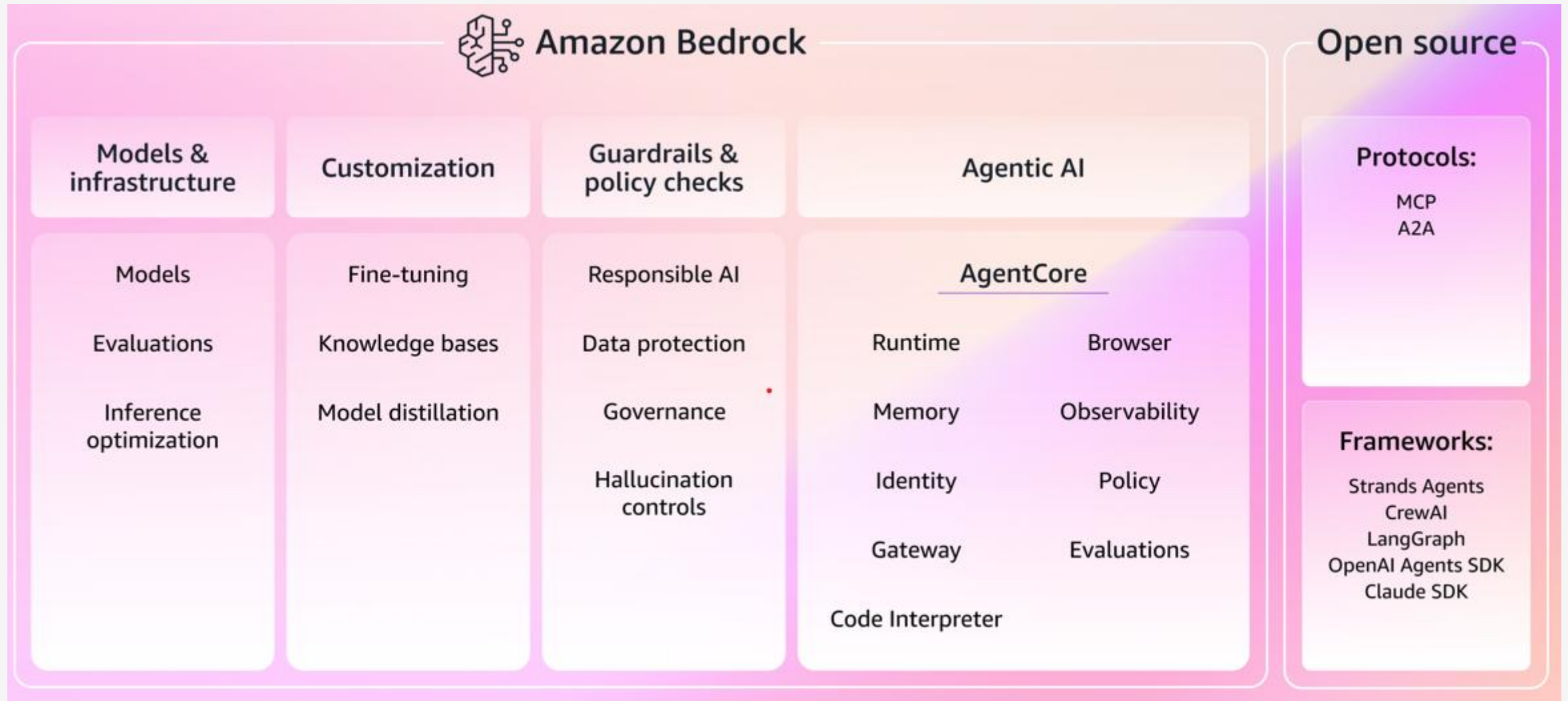
Services (2)

Search

bedrock X Clear filters

	Service Name	Type
☐	com.amazonaws.us-east-1.bedrock	Interface
☐	com.amazonaws.us-east-1.bedrock-runtime	Interface

Amazon Bedrock – Beyond the Model Hub



Amazon Bedrock – Console Walkthrough

The screenshot displays the Amazon Bedrock console interface. On the left is a navigation sidebar with sections: Amazon Bedrock, Discover (Overview, Model catalog, API keys), Test (Chat / Text playground, Image / Video playground, Watermark detection, Tokenizer), Infer (Cross-region inference, Batch inference, Provisioned Throughput, Custom model on-demand), Tune (Custom models, Prompt router models, Imported models, Marketplace model deployments), and Build (Agents, Flows, Knowledge Bases). The main content area is titled 'Overview' and includes: a 'Model catalog' section with a 'View Model catalog' link; a 'Model spotlight' for 'Mistral Large 3' with an 'Open in chat playground' link; a 'Get started by using API Keys' section with a 'View API keys' link and a code snippet; and a 'Test' section with three sub-playgrounds: 'Chat / Text playground', 'Image / Video playground', and 'Watermark detection', each with a corresponding 'Open playground' link. The footer contains links for CloudShell, Feedback, Console Mobile App, and copyright information for Amazon Web Services, Inc. or its affiliates, along with Privacy, Terms, and Cookie preferences links.

Amazon Bedrock

Amazon Bedrock

- Discover
 - Overview
 - Model catalog
 - API keys
- Test
 - Chat / Text playground
 - Image / Video playground
 - Watermark detection
 - Tokenizer [New](#)
- Infer
 - Cross-region inference
 - Batch inference
 - Provisioned Throughput
 - Custom model on-demand
- Tune
 - Custom models
 - Prompt router models
 - Imported models
 - Marketplace model deployments
- Build
 - Agents
 - Flows
 - Knowledge Bases

Overview [Info](#)

Model catalog

Amazon Bedrock supports over 100 foundation models from industry-leading providers and emerging leaders. Select a serverless model or Bedrock Marketplace model that is best suited for achieving your unique goals.

[View Model catalog](#)

Model spotlight

Mistral Large 3

Mistral AI is a small creative team with high scientific standards. We make efficient, helpful and trustworthy AI models through ground-breaking innovations.

[Open in chat playground](#)

Get started by using API Keys

Create an Amazon Bedrock API key in one click using the console and use it to authenticate API requests to Amazon Bedrock. After you create the key, you can integrate it into your setup to be automatically recognized when making requests.

[View API keys](#)

```
1 import { UserManager } from 'oidc-client'
2 const authConfig = {
3   authority: 'user_pool_arn',
4   client_id: 'client_id',
5   redirect_uri: 'redirect_uri',
6   scope: 'list of scopes',
7   extraQueryParams: {
```

Test

Chat / Text playground

Experiment with Bedrock models to generate high-quality text for a variety of language processing tasks. Choose between single-prompt generation or chat-style interactions that incorporate context from previous prompts.

[Open playground](#)

Image / Video playground

Use Bedrock models to create and edit high-quality images or generate videos using simple text prompts. For videos, choose between single-shot generation or try the storyboarding capability for more control over visual outputs.

Watermark detection

Verify whether an image was generated by a Titan Image Generator model by checking for the presence of an embedded watermark.

[View watermark detection](#)

CloudShell Feedback Console Mobile App

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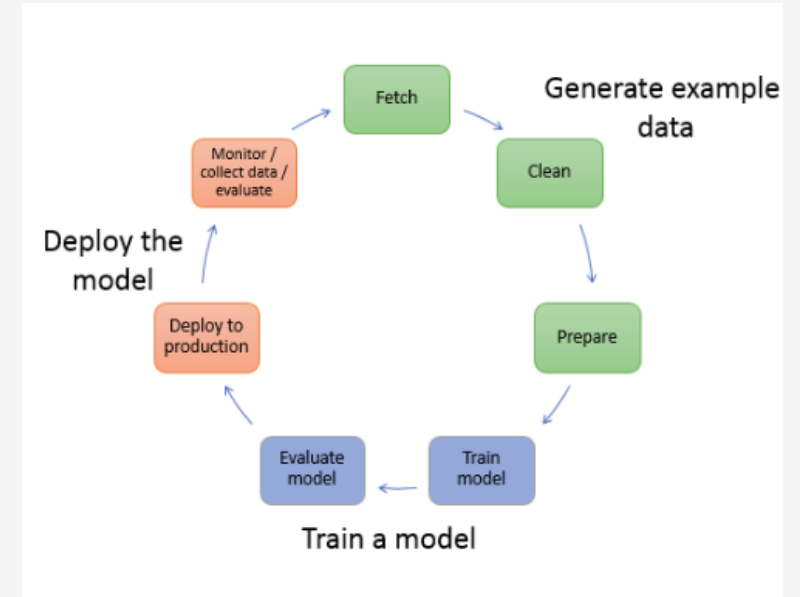
Select right AWS Generative AI Service

Option 2 - Amazon SageMaker JumpStart



Amazon SageMaker Jumpstart (part of Amazon SageMaker AI)

- Amazon SageMaker AI is **an AI/ML service to help you build, train and deploy Machine Learning Models.**
- Example – Gmail Spam Filter, Forecasting
- Unified platform for data, analytics, and AI.
- Amazon SageMaker AI capabilities will be discussed in the **Data and Responsible AI sections** later.
- **Amazon SageMaker Jumpstart** – Allows **deploy, fine-tune, and evaluate foundation** models.



Amazon SageMaker Jumpstart

1. AWS Service Description

AWS Sagemaker Jumpstart (Part of SageMaker AI focusing on GenAI)

- Allows access to different models in area of :
 - **Foundation Models** (Open source and proprietary models)
 - Computer Vision (image and video) & NLP Models (language)
 - Allows **fine-tune foundation models** (using SageMaker training jobs)
 - **Foundation Model Evaluation** - integrations with SageMaker Clarify Foundation Model Evaluations

2. Use Cases

- **Text Summarization, Code Generation** and any other Gen AI use case
- Allows you to **provision infrastructure to deploy and fine tune foundation models**
 - Open-source models and Proprietary models
- **Users of this service** – Machine Learning Engineers, Data Scientists, Developers

Amazon SageMaker AI

SageMaker Jumpstart

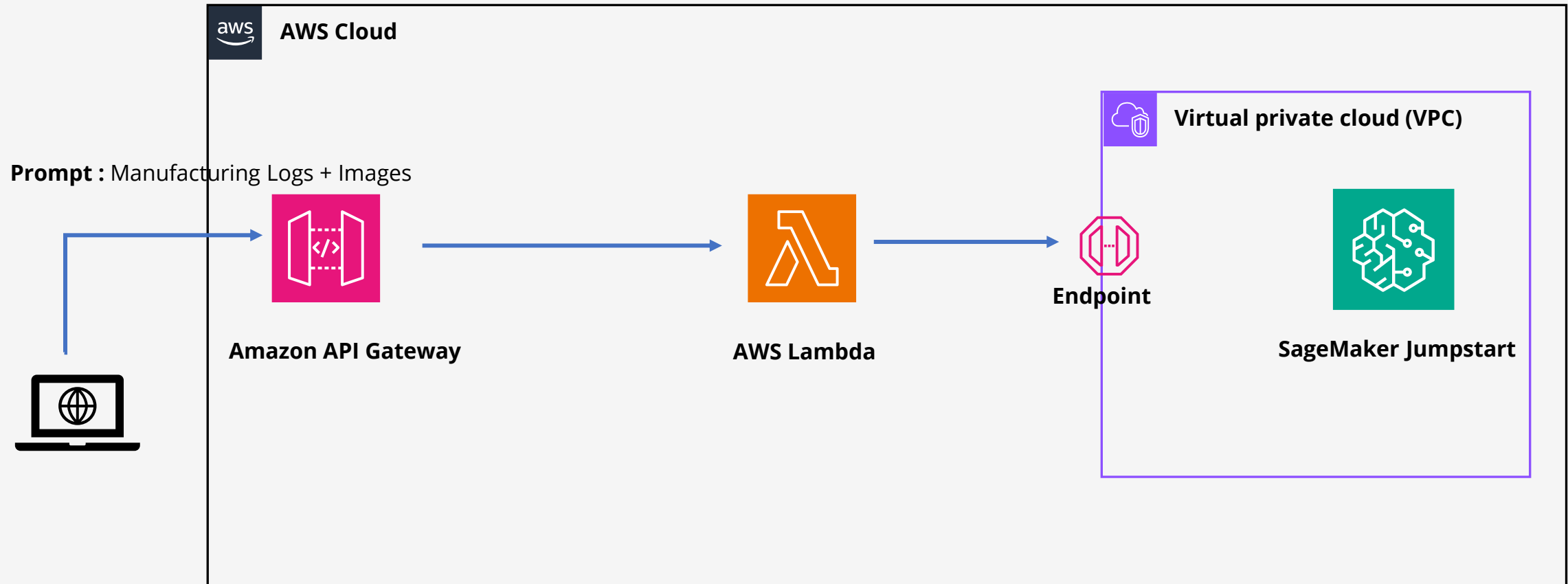
(GenAI focused)

Discover foundation models and deploy with SageMaker's enterprise-ready features

AVAILABLE ON SAGEMAKER JUMPSTART

AI21labs	Meta AI	cohere	Hugging Face	stability.ai	LightOn	databricks	alexa
Models Jurassic2 Ultra, Mid Contextual answers Summarize Paraphrase Grammatical error correction	Models Llama 2 7B, 13B, 70B Code Llama 7B, 13B, 34B Open LLaMA	Models Command Cohere Light	Models Falcon-7B, 40B, 180B, Mistral 7B RedPajama MPT-7B Bloom2 1.76B Flan T5 DallE GPT2 GPT NeoXT Bloom	Models Stable Diffusion XL 1.0 2.1 base Upscaling Inpainting	Models Lynx-Pr 10B, Mini	Models Dolly	Models AlexaTM 20B
Tasks Text generation Long-form generation Summarization Paraphrasing Chat Information extraction	Tasks Question answering Chat Summarization Paraphrasing Sentiment analysis Text generation	Tasks Text generation Information extraction Question answering Summarization	Tasks Text generation Machine translation Question answering Summarization	Tasks Generate photo-realistic images from text input Improve quality of generated images	Tasks Text generation Keyword extraction Information extraction Question answering Summarization Sentiment analysis Classification	Tasks Question answering Chat Summarization Paraphrasing Sentiment analysis Text generation	Tasks Machine translation Question answering Summarization Annotation Data generation

Amazon Sagemaker Jumpstart Deployment



Standard Instances	vCPU	Memory	Price per Hour
ml.t3.large	2	8 GiB	\$0.10

Amazon SageMaker Jumpstart

3. Architecture, Security & Pricing

Architecture & Security

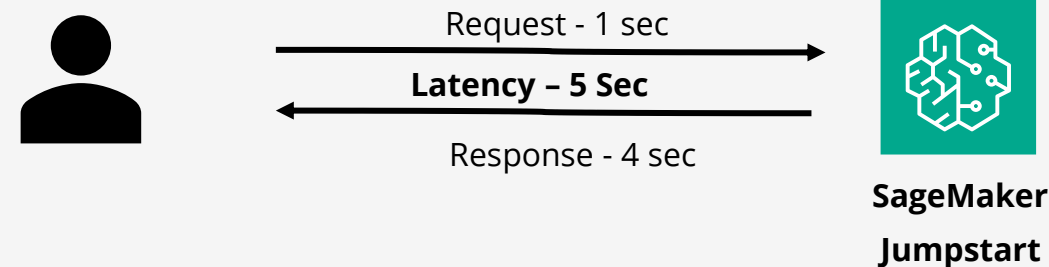
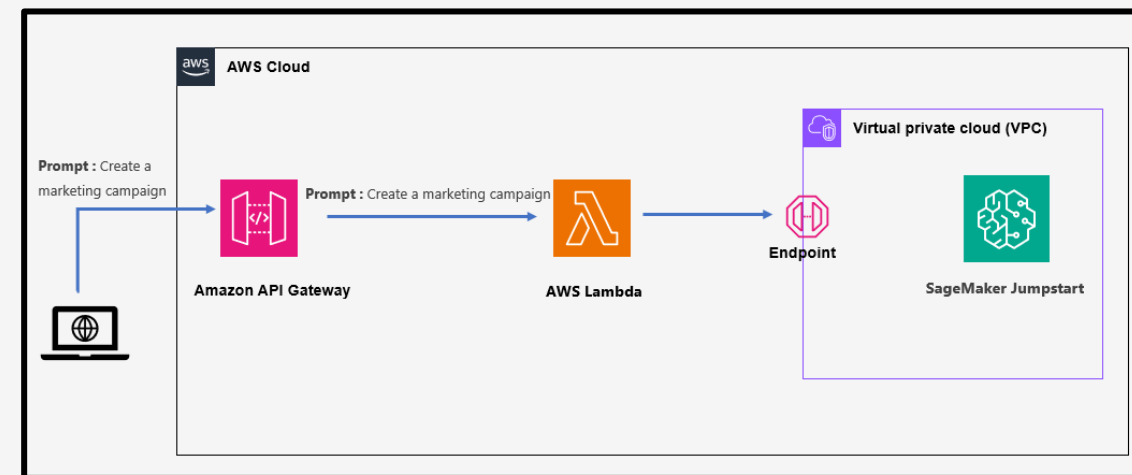
- **Deployed in Customer VPC**
- **Full control (with VPC and IAM)** over who has access to :
 - Prompt
 - Model Inference & Training Data

Pricing

- Based on Provisioning Capacity – [Link](#)

4. Performance

- Deploy latency and throughput sensitive workloads (performance controlled through instance sizing and auto scaling)
 - Inference latency
 - Throughput
- Enables high-performance inference with predictable cost for steady workloads



Amazon SageMaker Jumpstart

5. Scalability, Availability & Monitoring

- **AWS Managed services.**
- **AWS Manages**
 - Scalability - Autoscaling
 - Availability – Multi AZ
 - Monitoring – CloudWatch
- **Customer needs to provision the resources.**



Foundational Models available on SageMaker JumpStart

The screenshot displays the SageMaker JumpStart console interface. On the left is a navigation sidebar with categories: Domains (Images, Lifecycle configurations, Role manager), Applications and IDEs (SageMaker Studio, Canvas, RStudio, Notebooks, TensorBoard, Profiler, Partner AI Apps), Model training & customization (Training & tuning jobs, JumpStart model hub, Nova Forge, HyperPod clusters), Training plans, Deployments & inference (Deployable models, Endpoints, Endpoint configurations, Batch transform jobs, Shadow tests, Compilation jobs, Inference Recommender), Model governance (Model dashboard, Model cards), and AWS Marketplace resources (Marketplace model packages, All products).

The main content area is titled 'Foundation Models' and includes tabs for 'Foundation Models', 'Computer Vision Models', and 'Natural Language Processing Models'. Below the tabs, a search bar is present with the placeholder text 'Search for a model'. The models are displayed in a grid of cards, each with a logo, name, version, description, and a 'View model' button.

Model Name	Provider	Version	Description
Stable Diffusion XL 1.0	stability.ai	By Stability AI Ver 20230726	PROFESSIONAL: COMPARED TO PREVIOUS VERSIONS, SDXL 1.0 GENERATES MORE VIBRANT AND ACCURATE... The official foundation model for image generation from Stability. Deploy this optimized instance and serve generative AI within minutes.
Meta Llama 2 7B Chat	Meta	By Meta Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 7B dialogue use case optimized variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range...
Meta Llama 2 70B Chat	Meta	By Meta Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 70B dialogue use case optimized variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range...
AI21 Jurassic-2 Ultra	AI21 Labs	By AI21 Labs	RECOGNIZED AMONG STANFORD'S TOP-TIER LLM EVALUATIONS, JURASSIC-2 ULTRA ALLOWS USERS TO... Top-of-the-line LLM for complex text generation tasks that require the highest quality output.
LightOn Mini-instruct 40B	LightOn	By LightOn Ver v1.0	TEXT GENERATION, KEYWORD EXTRACTION, INFORMATION EXTRACTION, QUESTION ANSWERING,... 40B multilingual and instruction-tuned language model.
Meta Llama 2 7B	Meta	By Meta Ver 1.0.0	TEXT GENERATION, LLAMA 2, FINE-TUNABLE 7B variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range of parameter...
Meta Llama 2 13B	Meta	By Meta Ver 1.0.0	
Meta Llama 2 13B Chat	Meta	By Meta Ver 1.0.0	
Meta Llama 2 70B	Meta	By Meta Ver 1.0.0	

Amazon Bedrock vs Amazon SageMaker Jumpstart

Category	Amazon Bedrock 	Amazon SageMaker Jumpstart 
What it is	Managed GenAI service	- Amazon SageMaker AI - End-to-end AI/ML platform - GenAI capabilities via SageMaker Jumpstart
Primary use case	Use foundation models via API	Deploy and host foundation models in your VPC.
Infrastructure	Fully serverless - don't have to manage or provision infrastructure.	AWS Managed service but customers provisions compute capacity
Pricing	Pay-as-you-go pricing based on the number of API requests (+ tokens processed)	Per hour, based on the provisioning of compute, resources, storage.
Target Users	App and GenAI Developers	Data scientists, machine learning engineers & developers
Typical Workloads	Chatbots, RAG, Agents using Foundation Models	- Traditional ML Model - Fine-tuning of base foundation models.

Decision Point 2 :
Selection of Foundation Model

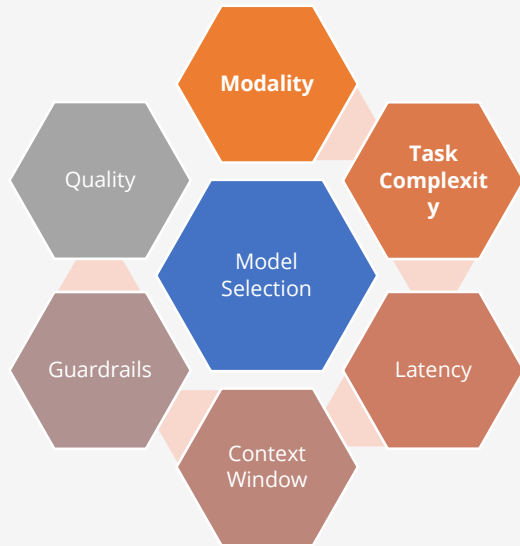


Selection of Foundation Model

1. Factors Impacting Choice of Foundation Models



Foundation Model Selection Strategy



1. **Modality** : Text | Image | Multi-Modal | Embedding – Manufacturing use case - multimodal
2. **Task Complexity** : Summarization vs reasoning ---> Small Language Model (SLM) vs LLM trade-offs
3. **Context Window** : 8K - 200K token size – Size of the user input (legal docs) --> Long context models
4. **Guardrails & Safety** : Built in model guardrails & additional capabilities – Toxicity, content moderation
5. **Quality & Accuracy of output** : Accuracy, reasoning, completeness ---> Bedrock model output evaluation
6. **Cost Predictability & Token Efficiency** : Cost optimization – cost per 1000 tokens
7. **Regional Availability, Quotas** – Service quotas & rate limits, region availability
8. **Latency** : Response time for requirements ---> Latency-optimized inference models
9. **Proprietary or Open Source** : Prefer open source
10. **Language Support** – Available in the particular language for Use Case

Select the smallest, safest & most cost-efficient model that meets accuracy, latency & governance requirements for the workload.

Complexity of Foundation Model for your application

Complexity of Foundation Models

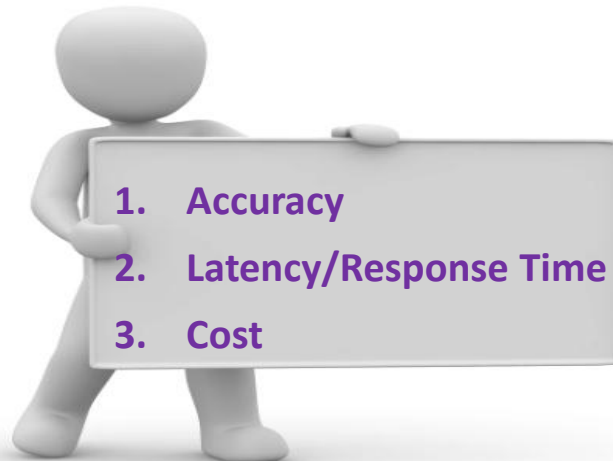
- Large Language Models (LLM) - 8B to 100's of Billions of parameters
- Small Language Model (SLM) – Less than 8B parameters

More number of Foundation Model Parameters

Complex Response + Higher response time + Costlier



Choice of Foundation Model for a Use Case



Lesser number of Foundation Model Parameters ==

Low complexity Response + Lower response time + Cheaper

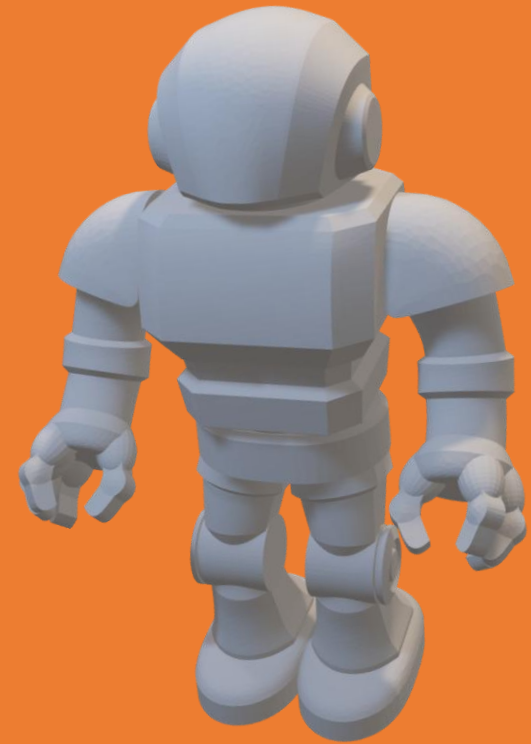


Selection of Foundation Model

2. How Inference Parameters Influence Foundation Model Behavior



How Inference Parameters Influence Foundation Model Behavior



Impact of Inference Parameters on Model Response

Inference parameters influence the response generated by the model.

Inference Parameters

1. Randomness and diversity

Temperature

Top K

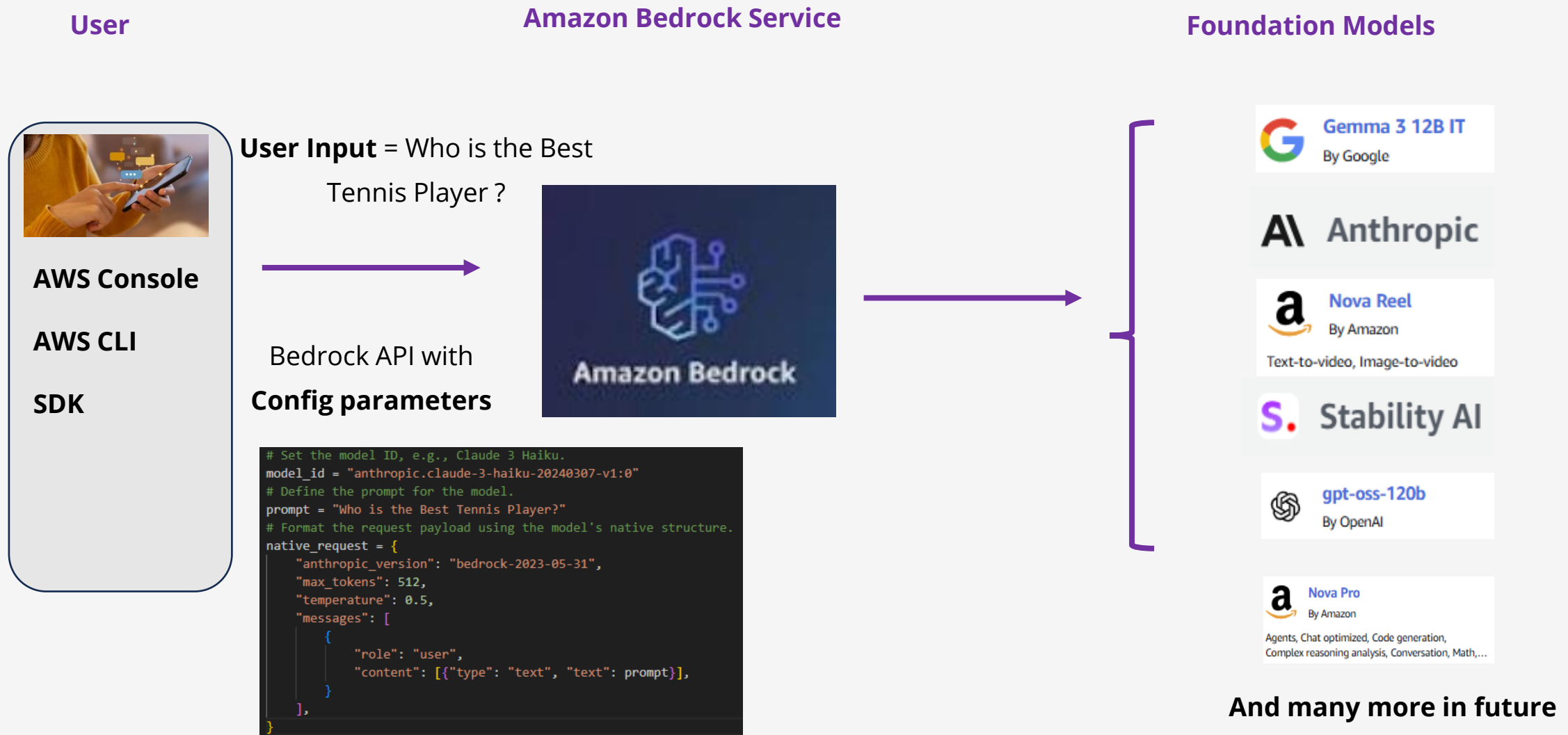
Top P

2. Length

Max Tokens
(max_gen_len)

Stop Sequence

How does Amazon Bedrock work – High Level ?



Inference Parameter – 1. Randomness and Diversity

Prompt : I hear the hoof beats of

Parameter	Possible Values	Description	Example
Temperature*	Foundation Models use : Probability to construct the words in a sequence		• Horse --- > 0.4 • Wind ----- > 0.2 • Unicorns -- - > 0.1 • Change in distance --- > 0.05
	Low value (closer to zero)	Model tends to select the higher-probability words .	• Horse
	High value (closer to 1)	Temperature further away from zero, the model may select a lower-probability word	• Change in distance
Top K	1-500	Cutoff where the model no longer selects the words .	if K=50, the model selects from 50 of the most probable words
Top P	0.01-0.99	Caps choices based on the sum of their probabilities .	If Top P=0.6, Horse and Wind selected

Inference Parameter – 2. Length

Prompt : Write an essay on horse

Parameter	Possible Values	Description
Length : Controls the length of the generated response . Helps control cost .		
Max Length	1- 64000 tokens (varies per model)	Specify the maximum number of tokens to use in the generated response.
Stop Sequence	Any keyword. Example : • '}' • "</answer>"	• Model-decided stop (end-of-sequence token) – Model training • System-enforced stop (stop sequences or max tokens) • After a stop sequence, the model stops generating further tokens. • The returned text doesn't contain the stop sequence. • {"firstName":"John", "lastName":"Doe"}

Decision Point 3:
Amazon Bedrock Guardrails &
Automated Reasoning

(Responsible AI will be discussed in detail in later section)

Amazon Bedrock Guardrails & Automated Reasoning

Challenges with Foundation Models & Responsible AI



Hallucinations – Foundation Model generates **factually incorrect** or **fabricated response**



Toxicity : Generation of content that is **offensive, discriminatory or harmful**



Privacy and Security : Model “leaks” private or confidential data (PII data).

Amazon Bedrock Guardrails



(Responsible AI will be discussed in detail in later section)

Guardrails for Amazon Bedrock

- Implements **safeguards customized to your application requirements** and **responsible AI policies**
- Guardrails for Amazon Bedrock provides safeguards on top of the native protections of FMs



Content filters - Detect **and filter harmful text or image content** in input prompts or model responses.



Denied Topics - Block undesirable topics in your generative AI applications



Word Filters - Block inappropriate content with a custom word filter



Sensitive information filters - Redact sensitive information (PII) to protect privacy



Contextual grounding checks – Detect and filter hallucinations in model responses based on grounding in a source and relevance to the user query.

How to use Bedrock Guardrails in GenAI apps

Request Syntax

```
response = client.invoke_model(  
    body=b'bytes'|file,  
    contentType='string',  
    accept='string',  
    modelId='string',  
    trace='ENABLED'|'DISABLED'|'ENABLED_FULL',  
    guardrailIdentifier='string',  
    guardrailVersion='string',  
    performanceConfigLatency='standard'|'optimized',  
    serviceTier='priority'|'default'|'flex'|'reserved'  
)
```

How to use Bedrock Guardrails in GenAI apps

```
response = client.create_guardrail(  
    name='string',  
    description='string',  
    topicPolicyConfig={  
        'topicsConfig': [  
            {  
                'name': 'string',  
                'definition': 'string',  
                'examples': [  
                    'string',  
                ],  
                'type': 'DENY',  
                'inputAction': 'BLOCK'|'NONE',  
                'outputAction': 'BLOCK'|'NONE',  
                'inputEnabled': True|False,  
                'outputEnabled': True|False  
            },  
        ],  
        'tierConfig': {  
            'tierName': 'CLASSIC'|'STANDARD'  
        }  
    },  
    contentPolicyConfig={  
        'filtersConfig': [  
            {  
                'type': 'SEXUAL'|'VIOLENCE'|'HATE'|'INSULTS'|'MISCONDUCT'|'PROMPT_ATTACK',  
                'inputStrength': 'NONE'|'LOW'|'MEDIUM'|'HIGH',  
                'outputStrength': 'NONE'|'LOW'|'MEDIUM'|'HIGH',  
                'inputModalities': [  
                    'TEXT'|'IMAGE',  
                ],  
                'outputModalities': [  
                    'TEXT'|'IMAGE',  
                ],  
                'inputAction': 'BLOCK'|'NONE',  
                'outputAction': 'BLOCK'|'NONE',  
                'inputEnabled': True|False,  
                'outputEnabled': True|False  
            },  
        ],  
        'tierConfig': {  
            'tierName': 'CLASSIC'|'STANDARD'  
        }  
    }  
)
```

https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/bedrock/client/create_guardrail.html

Amazon Bedrock Automated Reasoning

(Responsible AI will be discussed in detail in later section)

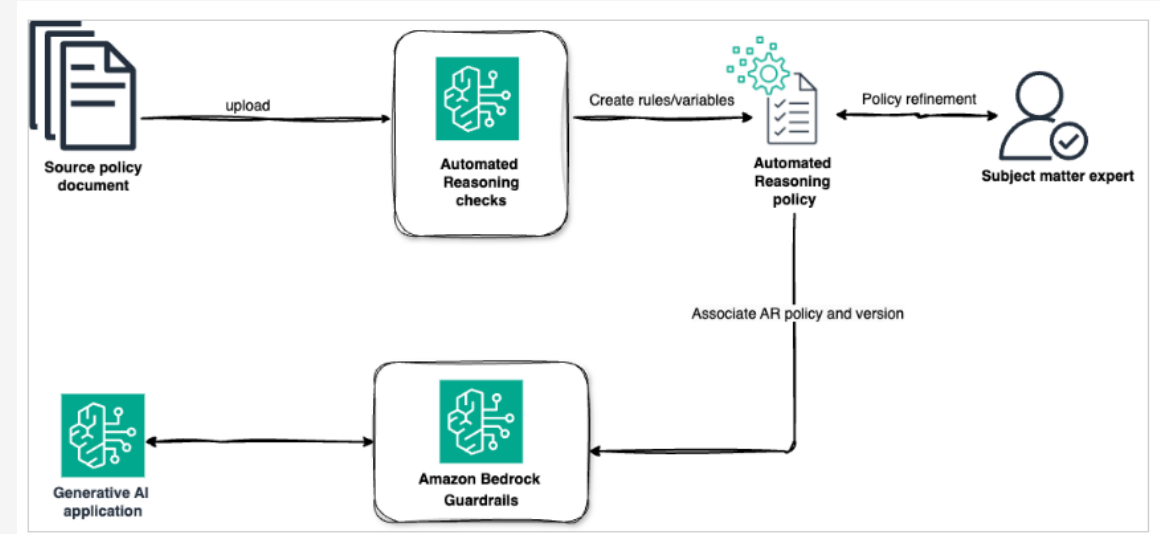
Amazon Bedrock Automated Reasoning

- Amazon Bedrock Automated Reasoning checks helps improve **factual accuracy of large language model (LLM)** responses.
- Automated reasoning checks is part of **Amazon Bedrock Guardrails**.
- Through logic-based algorithms and mathematical validation, Automated Reasoning Checks **validate LLM outputs against domain knowledge encoded in the Automated Reasoning policy** to help prevent factual inaccuracies.

Homeowners Insurance Policy Document

Retrieve Rules

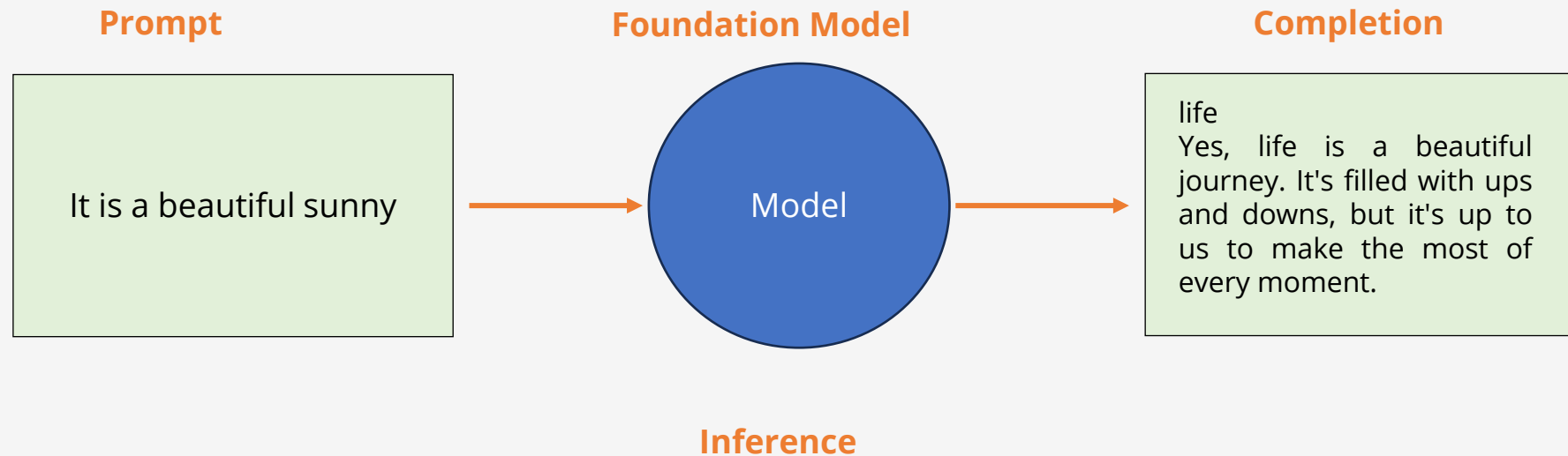
- Policy ID, Driving License



Customer Question : Which documents are required to file home insurance claim ?

Decision Point 4 :
Prompt Engineering

What is a Prompt ?



- **Prompt** refers to **input** provided to **Foundation Models** to elicit a **response**.
- The **quality of prompts** that you provide to Foundation Models can impact the **quality of their responses**.
- **Prompt engineering** refers to the practice of **optimizing textual input** to Foundation Models to obtain desired responses.

Elements of a Good Prompt Deign

"Lionel Messi can finally be called a world champion.

Messi scored twice in one of the most epic soccer games anyone has ever watched as Argentina won the 2022 FIFA World Cup Final over France on penalties.

The climactic match in Qatar finished 3-3 after extra time, with La Albiceleste claiming the shootout by a 4-2 margin.

Argentina held a comfortable 2-0 lead until the 80th minute courtesy of a Messi penalty and a sublime team goal finished by Ángel Di María in the first half. However, Kylian Mbappé converted from the spot and finished a sumptuous volley in a span of two minutes to send the game to extra time.

Messi was once again on hand to put Argentina in front in the 108th minute, but Mbappé kept his cool from the penalty spot once more to send the final to a shootout.

Messi knocked home the first penalty, then Argentina keeper Emiliano Martínez made a save in the second round from Kingsley Coman's effort. France's Aurélien Tchouaméni sent his spot-kick wide in the third round, leaving Gonzalo Montiel to seal the title in the fourth."

Summarize :

Elements of a Good Prompt Design

This is a review of football world cup Qatar 2022 :

**Contextual Information
about the task**

"Lionel Messi can finally be called a world champion.
Messi scored twice in one of the most epic soccer games anyone has ever watched as Argentina won the 2022 FIFA World Cup Final over France on penalties.
The climactic match in Qatar finished 3-3 after extra time, with La Albiceleste claiming the shootout by a 4-2 margin.
Argentina held a comfortable 2-0 lead until the 80th minute courtesy of a Messi penalty and a sublime team goal finished by Ángel Di María in the first half. However, Kylian Mbappé converted from the spot and finished a sumptuous volley in a span of two minutes to send the game to extra time.
Messi was once again on hand to put Argentina in front in the 108th minute, but Mbappé kept his cool from the penalty spot once more to send the final to a shootout.
Messi knocked home the first penalty, then Argentina keeper Emiliano Martínez made a save in the second round from Kingsley Coman's effort. France's Aurélien Tchouaméni sent his spot-kick wide in the third round, leaving Gonzalo Montiel to seal the title in the fourth."

Input text for Task

Summarize the above review in 2 lines:

- 1. Task Specification – Summarize**
- 2. Output Specification – 2 lines**

1. Clear concise instructions 2. Task to be accomplished at the end 3. Output specification

Sample Prompt - 1

Example prompt with clear output constraints indicator

User prompt:

"Charles Mingus Jr. was an American jazz upright bassist, pianist, composer, bandleader, and author. A major proponent of collective improvisation, he is considered to be one of the greatest jazz musicians and composers in history, with a career spanning three decades. Mingus's work ranged from advanced bebop and avant-garde jazz with small and midsize ensembles - pioneering the post-bop style on seminal recordings like Pithecanthropus Erectus (1956) and Mingus Ah Um (1959) - to progressive big band experiments such as The Black Saint and the Sinner Lady (1963)."

*Please summarize the above text **in one phrase**.*

Output:

Charles Mingus Jr. is considered one of the greatest jazz musicians of all time.

Example without clear output specifications

User prompt:

"Charles Mingus Jr. was an American jazz upright bassist, pianist, composer, bandleader, and author. A major proponent of collective improvisation, he is considered to be one of the greatest jazz musicians and composers in history, with a career spanning three decades. Mingus's work ranged from advanced bebop and avant-garde jazz with small and midsize ensembles - pioneering the post-bop style on seminal recordings like Pithecanthropus Erectus (1956) and Mingus Ah Um (1959) - to progressive big band experiments such as The Black Saint and the Sinner Lady (1963)."

Please summarize the above text.

Output:

Charles Mingus Jr. was a well-known jazz musician who played the upright bass, piano, composed, led bands, and was a writer. He was considered one of the most important jazz musicians ever, with a career that spanned more than

Sample Prompt - 2

- Output format - XML tags.
- Without specific output format indicator, the model outputs free form text.

Example with clear indicator, with output

User prompt:

Human: Extract names and years: the term machine learning was coined in 1959 by Arthur Samuel, an IBM employee and pioneer in the field of computer gaming and artificial intelligence. The synonym self-teaching computers was also used in this time period.

Please generate answer in <name></name> and <year></year> tags.

Assistant:

Output:

<name>Arthur Samuel</name> <year>1959</year>

Example without clear indicator, with output

User prompt:

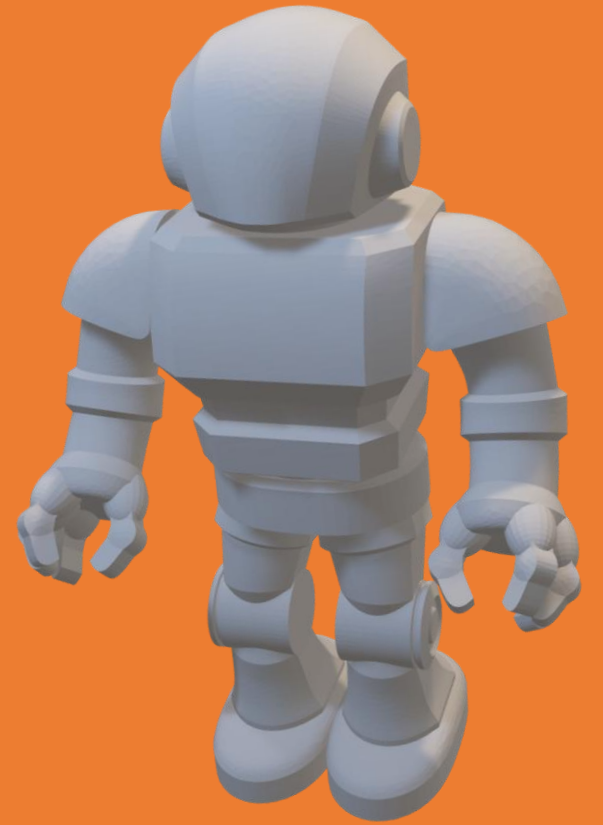
Human: Extract names and years: the term machine learning was coined in 1959 by Arthur Samuel, an IBM employee and pioneer in the field of computer gaming and artificial intelligence. The synonym self-teaching computers was also used in this time period.

Assistant:

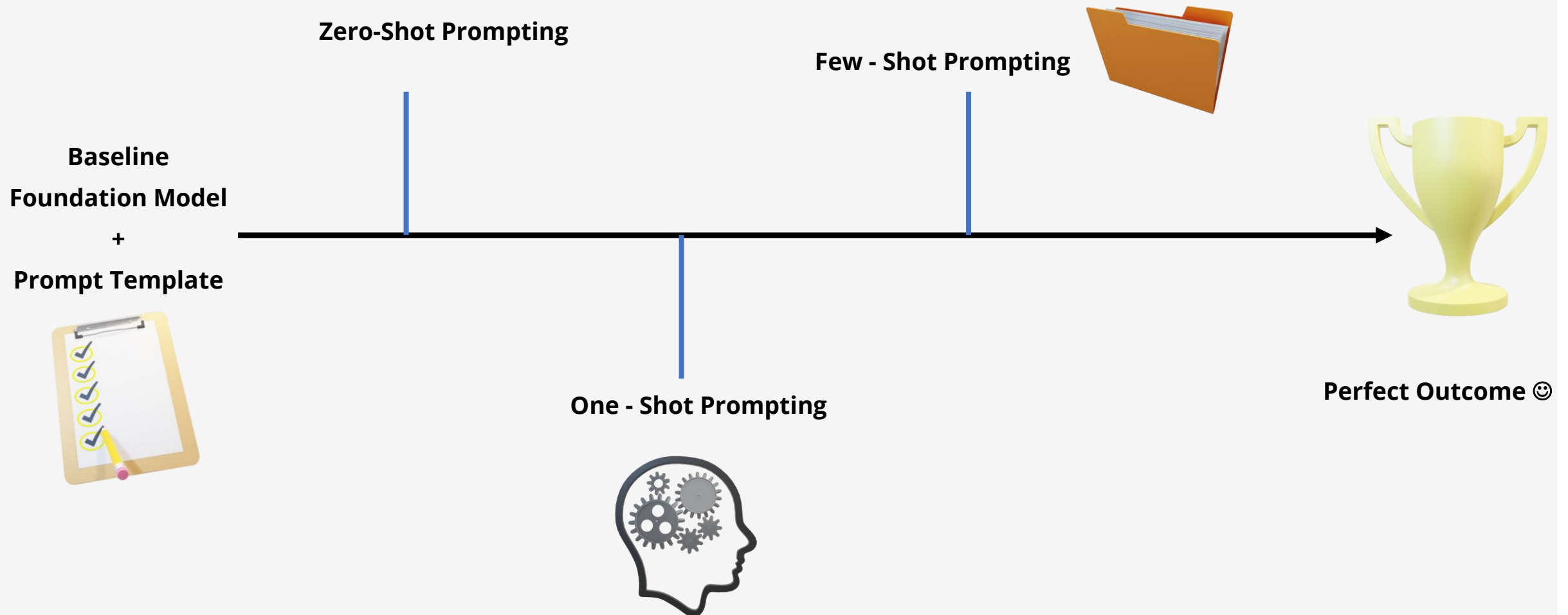
Output:

Arthur Samuel - 1959

Prompt Engineering Techniques



Prompt Engineering Techniques



Zero-Shot Prompting

Zero-Shot Prompting

- FMs are capable of **performing some simple tasks "zero-shot."**
- **No examples** needed to be provided to accomplish the task



PROMPT

Write an engaging product description for a T-shirt for a clothing eCommerce site:

Output

This T-shirt is perfect for any occasion. It's made of high-quality materials and it's comfortable to wear. It's also stylish and versatile, so you can wear it with anything.

One -Shot Prompting

One-Shot Prompting

- One-Shot Prompting provides **examples to the model** to assist the model to comprehend and generate responses

PROMPT

Write an engaging product description for a clothing eCommerce site.

Product: Women's Boho Beach Dress **Features:** - Midi dress - Swing hem - Slimming - Special for summer

Description: Women's Boho Beach Dress The style of midi dress with swing hem is slimming,

Write an engaging product description for a clothing eCommerce site.

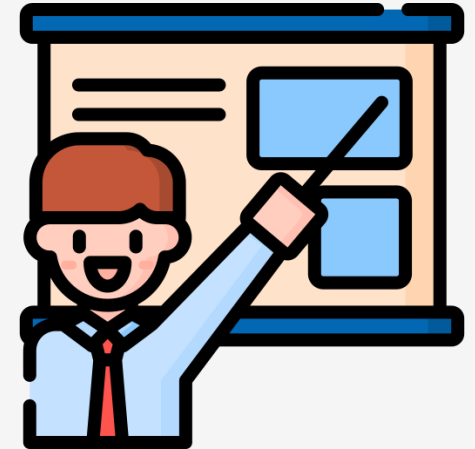
Make sure to include the following features in the description.

Product: Humor Men's Graphic T-Shirt **Features:** - Soft cotton - Short sleeve - Have a print of Einstein's quote: "artificial intelligence is

Description:

Output

This soft cotton t-shirt is perfect for fans of the classic scientist, Albert Einstein. The quote "artificial intelligence is no match for natural stupidity" is a reminder that sometimes the most intelligent people are the most foolish.....



Few - Shot Prompting

Few-Shot Prompting

- Few examples provided to the Model to generate high quality output

PROMPT

Write an engaging product description for a clothing eCommerce site. Make sure to include the following

Product: Camp Director Crossbody Bag

Features:

- Canvas purse
- Rainbow space print
- Leather trim
- Two securely-zipped compartments

Description:

This Camp Director Crossbody Bag is the perfect accessory for carrying your essentials in style. The canvas

##

Write an engaging product description for a clothing eCommerce site. Make sure to include the following

Product: Women's Boho Beach Dress

Features:

- Midi dress
- Swing hem
- Slimming
- Special for summer

Description:

Women's Boho Beach Dress The style of midi dress with swing hem is slimming, It will make your turn

##

Write an engaging product description for a clothing eCommerce site. Make sure to include the following

Product: Humor Men's Graphic T-Shirt

Features:

- Soft cotton
- Short sleeve
- Have a print of Einstein's quote: "artificial intelligence is no match for natural stupidity"

Description:



Output

This Humor Men's Graphic T-Shirt is perfect for anyone who wants to make a statement.

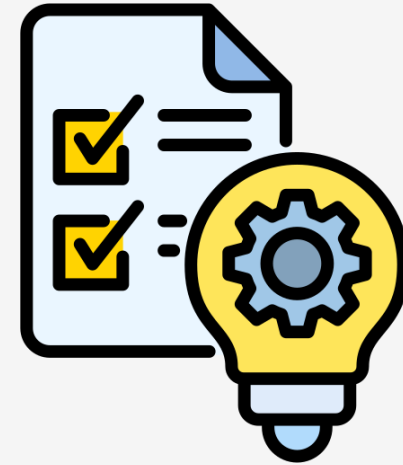
The soft cotton and short sleeve make it comfortable to wear, and the print of Einstein's quote is sure to get a chuckle.

Whether you're a fan of science or just looking for a funny shirt, this shirt is a great choice.

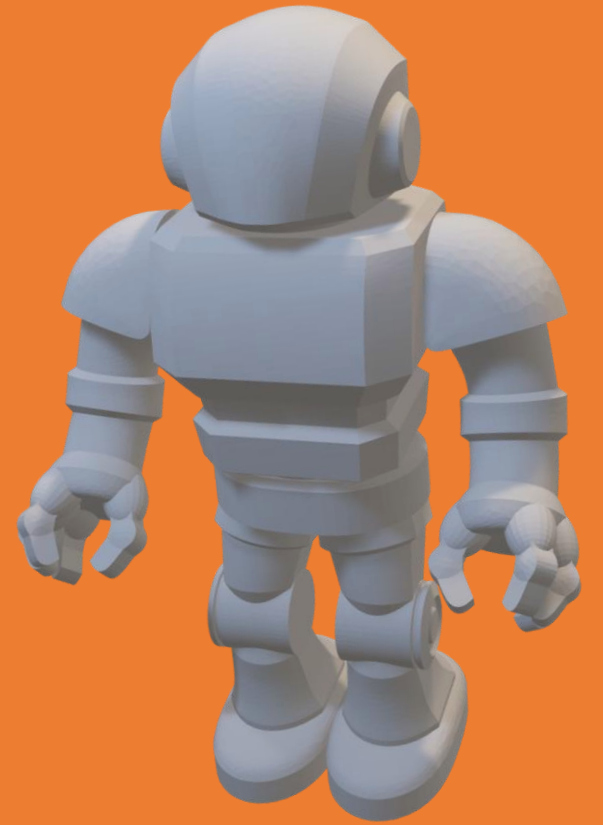
Additional Prompting Strategies

Additional Prompting Strategies – Will be discussed in Section under Agents

- Chain of Thought Reasoning
- ReACT – Reasoning and Action



Bedrock Prompt Management



Amazon Bedrock - Prompt Management

Amazon Bedrock Prompt Management simplifies the **creation, evaluation, versioning, and sharing of prompts** to get better responses from foundation models (FMs).

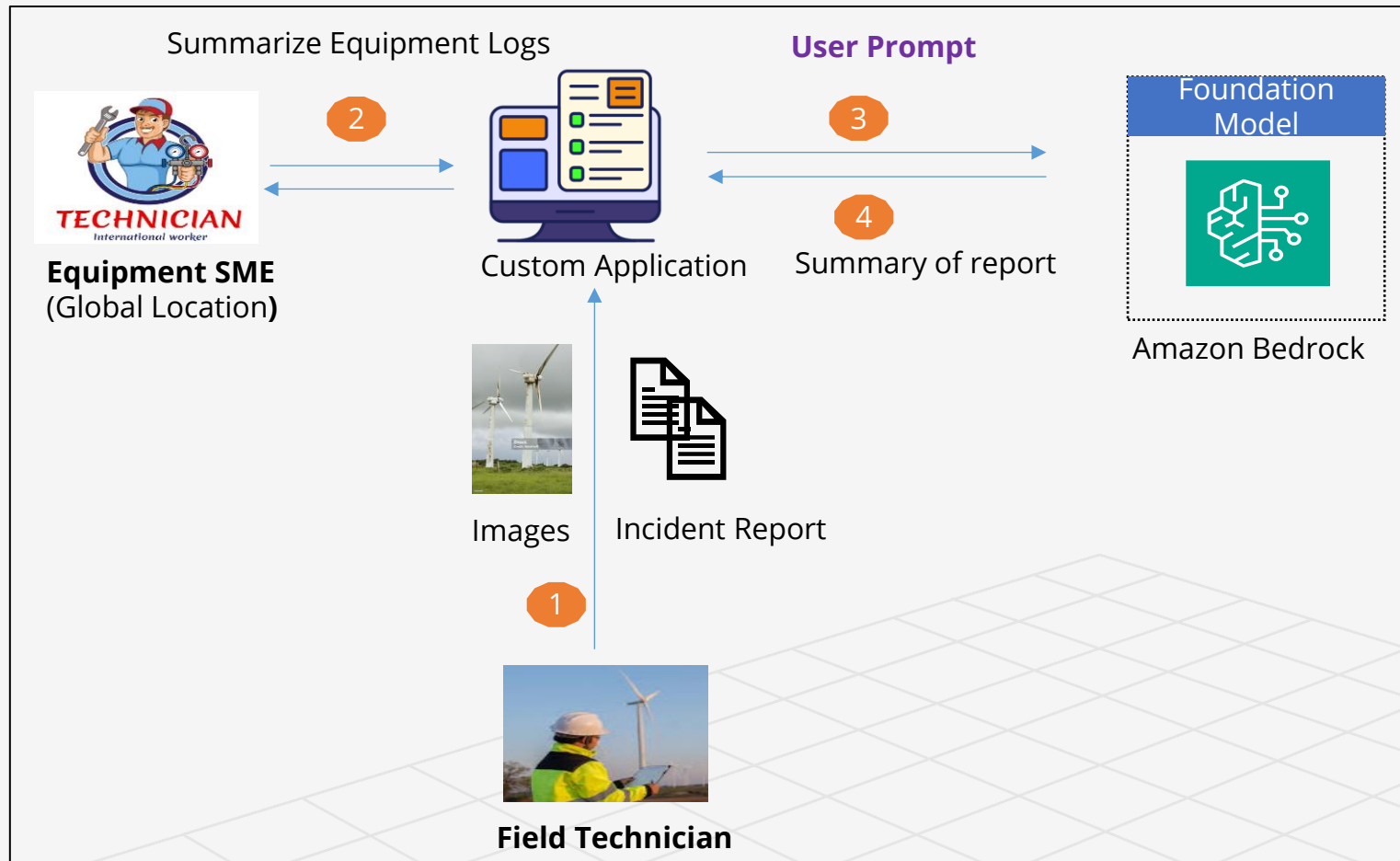
User Prompt : Equipment **Log Details**

System Prompt :

- You are a **Wind Turbine Subject Matter Expert (SME)** specializing in **equipment issue analysis and log-based troubleshooting**.
- Produce a **concise, one-page summary**
- Highlight **key anomalies** and failure patterns

Use Case 1 : GenAI-Powered Equipment SME Assistant

Business Use Case : Reduce Wind Turbine Downtime with a **GenAI-Powered Equipment SME Assistant**



Business Benefits

- Reduce wind turbine downtime
- Improve the productivity of the Equipment SME

Manufacturing Use Case - Prompt for Summarization Task

Log Details :

Issue Log Date – 25-12-2023, Model Number – TB-CL-7882, **Issue** - Cracks appeared in the part MR 7882-9571 next to the rotor hub. The nut connecting the rotor blade to the rotor hub seems to be damaged. The Anemometer readings seem to be within range. The electric braking seems to be unused. No indication of damage to any other component of the turbine except normal wear and tear.

Potential Root Cause – Seems due to reduced tensile strength of the nut connecting the blade to the rotor. Last Maintenance Date – 12-12-2023, **Last Maintenance Issues Recorded** - No known issues recorded and all the parameters were within range.

Manufacturing Use Case - Prompt for Summarization Task

System Prompt :

You are a **Wind Turbine Subject Matter Expert** (SME) specializing in equipment issue analysis and log-based troubleshooting.

- Produce a **concise, one-page summary**
- Highlight key anomalies and failure patterns

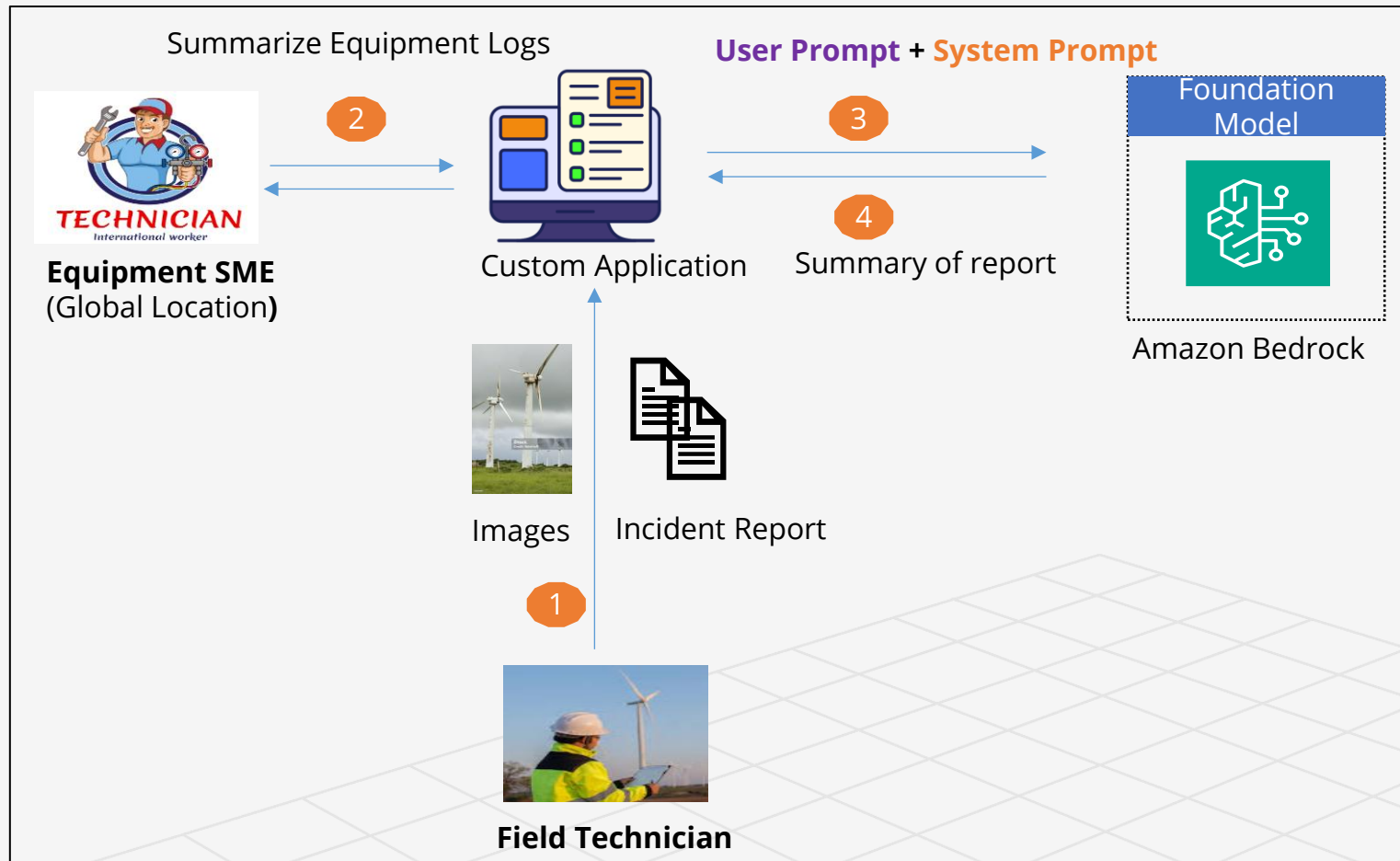
User Prompt :

Issue Log Date – 25-12-2023, Model Number – TB-CL-7882, **Issue** - Cracks appeared in the part MR 7882-9571 next to the rotor hub. The nut connecting the rotor blade to the rotor hub seems to be damaged. The Anemometer readings seem to be within range. The electric braking seems to be unused. No indication of damage to any other component of the turbine except normal wear and tear.

Potential Root Cause – Seems due to reduced tensile strength of the nut connecting the blade to the rotor. Last Maintenance Date – 12-12-2023, **Last Maintenance Issues Recorded** - No known issues recorded and all the parameters were within range.

Use Case 1 : GenAI-Powered Equipment SME Assistant

Business Use Case : Reduce Wind Turbine Downtime with a **GenAI-Powered Equipment SME Assistant**



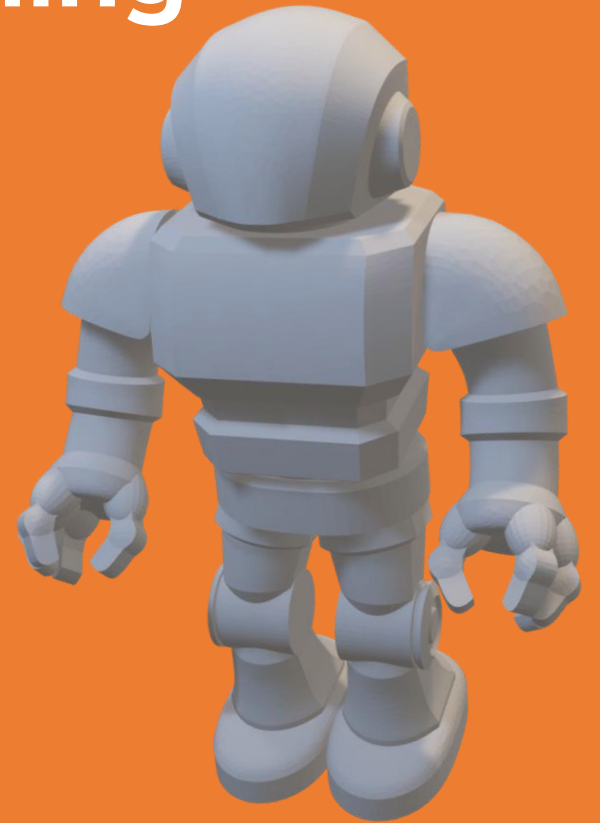
Business Benefits

- Reduce wind turbine downtime
- Improve the productivity of the Equipment SME

Architecture Decision Point 5 :
Bedrock Inference Optimization

Inference Optimization

1. Service Tiers & Capacity Planning



Service tiers for optimizing performance and cost

Amazon Bedrock offers four service tiers for model inference:

1. Reserved Tier

Best for: Mission-critical, always-on workloads

- **Reserve prioritized compute capacity** for your mission-critical applications
- Targets **99.5% uptime** for model response.
- Reserve capacity for **1 month or 3-month** duration.
- Flexibility to allocate different **input and output tokens-per-minute** capacities to match requirements
- Pay a fixed **price per 1K tokens-per-minute** and are billed monthly.



```
response = client.invoke_model_with_response_stream(  
    body=b'bytes'|file,  
    contentType='string',  
    accept='string',  
    modelId='string',  
    trace='ENABLED'|'DISABLED'|'ENABLED_FULL',  
    guardrailIdentifier='string',  
    guardrailVersion='string',  
    performanceConfigLatency='standard'|'optimized',  
    serviceTier='priority'|'default'|'flex'|'reserved'  
)
```

2. Priority Tier

Best for: Latency-sensitive, customer-facing applications that do not warrant 24X7 capacity reservation

- Delivers the **fastest response times for a price premium over standard** on-demand pricing.
- Set the "**service_tier**" optional parameter to "**priority**" to avail request level prioritization.



Service tiers for optimizing performance and cost

3. Standard Tier (Default)

Best for: General production workloads

- **Provides consistent performance for everyday tasks** - content generation, text analysis etc.
- By default, all inference requests are routed to the Standard tier

4. Flex Tier

Best for: Cost-optimized, non-urgent workloads

- Lowest cost inference option
- Uses spare capacity with variable latency
- **Suitable for batch, async, or background processing**
- Choose Flex when cost matters more than response time.

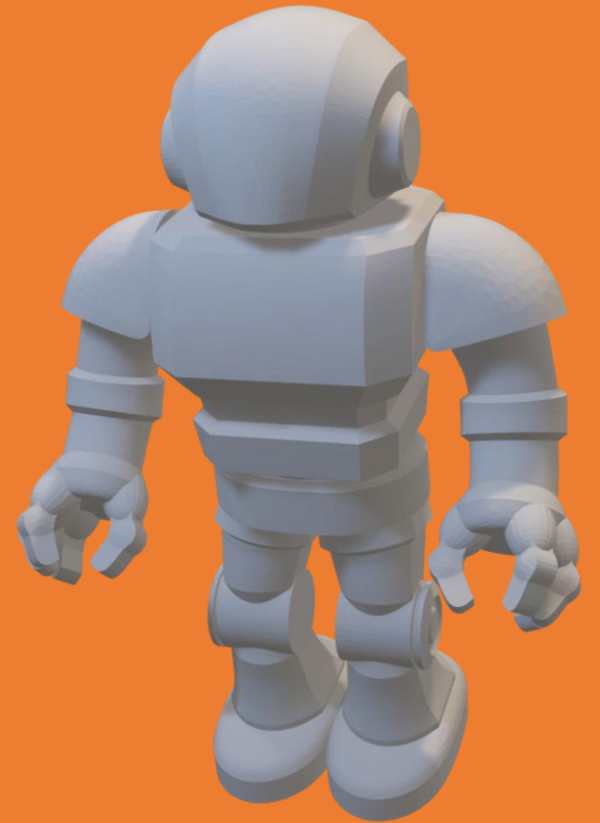
```
response = client.invoke_model_with_response_stream(  
    body=b'bytes'|file,  
    contentType='string',  
    accept='string',  
    modelId='string',  
    trace='ENABLED'|'DISABLED'|'ENABLED_FULL',  
    guardrailIdentifier='string',  
    guardrailVersion='string',  
    performanceConfigLatency='standard'|'optimized',  
    serviceTier='priority'|'default'|'flex'|'reserved'  
)
```



```
"service_tier" : "reserved | priority | default | flex"
```

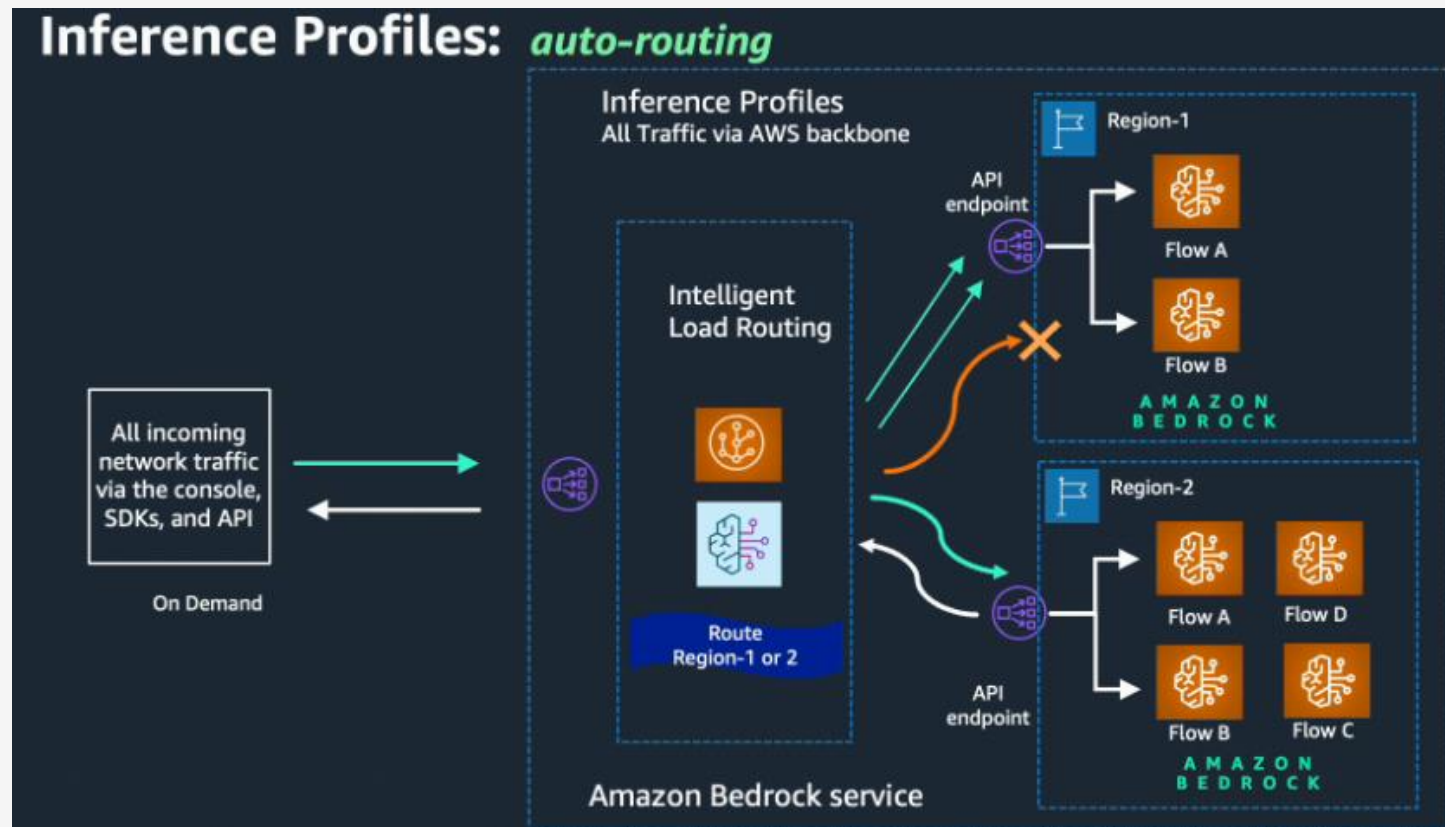
Inference Optimization

2. Cross-region Inferencing



Increase throughput with cross-region inference

Cross-region inference uses inference profiles to increase **throughput and improve resiliency** by **routing requests** across multiple **AWS Regions** during peak utilization bursts.



Increase throughput with cross-region inference

```
import boto3
import json

# Create a Bedrock Runtime client in the AWS Region of your choice.
client = boto3.client("bedrock-runtime", region_name="us-east-1")
# Set the model ID, e.g., Claude 3 Haiku.
model_id = "anthropic.claude-3-haiku-20240307-v1:0"
# Define the prompt for the model.
prompt = "Describe the purpose of a 'hello world' program in one line."
# Format the request payload using the model's native structure.
native_request = {
    "anthropic_version": "bedrock-2023-05-31",
    "max_tokens": 512,
    "temperature": 0.5,
    "messages": [
        {
            "role": "user",
            "content": [{"type": "text", "text": prompt}],
        }
    ],
}
# Convert the native request to JSON.
request = json.dumps(native_request)
# Invoke the model with the request.
streaming_response = client.invoke_model_with_response_stream(
    modelId=model_id, body=request
)
# Extract and print the response text in real-time.
for event in streaming_response["body"]:
    chunk = json.loads(event["chunk"]["bytes"])
    if chunk["type"] == "content_block_delta":
        print(chunk["delta"].get("text", ""), end="")
```

Model ID when cross-region inference **NOT** enabled.

Increase throughput with cross-region inference

Amazon Bedrock > Cross-region inference

Cross-region inference [Info](#)

Cross-region inference uses inference profiles to increase throughput and improve resiliency by routing your requests across multiple AWS Regions during peak utilization bursts.

How it works

Select a system-defined inference profile

Select a system-defined inference profile from the table to view information about the model and Regions to which it can route inference requests.

Use a system-defined inference profile

You can do cross-region inference by specifying a system-defined inference profile ID or its ARN when running model inference or using an Amazon Bedrock resource. You can also create a application inference profile from the system-defined one to track costs and usage.

Inference profiles (55) [Info](#)

Find inference profiles

Name	Model	Description	Inference profile ID	Inference profile ARN	AWS Regions
GLOBAL Amazon Nova 2 Lite	Nova 2 Lite	Routes requests to Amazon Nova 2 Lite globally across all supported AWS Regions	global.amazon.nova-2-lite-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
Global Anthropic Claude 4.5	Claude Haiku 4.5	Routes requests to Anthropic Claude Haiku 4.5 globally across all supported AWS Regions	global.anthropic.claude-haiku-4-5-20251001-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
GLOBAL Anthropic Claude 4.5	Claude Opus 4.5	Routes requests to Anthropic Claude Opus 4.5 globally across all supported AWS Regions	global.anthropic.claude-opus-4-5-20251101-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
Global Claude Sonnet 4	Claude Sonnet 4	Routes requests to Claude Sonnet 4 globally across all supported AWS Regions	global.anthropic.claude-sonnet-4-20250514-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
Global Claude Sonnet 4.5	Claude Sonnet 4.5	Routes requests to Claude Sonnet 4.5 globally across all supported AWS Regions	global.anthropic.claude-sonnet-4-5-20250929-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
Global Cohere Embed v4	Embed v4	Routes requests to Embed v4 globally across all supported AWS Regions	global.cohere.embed-v4:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
GLOBAL TwelveLabs Pegasus v1.2	Pegasus v1.2	Routes requests to Pegasus v1.2 globally across all supported AWS Regions	global.twelvelabs.pegasus-1-2-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	Commercial AWS Regions
US Amazon Nova 2 Lite	Nova 2 Lite	Routes requests to Amazon Nova 2 Lite in us-east-1, us-east-2 and us-west-2	us.amazon.nova-2-lite-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	US East (Virginia) us-east-1, US East (Ohio) us-east-2, US West (Oregon) us-west-2
US Anthropic Claude 3 Haiku	Claude 3 Haiku	Routes requests to Anthropic Claude 3 Haiku in us-east-1 and us-west-2	us.anthropic.claude-3-haiku-20240307-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	US East (Virginia) us-east-1, US West (Oregon) us-west-2
US Anthropic Claude 3 Opus	Claude 3 Opus	Routes requests to Anthropic Claude 3 Opus in us-east-1 and us-west-2	us.anthropic.claude-3-opus-20240229-v1:0	arn:aws:bedrock:us-east-1:196715057542:inference-profile/...	US East (Virginia) us-east-1, US West (Oregon) us-west-2

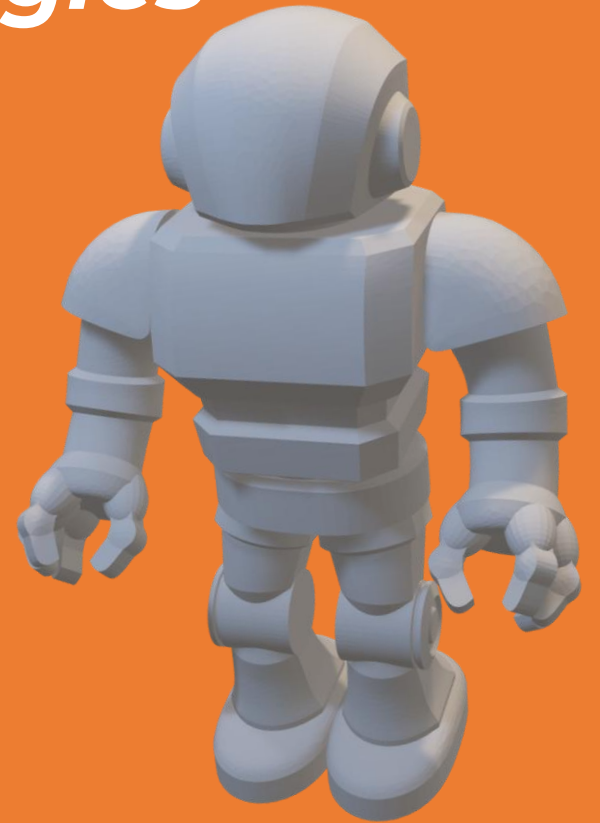
<https://docs.aws.amazon.com/bedrock/latest/userguide/inference-profiles-support.html>

Increase throughput with cross-region inference

Feature	Geographic Cross-Region Inference	Global Cross-Region Inference	Recommendation
Data residency	Within geographic boundaries (US, EU, APAC)	Any supported AWS commercial Region worldwide	Choose Geographic for compliance requirements
Throughput	Higher than single-region	Highest available	Choose Global for maximum performance
Cost	Standard pricing	Approximately 10% savings	Choose Global for cost optimization
SCP requirements	Allow all destination Regions in profile	Allow "aws:RequestedRegion": "unspecified"	Configure based on your organizational policies
Best suited for	Organizations with data residency regulations	Organizations prioritizing cost and performance	Assess your compliance and performance needs

Inference Optimization

3. Latency Optimization Strategies



Latency Optimization Strategies

- **Latency-optimized inference** for foundation models in Amazon Bedrock delivers faster **response times and improved responsiveness** for AI applications.
- Following models **well-suited for low-latency use cases**, balancing **speed and quality**:
 - Amazon Nova Pro
 - Anthropic's Claude 3.5 Haiku model
 - Meta's Llama 3.1 405B and 70B models
- Set **“Latency” parameter** to **“optimized”** while calling the Amazon Bedrock runtime API

```
import boto3
import json

def lambda_handler(event, context):
    bedrock = boto3.client('bedrock-runtime', region_name='us-east-1')

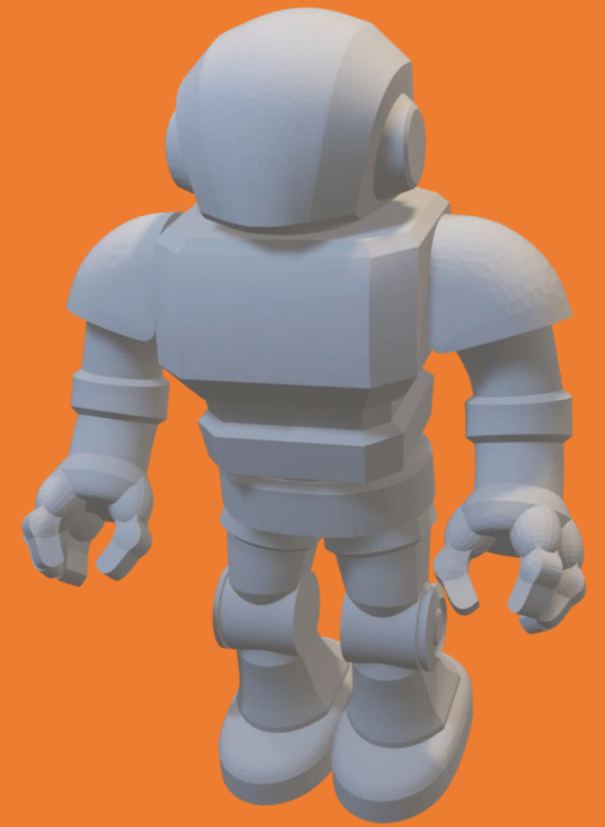
    response = bedrock.invoke_model(
        modelId='us.amazon.nova-pro-v1:0',
        body=json.dumps({
            "messages": [{"role": "user", "content": [{"text": event.get('prompt', 'What is Amazon Bedrock')}]}],
            "inferenceConfig": {
                "max_new_tokens": 1000,
                "temperature": 0.7
            }
        })),
        contentType='application/json',
        accept='application/json',
        guardrailIdentifier='string',
        guardrailVersion='string',
        performanceConfigLatency='standard'|'optimized',
        serviceTier='priority'|'default'|'flex'|'reserved',
        guardrailIdentifier=('7f13opvjaato'),
        guardrailVersion='DRAFT',
        performanceConfigLatency='optimized'
    )

    result = json.loads(response['body'].read())
    print(response)
    print(json.dumps(result))
    return {
        "statusCode": 200,
        "body": json.dumps({
            "answer": result["output"][0]["message"][0]["content"][0]["text"],
            "performanceConfigLatency": response.get("performanceConfigLatency"),
            "serviceTier": response.get("serviceTier")
        })
    }
```

```
response = client.invoke_model_with_response_stream(
    body=b'bytes'|file,
    contentType='string',
    accept='string',
    modelId='string',
    trace='ENABLED'|'DISABLED'|'ENABLED_FULL',
    guardrailIdentifier='string',
    guardrailVersion='string',
    performanceConfigLatency='standard'|'optimized',
    serviceTier='priority'|'default'|'flex'|'reserved'
)
```

Decision Point 6 :
Cost Optimization

Bedrock Pricing Models



Amazon Bedrock – Pricing Models

Modes - On-Demand, Provisioned Throughput & Batch

1. On-Demand Mode

- Pay for what you use, with **no time-based term commitments**
- **Text generation models**
 - Charged for every input token processed and every output token generated [1000 Tokens ~ 750 Words]
- **Image generation models**
 - Charged for every image generated.
- **Embeddings models**
 - Charged for every input token processed

For latest pricing, refer to following link - <https://aws.amazon.com/bedrock/pricing/>

Amazon Bedrock – Pricing Models

2. Provisioned Throughput

- Primarily designed for **large consistent inference workloads** that need guaranteed throughput.
- Can purchase model units for a **specific base or custom model with time commitment** (1-6 months)

Amazon Bedrock – Pricing Models

3. Batch Inferencing

- Run model inferences asynchronously in batches to efficiently process many requests at once.
- Example – Summarize 100's of large legal documents
- The job will output a JSON file in the assigned S3 location.
- Batch inference **speeds up inferences** on large datasets.
- In some cases batch inference available **at a 50% lower price** compared to on-demand inference pricing.

Load Input



S3

Create Batch inference job



Amazon Bedrock Select Model

Output



S3

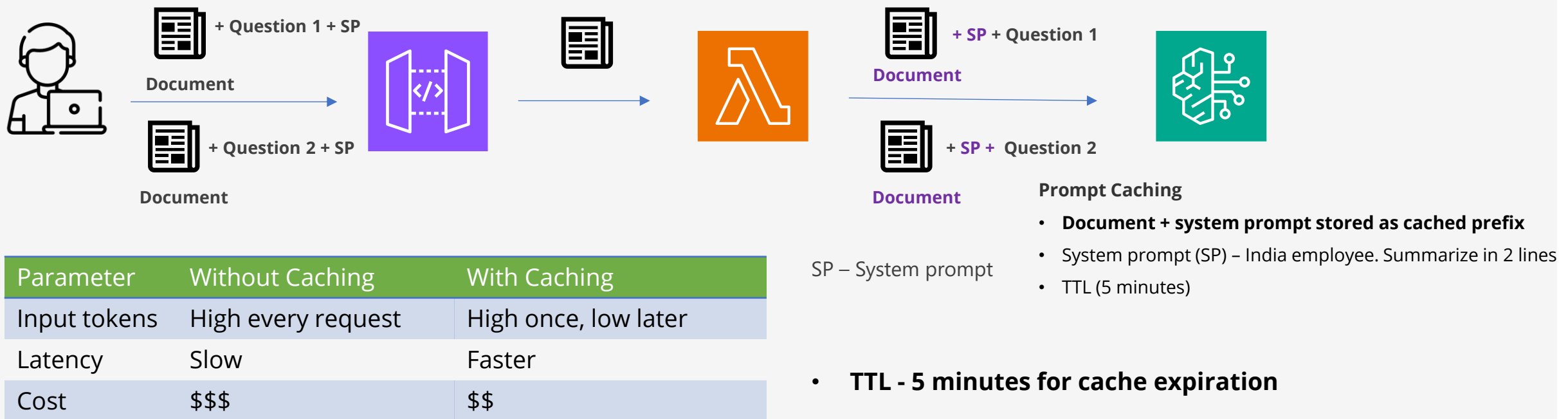
```
{ "recordId" : "alphanumeric string", "modelInput" : {JSON body} }  
...
```

Amazon Bedrock Prompt Caching



Amazon Bedrock Prompt Caching

- Prompt caching in Amazon Bedrock **reduces inference response latency and input token costs**.
- Most effective for **workloads with long and repeated contexts that are frequently reused** for multiple queries.
- Bedrock creates a cache using **cache checkpoints/ Prompt Prefix** – [**Bookmarks Static part**] – Document + SP
- Each model has a **minimum token size for caching** - Claude 3.7 Sonnet (Minimum: 1,024 tokens per checkpoint – first checkpoint)



Amazon Bedrock Intelligent Prompt Routing



Amazon Bedrock Intelligent Prompt Routing

Dynamically **routes** (using inbuilt routing logic) **requests between foundational models from the same family**, optimizing for **response quality and cost**.

- Anthropic (Haiku, Haiku 3.5, Claude Sonnet 3.5 v1, Claude Sonnet 3.5 v2)
- Meta Llama (3.1 8b, 70b, 3.2 11B, 90B and 3.3 70B)
- Amazon Nova (Nova Lite & Nova Pro).

Scenario 1 : "Summarize this paragraph in one sentence."

Predicted quality difference: --- > Nova Lite vs Nova Pro = 4% --- > Routed to **Nova Lite**

Scenario 2 : "Analyze these logs and identify the root cause of a cascading failure across distributed manufacturing systems."

Predicted quality difference: --- > Nova Lite vs Nova Pro = 32%--- > Routed to **Nova Pro**

Priority Tier

Input modalities Text, Image, Video

Amazon Nova models	Price per 1,000 input tokens
Amazon Nova 2 Lite	\$0.000525

Input modalities Text, Image, Video, Audio

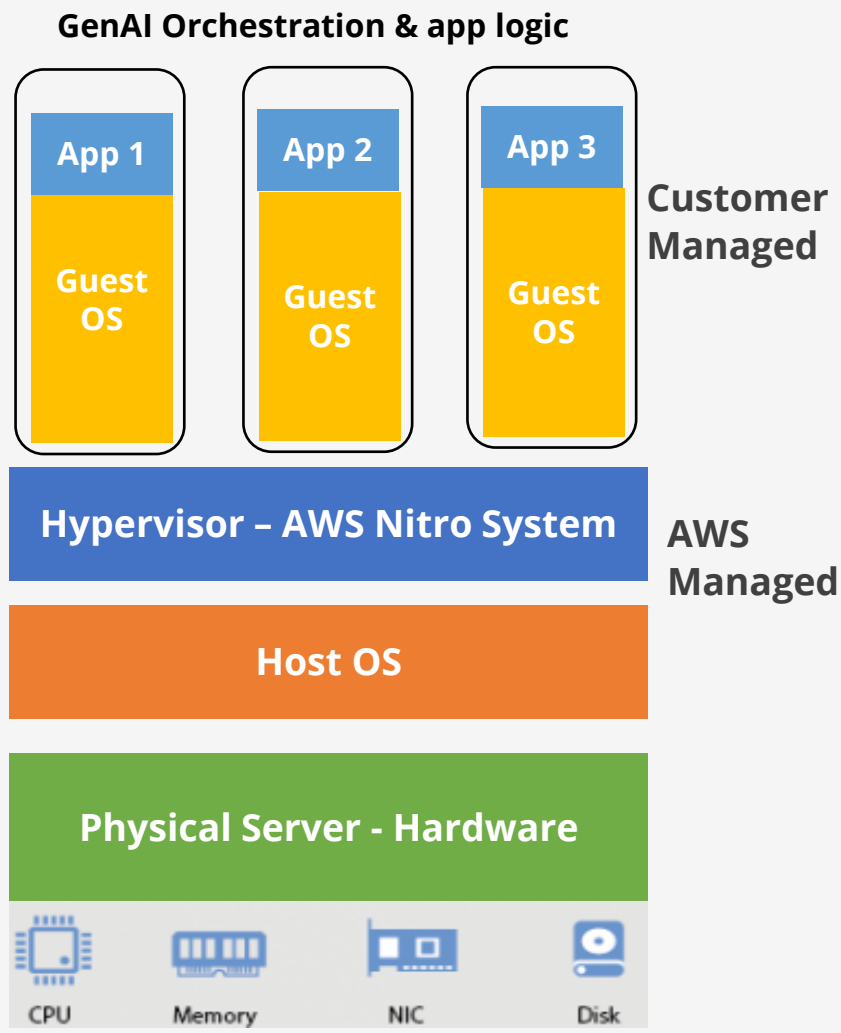
Amazon Nova models	Price per 1,000 input tokens (text)	Price per 1,000 input tokens (image)	Price per 1,000 input tokens (video)
Amazon Nova 2 Omni (Preview)	\$0.0005	\$0.0005	\$0.0005
Amazon Nova 2 Pro (Preview)	\$0.0021875	\$0.0021875	\$0.0021875

Decision Point 7 :

Selection of GenAI Compute & Orchestration Layer

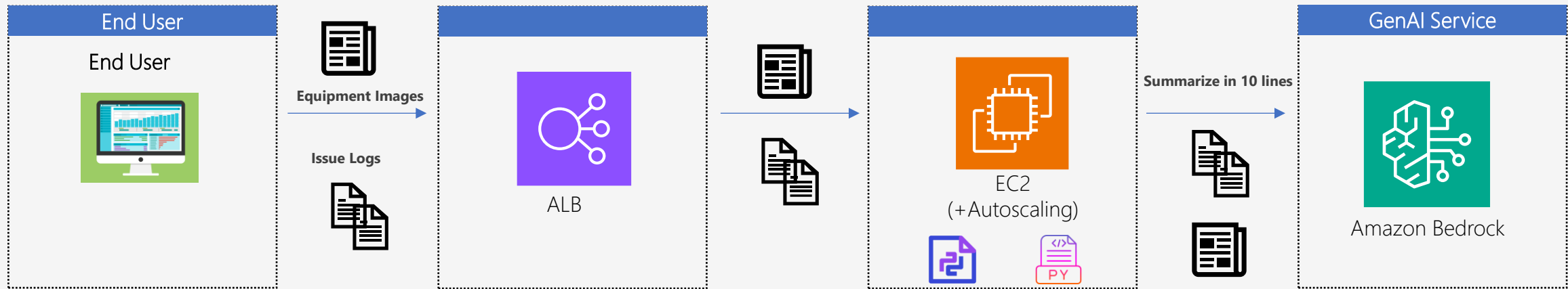
AWS Lambda vs EC2 vs ECS

GenAI Compute & Orchestration Layer – EC2



Feature	Amazon EC2 (Elastic Compute Cloud)
Architecture	Virtual Machines
Management	Customer manages guest OS - patching, security, scaling etc.
Pricing Model	Per hour or second for running provisioned instances (on-demand, reserved, spot options).
Execution Limit	Designed for long-running applications
Scaling	Manual scaling or with Auto Scaling Group (ASG) (with ALB).
Use Cases	• GenAI Orchestration & prompt assembly • LLM Training and hosting (LLaMA, Mistral, Falcon) • Model fine-tuning & experimentation • RAG pipelines with vector databases (host vector DB, Chunking etc.) • Long-running inference services

Common arch. pattern for EC2 based GenAI apps



Orchestration logic for long running GenAI apps

– machine logs in S3, store output



Prepares prompts

System prompts + Inference parameters (Bedrock, Jumpstart)

AWS Trainium & Inferentia

Use Cases

- Build your own **domain specific** Foundation Model (**FM**) from **scratch**
 - **BloombergGPT** (50B)
- Take an **open-source model** such as Llama and further **train & deploy it**.

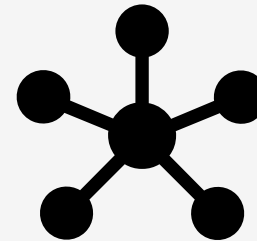
AWS Service Description

AWS Trainium

- Type of **EC2 instance** but **purpose-built ML chip by AWS** for Deep Learning Models
- **Reduce Training time** for **LLM's** being built from scratch - **100B+ parameter** models.

AWS Inferentia

- Optimized to deploy **LLM's and diffusion models for Inferencing (Output)**

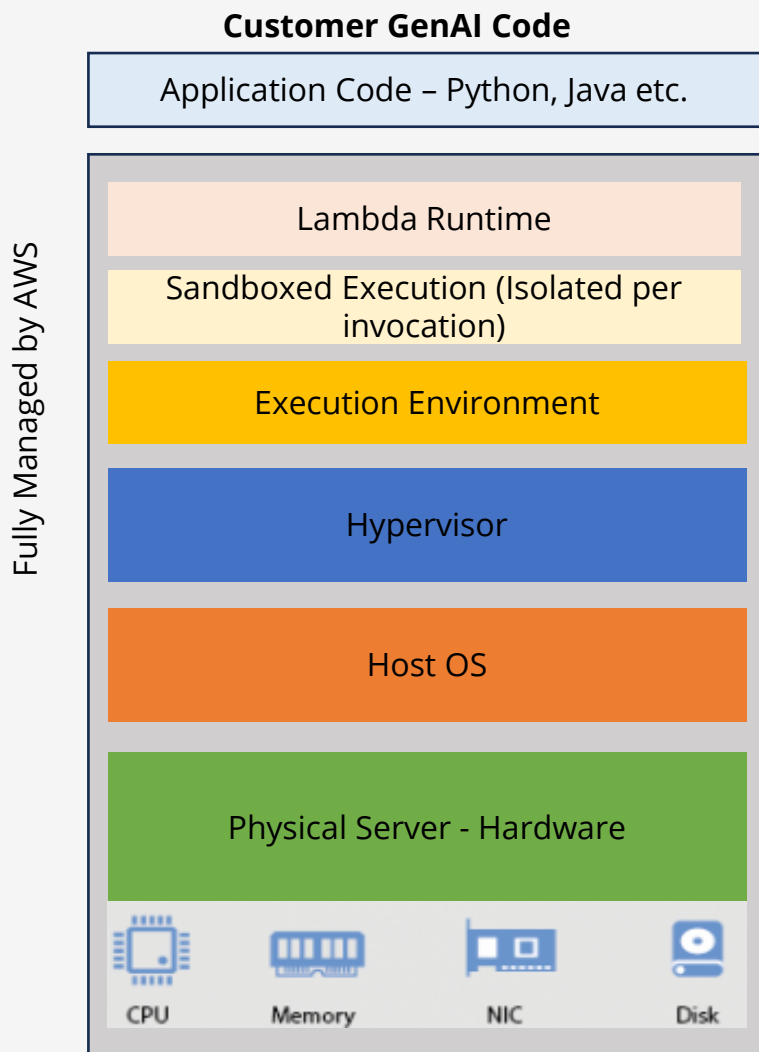


Build, Train & Deploy
your own Model



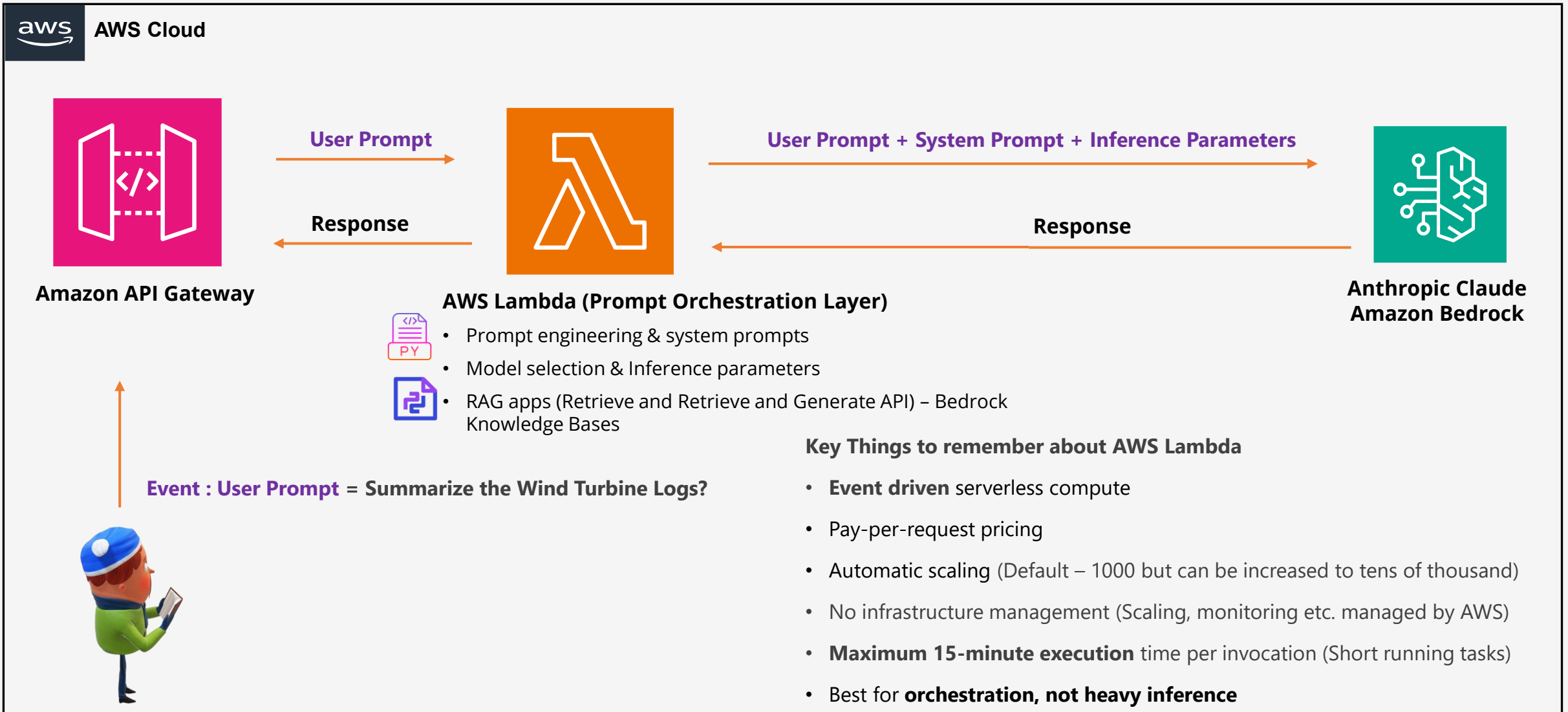
Deploy & Fine Tune
Open-Source Model

GenAI Compute & Orchestration Layer – AWS Lambda

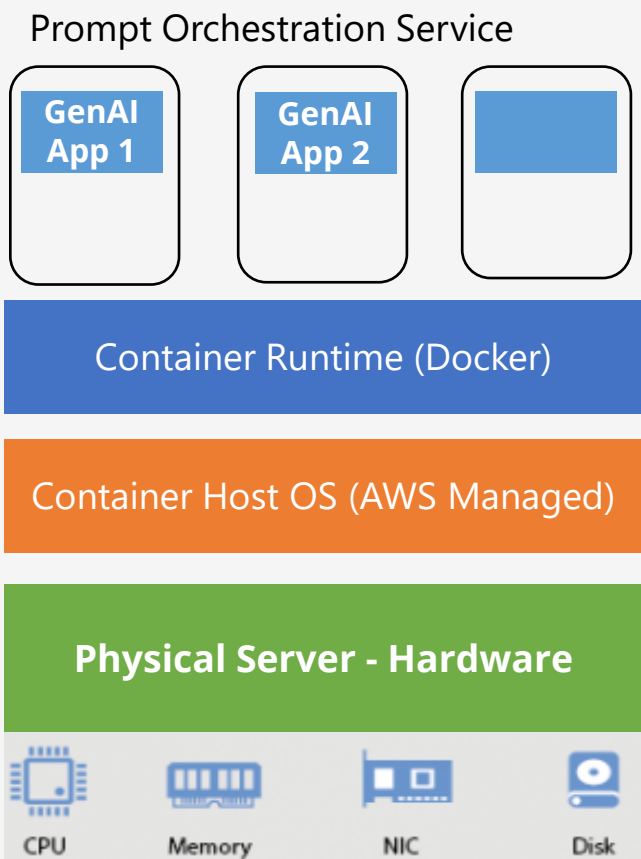


Feature	AWS Lambda
Architecture	Serverless Service
Management	Fully managed - AWS manages everything, including infrastructure, scaling, and maintenance.
Pricing Model	Pay as you use Memory allocated x Execution time × Number of requests
Execution Limit	Maximum 15-minute execution time per invocation. (Short running tasks)
Scaling	Automatic and dynamic based on request volume Default – 1000 but can be increased to tens of thousand
Use Cases	1. Orchestration logic for GenAI app • Prompt engineering & system prompts • Model selection & Inference parameters 2. RAG apps (Retrieve and Generate API) – Knowledge Bases3. Amazon Bedrock Agent Tools
Remember – No model hosting, not for long running tasks (>15 mins)	

Serverless GenAI Application Arch – API Gateway, AWS Lambda, and Amazon Bedrock



GenAI Compute & Orchestration Layer – Amazon ECS



Feature	Amazon ECS (Elastic Container Service)
Architecture	Fully managed, highly scalable container orchestration service
Management	Compute runs on AWS Fargate (no servers) or Amazon EC2 (customer-managed instances).
Scaling	Automatic scaling based on container/task metrics. Scales containers quickly without provisioning new virtual machines
Pricing Model	- Fargate - Based on pay-as-you-use (vCPU + memory per second usage) - For underlying EC2 instance provisioned (EC2-ECS)
Execution Limit	Suitable for long-running GenAI inference . No hard execution limit (unlike AWS Lambda 15-minute limit)
Use Cases	- Custom GenAI orchestration - RAG pipelines (Chunking, retrieval etc.) – Open-Source frameworks - Microservices architecture - Model Hosting - GPU-based inference using ECS on EC2 (G5, P4d etc.) - Long-running inference workloads (Inference > 15 minutes)

Containers – Any of the following Services : ECS (Fargate or EC2) – AWS native , Managed EKS

GenAI Compute & Orchestration Layer - EC2 vs ECS vs Lambda



AWS Lambda

- **Use Case - Event driven, Bursty, short running tasks** (15 mins) – Low complexity GenAI apps.
- **Serverless** – Don't want to manage infrastructure, scaling, monitoring, **AWS manages infrastructure**
- **Pricing based on usage** (No invocation, no charges)
- **Avoid - Long running tasks**, Model training or Inference



Containers (ECS)

- **Use Case - Long running tasks** - Document analysis, Agents, RAG, Model Training
- Serverless (Fargate) and Provisioned Mode (EC2 mode)
- **Avoid** – Bursty low traffic use cases



EC2

- **Use Case** – Model training, custom inference - Trainium and Inferentia
- **Avoid:** Simple Bedrock API calls (use Lambda)



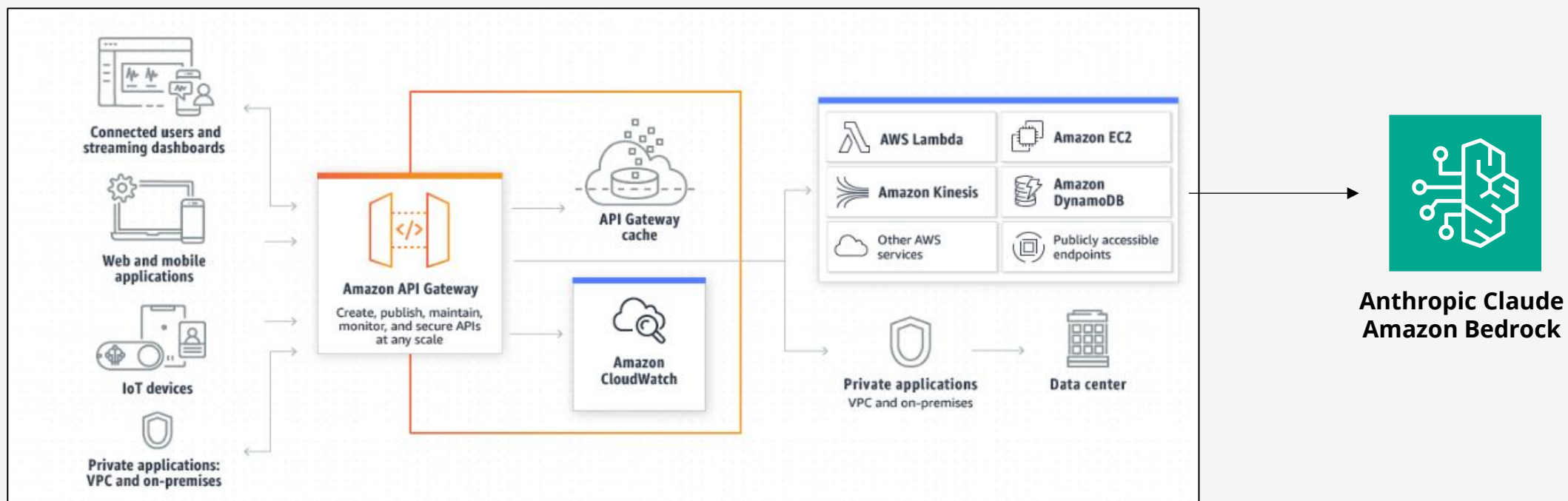
Decision Point 8 :

Selection of GenAI API Layer

API Gateway vs AWS AppSync vs ALB

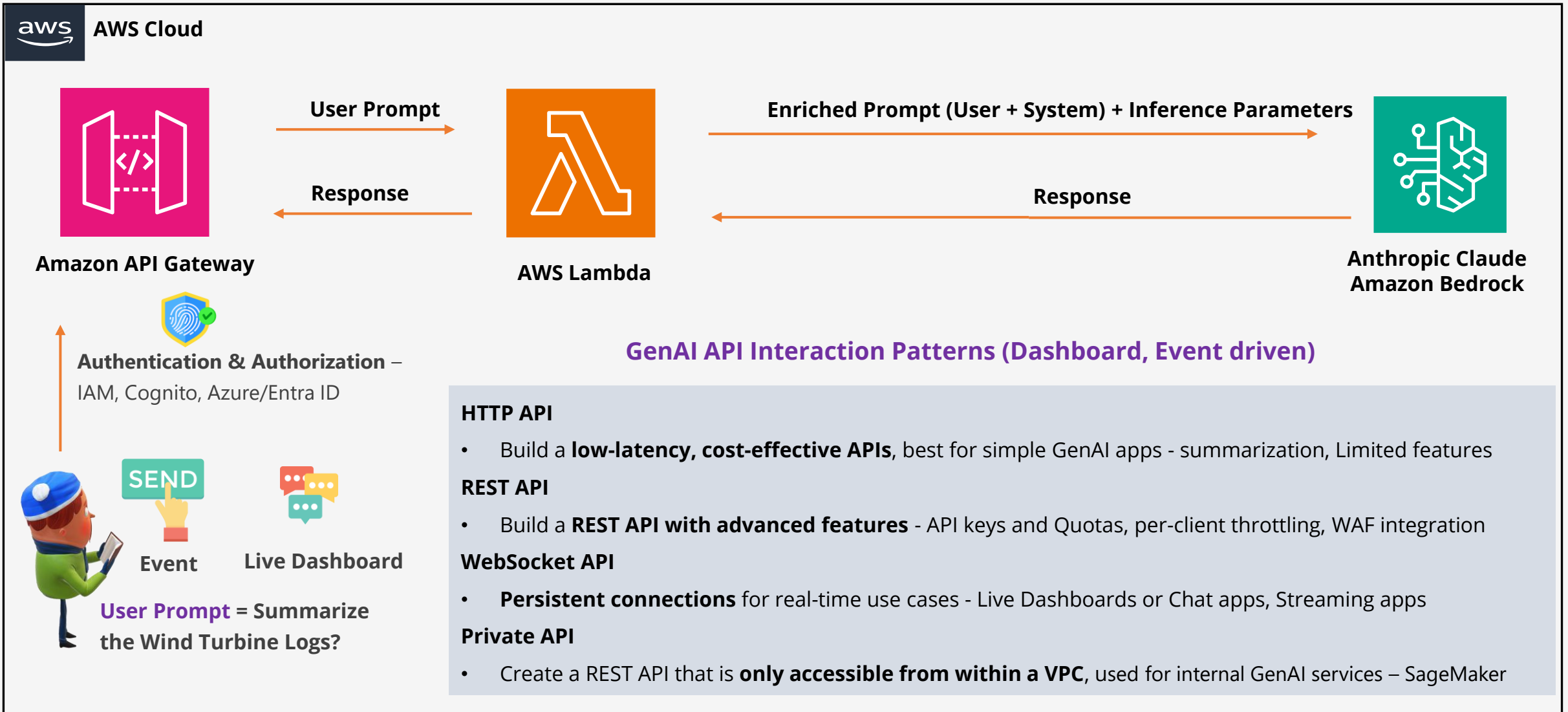
GenAI API Layer – Amazon API Gateway

Amazon API Gateway is an AWS service for **creating, publishing, maintaining, monitoring, and securing REST, HTTP, and WebSocket APIs** at any scale.



In GenAI architectures, API Gateway acts as the secure entry point for prompts, streaming responses, and model-backed APIs.

Serverless GenAI Application Arch – Type of API Gateway



Pattern 1 - Lambda Execution Models for GenAI Workloads

1. Synchronous Invocation (Request/Response)

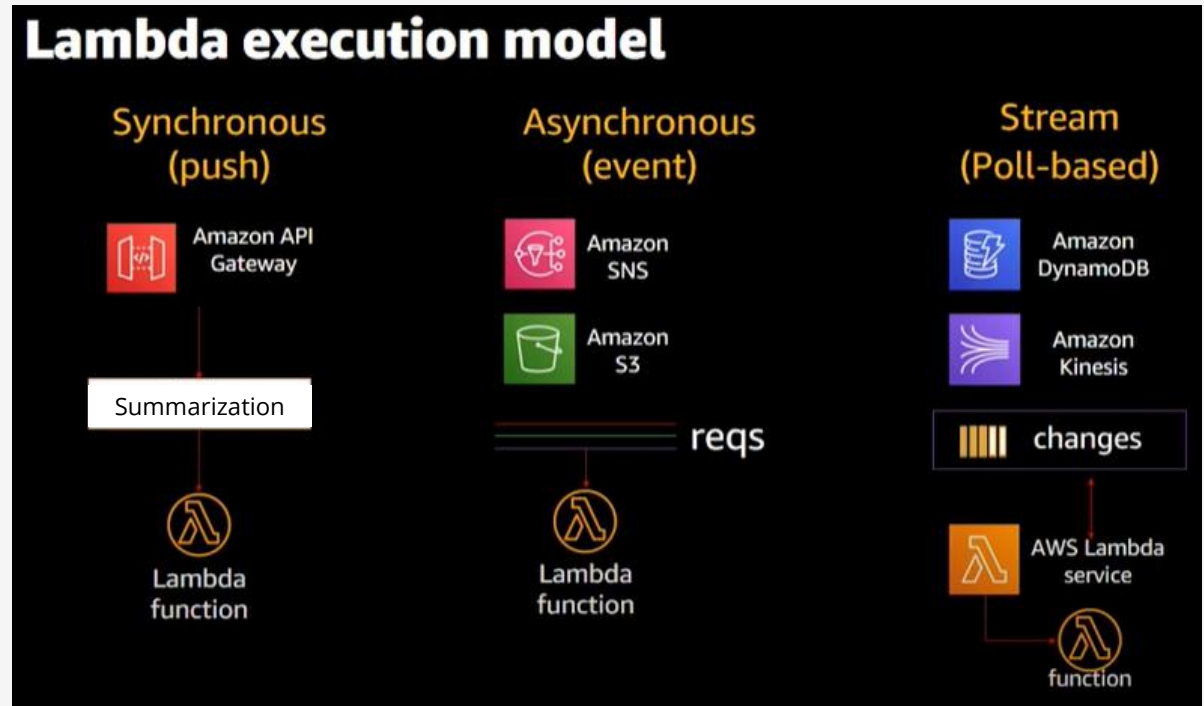
- API Gateway with AWS Lambda
- **Triggered by:** Amazon API Gateway (REST/HTTP or WebSocket)

Key GenAI Characteristics

- Client waits for model response
- Used for **low-latency GenAI inference**
- Common with Amazon Bedrock

GenAI Use Cases

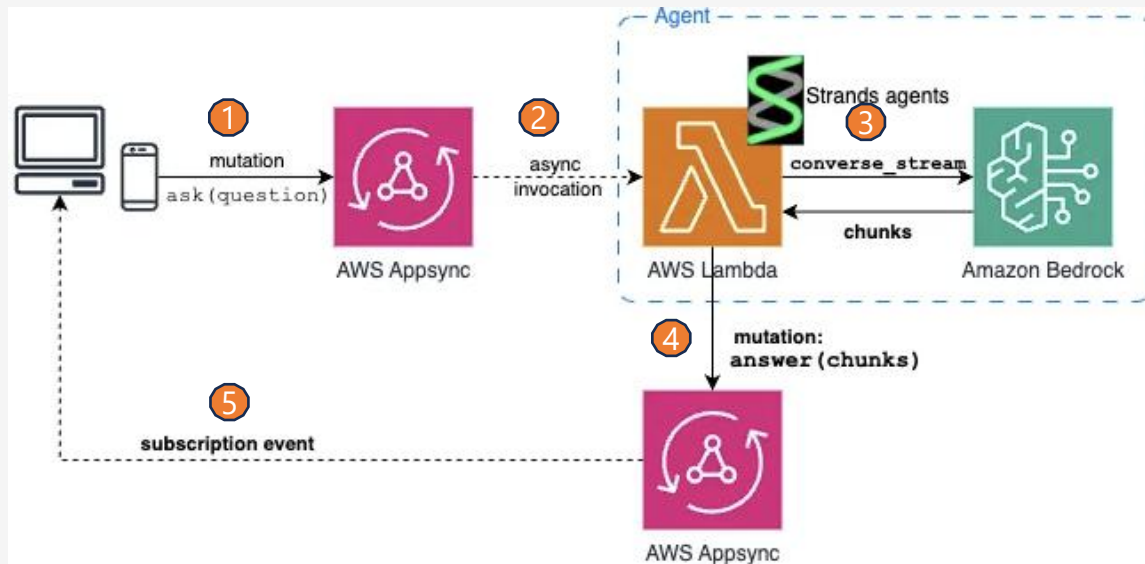
- GenAI Apps – Summarization (Prompt → response flows)
- Use **synchronous Lambda invocation** for interactive GenAI APIs that require immediate responses.
- Chatbot APIs → REST/HTTP/WebSocket



Challenges and Issues

- REST/HTTP API has a hard timeout of ~29 seconds for sync
- REST/HTTP API works only if the full GenAI response is returned within 29 seconds.
- Solution Option 1: **WebSocket API** – Persistent Connection & Streaming
- Solution Option 2: **Asynchronous pattern**

GenAI API Layer – AWS AppSync



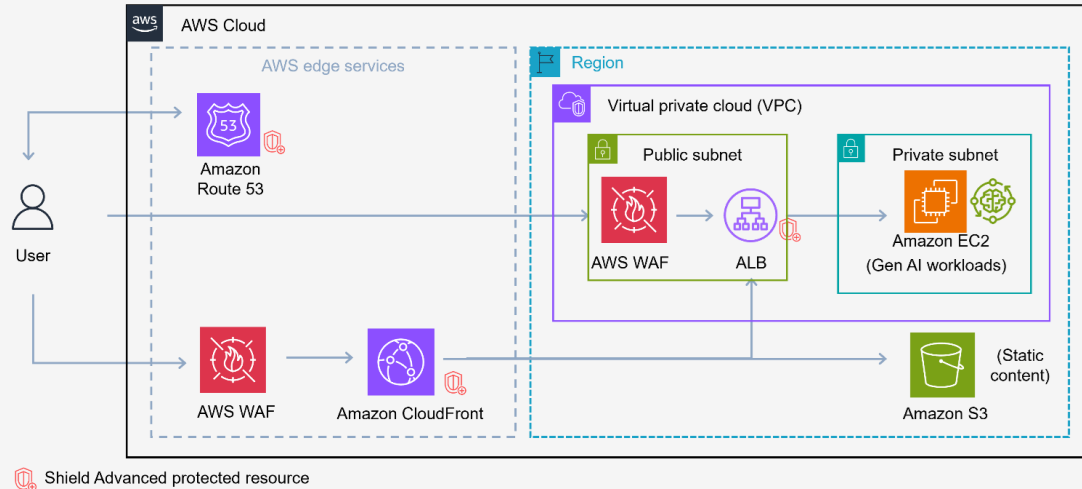
Feature	AWS AppSync
Primary Function	- Fully managed service for building GraphQL - GraphQL is a query language for APIs that allows clients to request exactly the data they require - With GraphQL, you can retrieve data from multiple related data sources in one API call, instead of making separate requests for each resource.
API Type/Protocol	GraphQL (over HTTP/HTTPS and WebSockets).
Core Features	- Real time subscription - Offline support (mobile/web)
Integration	Lambda for orchestration, Bedrock for inference
Primary Use Case	Chatbots, Dashboards requiring live updates

AppSync is ideal for **GenAI Chat and Live dashboard**

GenAI API Layer - When to select which service ?

Requirement	Best Choice	Challenges
Simple GenAI API – Low cost and latency	Amazon API Gateway – HTTP API	29 Sec response time limit
Public GenAI API with throttling and additional security features	Amazon API Gateway – REST API	29 Sec response time limit
Streaming GenAI responses	Amazon API Gateway – WebSocket API	Manage connections and routing (\$connect, \$disconnect, message routes)
Streaming GenAI + no WS management	AppSync (Fully Managed)	Requires GraphQL, Not ideal for simple request/response APIs
Real-time live dashboards (Updates automatically)	AppSync	
Token-by-token chat (ex ChatGPT)	WebSocket / AppSync	

GenAI API Layer –Application Load Balancer



Primary GenAI Use Cases

- Self-hosted open-source models (LLaMA, Mistral)
- User --- > ALB ---- > Model running on EC2
- Long-running LLM inference
- High-throughput RAG – RAG DB, Chunking etc.
- Multi-model routing (A/B testing)

Feature	Application Load Balancer
Primary Function	Distributes incoming application traffic across compute targets to provide high availability, scalability, and low-latency inference for GenAI workloads. Supported Targets • EC2 instances (most common for GenAI) • ECS / EKS containers, IP Addresses • Lambda functions
API Type/Protocol	ALB (Layer 7): HTTP, HTTPS, WebSocket NLB (Layer 4): TCP, UDP (ultra-low latency)
Core Features	• No API Gateway 29-second timeout • Ideal for long-running inference • Works well with GPU-backed EC2

Architecture Decision Point 9 :
Observability & Monitoring

Observability & Monitoring

Amazon Bedrock Logging and Monitoring

- Monitor all aspects of your Amazon Bedrock application using **Amazon CloudWatch**
- **Invocation logs** - full request data, response data, and metadata associated with all calls performed in your account
- **Metrics** - monitor the GenAI app performance using CloudWatch near real-time metrics.
- Common metrics – Token count for cost, Latency and Errors
- Set alarms that watch for certain thresholds, and send notifications or take actions when values exceed those thresholds.

Monitor model invocation using CloudWatch Logs & S3

Metric name	Unit	Description
Invocations	SampleCount	Model invocation logging to collect invocation logs, model input data, and model output data for all invocations
InvocationLatency	Milliseconds	Latency of the invocations.
InvocationClientErrors	SampleCount	Number of invocations that result in client-side errors.
InvocationServerErrors	SampleCount	Number of invocations that result in AWS server-side errors.
InvocationThrottles	SampleCount	Number of invocations that the system throttled .
InputTokenCount	SampleCount	Number of tokens in the input.
LegacyModelInvocations	SampleCount	Number of invocations using <u>Legacy</u> models
OutputTokenCount	SampleCount	Number of tokens in the output.
OutputImageCount	SampleCount	Number of images in the output (only applicable for image generation models).

Link : <https://docs.aws.amazon.com/bedrock/latest/userguide/monitoring.html>

Monitor Amazon Bedrock Guardrails using CloudWatch metrics

Metric name	Unit	Description
Invocations	SampleCount	Number of requests to the ApplyGuardrail API operation
InvocationLatency	Milliseconds	Latency of the invocations
InvocationClientErrors	SampleCount	Number of invocations that result in client-side errors
InvocationServerErrors	SampleCount	Number of invocations that result in AWS server-side errors
InvocationThrottles	SampleCount	Number of invocations that the system throttled. Throttled requests don't count as invocations or errors
TextUnitCount	SampleCount	Number of text units consumed by the guardrails policies
InvocationsIntervened	SampleCount	Number of invocations where the guardrails intervened
FindingCounts	SampleCount	Counts for each type of finding from InvokeAutomatedReasoningCheck
TotalFindings	SampleCount	Counts number of findings produced for each InvokeAutomatedReasoningCheck request
Invocations	SampleCount	Number of requests to InvokeAutomatedReasoningCheck
Latency	Milliseconds	Latency of verification using automated reasoning policy

Monitoring GenAI apps using CloudWatch Logs & Metrics

Data Type	CloudWatch Feature	Purpose
Metrics	CloudWatch Metrics	Numeric time-series (latency, invocations, errors)
Logs	CloudWatch Logs	Raw event data (requests, responses, errors)
Log Queries	CloudWatch Logs Insights	Query & analyze logs, not metrics

CloudWatch Logs Insights for querying and log analysis

The screenshot displays the AWS CloudWatch Logs Insights interface. At the top, the breadcrumb navigation shows 'CloudWatch > Logs Insights'. The main section is titled 'Query definition' with an 'Info' link. It includes a time range selector (5m, 30m, 1h, 3h, 12h, Custom) and buttons for 'Compare (Off)', 'UTC timezone', and 'Start tailing'. Below this, the 'Query scope' section has tabs for 'Log groups' and 'Property selector'. A dropdown menu for 'Log group name' is open, showing 'bedrock_demo' with the monitoring account '196715057542'. A 'Clear all' button is also present. The query editor contains the following SQL query:

```
1 fields modelId, inputTokenCount, outputTokenCount
2 | stats sum(inputTokenCount) as inputTokens,
3 | sum(outputTokenCount) as outputTokens
4 | by modelId
```

Below the query editor, there are tabs for 'Logs Insights QL', 'Query generator', 'Fields', 'Saved and sample queries', and 'Query commands'. A 'Run query' button is highlighted in orange, with 'Cancel', 'Save', 'Schedule query', and 'History' buttons next to it. A status message indicates 'Completed. Query executed for 1 log group.' Below this, the 'Logs (1)' tab is selected, showing a summary of the results. The summary text states: 'The logs show that only one model, amazon.nova-pro-v1:0, appears in the results from the bedrock_demo log group during the three-hour period from 21:45:45 UTC to 00:45:45 UTC on January 10-11, 2026. The aggregation query returned model identification data but no corresponding input or output token counts, indicating that while the model was referenced in the logs, no actual token consumption metrics were recorded during this timeframe. The absence of token count data means no computational usage statistics are available for analysis despite the model's presence in the log entries.' At the bottom, there are buttons for 'Summarize results', 'Investigate', 'Export results', and 'Add to dashboard'. A footer indicates 'Showing 1 of 1 records matched'.

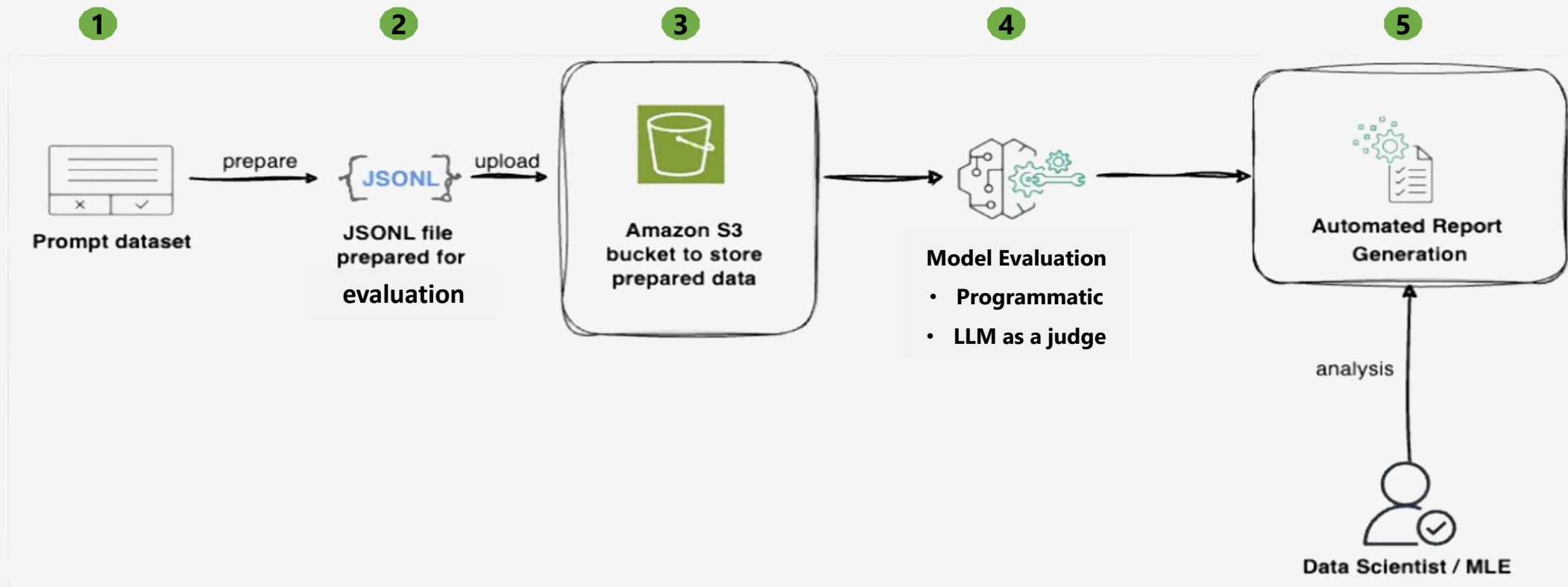
Architecture Decision Point 10 :

Model Evaluation

Model Evaluation

Bedrock Model Evaluation Service

Amazon Bedrock provides evaluation tools to evaluate, compare, and select the foundation model for your use case through Amazon Bedrock Model Evaluation capability.

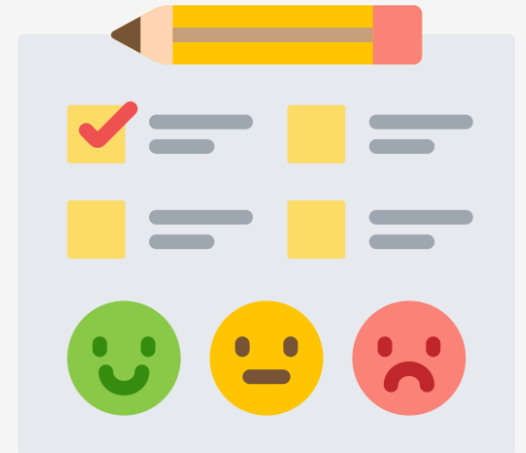


Bedrock Model Evaluation Service

Three evaluation approaches available – Programmatic, LLM-as-a-Judge & Human Evaluation.

1. Programmatic

- Evaluate performances using just the model and metrics you select.
 - Task Type ----> Text Generation, Text Summarization, Q&A, Text Classification
 - Metrics --- > Accuracy, Toxicity and Robustness
 - Datasets --- > Built in - Gigaword, XSUM or Bring your own dataset



Bedrock Model Evaluation Service

Evaluation name
Name must be unique within your account's AWS region.

evaluation-job-quick-start-20260113104512

Valid characters are a-z (lowercase), 0-9, and - (hyphen). The name can have up to 50 characters.

Description - optional

Provide a description

The description can have up to 200 characters.

► **Tags - optional**

Model selector [Info](#)
Choose the model you want to evaluate. To change the hyperparameters of the model, choose [model access](#).

 **Nova Pro 1.0** 
 On-demand

Inference configuration: Default [Update inference configurations](#)

Task type [Info](#)
Choose a Model Evaluation task type to define the model evaluation criteria.

Task type

☐ **General text generation**
The model performs natural language processing and text generation tasks.

☐ **Text summarization**
The model summarizes text based on the prompts that you provide.

☐ **Question and answer**
The answers that models provide are based on your prompts.

☐ **Text classification**
The model categorizes text into predefined classes based on the input dataset.

Evaluation results [Info](#)

Specify the S3 location
This is the folder in the S3 bucket where the results of the model evaluation job are stored.

S3 URI

[View](#)

[Browse S3](#)

Bedrock Model Evaluation Service

2. LLM as a judge

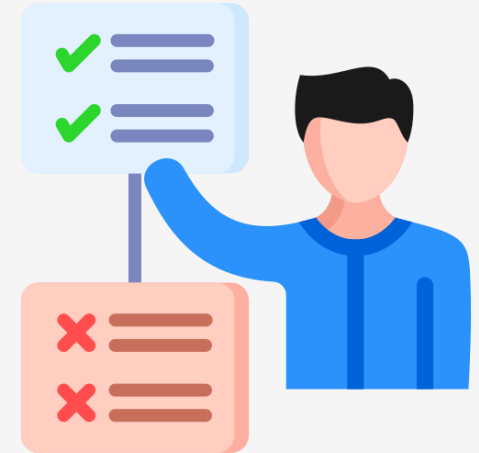
- A pre-trained model evaluates your model's responses using metrics you've selected.
 - Evaluator Model
 - Inference source --- > Bedrock Model (Select Model)
 - Metrics (Several) --- > Quality and Responsible AI or Custom metrics
 - Datasets --- > Built in - Gigaword, XSUM or Bring your own dataset



Bedrock Model Evaluation Service

3. Human: AWS Managed work team

- Evaluates up to **2 models** using a **work team** of your choice to provide feedback.
- Provides **results based** on the parameters that are specified while creating the evaluation.



4. Human: Bring your own work team

- Evaluate responses from up to 2 models using your own work team.
- You can define evaluation metrics specific to your job.

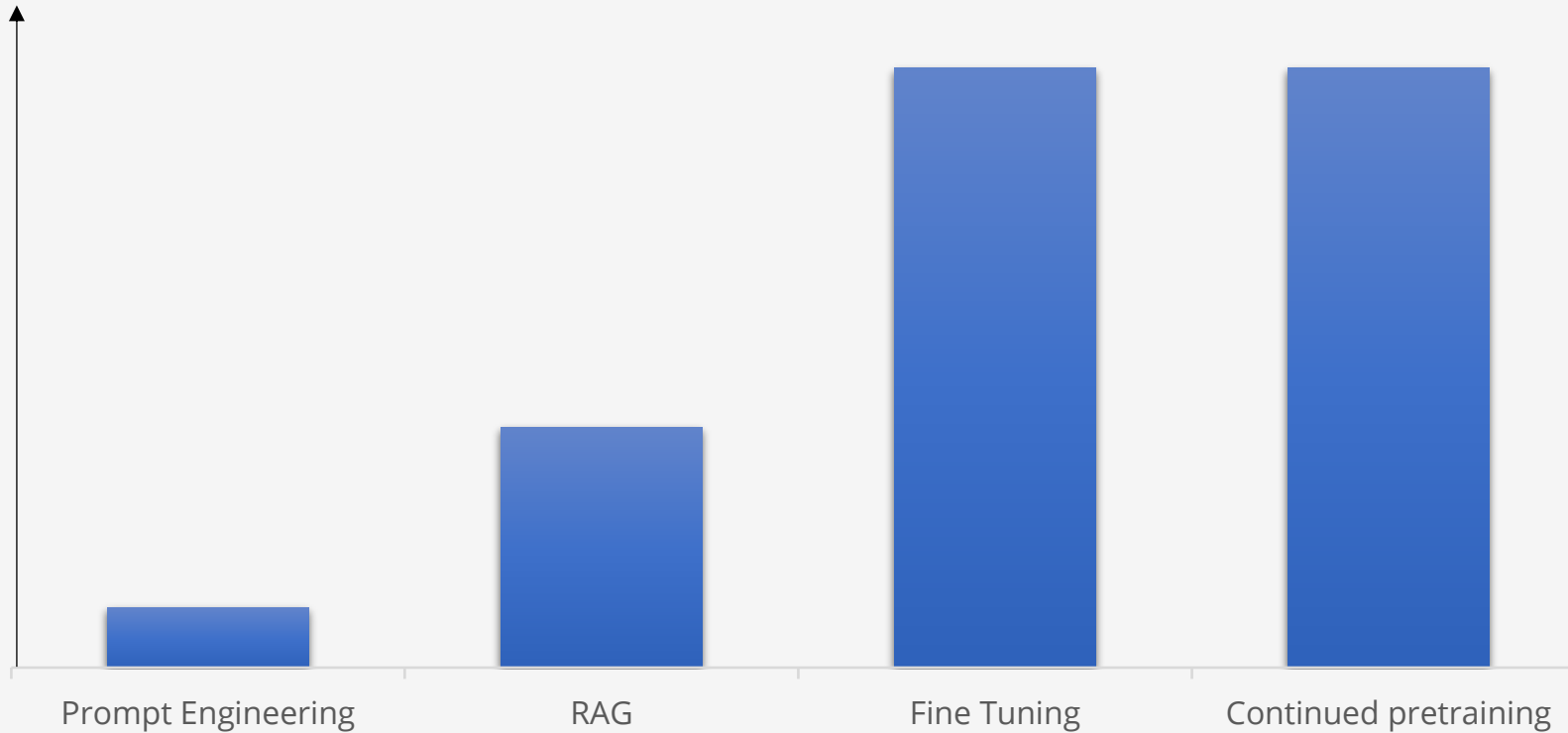


Architecture Decision Point 11 :
Model Customization



Model Adaptation Approaches in Amazon Bedrock

Cost & Time



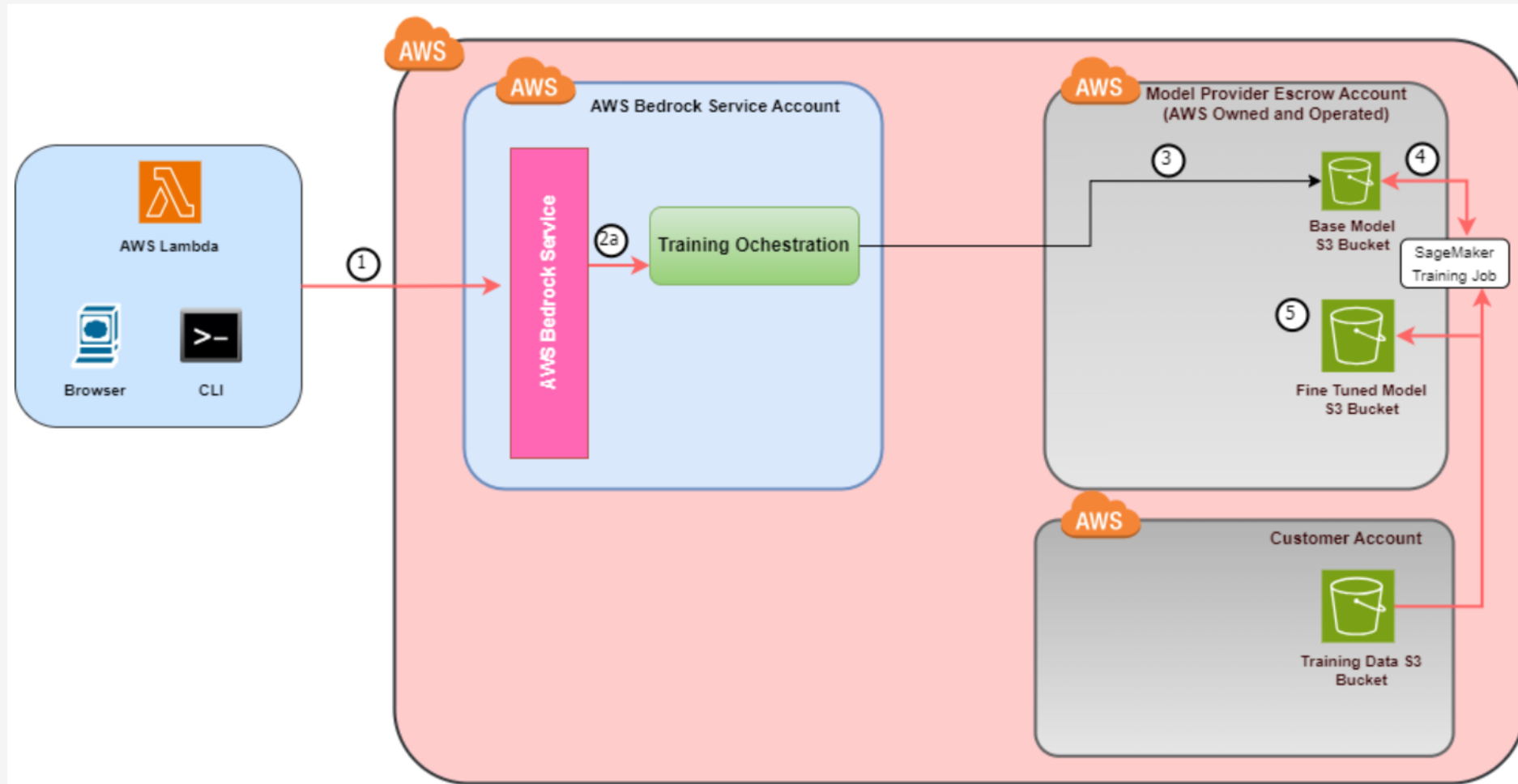
Model Customization in Amazon Bedrock

Model customization in Amazon Bedrock is done to improve a model's performance, accuracy, and relevance for specific, proprietary business use cases and tasks.

Model Customization Approaches:

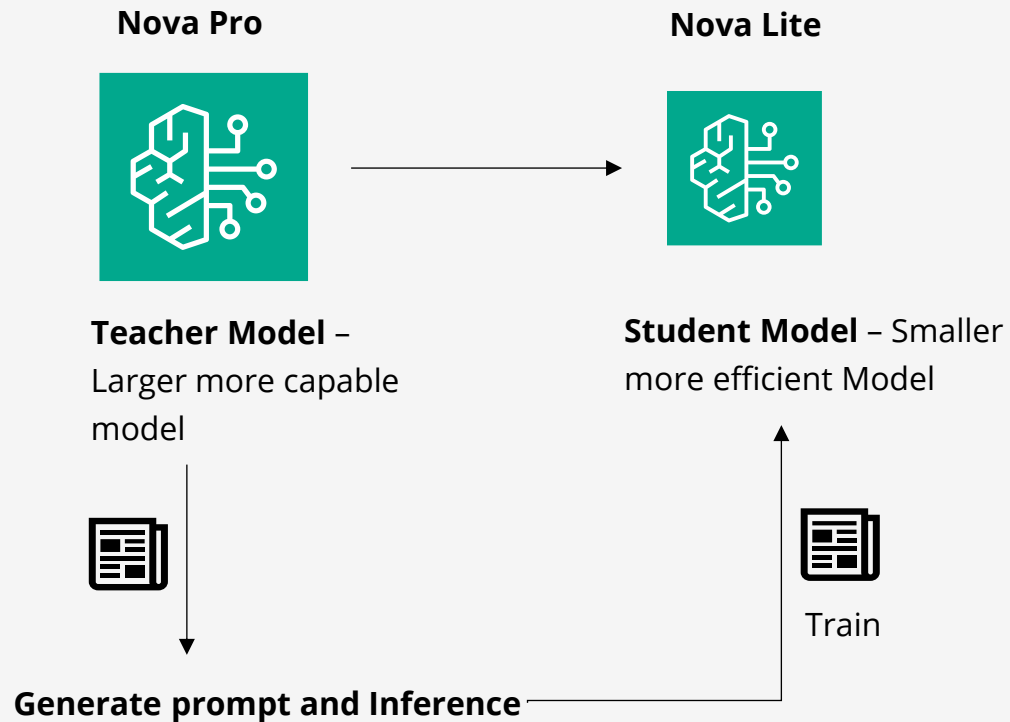
1. Model Distillation
2. Fine Tuning
3. Continued Pre-Training
4. Reinforcement fine-tuning

Customization Approach - Architecture



- Base Model : Baseline model provided to every AWS account
- Fine Tuned Model : When Base Model is trained using custom labelled data to generate Fine Tuned Model

Model Customization - 1. Model Distillation



Transfers knowledge from a large, accurate **teacher model** to a smaller, faster, and cheaper **student model**.

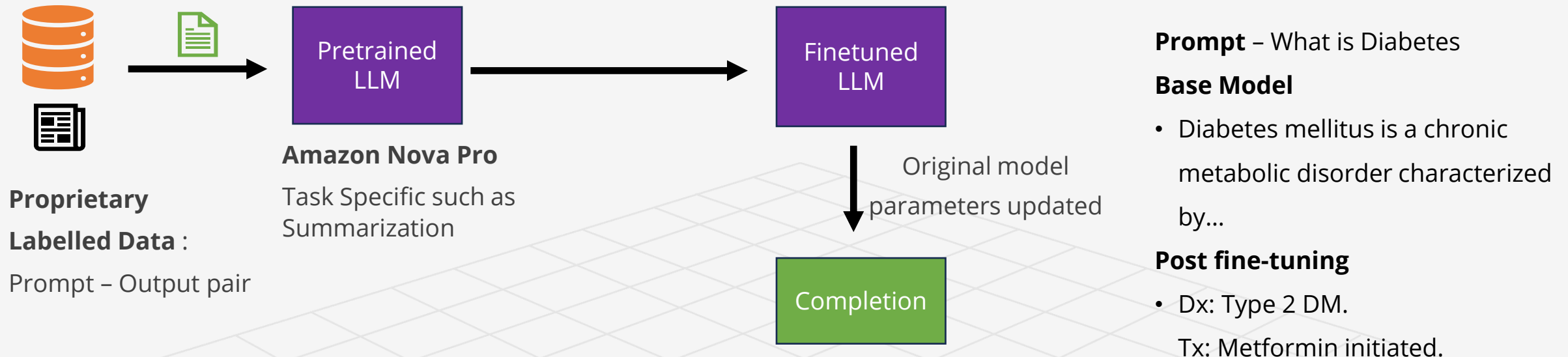
- You provide prompts (and optionally labeled examples).
- Bedrock generates high-quality responses using the teacher model.
- These responses are used to fine-tune the student model.

Best for: Achieving near-large-model accuracy at lower latency and cost.

Model Customization– 2. Supervised fine-tuning

- Used to improve **performance on specific tasks** – Summarization, Q&A - Style, format, or depth
- Training dataset of **labeled examples** used for fine tuning.
- Model learns to associate what types of outputs should be generated for certain types of inputs.

```
{  
  "prompt": "What is Type 2 diabetes?",  
  "output": "Dx: Type 2 DM"  
}  
[Abbreviated, clinical style]
```

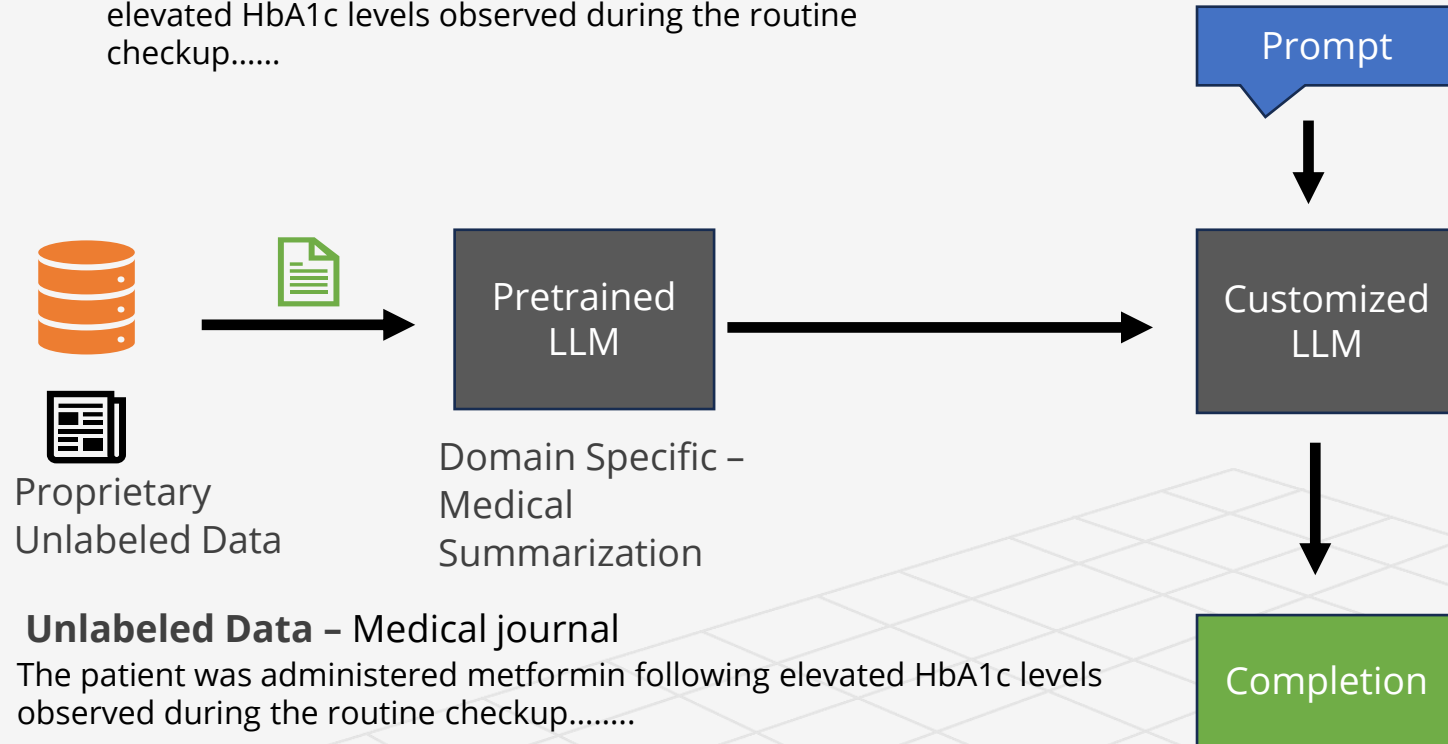


Model Customization – 3. Continued pre-training

- Process of further training a **pre-trained language model to improve domain knowledge of the model.**
- Trained on a large corpus **of unlabeled domain text.**

Transcript from medical journal :

The patient was administered metformin following elevated HbA1c levels observed during the routine checkup.....



Example : Train a model with private data, such as medical journals/documents, that are not publicly available for training large language models.

Best for: Domain adaptation when labeled data is limited.

Fine Tuning Approaches – LoRA [How?]

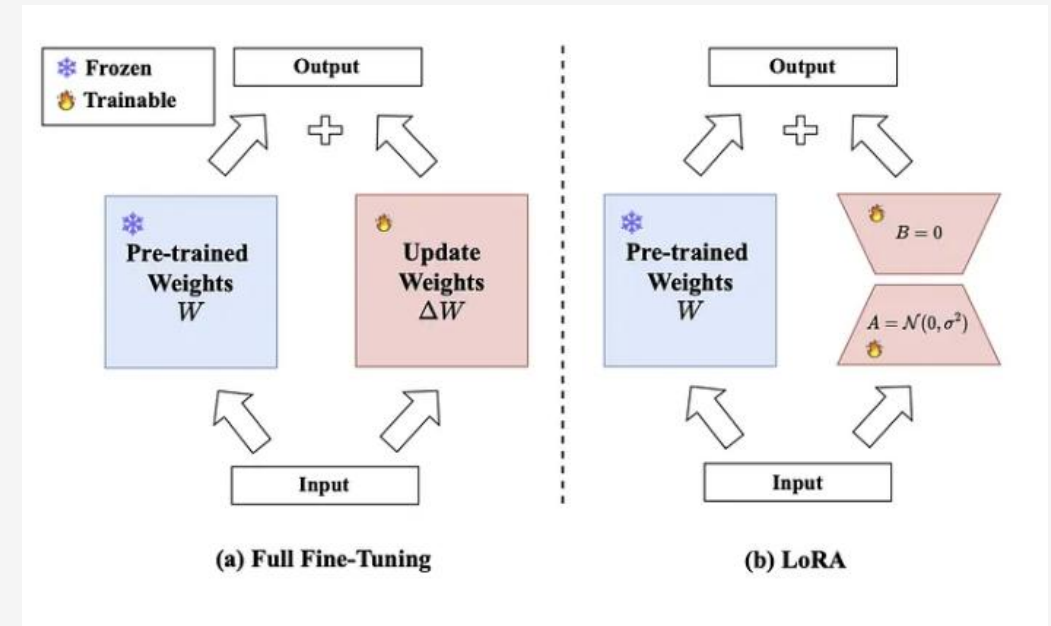
Fine-Tuning Approaches

1. Full Fine-Tuning

- Updates **all model parameters** across every transformer layer.
- Better results but requires **high compute, large memory** etc.

2. PEFT (Parameter-Efficient Fine-Tuning)

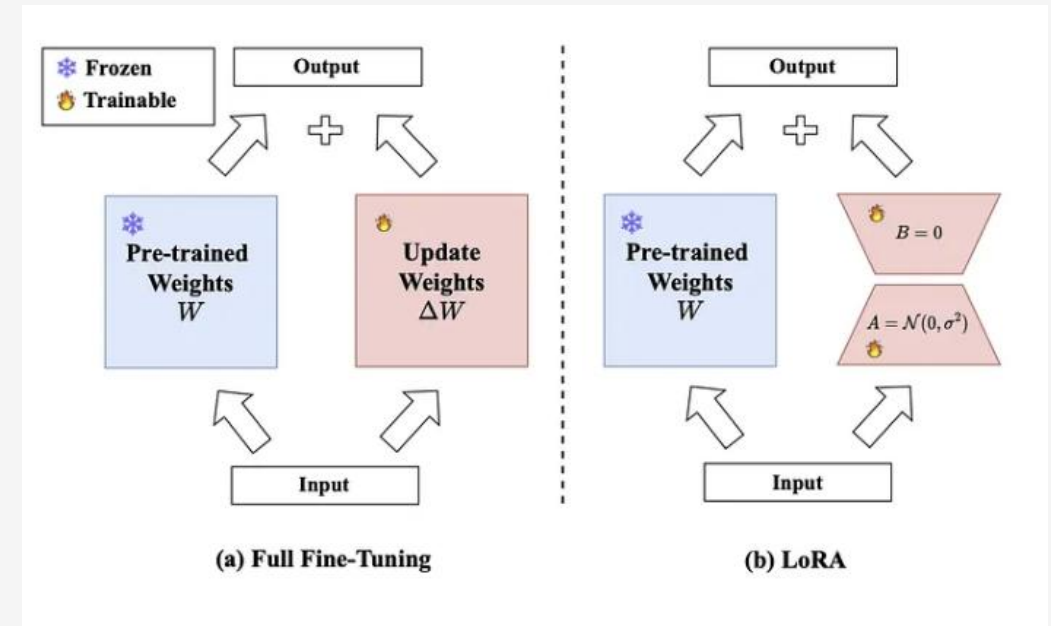
- **Freezes the base model weights** and **trains only a small set of added parameters.**
- Reduces **training cost** while preserving the model's pre-trained knowledge.
- LoRA, Adapters, Prefix / Prompt Tuning, QLoRA, etc.



Fine Tuning Approaches - LoRA

LoRA (Low-Rank Adaptation)

- **Domain adaptation for enterprise apps** - healthcare provider might use LoRA to adapt a base model to medical terminology & clinical documentation standards.
- **Multi-language adaptation** - global ecommerce platform might employ LoRA to adapt their customer service model to different regional languages & cultural contexts.
- Improves model performance while keeping **training cost** low and preserving the model's pre-trained knowledge.

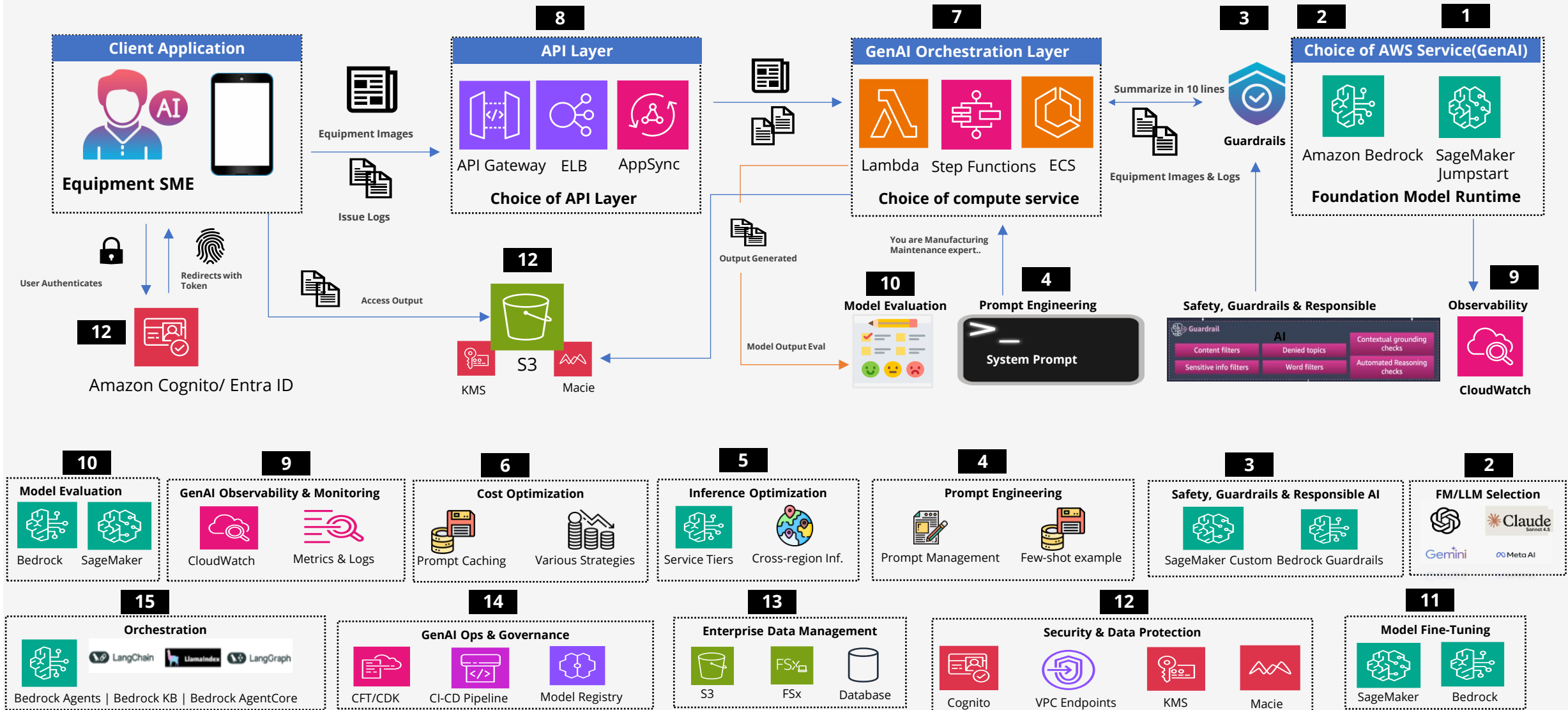


Decision Point 12-15 :

***Will be discussed in separate sections
in the course later***



Decision Point 12-15 (Will be discussed in a separate section in the course)



Section on :

Retrieval Augmented Generation (RAG)

10 Key Architecture considerations to

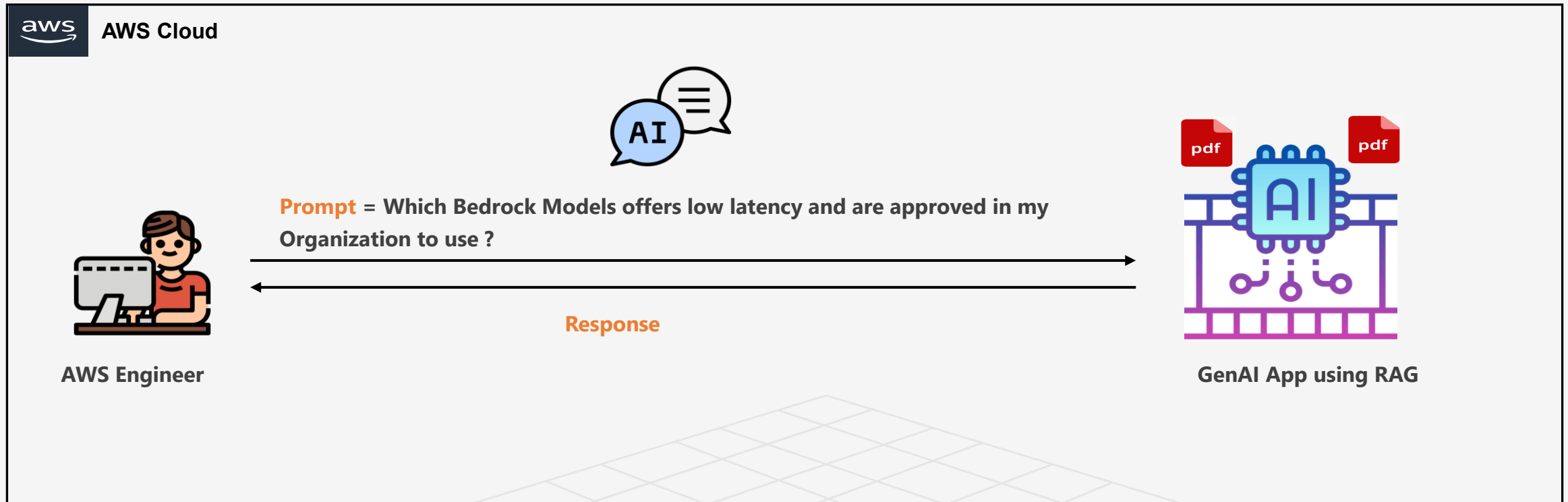
build Production Ready RAG Apps

***Business Use Case 2 –
e-Learning GenAI App (RAG)***



Use Case 2 : **e-Learning** Q&A App using Generative AI

Use Case : Build an e-Learning Generative AI application to accelerate organization learning on emerging AI technologies.



Retrieval Augmented Generation(RAG)

Need for RAG ?



Problem RAG is trying to solve ?

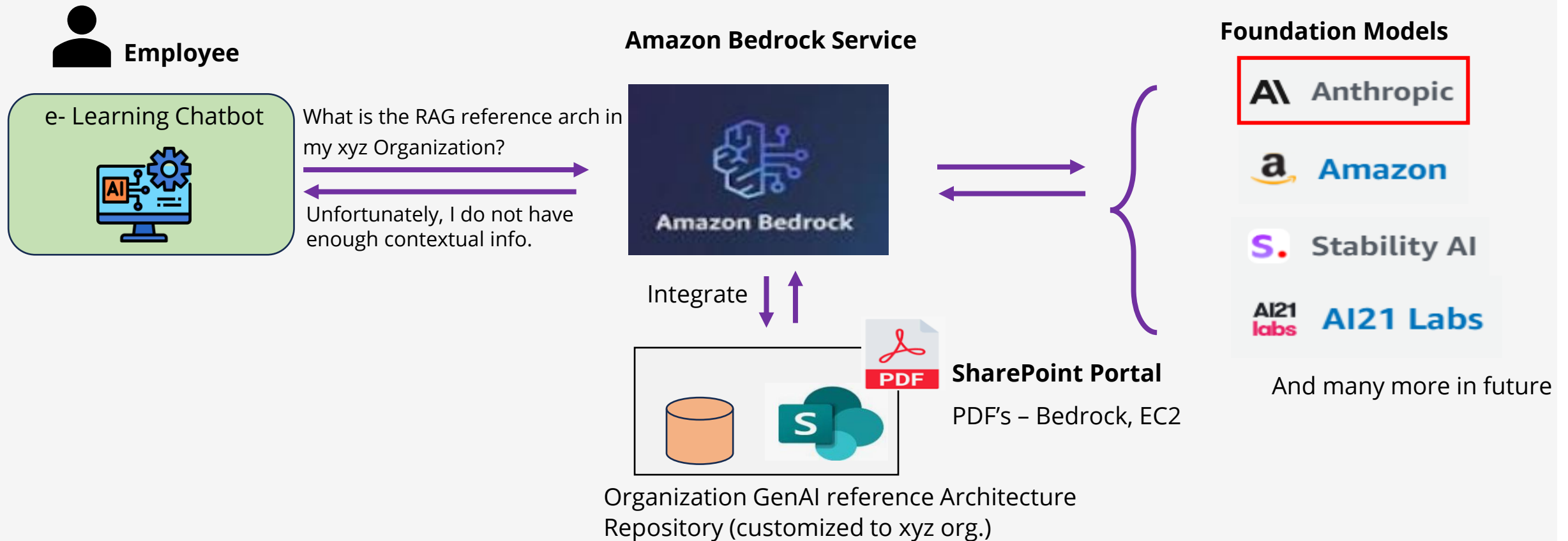
Challenges with Foundation Models

- Lack of proprietary information – Enterprise Information
- No up-to-date information post training cut off date

Retrieval Augmented Generation or RAG equips foundation models (FMs) :

- Proprietary information by fetching data from company data sources
- Up-to-date knowledge (beyond model training cut-off date)

What is Retrieval Augmented Generation (RAG)



What is RAG ?

RAG is the process of optimizing the output of a large language model by supplementing the knowledge of LLM with internal organization data source.

Retrieval Augmented Generation(RAG)

Basic Introduction to

Vectors, Embeddings, Vector DB & Search



Why do we need Vectors ?



Structured Data

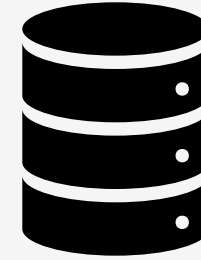
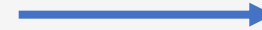
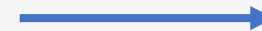


Image Search or Unstructured Data



???



What are Vectors ?

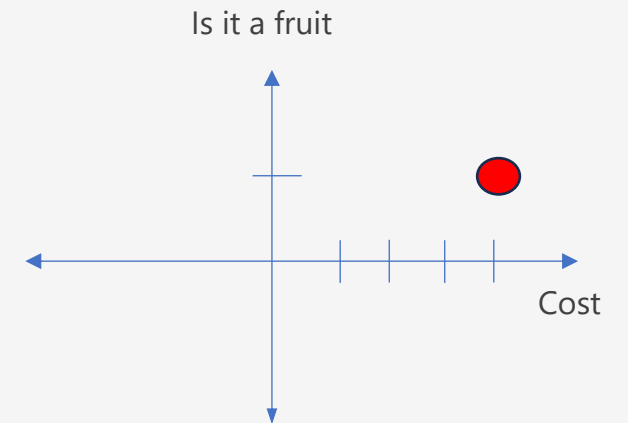
Vectors are **mathematical representation of words, sentences** and documents

Vectors are represented by **List of numbers** or sequence of numbers



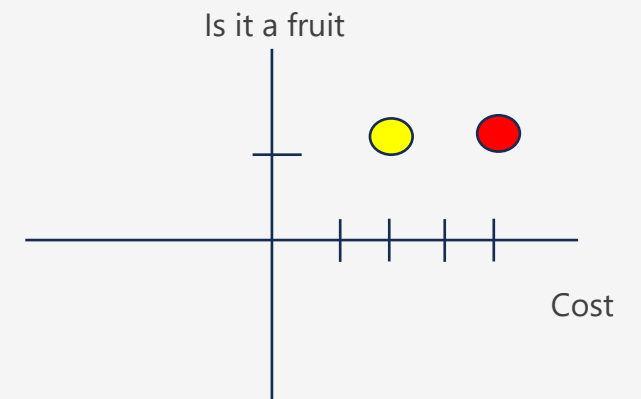
Is it a fruit	Cost
1	\$4

$[1, 4]$



Is it a fruit	Cost
1	\$2

$[1, 2]$

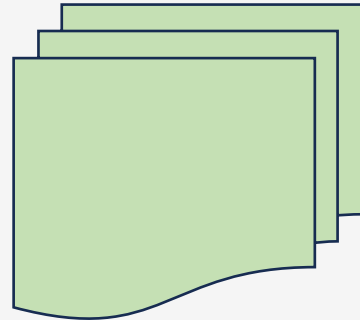


Real world : 512-1024 Dimensions

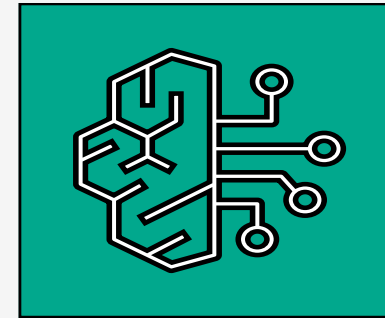
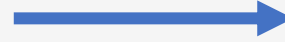
1. Text Splitting or Data Chunking



Source Data - PDF



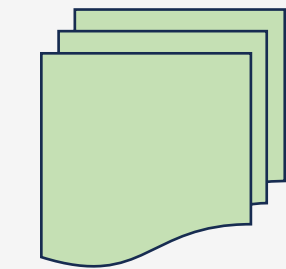
- Split by Character
- Split by Token
- Split by Code



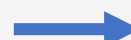
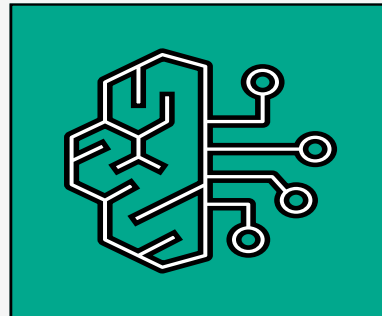
Embedding Model

2. Embedding Models

Output from Embedding Models



One Chunk (Split to few lines)



```
[
  0.022321066, -0.027544279, -0.006137953, 0.0024092742, -0.0003515296,
  -0.007889225, -0.01853968, -0.037405808, -0.004294867, -0.012602357,
  0.009385457, 0.02421176, -0.004383281, -0.011901848, -0.020702416,
  0.00431187, 0.055986296, 0.0040738326, 0.021300908, -0.03713377,
  -0.004944368, 0.012269106, 0.02614326, -0.019777471, 0.0017206672,
  0.014867109, 0.004597514, -0.021913003, -0.0042336574, 0.013180447,
  0.032699477, -0.027245032, -0.0112965545, -0.009358253, -0.004951169,
  -0.015030335, -0.007501565, -0.018906936, 0.0074675595, -0.0067568496,
  0.0011952856, 0.0035501514, 0.012316713, 0.016132105, -0.014839904,
  -0.0044002836, 0.010405616, -0.0066276295, -0.015751246, -0.007141109,
  0.02729944, -0.00031348618, -0.015669633, -0.015656032, 0.015696838,
  0.013554505, -0.001635654, 0.006069943, 0.009167824, -0.01917898,
  -0.006059741, 0.013969369, -0.03006067, 0.007691995, 0.0005028531,
  -0.005468049, -0.007263528, 0.009970348, -0.016254524, 0.0041248407,
  0.01964145, 0.037324198, -0.0011340762, -0.0013296065, 0.010392014,
  -0.01268397, -0.013037625, -0.006811258, -0.0045601083, 0.0028955496,
  0.027435461, -0.024320576, -0.006161757, 0.014350229, 0.009392259,
  -0.006328383, -0.025490358, 0.023273215, 0.0083925035, -0.003934411,
  0.023722084, -0.0072499258, -0.0053626327, 0.017818768, -0.03327077,
  0.019097365, 0.0022086431, 0.046709653, -0.016825814, -0.032182597,
  ... 1436 more items
]
```

Multiple chunks

Amazon Titan Embeddings

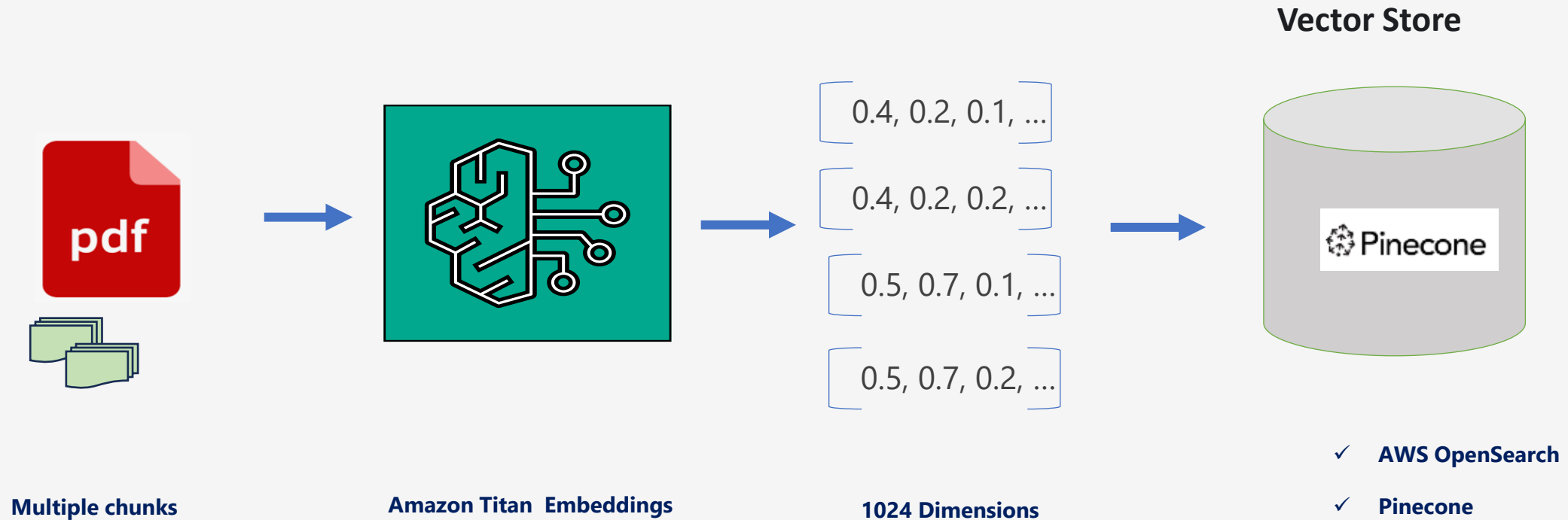
1024 Dimensions

Word Embeddings
Ex : great

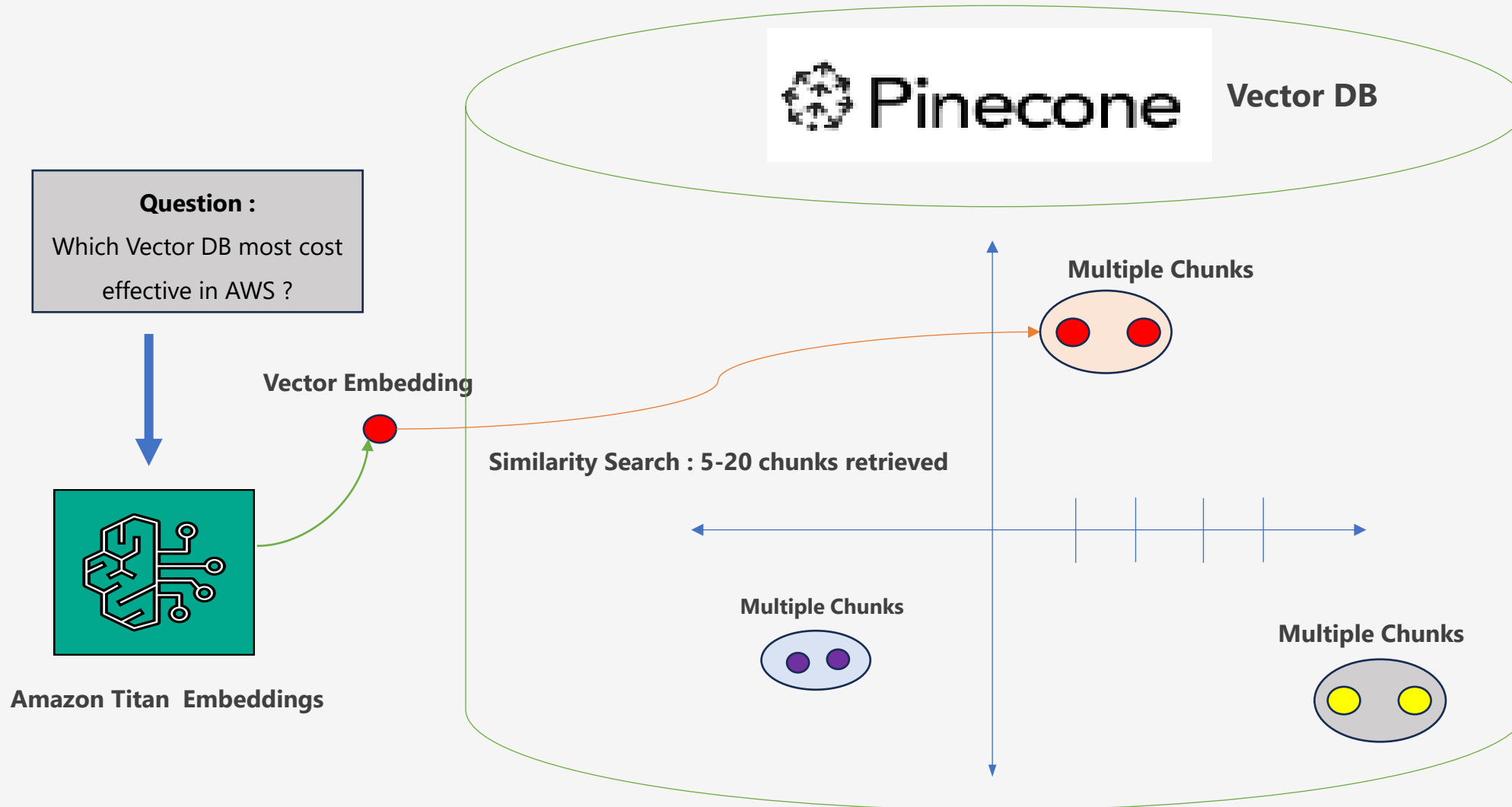
Sentence Embeddings
Ex. : Price of apple ?

Doc Embeddings

3. Vector Store?



4. Vector Search (Similarity Search)



5. Similarity Search : K-Nearest Neighbor Algorithm or ANN

KNN is a ***supervised machine learning algorithm*** used **classification and regression** tasks.

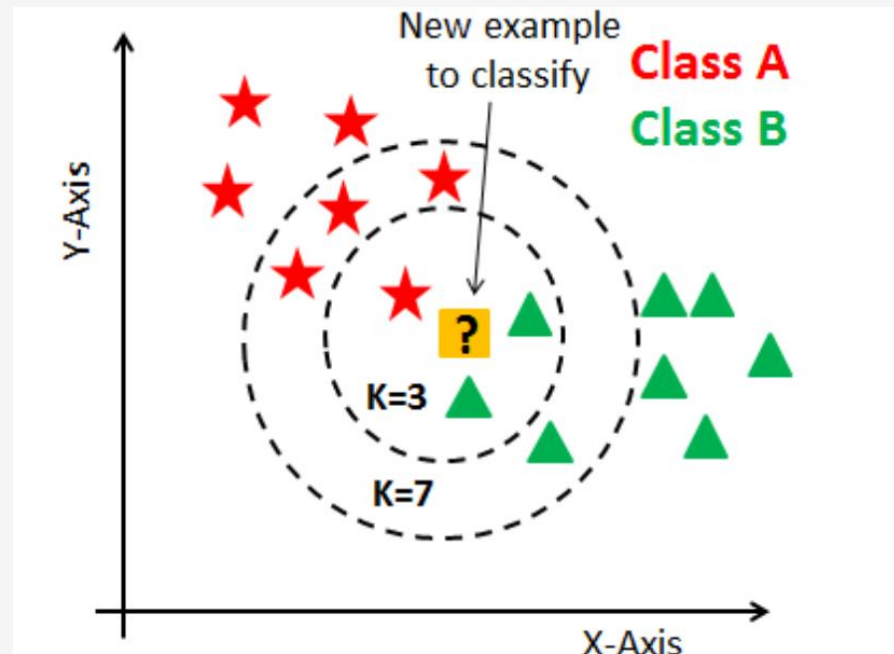


Image Source : Internet

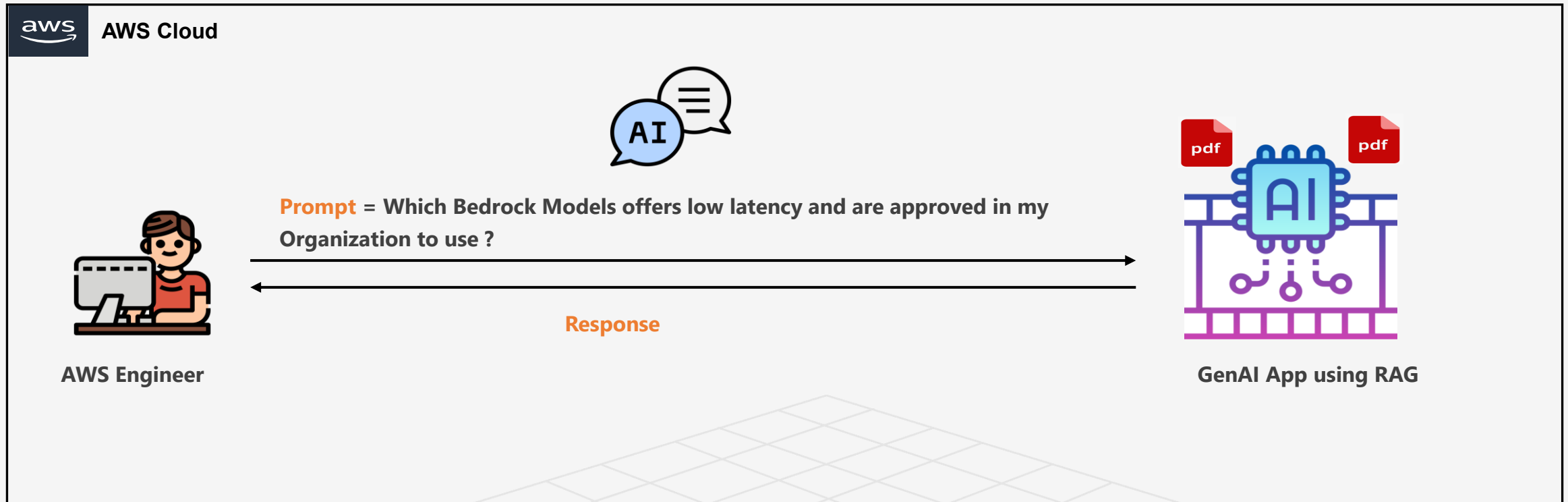
ANN - Approximate Nearest Neighbour

e-Learning App Architecture :

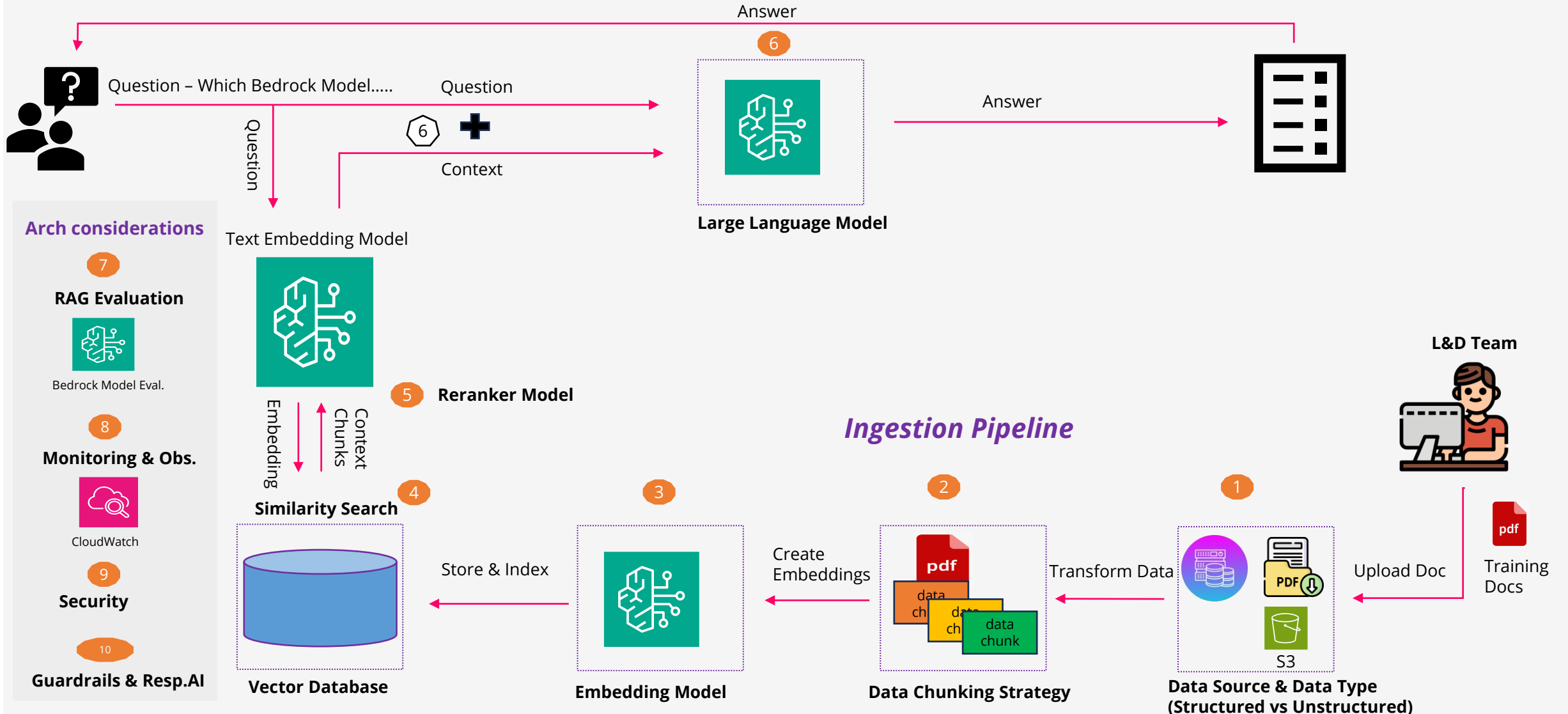
***10 Key Architecture considerations to build
Production Ready RAG Apps***

Use Case 2 : e-Learning Q&A App using Generative AI

Use Case : Build an e-Learning Generative AI application to accelerate organization learning on emerging AI technologies.



10 Architecture Considerations for building any production grade (RAG app) : e-Learning app



Amazon Bedrock Knowledge Bases

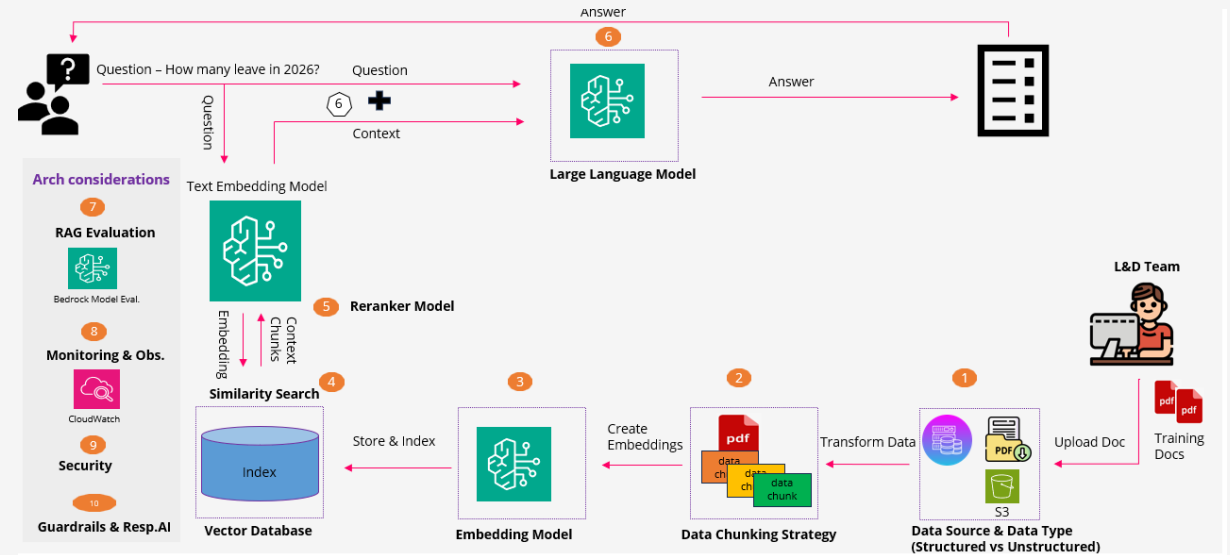
Overview



What is Amazon Bedrock Knowledge Bases

What it is Amazon Bedrock Knowledge Bases

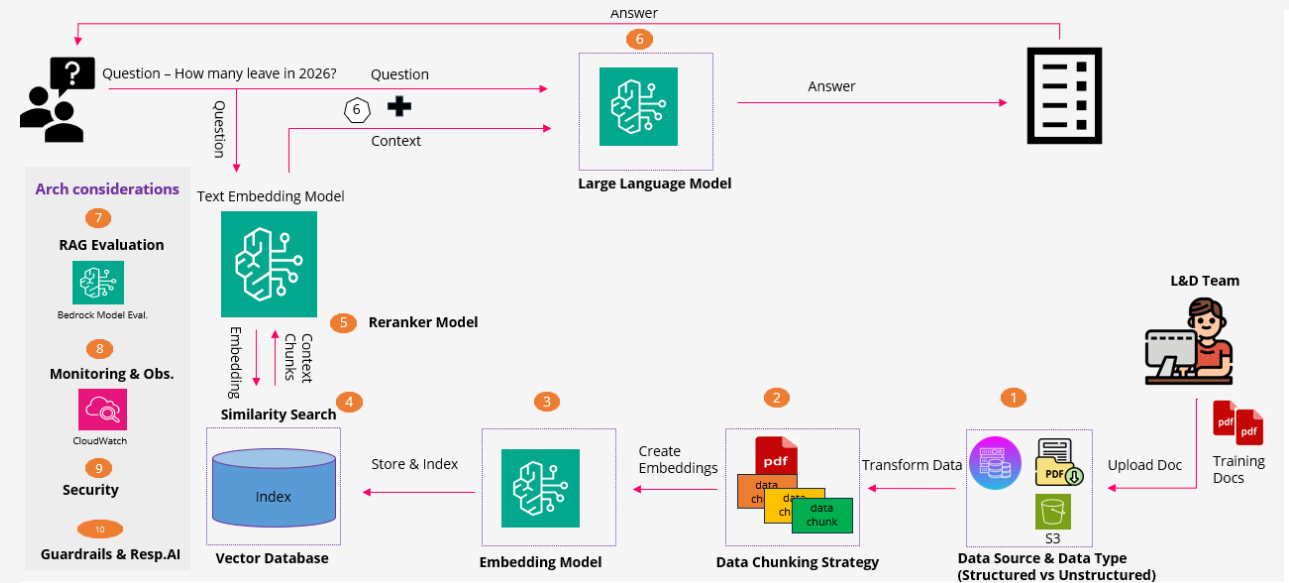
- **Fully managed RAG capability** in Amazon Bedrock
- Bedrock Knowledge Bases gives foundation models and agents **contextual information from company's private data sources** to deliver more relevant, accurate, and customized responses



What is Amazon Bedrock Knowledge Bases

How Amazon Bedrock Knowledge Bases Works

- Ingest data from multiple data sources (ex. PDF in Amazon S3 etc.)
- Chunk & embed documents automatically (Different chunking strategies – Fixed, Semantic)
- Store embeddings in vector store (multiple vector stores)
- Retrieve relevant chunks at query time and send to FM with user query.
- Foundation Model generates the answer



What is Amazon Bedrock Knowledge Bases

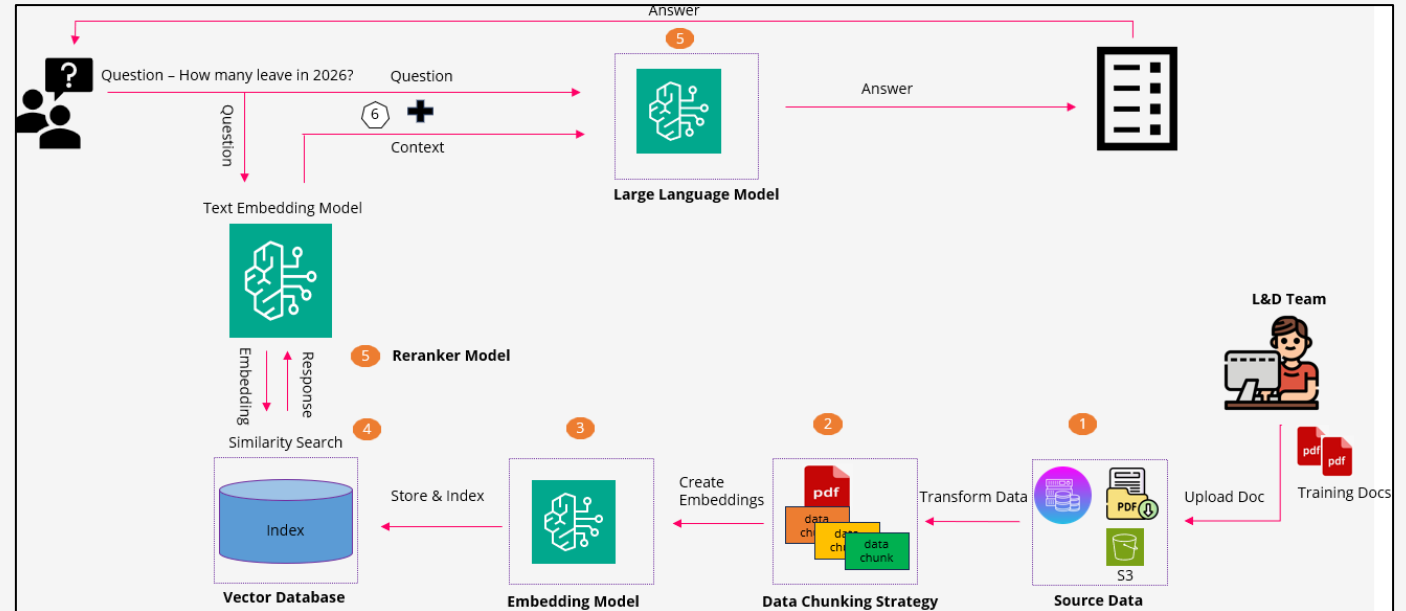
Retrieve API

- **Retrieval only**

- Only performs vector search
- Returns **retrieved chunks**
- No LLM response generation
- Used when you want **search results only**

RetrievalAndGenerate API

- Retrieves relevant chunks **AND**
- Sends them to a **Foundation Model** to generate an answer
- This is full RAG



Amazon Bedrock Knowledge Bases

Demo



Architecture Consideration 1 :
***Data Sources and Data Type (Structured &
Unstructured Data) for RAG***



RAG – Data Sources and Data Type – Structured & Unstructured

- Bedrock Knowledge Bases supports structured and unstructured data
- Supports a variety of documents- text, images, or multimodal documents that contain tables, charts, diagrams, and other images.

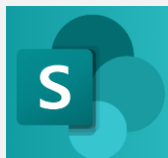


Supported Data Sources

Unstructured data from different Services/Data Sources



Amazon S3



Web Crawler



Custom

Structured Data from Redshift and other data sources via Glue catalogue



Amazon Redshift



AWS Glue Data Catalog
(AWS Lake Formation)

Amazon Bedrock Knowledge Bases – Structured Data

- Natural Language to SQL (NL2SQL)
- NL2SQL is hard
- Needs context to generate right SQL query



List the product descriptions of the transactions in gas stations in the Czech Republic

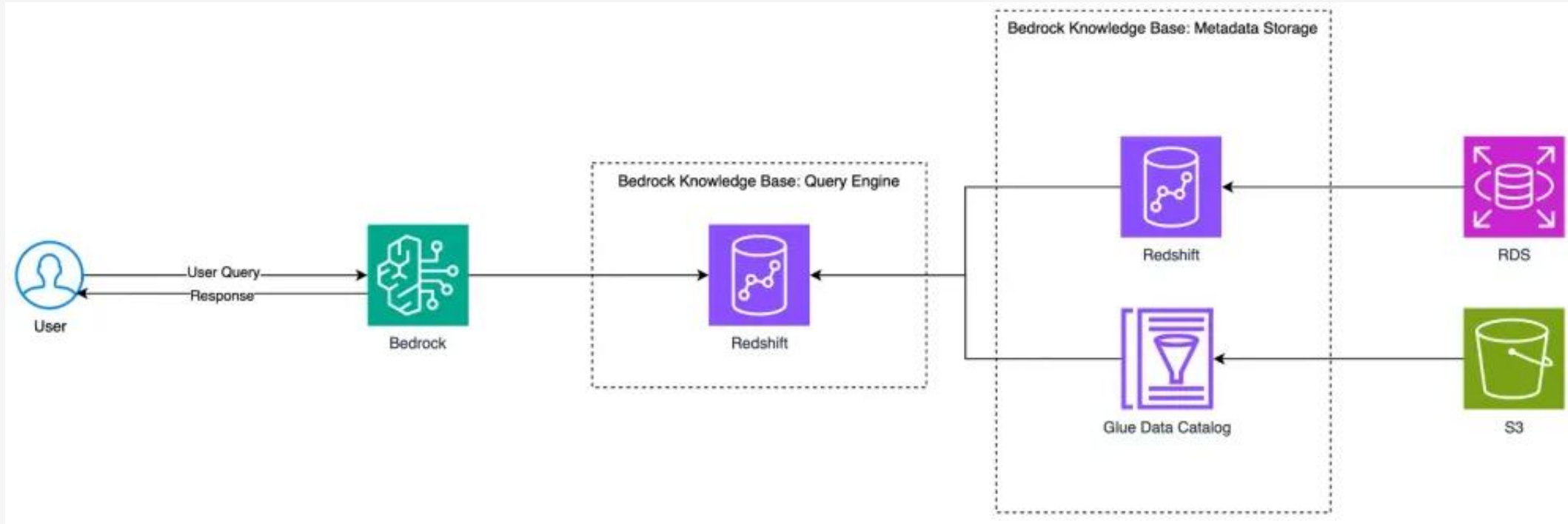
LLM output

```
SELECT DISTINCT T3.Description
FROM transactions_1k AS T1
INNER JOIN gasstations AS T2
ON T1.GasStationID = T2.GasStationID
INNER JOIN products AS T3
ON T1.ProductID = T3.ProductID
WHERE T2.Country = 'Czech Republic';
```

Expected output

```
SELECT DISTINCT T3.Description
FROM transactions_1k AS T1
INNER JOIN gasstations AS T2
ON T1.GasStationID = T2.GasStationID
INNER JOIN products AS T3
ON T1.ProductID = T3.ProductID
WHERE T2.Country = 'CZE';
```

Amazon Bedrock Knowledge Bases – Structured Data



- AWS Glue Data Catalog: A centralized metadata repository that stores metadata about data sources

Glue Data Catalogue Table

AWS Glue

Data catalog

Databases

Tables

Connections

Crawlers

Classifiers

ETL

Jobs

Triggers

Dev endpoints

Tutorials

Add crawler

Explore table

Add job

Resources 

Tables > data

Last updated 18 Jan 2018 Table Version (Current version) 

Add column

Edit schema

Cancel Save

Showing: 1 - 11 of 11  

	Column name	Data type	Key	
1	id	string		×
2	type	string		×
3	actor	struct		×
4	repo	struct		×
5	payload	struct		×
6	public	boolean		×
7	created_at	string		×
8	org	struct		×
9	year	string	Partition (0)	×
10	month	string	Partition (1)	×
11	day	string	Partition (2)	×

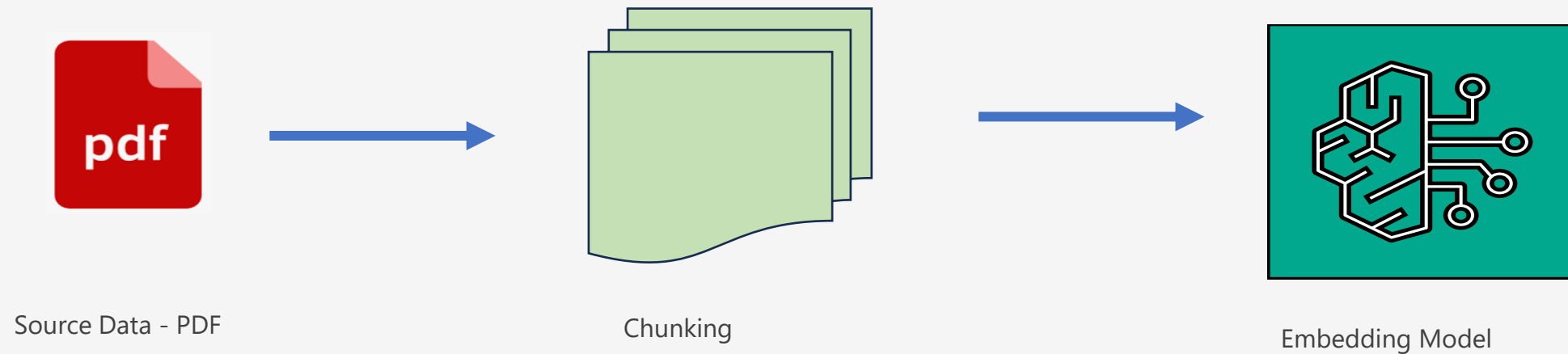
Architecture Consideration 2 :

Chunking Strategy



What is Data Chunking?

Chunking is the process of **splitting the documents or content into manageable text segments** (chunks) for efficient data retrieval.



Chunking Strategy

Lionel Messi - Born in Rosario, Argentina, in 1987, Lionel Messi is widely regarded as one of the greatest football players of all time, and his illustrious career proves why. The Argentinean footballer, who holds a record eight Ballon d'Or awards, has displayed his talent early on and has always been destined for greatness. When Messi was 13 years old, he and his family moved to Barcelona, where the club assisted him in treating his growth hormone deficiency. He started playing for FC Barcelona's U14 team. The Argentinean quickly rose through the ranks, impressing everyone with exceptional skills and talent. At 17, he made his first senior appearance for the club and became a vital player for the Blaugranas.....



1. No Chunking

- Entire document as ONE embedding
- Retrieval precision is poor
- **UC : Used only for very small docs**

Lionel Messi - Born in Rosario, Argentina, in 1987, Lionel Messi is widely regarded as one of the greatest football players of all time, and his illustrious career proves why. The Argentinean footballer, who holds a record eight Ballon d'Or awards, has displayed his talent early on and has always been destined for greatness.

2. Fixed Chunking (Default)

Chunk size: ~120 words (varies)

Overlap: Can be defined

Lionel Messi - Born in Rosario, Argentina, in 1987, Lionel Messi is widely regarded as one of the greatest football players of all time, and his illustrious career proves why. The Argentinean footballer, who holds a record eight Ballon d'Or awards, has displayed his talent early on and has always been destined for greatness.

Chunking Strategy

3. Semantic Chunking (Best Meaning-Based)

Chunks split by topic/meaning , not size.

Chunk 1 – Introduction & Greatness

Born in Rosario, Argentina, in 1987, Lionel Messi is widely regarded as one of the greatest football players of all time, and his illustrious career proves why. The Argentinean footballer, who holds a record eight Ballon d'

Chunk 2 – Childhood & Move to Barcelona

When Messi was 13 years old, he and his family moved to Barcelona, where the club assisted him in treating his growth hormone deficiency.

4. Hierarchical Chunking (Parent → Child) Example – Technical Manual

Parent 1 – Early Life

Born in Rosario, Argentina, in 1987... destined for greatness.

Parent 1 → Child

Child 1.1: Born in Rosario, Argentina, in 1987, Lionel Messi... eight Ballon d'Or awards.

Parent 2 – Barcelona Rise

When Messi was 13 years old...vital player for the Blaugranas.

Child 2.1: When Messi was 13 years old... growth hormone deficiency.

Chunking Strategy Decision Points

Strategy	Pros	Cons	Best Use Cases	When to Choose (Decision Rule)
No Chunking	- Simplest to implement - Lowest embedding cost	Very low retrieval precision	Tiny documents Short FAQs	- Choose only if documents are small (Few hundred words) - Precision is not critical
Fixed Chunking	- Easy to implement - Good cost-quality balance	Can split topics mid-way	- Default for RAG apps - Blogs, guides, mixed docs	Choose as default when you need fast setup and acceptable accuracy
Hierarchical Chunking	High precision for structured/hierarchical docs	More complex & higher cost	- Technical manuals - Legal/compliance docs	Choose when documents are large and structured and accuracy matters
Semantic Chunking	- Highest retrieval accuracy - Best answer quality	More complex logic & Higher preprocessing effort	- Customer support bots - Medical/legal/finance - High-stakes RAG	Choose when answer quality > simplicity and errors are costly

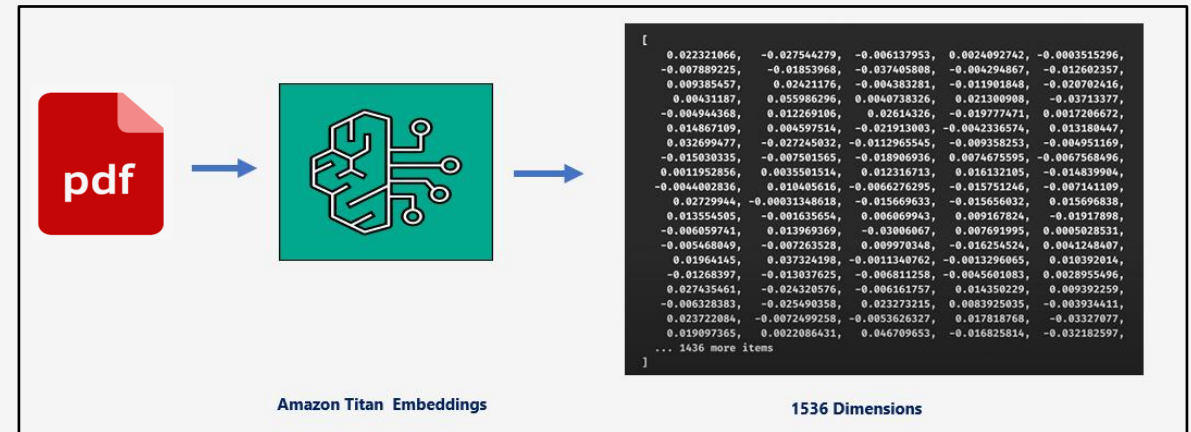
Architecture Consideration 3 :

Embedding Model



What are Embedding Models & How to select them ?

- Text embeddings represent **vector representations of unstructured text** and images such as documents, sentences etc.
- Input Modality – Text and Text + Image
- Output – Various dimensions 256, 512 and 1024
- How to select the right Embedding Model for Use Case**
- Quality vs Cost vs Latency
 - Higher-dimensional embeddings → better recall, higher cost
 - Lower-dimensional embeddings → faster, cheaper, lower accuracy
- Match Language & Multilingual Needs
- Domain fit – Fine Tuning if highly domain specific



Amazon Titan Text Embeddings V2 model

- Model ID** – `amazon.titan-embed-text-v2:0`
- Max input text tokens** – 8,192
- Max input text characters** – 50,000
- Languages** – English (100+ languages in preview)
- Output vector size** – 1,024 (default), 512, 256
- Inference types** – On-Demand, Provisioned Throughput
- Supported use cases** – RAG, document search, reranking, classification, etc.

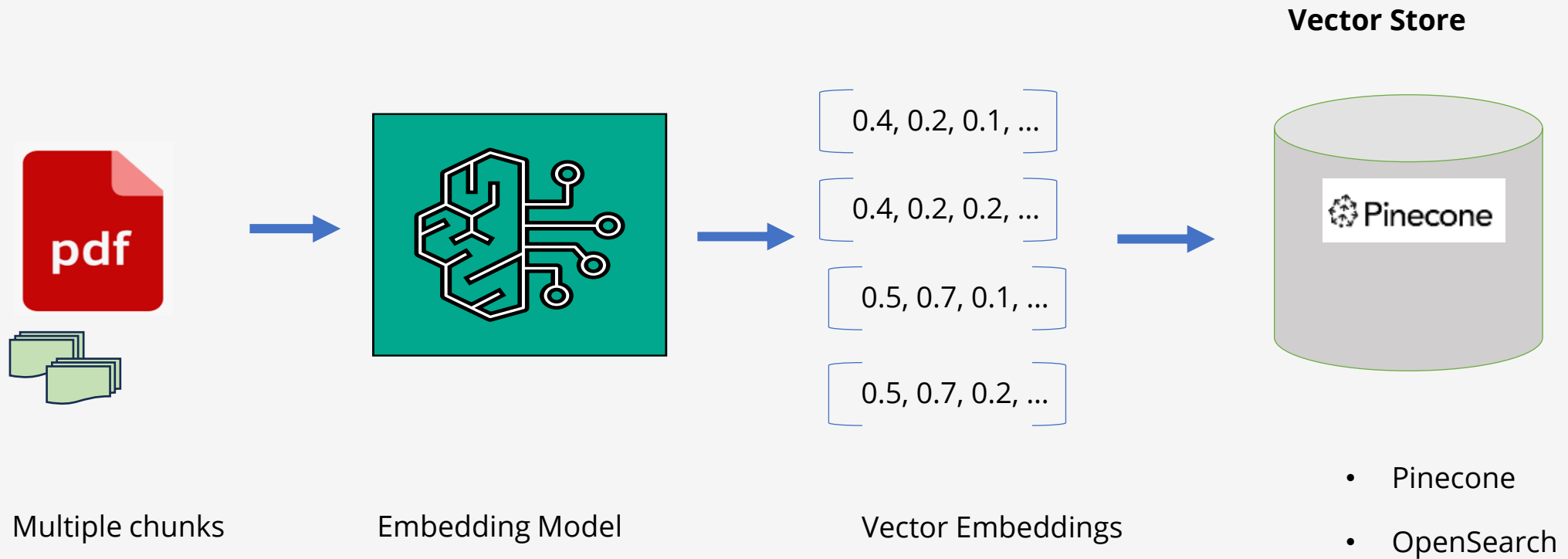
- Model ID - Titan Multimodal Embeddings G1**
- Input modalities - Text, Image

Architecture Consideration 4 :

Vector Database Strategy



What is Vector DB ?



AWS Vector DB Options - 1

Amazon Aurora PostgreSQL with pgvector(plugin/extension)

- Relational database with vector RAG
- Integrated with existing Aurora databases
- Cost-effective for smaller deployments
- **UC : Best for applications needing both traditional SQL and vector capabilities**



Amazon Aurora

Amazon Neptune Analytics

- Neptune Analytics – **Graph-based vector search**
- Ideal for connected data use cases – Social Media
- **UC : Neptune Analytics – Ideal for complex relationship-based queries and knowledge graphs**

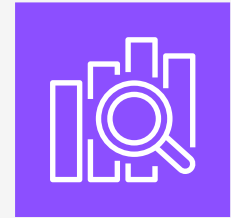


Amazon Neptune

AWS Vector DB Options - 2

Amazon OpenSearch Serverless

- **Amazon OpenSearch - Serverless and Provisioned mode** OCUs (OpenSearch Compute Units)
- **Search and analytics focus** - log analytics and monitoring, Enterprise search, E-commerce Product Search
- Flexible search options including traditional **key word search, vector search**, and hybrid approaches
- Store and search billions of high-dimensional vectors in milliseconds using advanced algorithm
- **UC - Strong in full-text search, analytics and vector search with low latency (sub-second) response.**



Amazon OpenSearch Service

Pinecone

- Purpose-built vector database
- Optimized for **high-performance similarity search**
- **UC - High-performance vector search and RAG**



AWS Vector DB Options - 3

Redis Enterprise Cloud

- In-memory data platform with vector search support (Redis Stack / RediSearch)
- Combines vector search + caching
- In-memory vector search for RAG workloads
- Delivers ultra-low-latency, real-time vector similarity search
- **UC** : Best suited for real-time GenAI applications requiring sub-millisecond to millisecond responses



In summary

- OpenSearch → search + analytics + vectors
- Pinecone → large-scale, purpose-built vector DB
- Aurora pgvector → relational + vectors
- Redis → fastest, in-memory, real-time

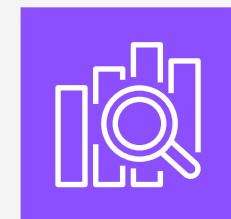
Amazon S3 Vectors



S3 Vectors and OpenSearch

Cost-optimized **AI-ready storage** with native support for storing vector embeddings at scale, reducing total costs by up to 90%

- **Use Case:** Dealing with **large-scale vector embeddings** and **latency is low priority**
- Search - **OpenSearch performs low-latency similarity search**
- Benefits: Cost-effective storage, efficient similarity search, scales well
- Amazon S3 is used for cost-optimized vector storage, while OpenSearch performs vector and hybrid similarity search.



Amazon OpenSearch Service

Amazon Kendra



Amazon Kendra

- Amazon Kendra is a fully managed **enterprise-grade intelligent search service and RAG**
- Uses natural **language processing (NLP) and machine learning for semantic search**
- Primary use case: Enterprise search and question answering
- Provides native connectors to common enterprise data sources (e.g., S3, SharePoint, Confluence, Salesforce)
- Supports quick setup without requiring deep search or ML expertise
- Built in vector storage - Automates ingestion, document processing, and relevance ranking
- **UC : Primary goal is an internal company search for documents, FAQs, or procedural information and RAG**



Amazon Kendra

Architecture Consideration 5 :
Retrieval Optimization – Reranking



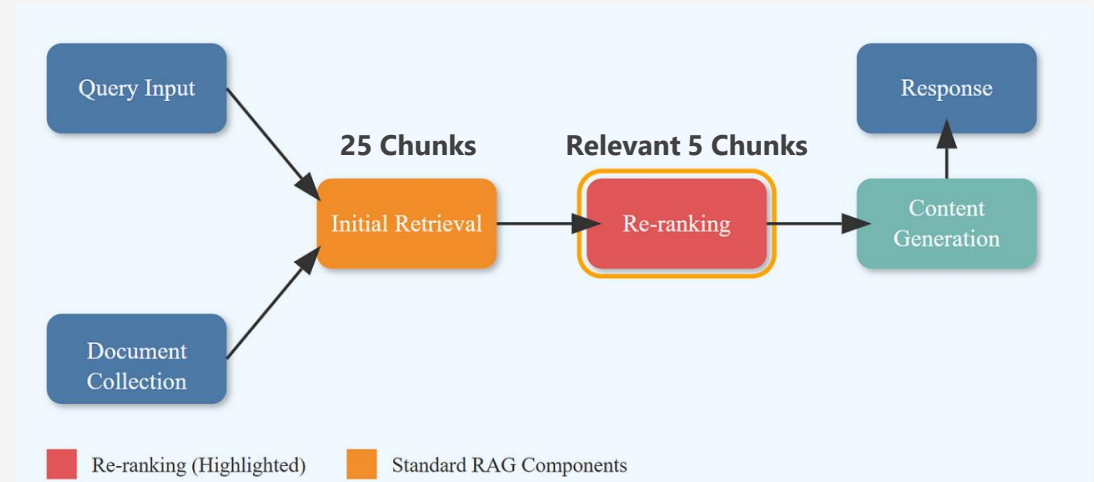
Reranker Model for improving the accuracy of the response

- **Reranker models are used to improve the relevance of the retrieved results** for any query.
- Reranker model calculates **relevance of chunks to a query and reorders the results** based on the scores that it calculates.

Reranking Example

User asks a question - “**What is the enterprise refund policy?**”

- **Retriever fetches chunks**
Vector search returns top 20–50 chunks (fast, approximate)
- **Reranker evaluates relevance**
Reranker model scores each (*query, chunk*) pair for true relevance
- **Results are reordered** - Only the top 5–8 most relevant chunks remain
- **LLM generates answer**
The model sees cleaner, more accurate context



Reranking Models

Provider	Model	Model ID
Amazon	Rerank 1.0	amazon.rerank-v1:0
Cohere	Rerank 3.5	cohere.rerank-v3-5:0

Benefits

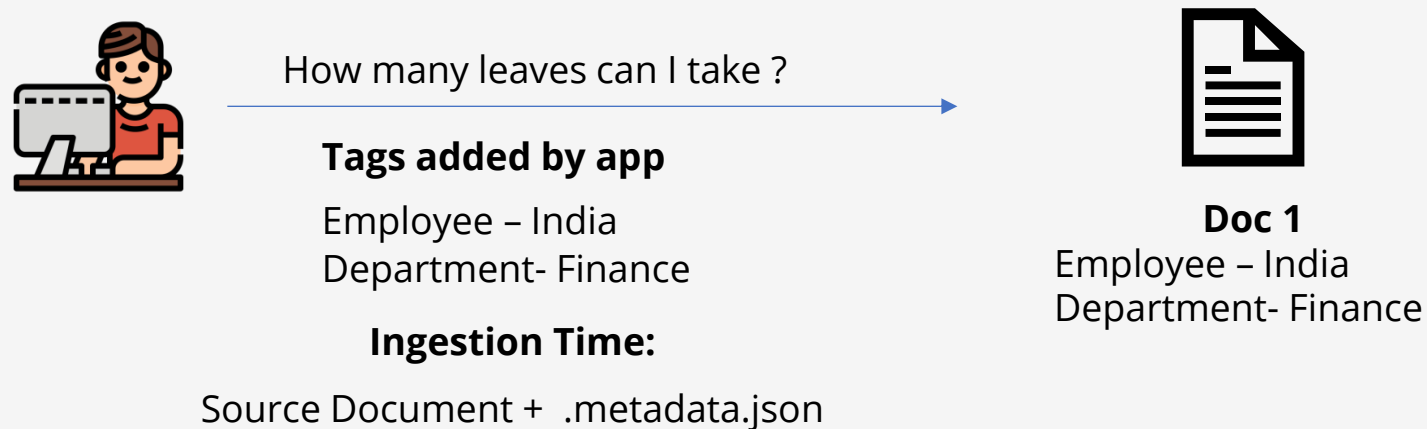
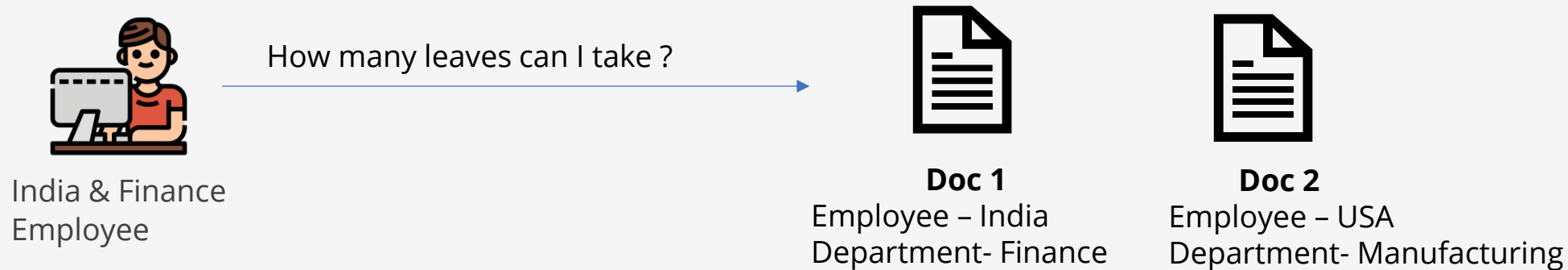
- **Improves answer accuracy** by selecting the **most relevant context**
- **Reduces hallucinations** by removing weak or irrelevant chunks
- **Lowers token usage** by sending fewer, higher-quality chunks to the LLM

Retrieval Optimization - Metadata Filtering



Metadata Filtering

- Metadata filtering allows you to **filter query results based on specific attributes** attached to your vectors.



Benefits of Metadata Filtering:

- Reduces number of chunks that are being searched
- Reduction in cost of querying the vector store
- Improvement in accuracy.

- Use metadata filters with query operations to find vectors that match both **similarity criteria and specific metadata conditions**

Metadata Filtering - .metadata.json

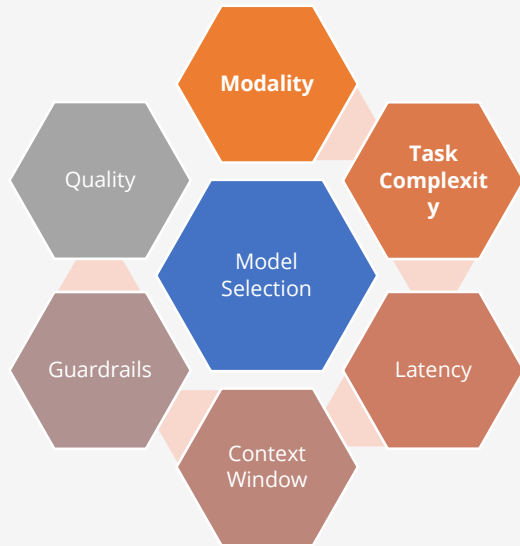
```
{  
  "metadataAttributes" : {  
    "tag" : "project EVE",  
    "employee" : India,  
    "department": "finance"  
  }  
}
```

Architecture Consideration 6 :

LLM Selection



Foundation Model Selection Strategy



1. **Modality** : Text | Image | Multi-Modal | Embedding – Manufacturing use case - multimodal
2. **Task Complexity** : Summarization vs reasoning ---> Small Language Model (SLM) vs LLM trade-offs
3. **Context Window** : 8K - 200K token size – Size of the user input (legal docs) --> Long context models
4. **Guardrails & Safety** : Built in model guardrails & additional capabilities – Toxicity, content moderation
5. **Quality & Accuracy of output** : Accuracy, reasoning, completeness ---> Bedrock model output evaluation
6. **Cost Predictability & Token Efficiency** : Cost optimization – cost per 1000 tokens
7. **Regional Availability, Quotas** – Service quotas & rate limits, region availability
8. **Latency** : Response time for requirements ---> Latency-optimized inference models
9. **Proprietary or Open Source** : Prefer open source
10. **Language Support** – Available in the particular language for Use Case

Select the smallest, safest & most cost-efficient model that meets accuracy, latency & governance requirements for the workload.

Architecture Consideration 7 :

RAG Observability & Monitoring



Bedrock Knowledgebase Monitoring

Bedrock Knowledge Base Logging – **Ingestion Logs & Resource Logs**

1. Ingestion logs = “Did we load the knowledge correctly?”

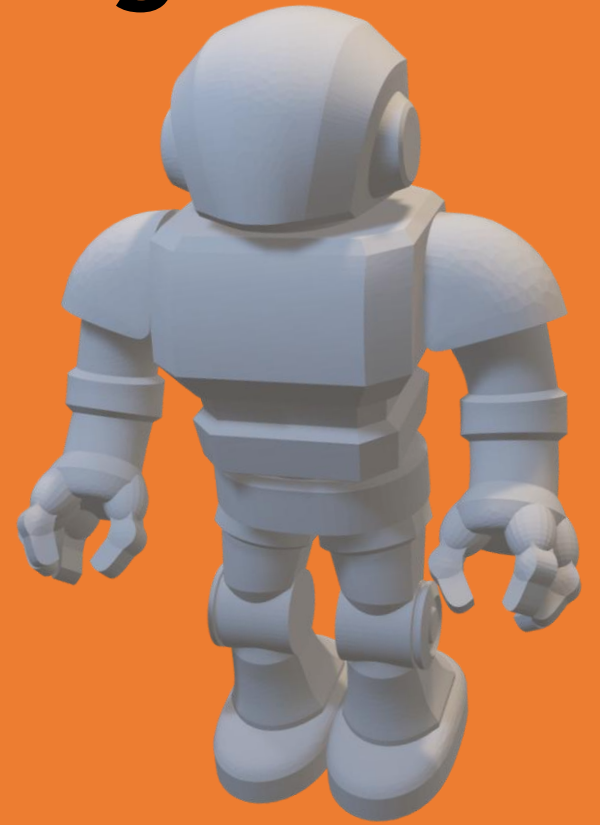
- Logs record **what happens when data is added or updated** in a Knowledge Base.
- Track the **offline pipeline that prepares your data** for RAG.
- Document **ingestion start / completion****Chunking operations & Embedding generation**

2. Resource logs = “How is the knowledge being queried and used?”

- Logs record **runtime usage of the Knowledge Base** and track live query-time behavior.
- Logged activities
- Retrieve API calls
- RetrieveAndGenerate calls
- Request metadata (timestamps, KB ID)
- Response status (success / failure)

RAG Observability & Monitoring

Demo



Architecture Consideration 8 :

RAG Evaluation



Amazon Bedrock Evaluations - RAG

- Bedrock evaluations can be used to **evaluate the performance of Bedrock models and knowledge bases**, as well as **models and Retrieval Augmented Generation (RAG) sources outside of Amazon Bedrock**.
- Amazon Bedrock can be evaluated based on different **performance metrics for a model** and the **correctness** of a **knowledge base in retrieving information and generating responses**.



Inputs

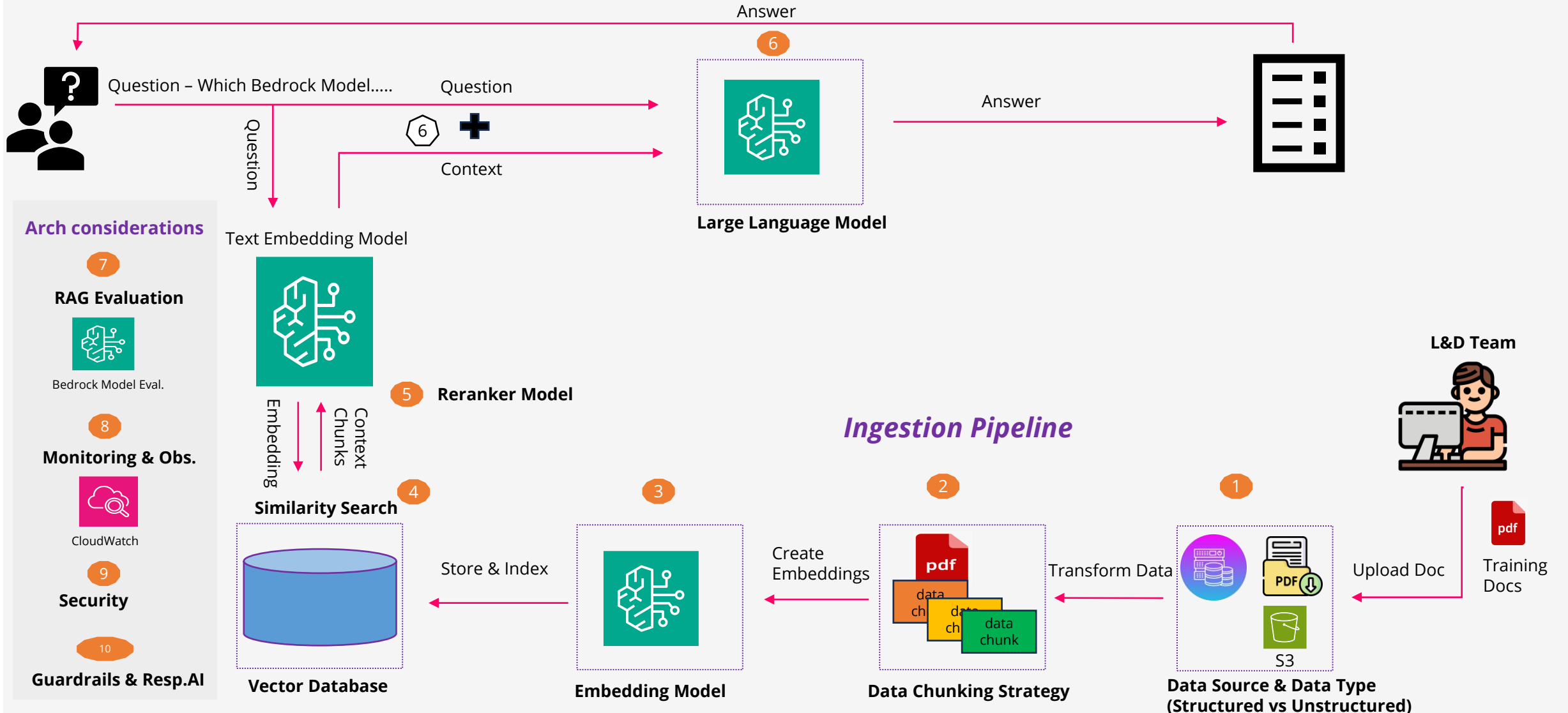
- Prompt
- Reference Response



Architecture Consideration 9 & 10 :
Security, Guardrails and Responsible AI
will be discussed in separate sections



9 & 10 Architecture Decision will be discussed in separate section.



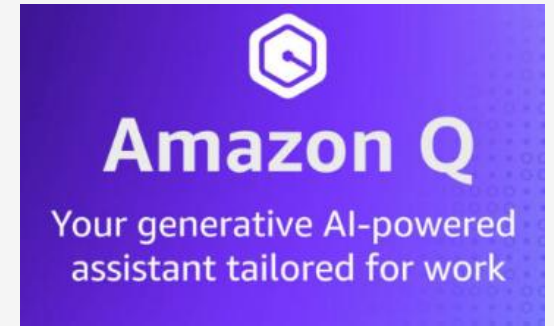
Section on :

Amazon Q - Business



What is Amazon Q Business?

Amazon Q Business is a **fully managed, generative-AI powered assistant**



- Generate Content



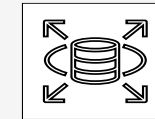
- Summarize Data



- Chat with Enterprise Data (40+ connectors)



S3



RDS



GDrive

40+ Pre Built
Connectors

- Assign Tasks (Plugins such as Jira)



ChatGPT with Enterprise Controls

Amazon Q for Business – A typical day for a Marketing Manager



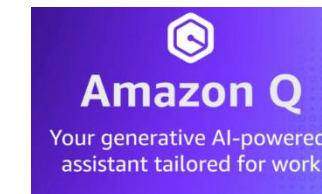
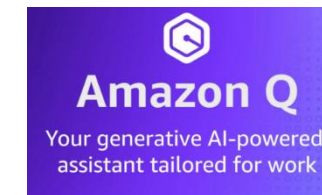
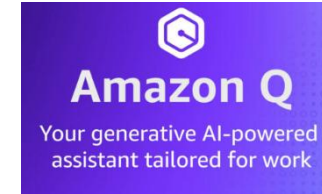
Improve employee productivity with Amazon Q

Feedback report on the impact of marketing campaign (20 Pages)

Draft e-mail to prospective client outlining steps to launch campaign for a juice drink

Chat with internal documents about Brand Guidelines

Assign Task in Jira to direct reports



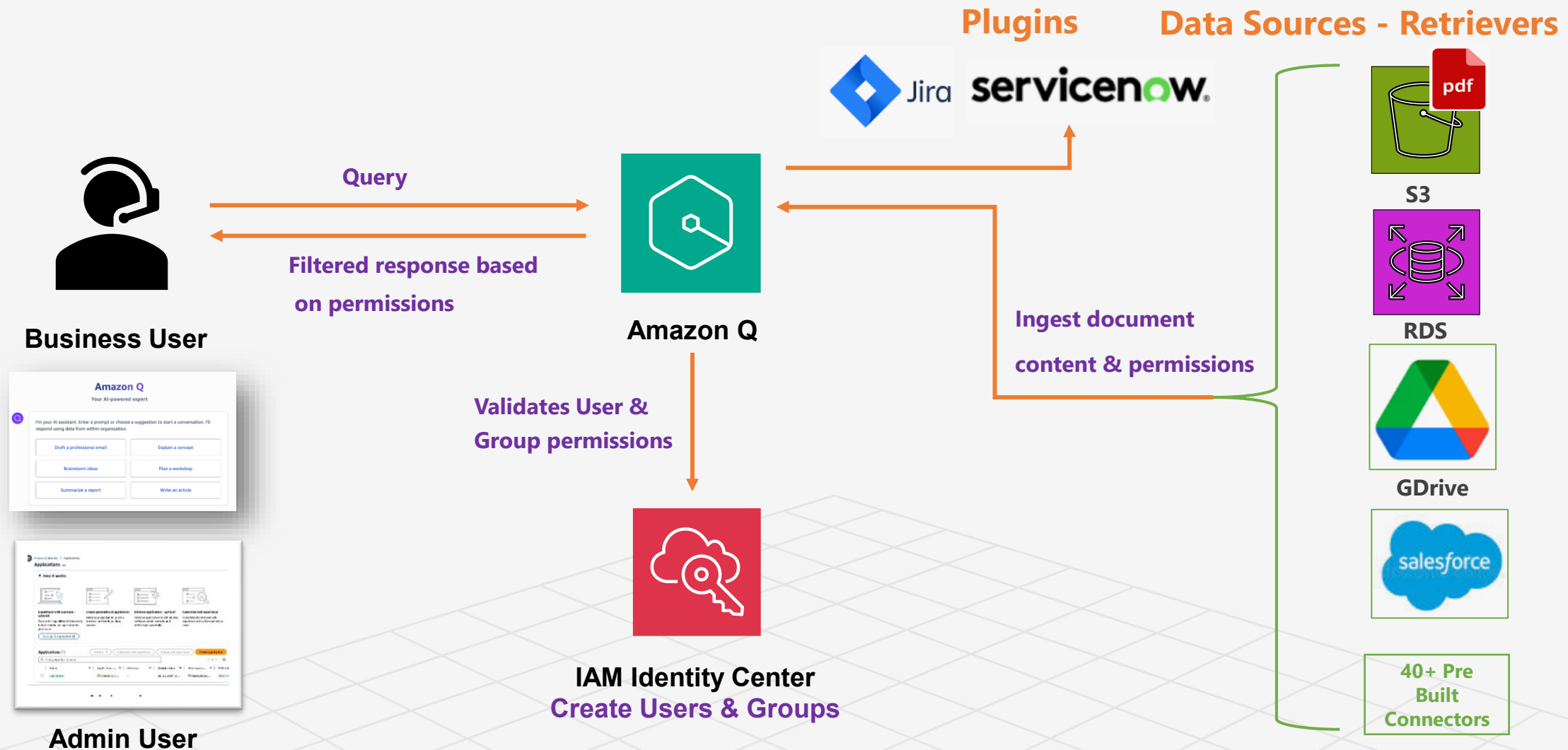
Summarize

Generate Content

Enterprise Data Search

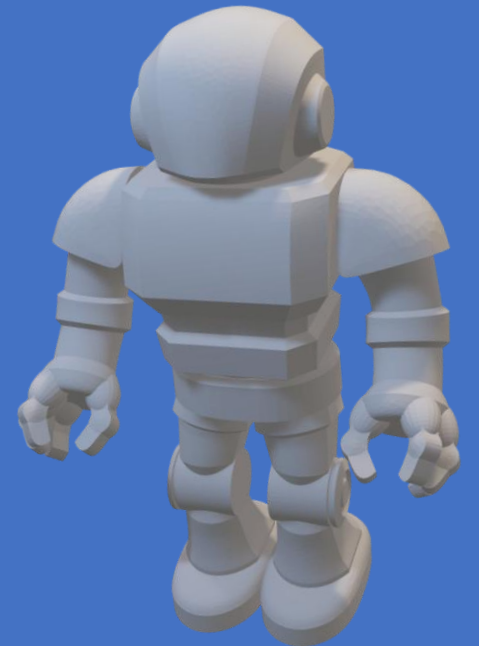
Task Mgmt.

What is Amazon Q for Business - Architecture



Section on :

AI Agents and Agentic AI Architectures



***Section on :
Evolution of AI Agents***



What are AI Agents – Lets understand through a Use Case



Hey I am John

I work at Micro-cool Corporation

I am planning a 2-week Skiing vacation from mid of December if I have leaves

Need help with vacation planning.

Can a GenAI App built using Large Language Model help John ?

John's Ask :

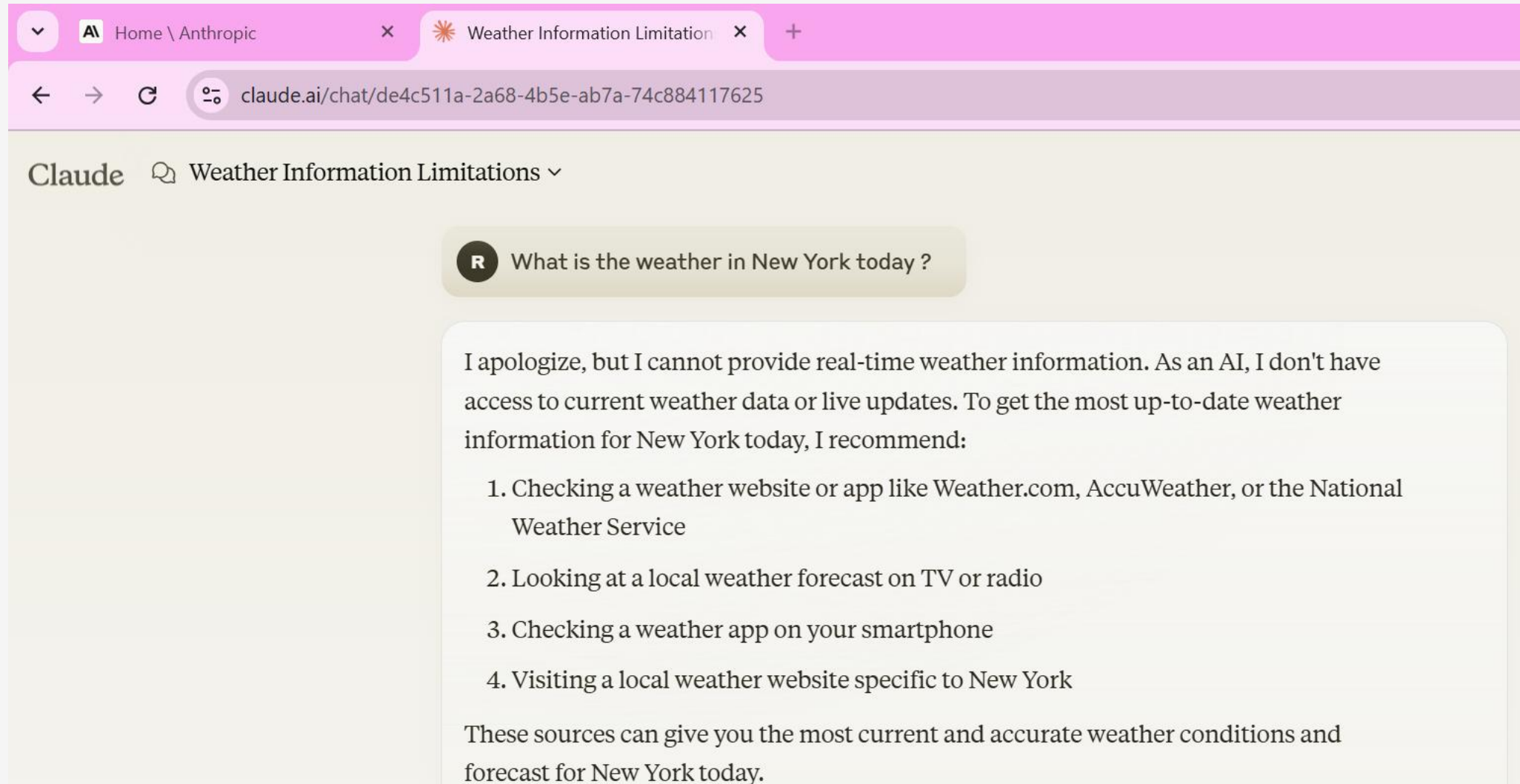
I am planning a 2-week Skiing vacation from mid of December if I have sufficient leaves

Collection of Tasks & Decisions

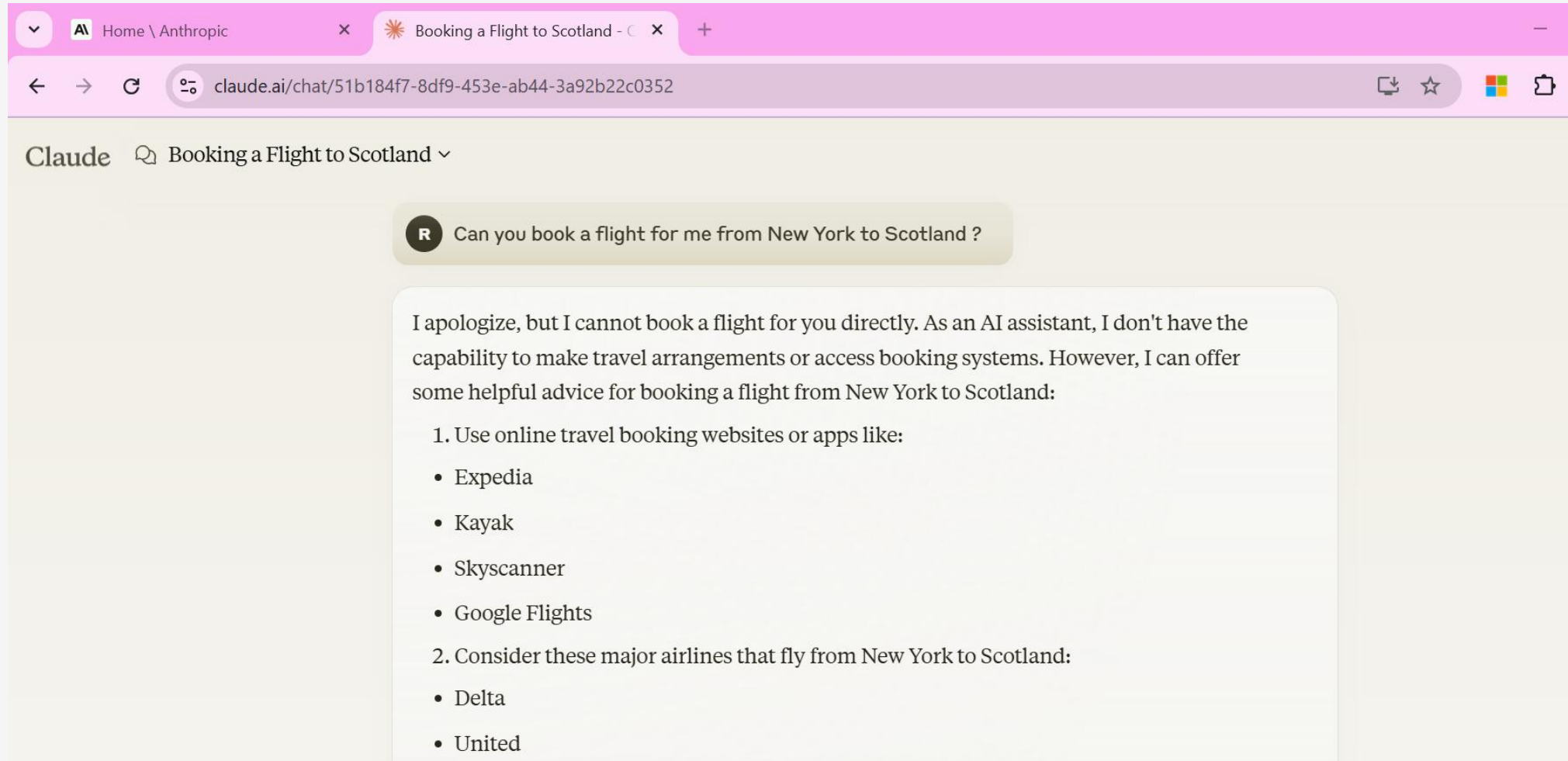
☒	_____
☒	_____
☐	_____
☐	_____
☐	_____
☐	_____

- ❖ Check my Leave balance
- ❖ Research locations for skiing
- ❖ Check weather report for location
- ❖ Book flight
- ❖ Book hotel

Limitations of LLM's to handle complex user requests



Limitations of LLM's to handle complex user requests

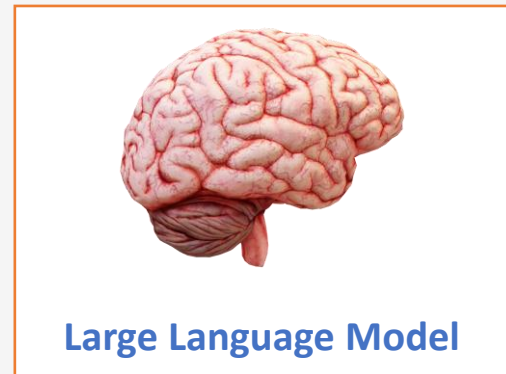


Limitations of LLM's to handle complex user requests

User Request

I am planning a 2-week Skiing
vacation from mid of December
if I have sufficient leaves

Micro-cool GenAI App



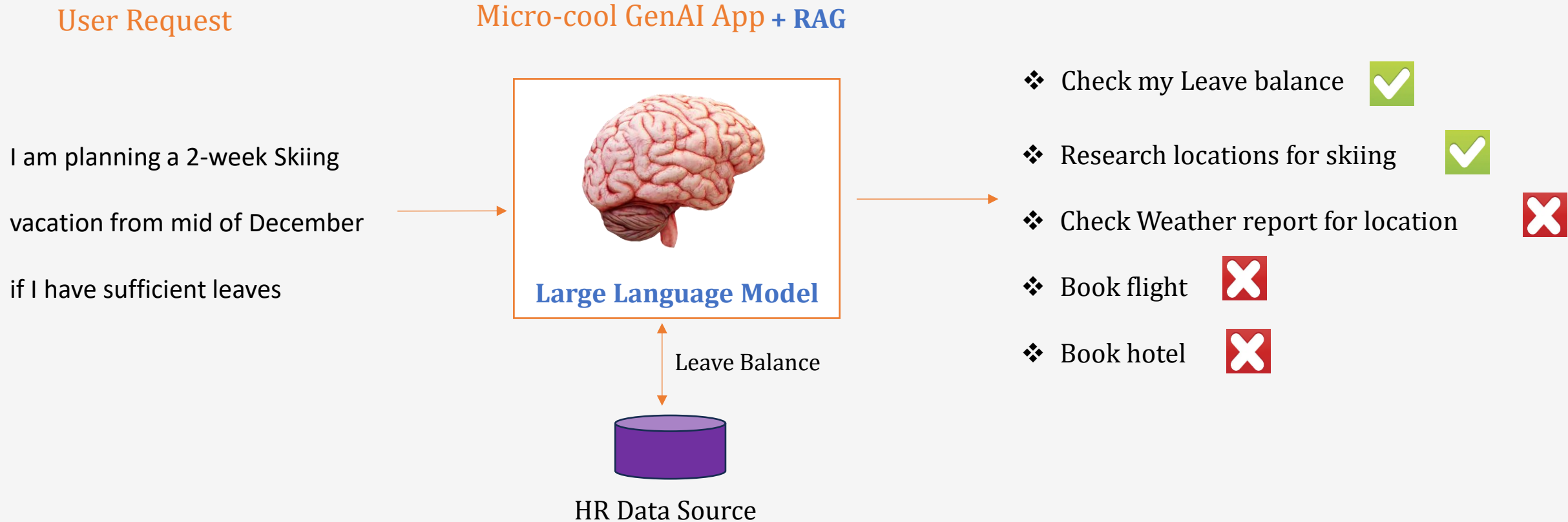
- ❖ Check my Leave balance ❌
- ❖ Research locations for skiing ✅
- ❖ Check Weather report for location ❌
- ❖ Book flight ❌
- ❖ Book hotel ❌

Limitations and Challenges of LLM based Applications

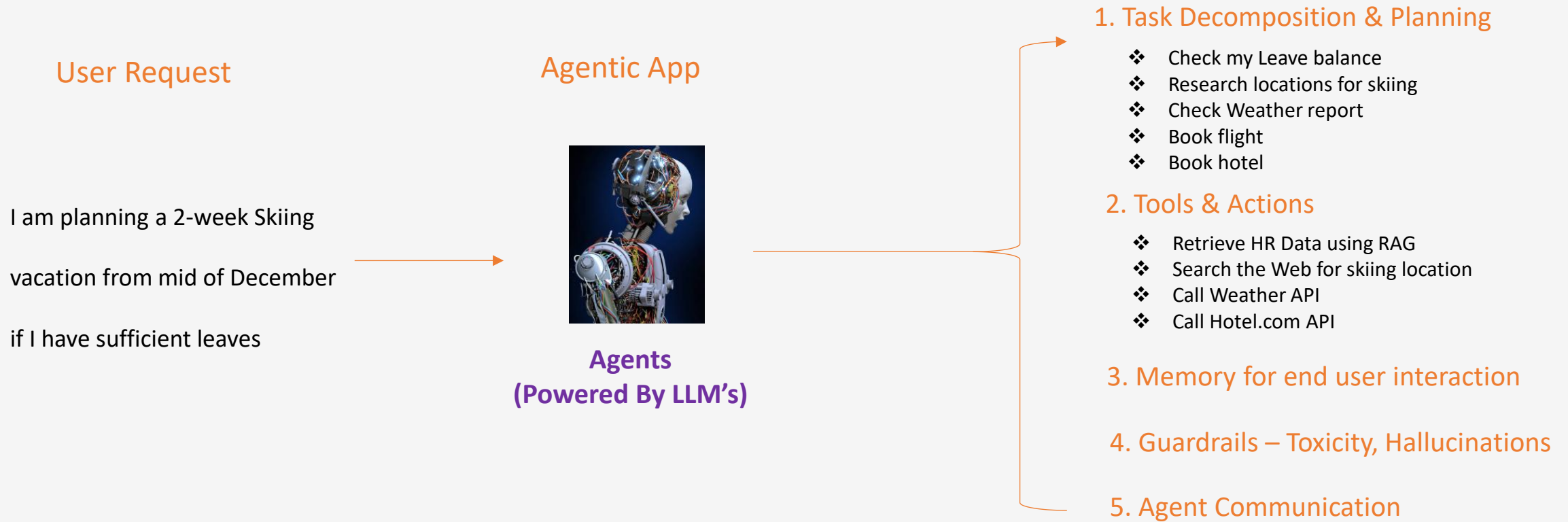
- Answer generic questions based on publicly available training data **but not company specific data**
- No way to **access current data or external API's** – Today's weather, book hotel etc.
- Answer only questions based **on training data with cut off date**



Limitations of **LLM's + RAG** to handle complex user requests



How **AI Agent** come to John's rescue



Core Capabilities of AI Agents

1. Planning

Dynamic Task Planning

2. Tools & Action

Access any data via API's

3. Memory

User Interaction

4. Guardrails

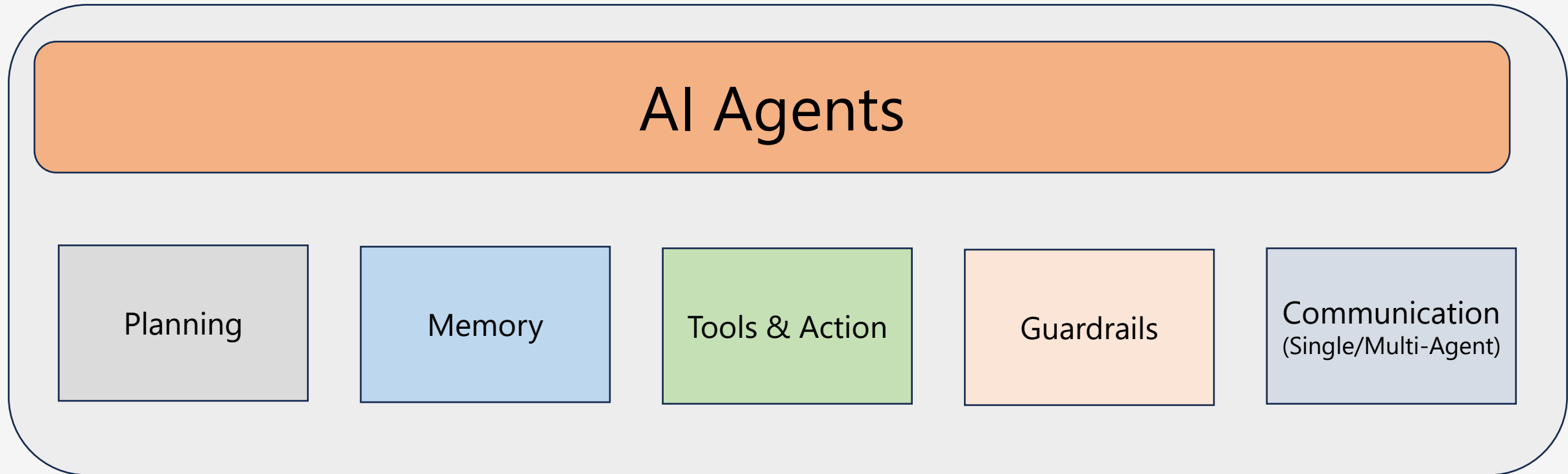
Toxicity & Hallucinations

5. Communication

Single or Multi-Agent

AI Agents – Key Characteristics

AI Agents have generally 5 Key characteristics



What does Agentic Mean ?



Agentic Definition

Stanford psychologist Albert Bandura coined the term to describe **people** who are **self-organizing, proactive, self-reflective, and self-regulating.**

What is Agentic AI ?



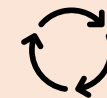
What Is Agentic AI?



Agentic AI uses **sophisticated reasoning and iterative planning** to **autonomously** solve complex, multi-step problems.



Intelligent Autonomous Systems



Plan, Reason and Act



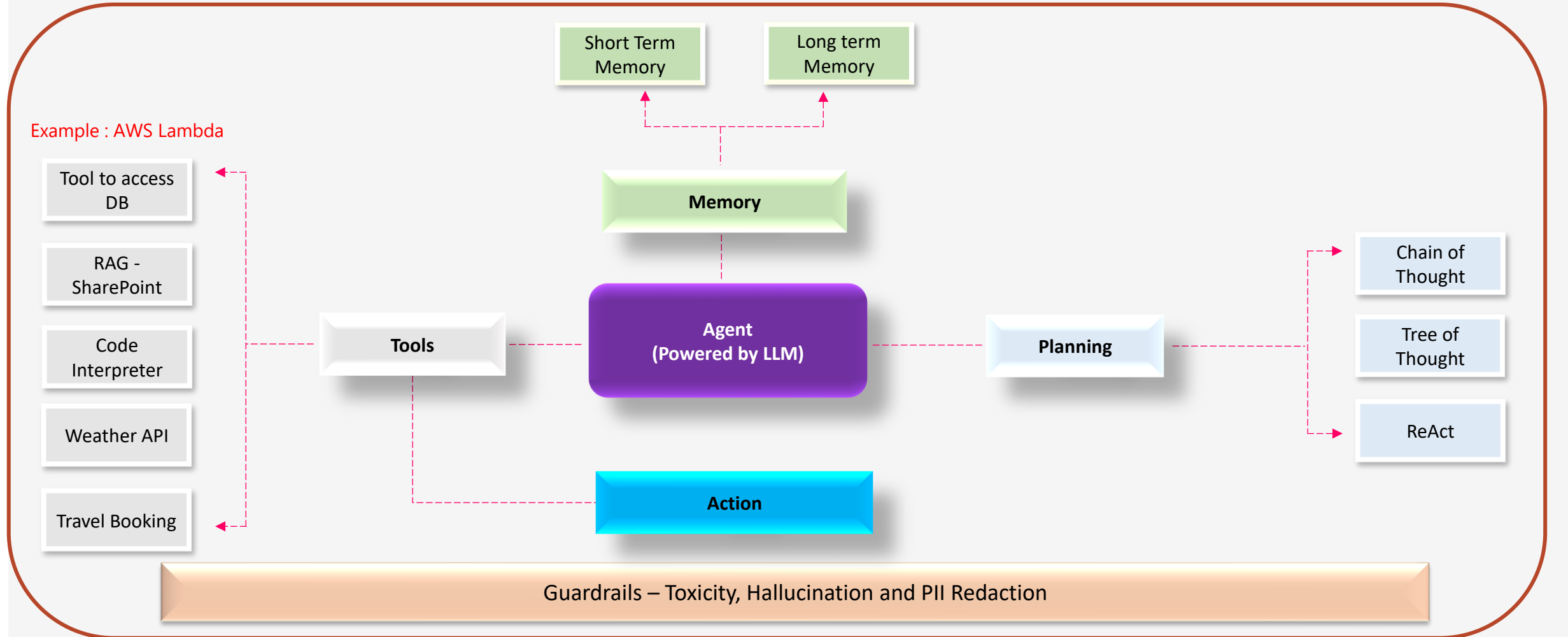
Are able to access enterprise data



Are able to use Tools

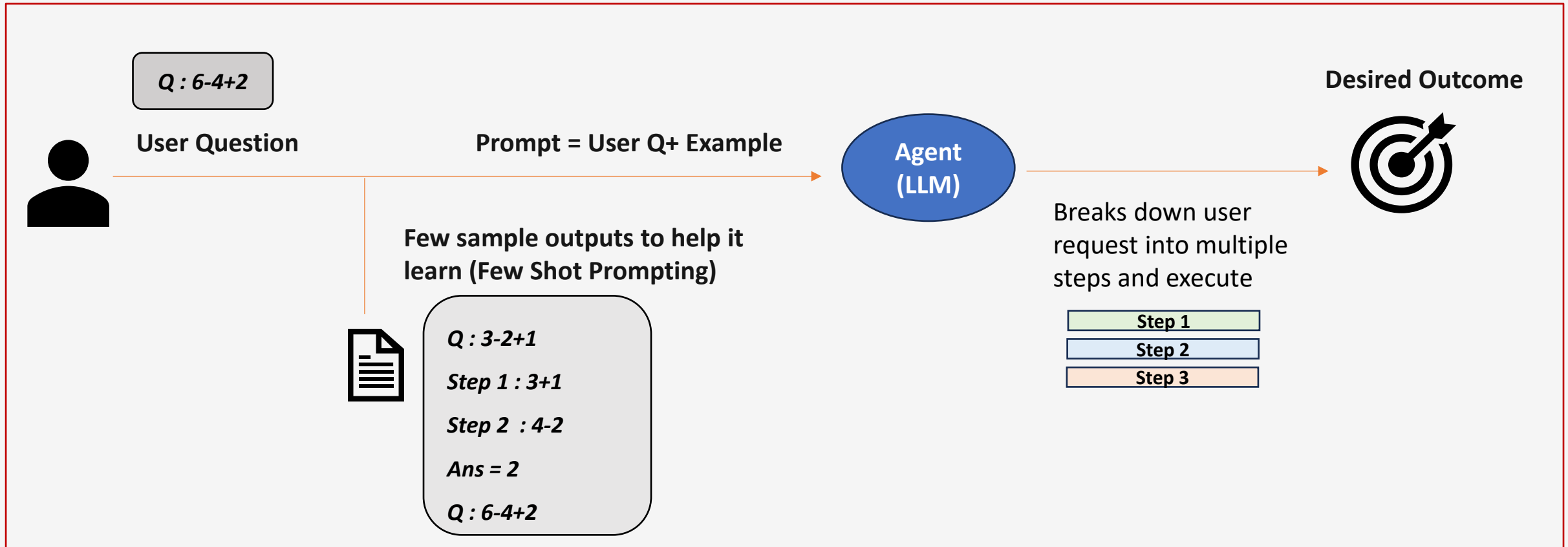
AI Agents – Key Characteristics

AI Agents



Planning by Agents

- Agent is powered by a Large Language Model(LLM) – Brain behind the Agents
- **How and why does Agent do the planning ?**



AI Agent - Planning

Two key approaches AI Agent planning

1. Planning without Feedback

- Chain-of-Thought (CoT) 
- Tree of Through Reasoning (ToT)

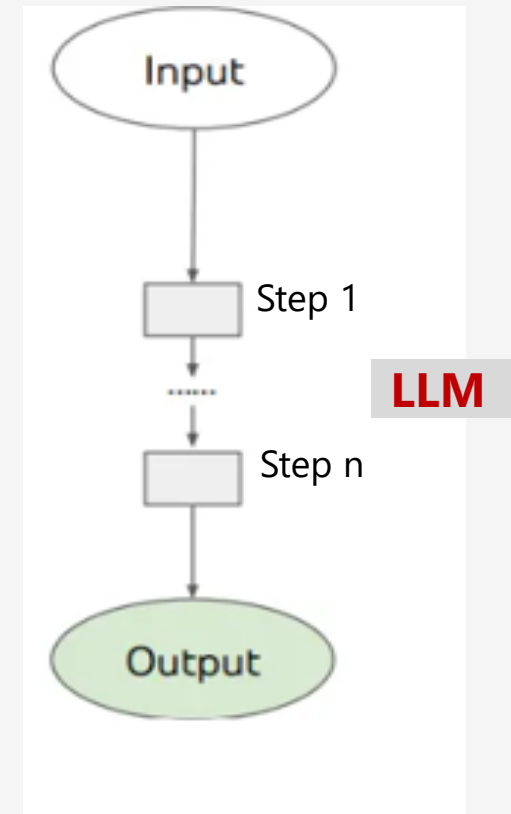
2. Planning with Feedback

- ReAct or Reasoning + Action 
- Reflexion

1. Planning - Planning without Feedback

Chain-of-thought prompting (CoT)

- CoT prompting enables complex reasoning capabilities for the LLM through intermediate reasoning steps
- CoT prompting has shown to **improve capability of LLM** to answer questions involving **arithmetic and commonsense reasoning**



Read More - <https://arxiv.org/pdf/2201.11903>

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Planning without Feedback – Chain of Thought Reasoning (CoT)

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

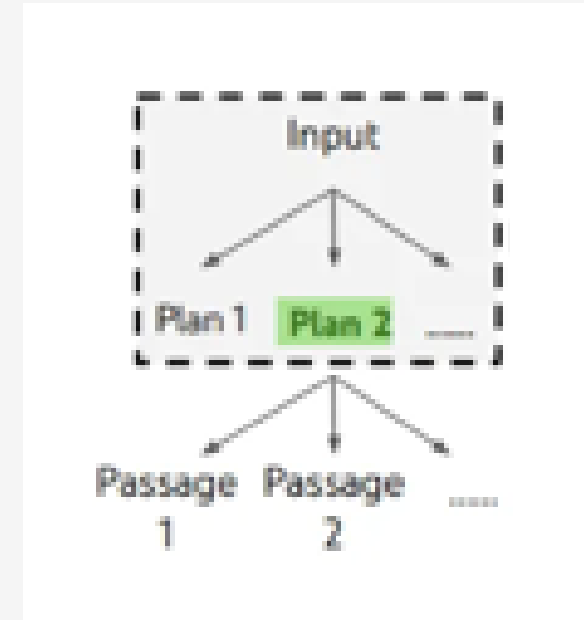
Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

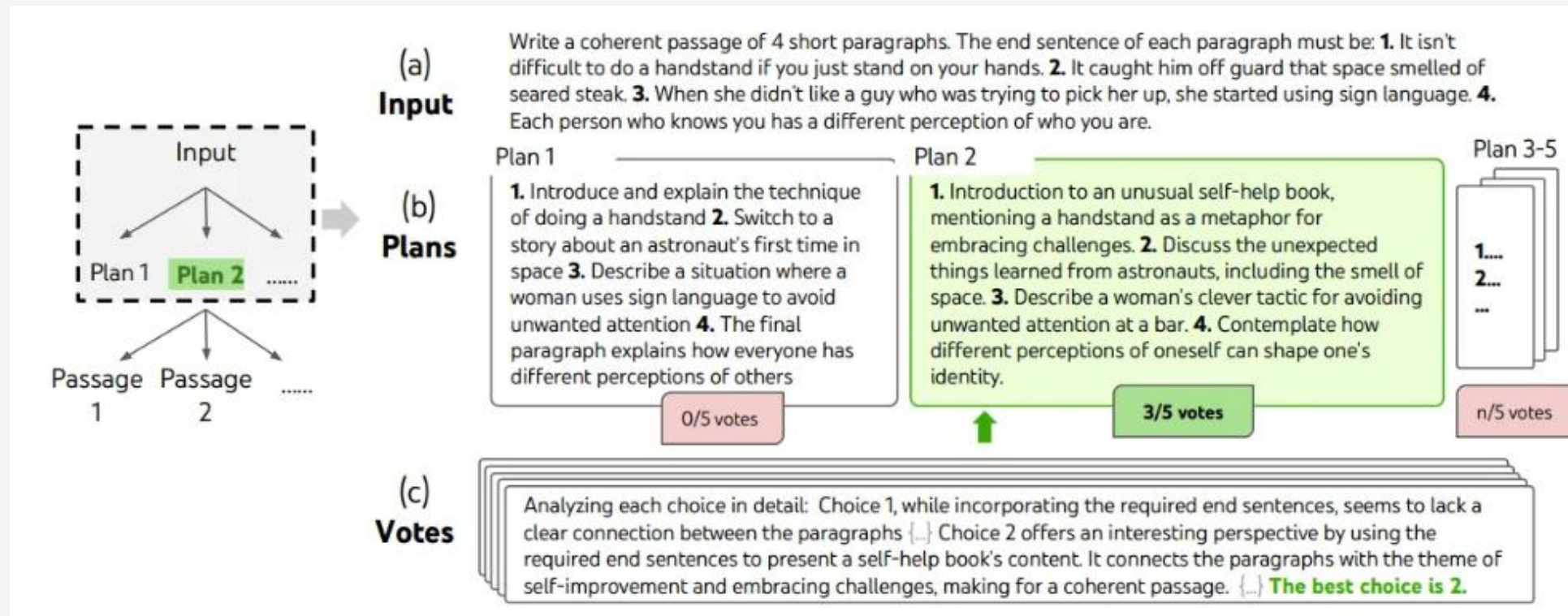
1. Planning without Feedback – Tree of Thought

Tree of Through Reasoning

- First decomposes the problem into multiple thought steps
- Generates multiple thoughts per step, creating a tree structure.



1. Planning without Feedback - Tree of Thought

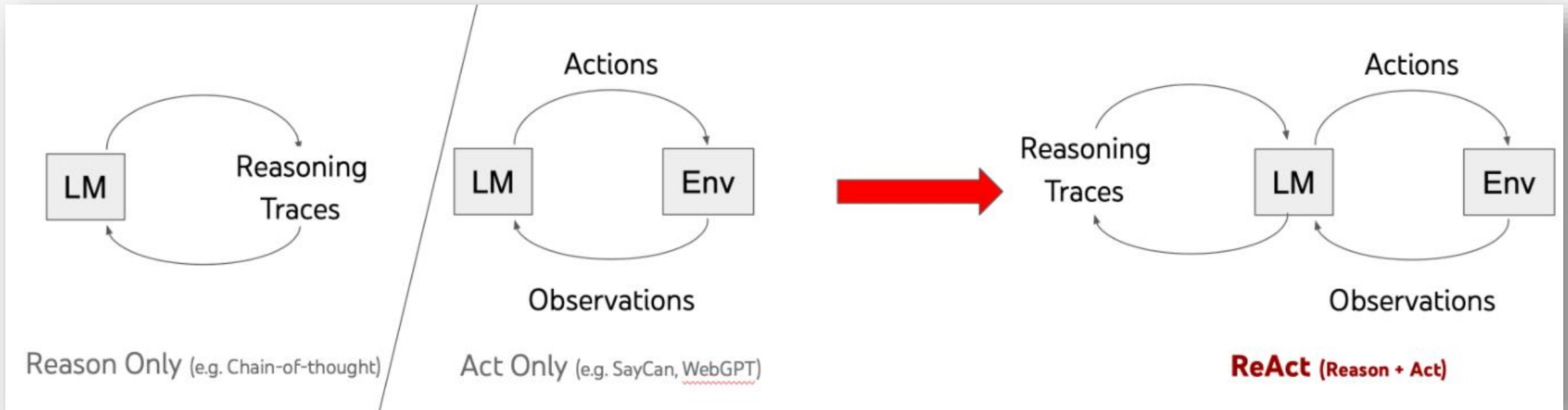


Read More - <https://arxiv.org/pdf/2201.11903>

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2. Planning with Feedback – ReAct (Reason + Act)

- ReAct is an advanced **prompting technique** that combines **Reasoning** and **Action** within a prompt.
- **Good for scenarios where external or internal data** is required to execute tasks.
- Dynamic execution - Replanning

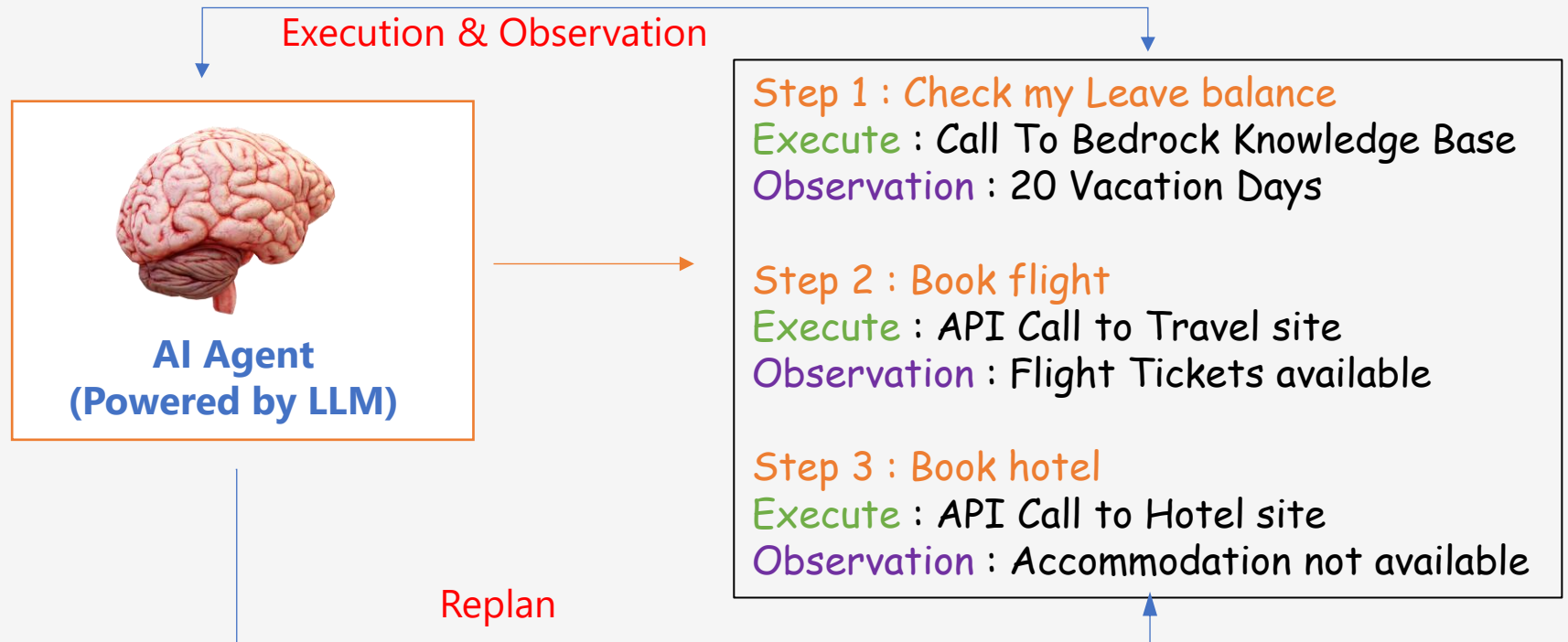


ReAct Example

User Request

I am planning a 2-week
Skiing vacation from
mid of December

Micro-cool AI Agent



Agents – Planning Capability Demo



AI Agent **Planning** Demo

User Request



I am planning a 5-day
Beach vacation from
mid of December

Micro-cool AI Agent



Execution & Observation

Step 1 : Check my Leave balance
Execute : **API Call to HR System**
Observation : 5 Vacation Days

Step 2 : Book hotel
Execute : **API Call to Hotel site**
Observation : Accommodation not available

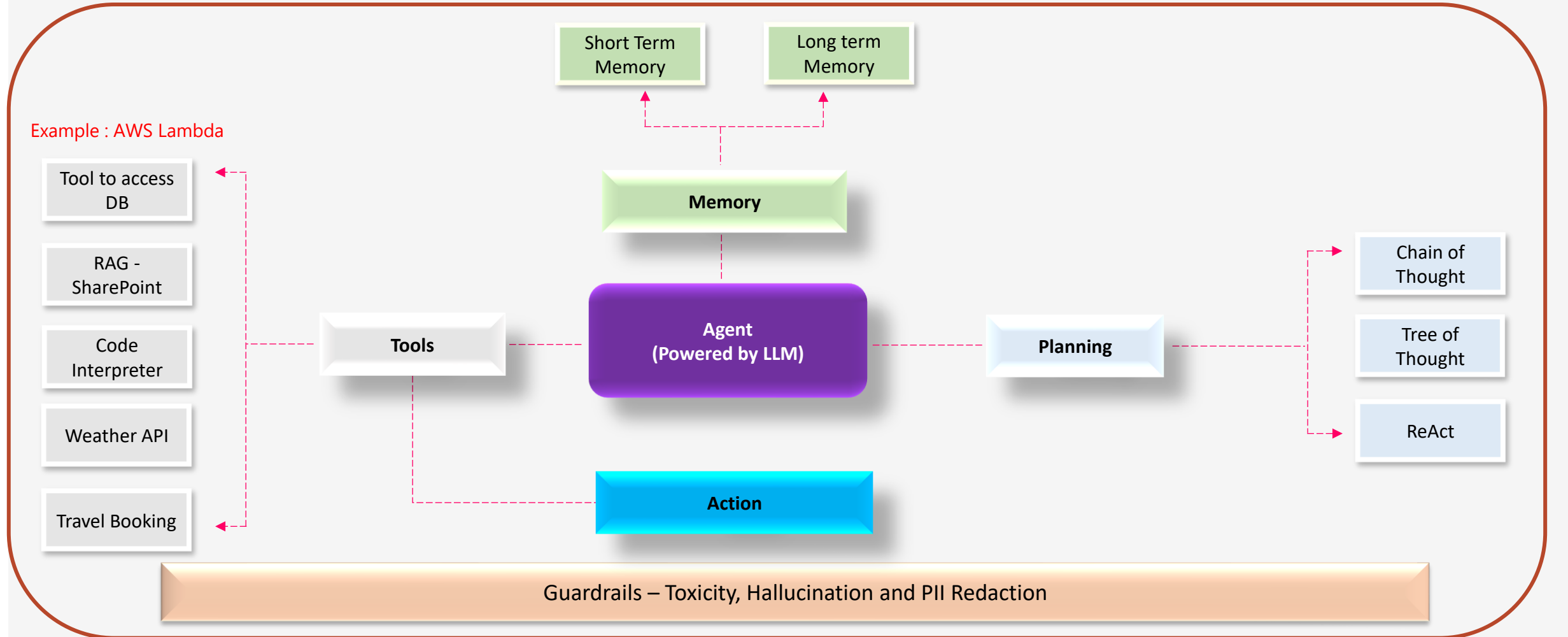
Replan

Questions

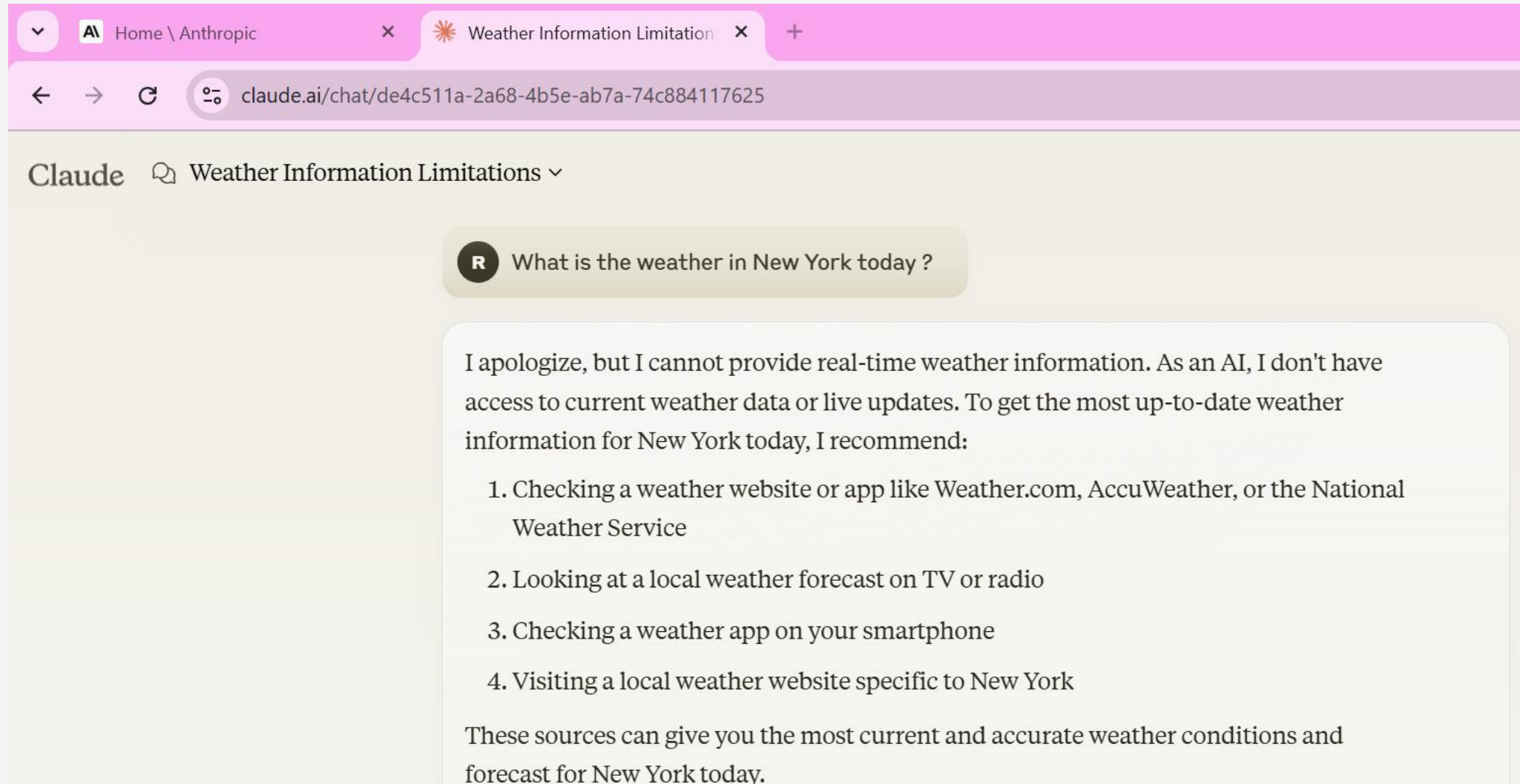
1. Book a hotel room for me with **employee id - 251205 if my leave balance is more than 5 days** with following details : Check-in date - 2025-12-26, 2. Room Type - Sea View 3. Guest Name – John Doe 4. Number of Nights – 5
2. Book a hotel room for me **with employee id - 251205 if my leave balance is more than 3 days** : 1. Check-in date - 2025-12-26, 2. Room Type - Sea View, 3. Guest Name – Jane Doe, 4. Number of Nights - 3

AI Agents – Key Characteristics

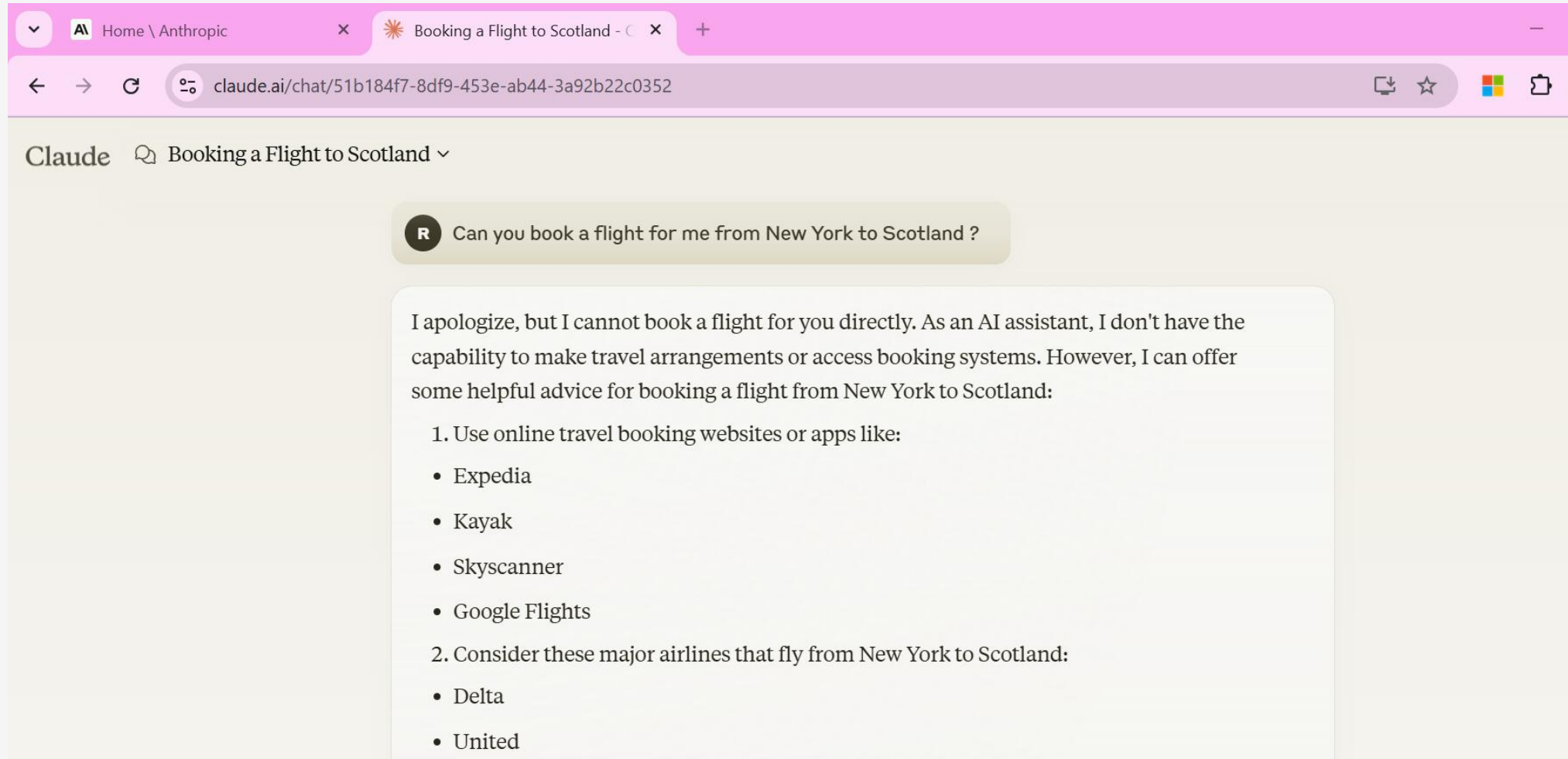
AI Agents



Limitations of LLM's to access internal or external data sources



Limitations of LLM's to access internal or external data sources

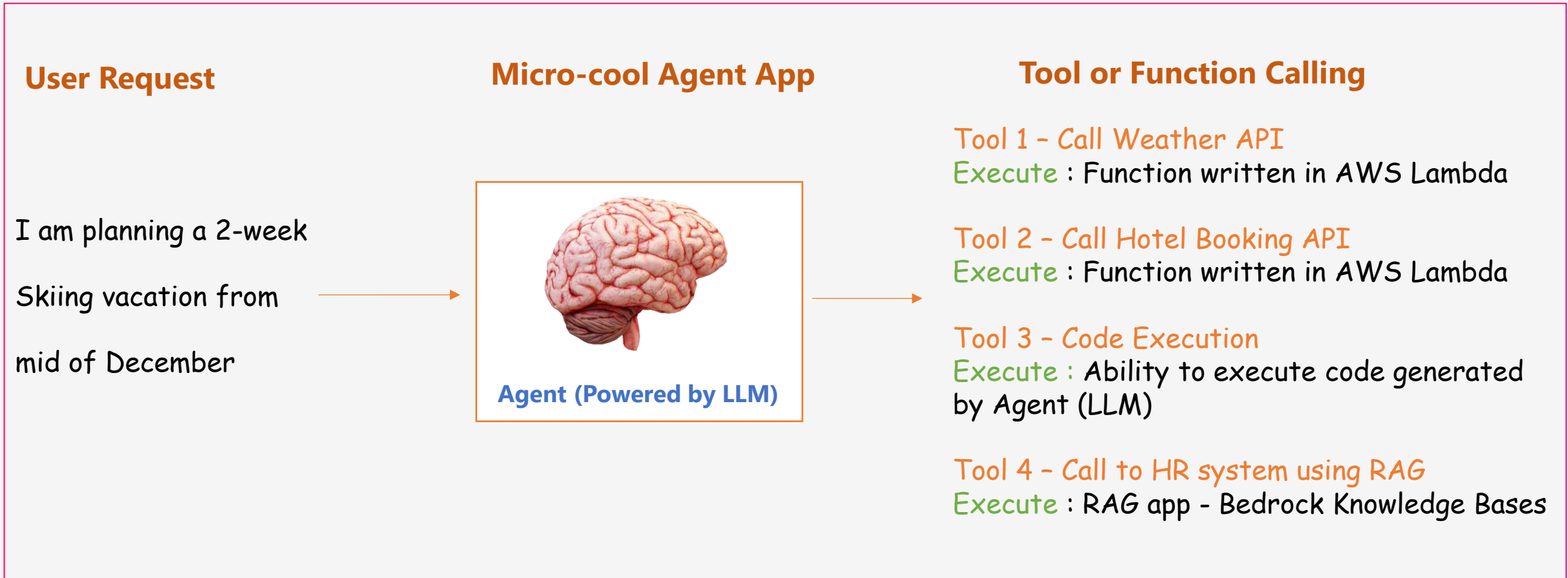


2. Tools Calling or Function Calling

- Function calling or Tools Calling enables Agents to connect and retrieve data from **external and internal source data and systems** to enhance the capabilities of Agents.
 - Google Search
 - Weather API
 - Extract data from enterprise Database such as MySQL
 - Retrieval Augmented Solution(RAG) based app
 - Code Interpreter for running custom code



2. Tools Calling or Function Calling



How Tools calling works ?

Prompt template editor

```
1 $instruction$
2
3 You have been provided with a set of tools to
  answer the user's question.
4 You may call them like this:
5 <function_calls>
6   <invoke>
7     <tool_name>$TOOL_NAME</tool_name>
8     <parameters>
9       <$PARAMETER_NAME>$PARAMETER_VALUE
        </$PARAMETER_NAME>
10     ...
11   </parameters>
12 </invoke>
13 </function_calls>
14
15 Here are the tools available:
16 <tools>
17   $tools$
18 </tools>
19
20
21 You will ALWAYS follow the below guidelines when
  you are answering a question:
22 <guidelines>
23   - Never assume any parameter values while
    invoking a function.
24 $ask_user_missing_information$
25   - Provide your final answer to the user's
    question within <answer></answer> xml tags.
26   - Think through the user's question, extract all
```

- Define the tools available to the Agent (powered by LLM)
- Automatically construct a **special system prompt from the tool definitions, tool configuration, and any user-specified system prompt.**

```
import anthropic

client = anthropic.Anthropic()

response = client.messages.create(
    model="claude-3-5-sonnet-20241022",
    max_tokens=1024,
    tools=[
        {
            "name": "get_weather",
            "description": "Get the current weather in a given location",
            "input_schema": {
                "type": "object",
                "properties": {
                    "location": {
                        "type": "string",
                        "description": "The city and state, e.g. San Francisco, CA"
                    }
                },
                "required": ["location"],
            }
        },
    ],
    messages=[{"role": "user", "content": "What's the weather like in San Francisco"}])
```

AI Agents – Tool Calling Demo

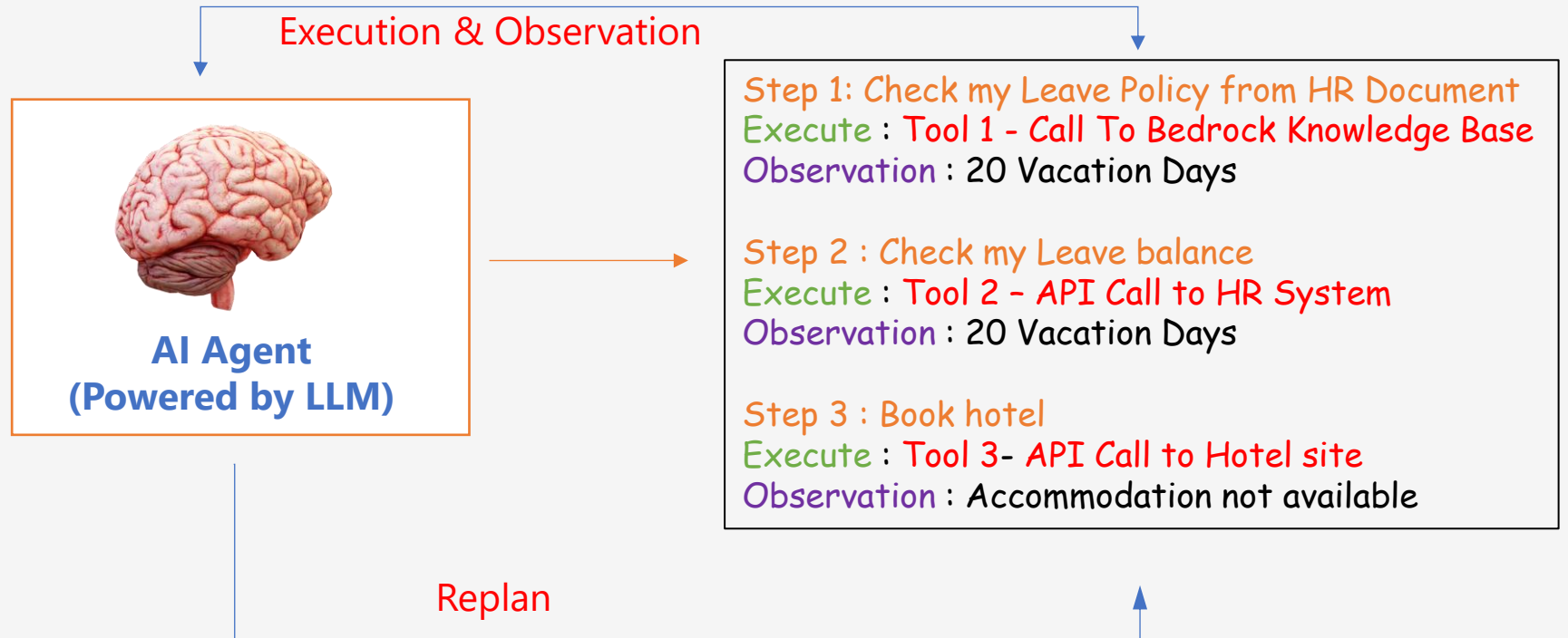


AI Agent Tool Calling Demo

User Request

I am planning a 5-day
Beach vacation from
mid of December

Micro-cool AI Agent

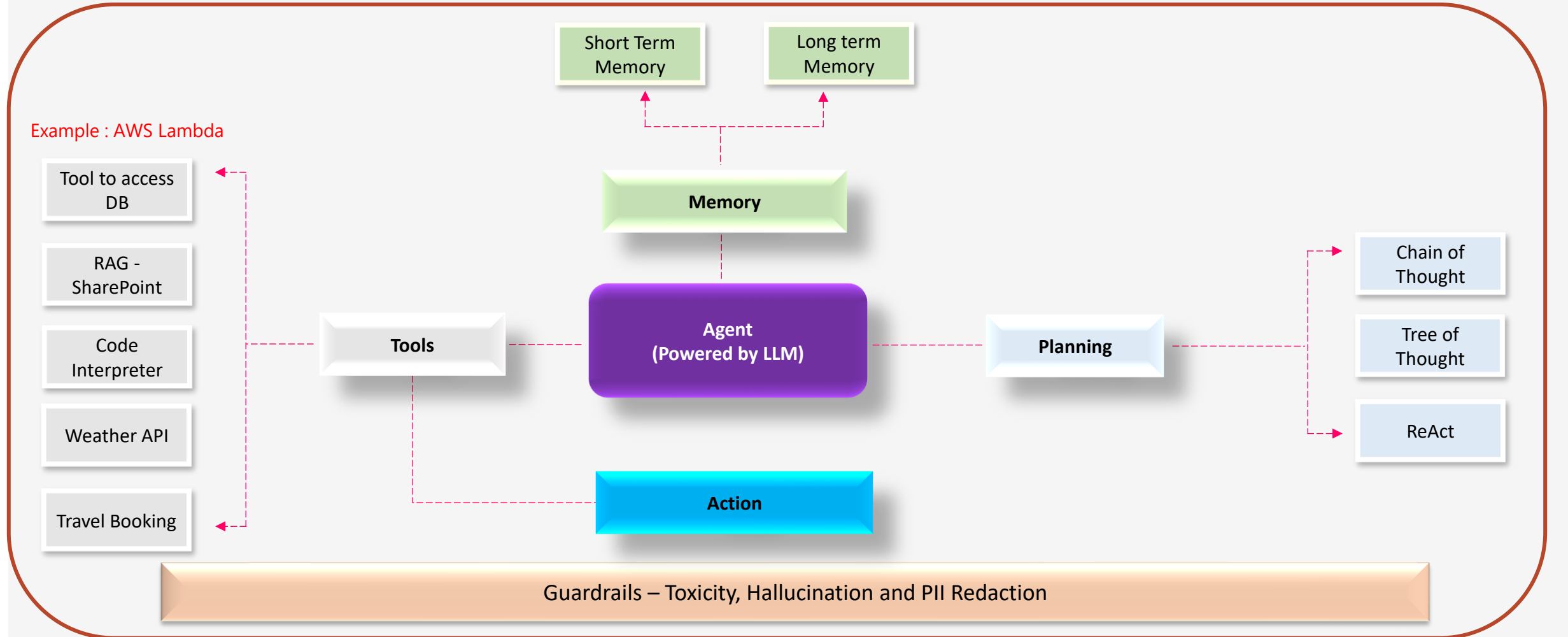


Questions

1. How many Paternity Leaves are available ?
2. Book a hotel room for me with employee id - 251205 if my leave balance is more than 3 days with following details : 1. Check-in date - 2025-12-27, 2. Room Type - Sea View, 3. Guest Name – Jane Doe, 4. Number of Nights - 3

AI Agents – Key Characteristics

AI Agents

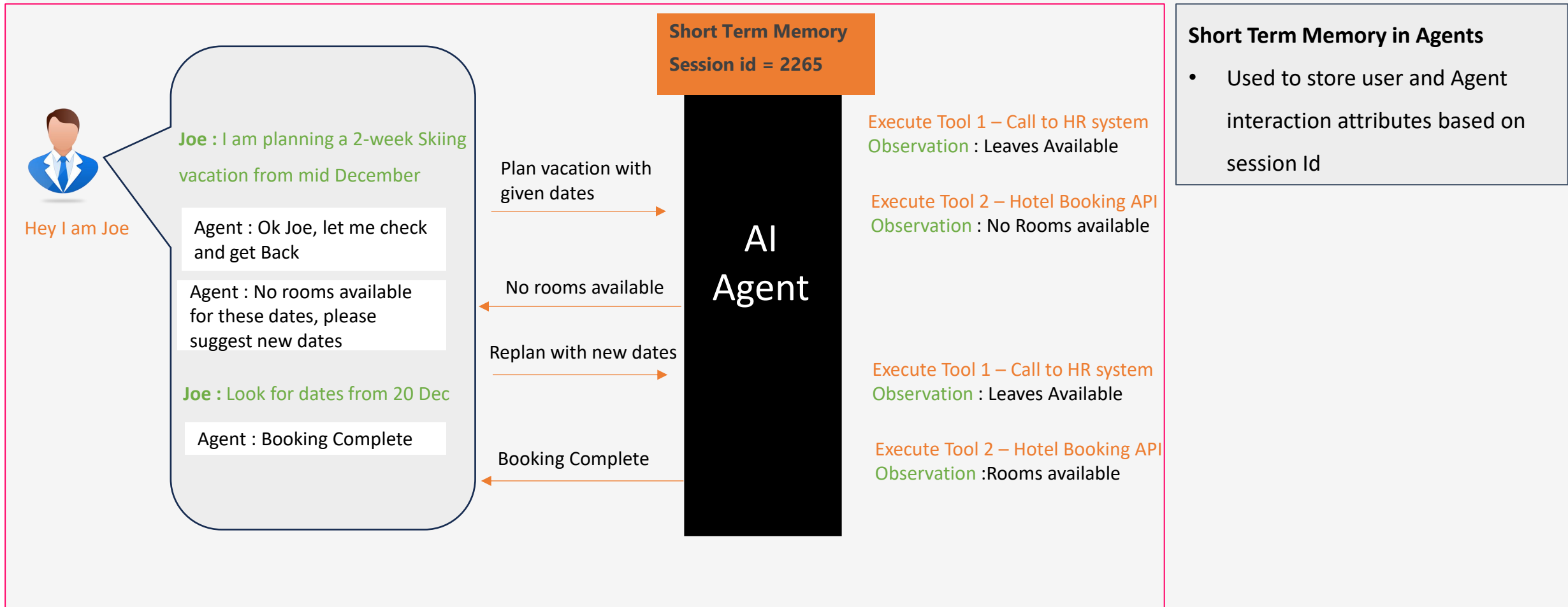


4. Agentic Memory – Short Term and Long Term

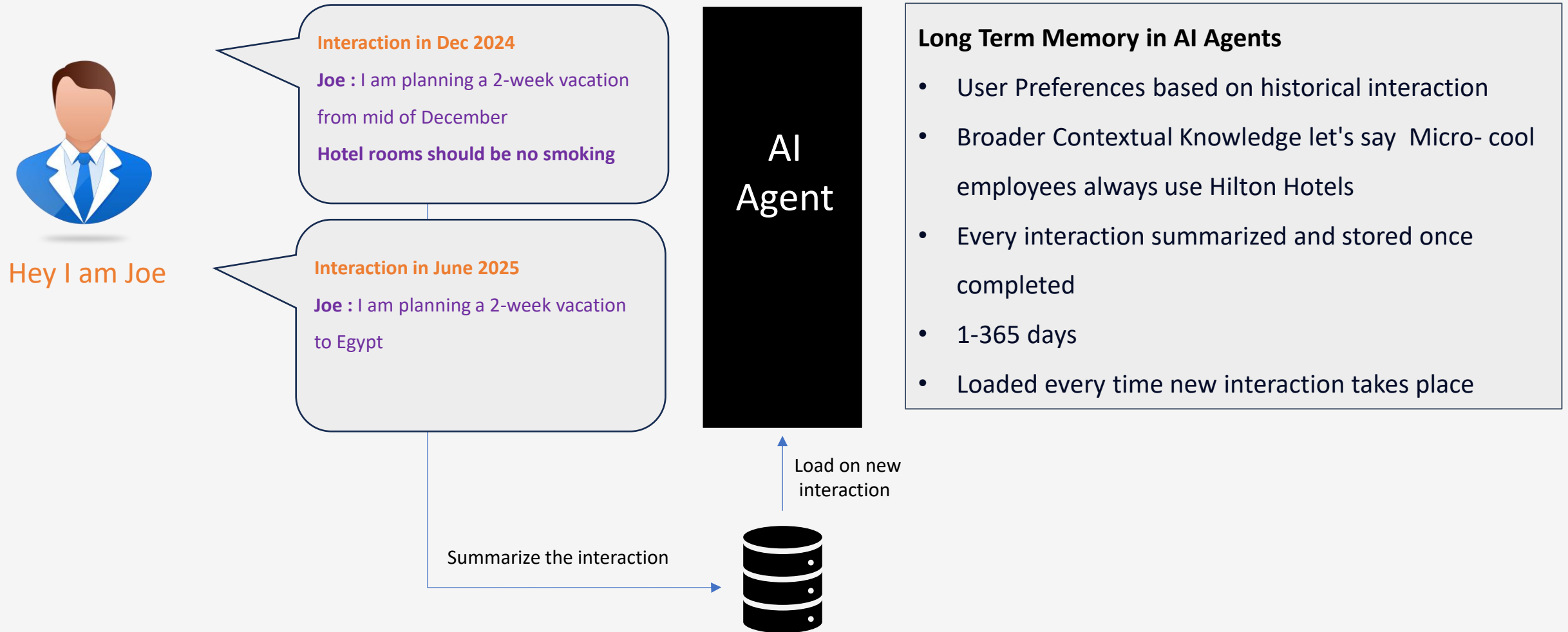
- Remember – LLM's are Stateless (No memory)
- Short term memory
 - Agent retains the conversation and user input during the particular session
- Long term memory
 - Allows agent to retain conversation across all previous interactions
 - Summarize interaction across all previous sessions
 - 1-365 days



4. Agent Memory : Short Term Memory



4. Agent Memory : Long-Term Memory



Amazon Bedrock Agents – Architecture and Capabilities Overview

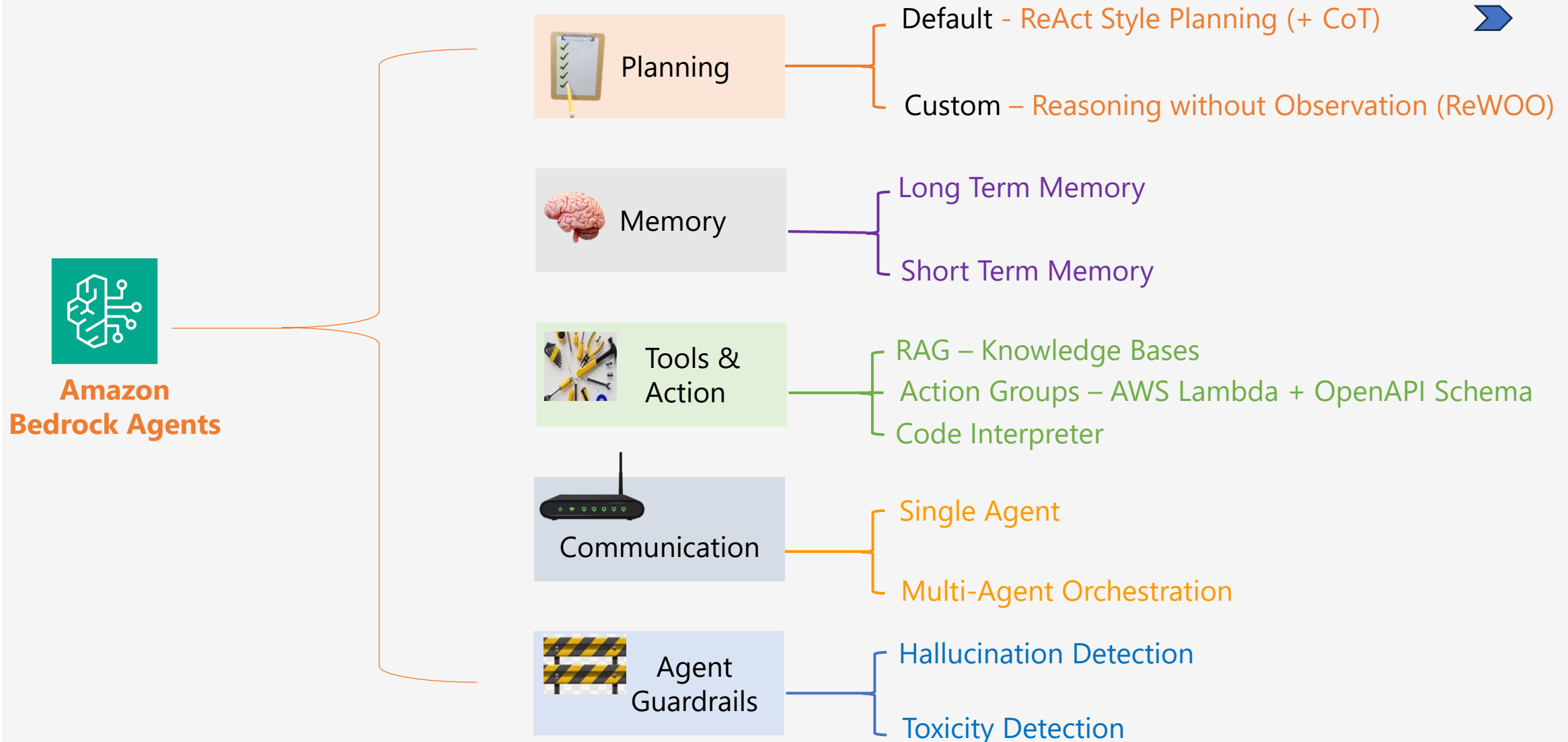


What are Amazon Bedrock Agents ?

Amazon Bedrock Agents is **fully managed service** that **enable generative AI applications** to **automate multistep tasks** by seamlessly **connecting with company systems, APIs, and data sources**



Core Components of Amazon Bedrock Agents



AI Agent **Planning** Demo

User Request



I am planning a 5-day
Beach vacation from
mid of December

Micro-cool AI Agent



Execution & Observation

Step 1 : Check my Leave balance
Execute : **API Call to HR System**
Observation : 5 Vacation Days

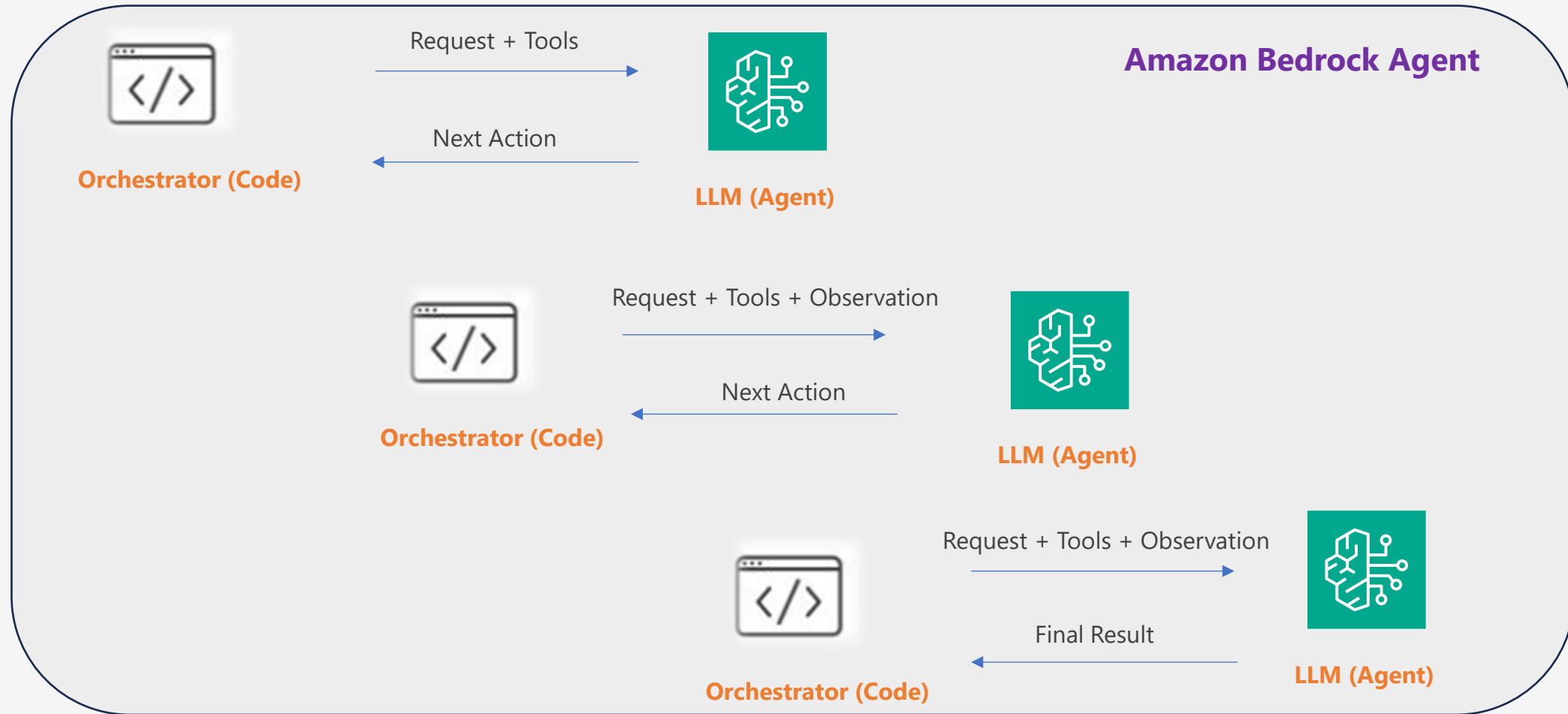
Step 2 : Book hotel
Execute : **API Call to Hotel site**
Observation : Accommodation not available

Replan

Questions

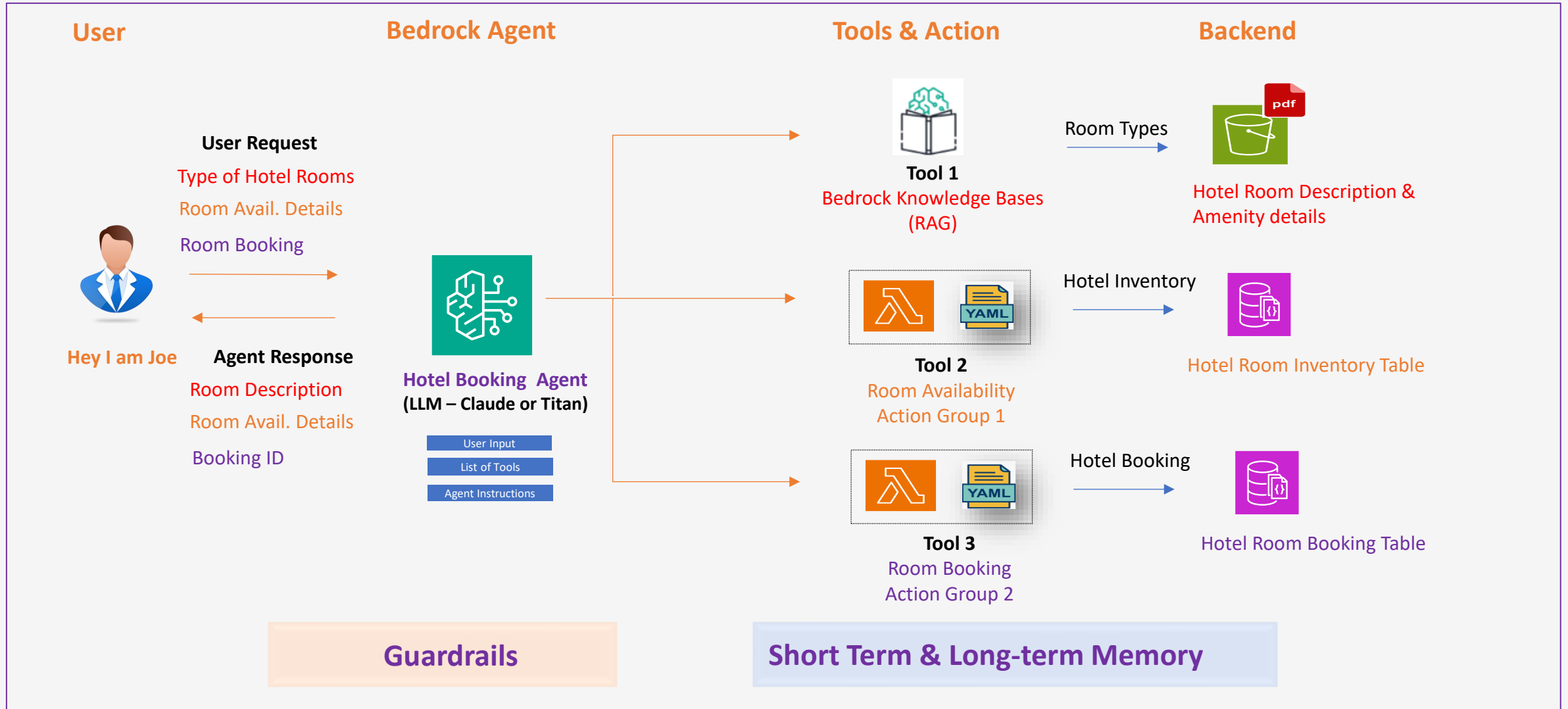
1. Book a hotel room for me with **employee id - 251205 if my leave balance is more than 5 days** with following details : Check-in date - 2025-12-26, 2. Room Type - Sea View 3. Guest Name – John Doe 4. Number of Nights – 5
2. Book a hotel room for me **with employee id - 251205 if my leave balance is more than 3 days** : 1. Check-in date - 2025-12-26, 2. Room Type - Sea View, 3. Guest Name – Jane Doe, 4. Number of Nights - 3

Amazon Bedrock Agents – ReAct Orchestration Style

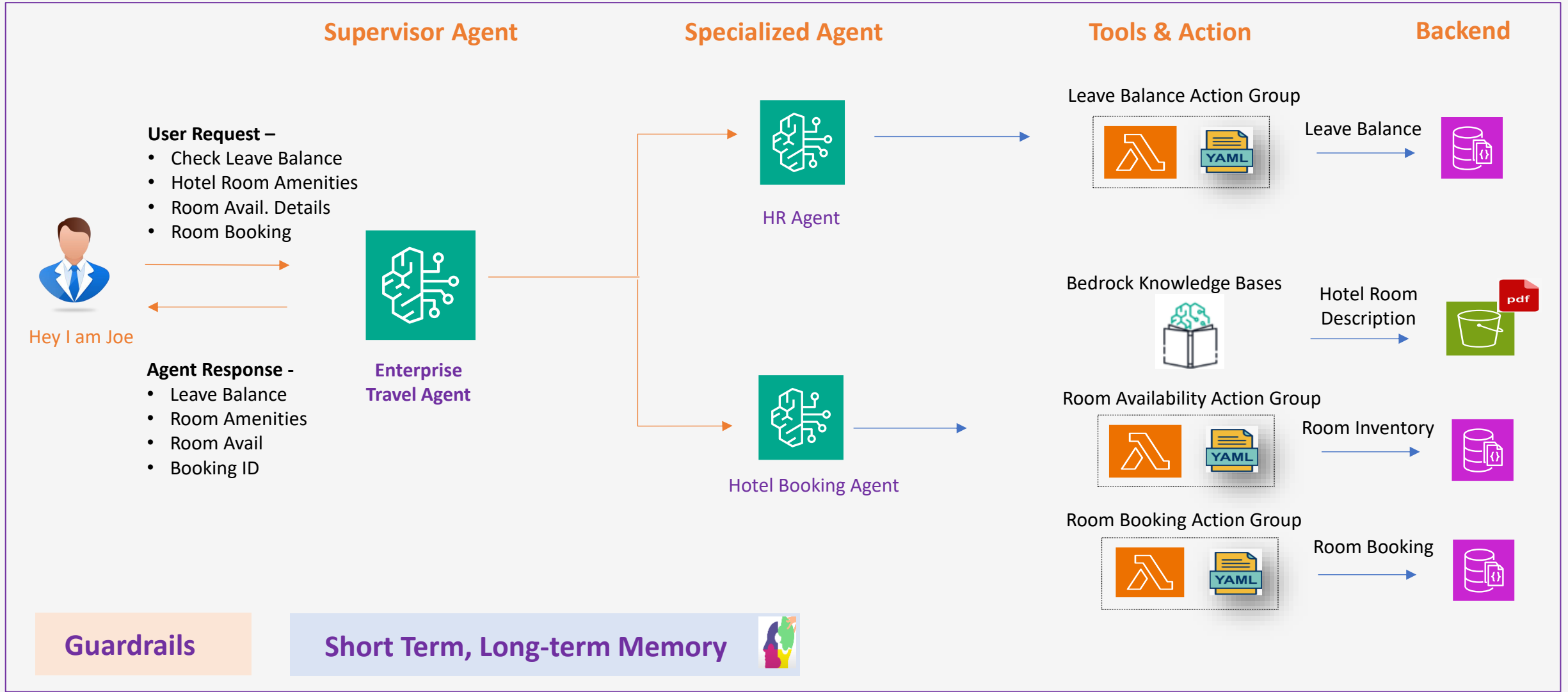


- Orchestrator refers to internal code
- ReAct – Orchestrator style or prompt style - Defines the plan and then modify as we go...but **Sequential in nature**

Hotel Booking Agent – Single Agent



Enterprise Travel Agent (Multi-Agent)

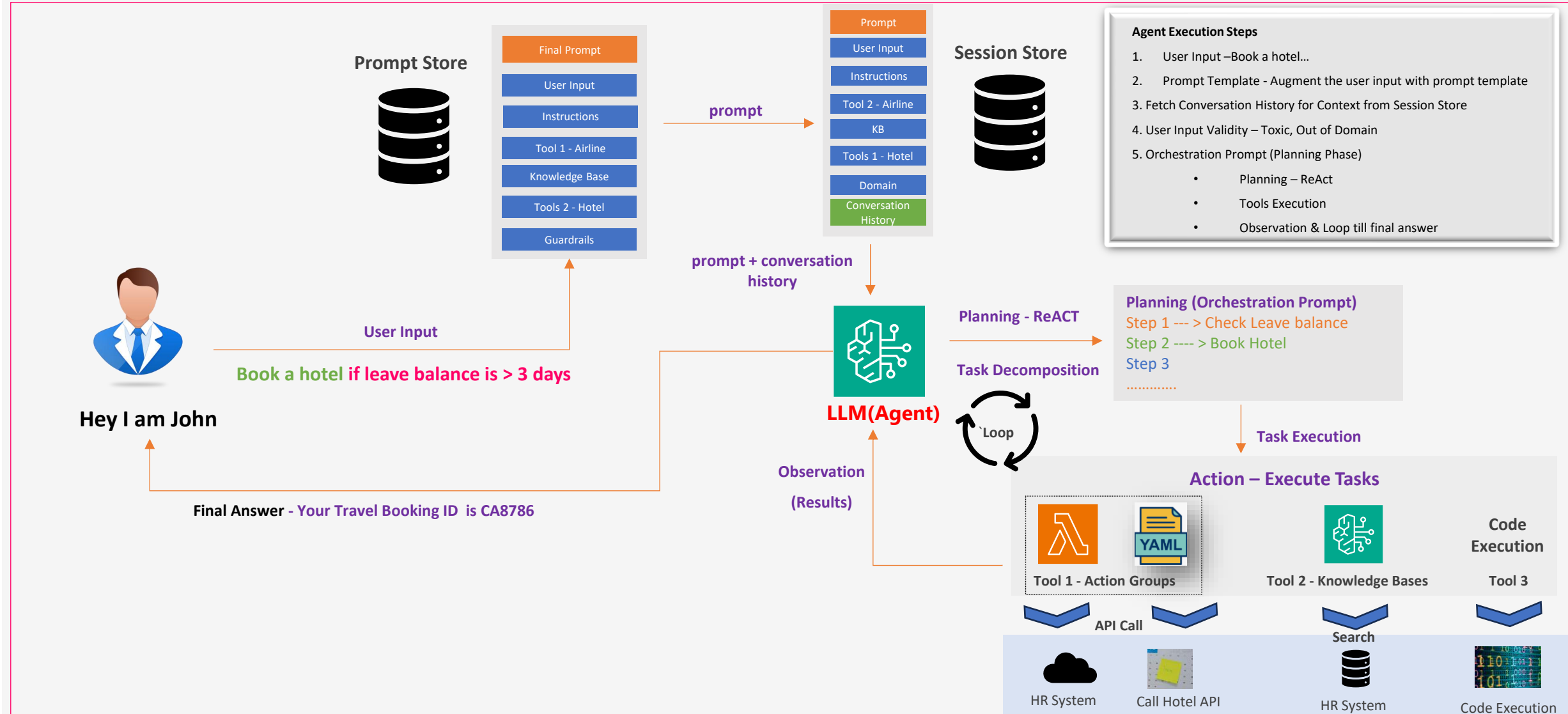


Amazon Bedrock Agents – Console Walkthrough



***Amazon Bedrock Agents –
How Amazon Bedrock Agents Work ?***

How do Bedrock Agents work – Step by Step



Model Context Protocol (MCP)

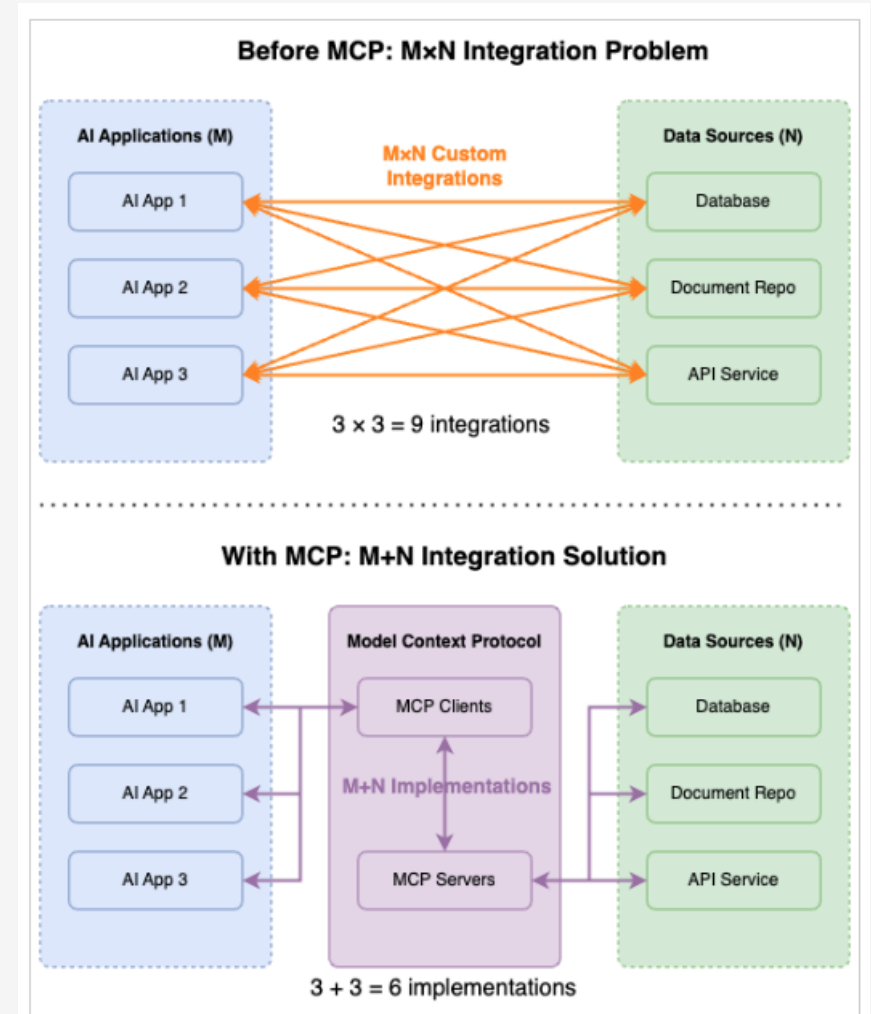
What is MCP and Problem MCP is trying to Solve

What is MCP ?

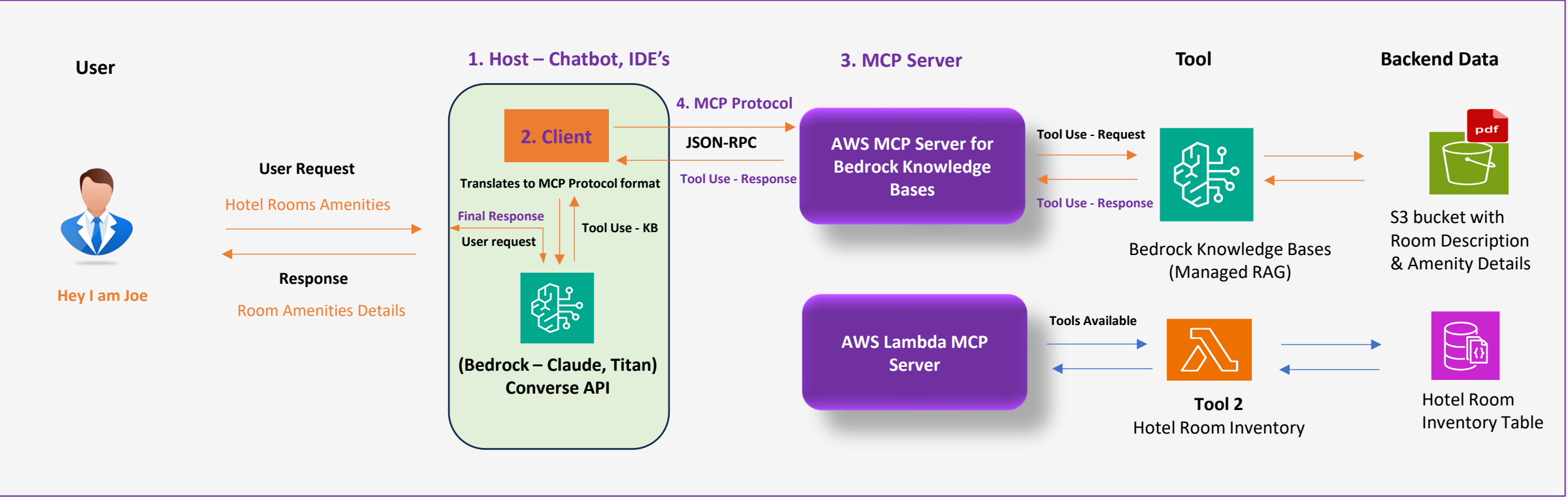
- MCP - Model Context Protocol
- MCP is an open protocol that standardizes how applications provide context to LLMs.

Problem MCP is trying to solve

- Solves MXN Integration problem
- Standardizes how LLM access data from different data sources



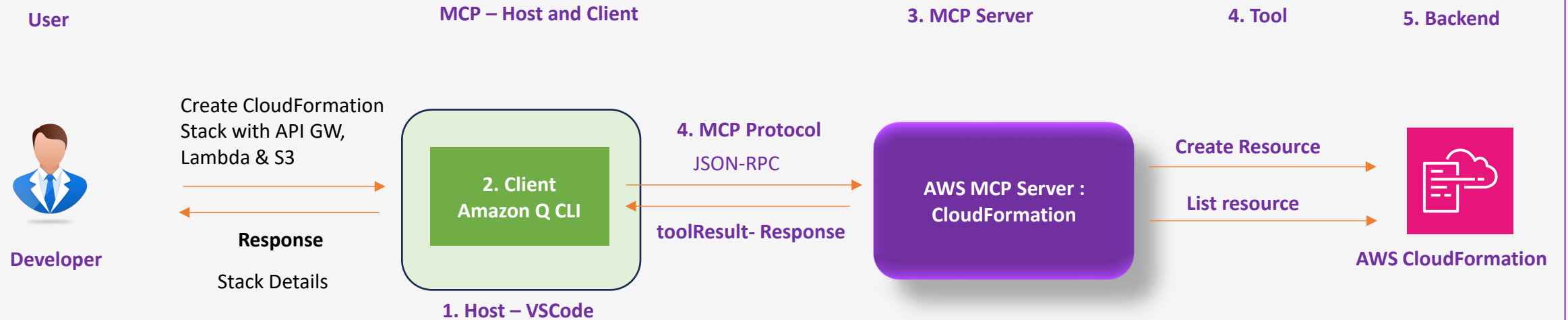
Vacation Planner AI Application using MCP



Components		
Host	IDE's, Chatbot, LLM Apps	Deployed Locally
Client	Cline, Amazon Q CLI, Windsurf, Claude Desktop - Link	1:1 connections with MCP Server, Embedded in the Host VSCode, Locally or Cloud
MCP Server	Built by Tool /Service Provider - AWS. Hosted by AWS as managed server or Local System or Docker	Lightweight Program exposes specific capabilities through the standardized Model Context Protocol
Transport Layer	STDIO, HTTP (JSON-RPC)	

***AWS CloudFormation MCP Server
& Amazon Q CLI (Client)
Demo***

AWS CloudFormation MCP Server - Demo



Basic Usage

Examples of how to use the AWS Infrastructure as Code MCP Server:

- "Create a new S3 bucket with versioning and encryption enabled"
- "List all EC2 instances in the production environment"
- "Update the RDS instance to increase storage to 500GB"
- "Delete unused NAT gateways in VPC-123"
- "Set up a three-tier architecture with web, app, and database layers"
- "Create a disaster recovery environment in us-east-1"
- "Configure CloudWatch alarms for all production resources"
- "Implement cross-region replication for critical S3 buckets"
- "Show me the schema for AWS::Lambda::Function"
- "Create a template for all the resources we created and modified"

Tools

create_resource

Creates an AWS resource using the AWS Cloud Control API with a declarative approach. **Example:** Create an S3 bucket with versioning and encryption enabled.

get_resource

Gets details of a specific AWS resource using the AWS Cloud Control API. **Example:** Get the configuration of an EC2 instance.

update_resource

Updates an AWS resource using the AWS Cloud Control API with a declarative approach. **Example:** Update an RDS instance's storage capacity.

delete_resource

Deletes an AWS resource using the AWS Cloud Control API. **Example:** Remove an unused NAT gateway.

list_resources

Lists AWS resources of a specified type using AWS Cloud Control API. **Example:** List all EC2 instances in a region.



Section on :
***Bedrock AgentCore - Basic
Concepts & Architecture***

What is Amazon Bedrock AgentCore & the Problem it is trying to solve ?

Enterprise Challenge with Agentic AI

- How to transform Agentic AI apps from “cool demos” into “production-ready business solutions”



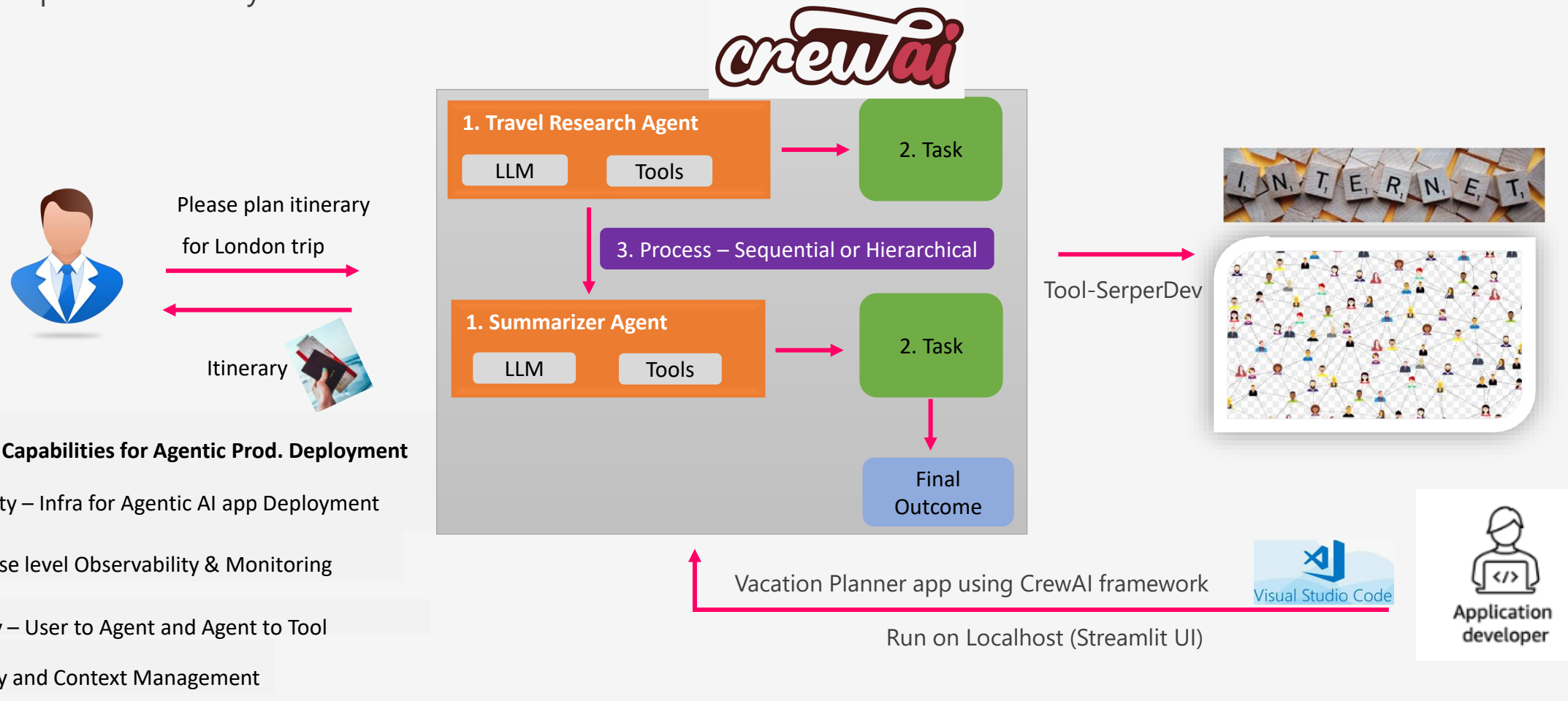
Amazon Bedrock AgentCore Service

- Provides a **set of capabilities** to transition your agentic AI applications from experimental proof of concept to production-ready systems.

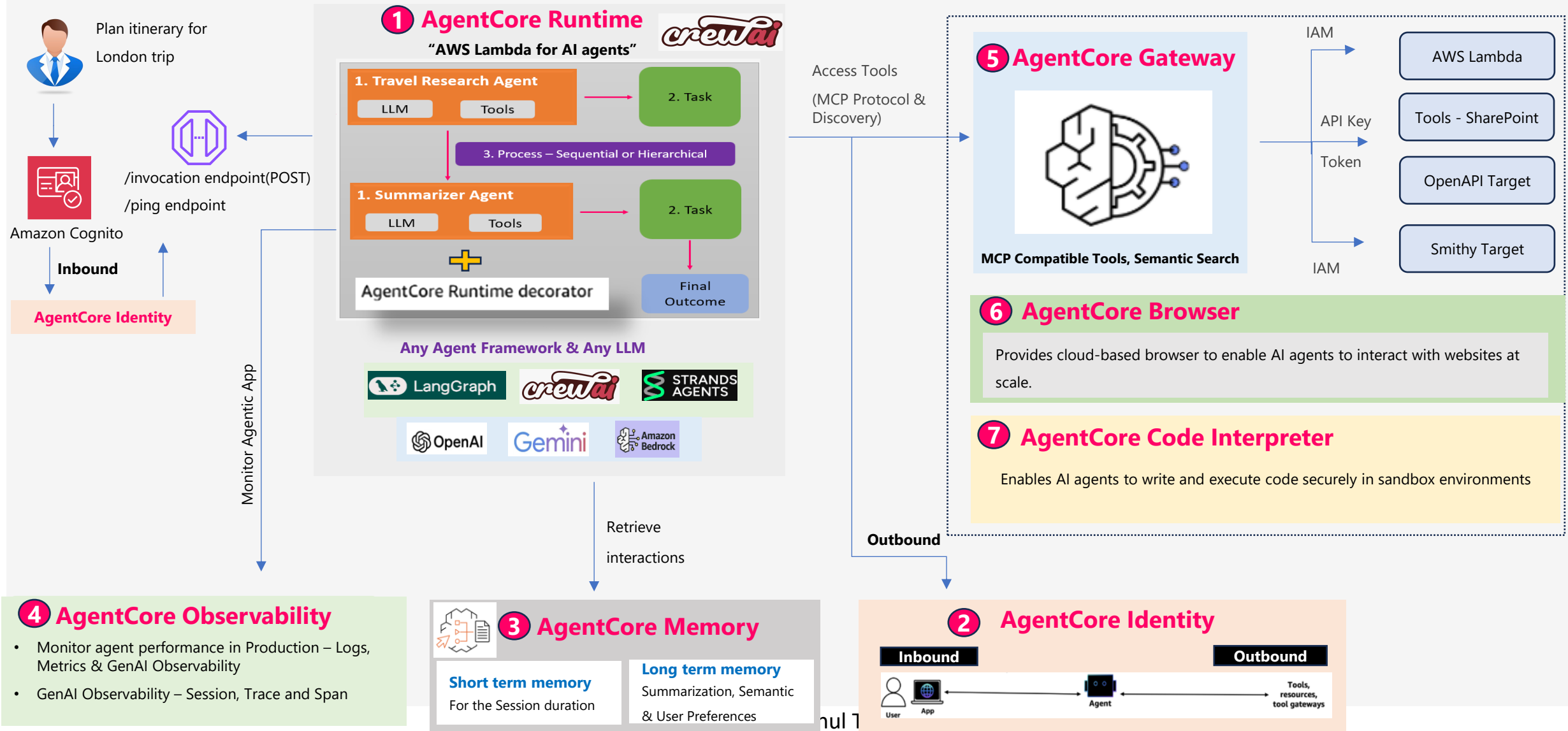


Bedrock AgentCore – Business Scenario : PoC to Production

Business Use Case : Dream Vacation Planner travel agency wants to build and test (PoC) an Agentic AI app which allows its users to plan an itinerary for their dream destination on their website.

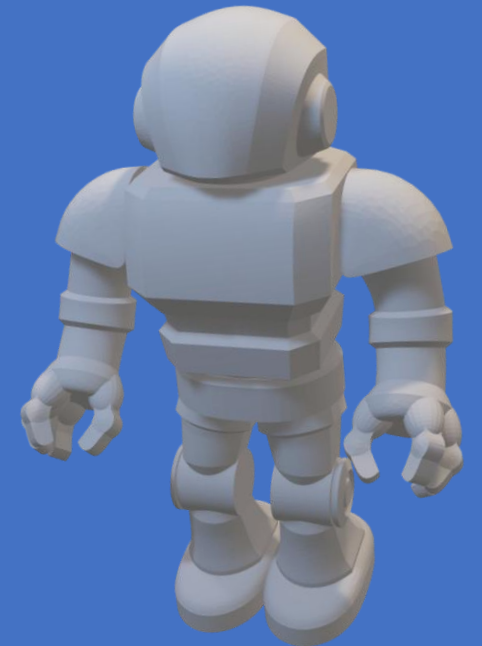


Bedrock AgentCore Capabilities – Overview



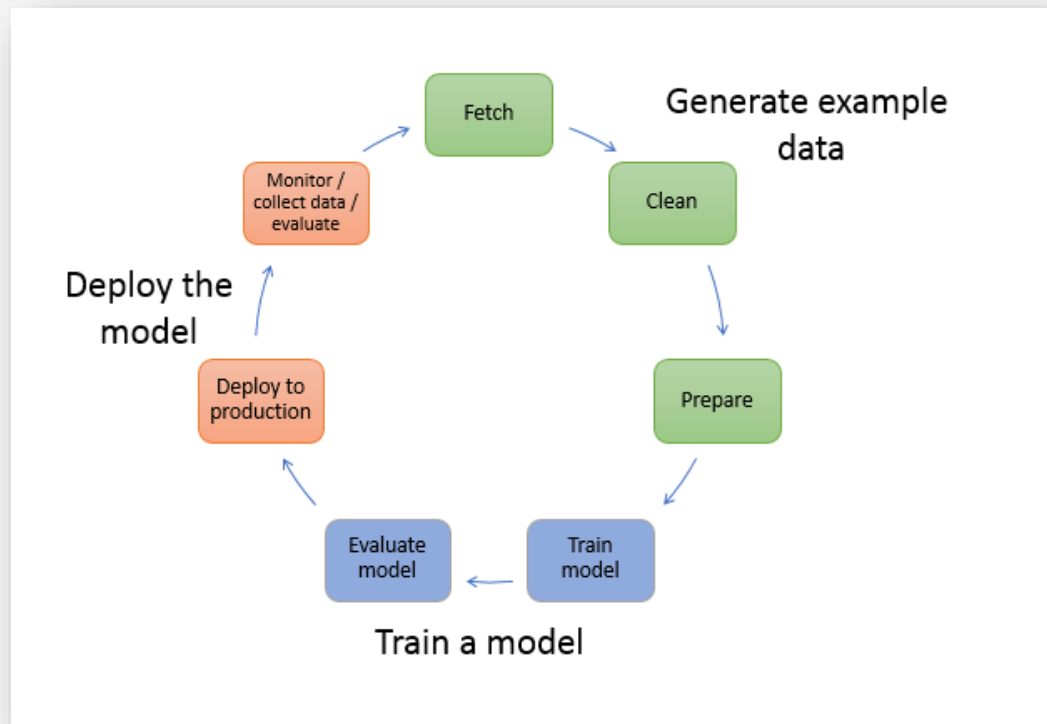
Section on :

Data Foundations for Generative AI on AWS

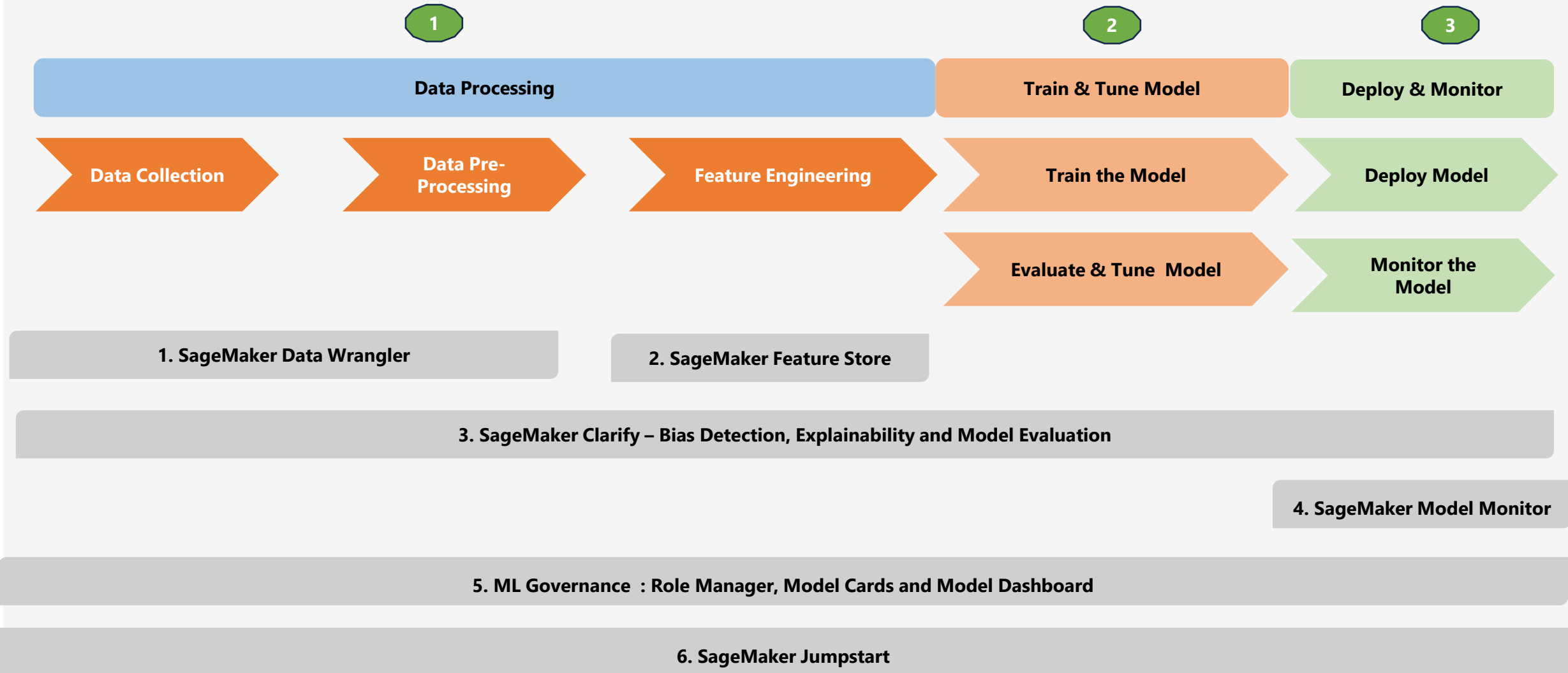


Amazon SageMaker - Overview

- Amazon SageMaker is a **fully managed machine learning service**.
- Used by data scientists and developers to **build, train, and deploy ML models**
- Collection of many different services



Amazon SageMaker mapping to ML Lifecycle



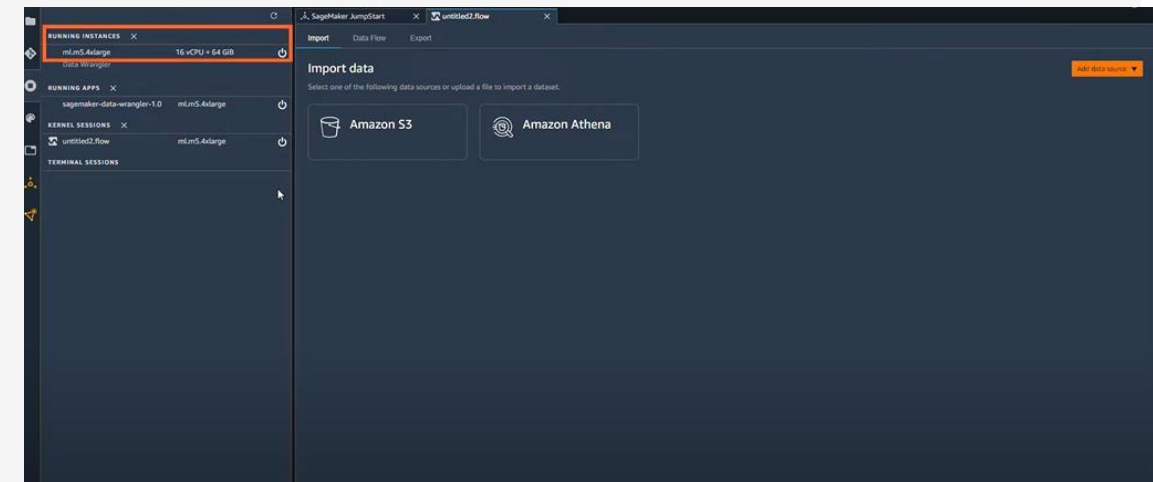
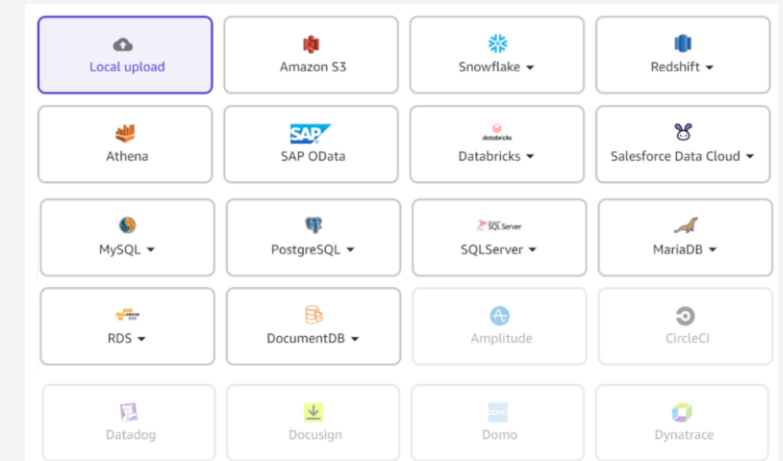
1. Amazon SageMaker Data Wrangler

1

Data Processing – Data Collection and Data Pre-Processing



1. Import data from Amazon services such as S3, Athena, Redshift, 50+ third-party sources.
2. Understand data quality and generate data insights - **Exploratory Data Analysis (EDA)**
 - Verifies data quality - missing values, duplicate rows
 - Detect anomalies - outliers, class imbalance, and data leakage
3. Transform data more efficiently and Analyse
 - Drop column, Handle outliers
 - Prebuilt or custom transformation
4. Export the Data
 - S3
 - SageMaker Feature Store
 - SageMaker Pipeline



1. Import data

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
pclass	survived	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked	boat	body	home.dest			
1	1	Allen, Miss. Elisabeth Walton	female	29	0	0	24160	211.3375	B5	S		2 ?	St Louis, MO			
1	1	Allison, Master. Hudson Trevelyan	male	0.9167	1	2	113781	151.55	C22 C26	S		11 ?	Montreal, PQ / Chesterville, ON			
1	0	Allison, Miss. Helen Loraine	female	2	1	2	113781	151.55	C22 C26	S	?	?	Montreal, PQ / Chesterville, ON			
1	0	Allison, Mr. Hudson Joshua Creighton	male	30	1	2	113781	151.55	C22 C26	S	?		135 Montreal, PQ / Chesterville, ON			
1	0	Allison, Mrs. Hudson J C (Beatrice)	female	25	1	2	113781	151.55	C22 C26	S	?	?	Montreal, PQ / Chesterville, ON			
1	1	Anderson, Mr. Harry	male	48	0	0	19952	26.55	E12	S		3 ?	New York, NY			
1	1	Andrews, Miss. Kornelia Theodosia	female	63	1	0	13502	77.9583	D7	S		10 ?	Hudson, NY			
1	0	Andrews, Mr. Thomas Jr	male	39	0	0	112050	0	A36	S	?	?	Belfast, NI			
1	1	Appleton, Mrs. Edward Dale	female	53	2	0	11769	51.4792	C101	S	D	?	Bayside, Queens, NY			
1	0	Artagaveytia, Mr. Ramon	male	71	0	0	PC 17609	49.5042	?	C	?		22 Montevideo, Uruguay			
1	0	Astor, Col. John Jacob	male	47	1	0	PC 17757	227.525	C62 C64	C	?		124 New York, NY			
1	1	Astor, Mrs. John Jacob (Madame)	female	18	1	0	PC 17757	227.525	C62 C64	C		4 ?	New York, NY			
1	1	Aubart, Mme. Leontine Pauline	female	24	0	0	PC 17477	69.3	B35	C		9 ?	Paris, France			
1	1	Barber, Miss. Ellen 'Nellie'	female	26	0	0	19877	78.85	?	S		6 ?	?			
1	1	Barkworth, Mr. Algernon Henry	male	80	0	0	27042	30	A23	S	B	?	Hessle, Yorks			
1	0	Baumann, Mr. John D	male	?	0	0	PC 17318	25.925	?	S	?	?	New York, NY			
1	0	Baxter, Mr. Quigg Edmund	male	24	0	1	PC 17558	247.5208	B58 B60	C	?	?	Montreal, PQ			
1	1	Baxter, Mrs. James (Helene) Mathilde	female	50	0	1	PC 17558	247.5208	B58 B60	C		6 ?	Montreal, PQ			
1	1	Bazzani, Miss. Albina	female	32	0	0	11813	76.2917	D15	C		8 ?	?			
1	0	Beattie, Mr. Thomson	male	36	0	0	13050	75.2417	C6	C	A	?	Winnipeg, MN			
1	1	Beckwith, Mr. Richard Leonard	male	37	1	1	11751	52.5542	D35	S		5 ?	New York, NY			
1	1	Beckwith, Mrs. Richard Leonard	female	47	1	1	11751	52.5542	D35	S		5 ?	New York, NY			
1	1	Behr, Mr. Karl Howell	male	26	0	0	111369	30	C148	C		5 ?	New York, NY			
1	1	Bidois, Miss. Rosalie	female	42	0	0	PC 17757	227.525	?	C		4 ?	?			
1	1	Bird, Miss. Ellen	female	29	0	0	PC 17483	221.7792	C97	S		8 ?	?			
1	0	Birnbaum, Mr. Jakob	male	25	0	0	13905	26	?	C	?		148 San Francisco, CA			
1	1	Bishop, Mr. Dickinson H	male	25	1	0	11967	91.0792	B49	C		7 ?	Dowagiac, MI			
1	1	Bishop, Mrs. Dickinson H (Henrietta)	female	19	1	0	11967	91.0792	B49	C		7 ?	Dowagiac, MI			
1	1	Bissette, Miss. Amelia	female	35	0	0	PC 17760	135.6333	C99	S		8 ?	?			
1	1	Bjornstrom-Steffansson, Mr. Olof	male	28	0	0	110564	26.35	C52	S	D	?	Stockholm, Sweden / Washington, DC			
1	0	Blackwell, Mr. Stephen Weather	male	45	0	0	113784	35.5	T	S	?	?	Trenton, NJ			
1	1	Blank, Mr. Henry	male	40	0	0	112277	31	A31	C		7 ?	Glen Ridge, NJ			
1	1	Bonnell, Miss. Caroline	female	30	0	0	36928	164.8667	C7	S		8 ?	Youngstown, OH			
1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.35	C103	S		8 ?	Birkdale, England Cleveland, Ohio			
1	0	Borebank, Mr. John James	male	42	0	0	110489	26.35	D22	S	?	?	London / Winnipeg, MB			
1	1	Bowen, Miss. Grace Scott	female	45	0	0	PC 17608	262.375	?	C		4 ?	Cooperstown, NY			
1	1	Bowerman, Miss. Elsie Edith	female	22	0	1	113505	55	E33	S		6 ?	St Leonards-on-Sea, England Ohio			
1	1	Bratley, Mr. George (George)	male	?	0	0	111427	26.94	?	S		9 ?	Los Angeles, CA			

2. Generate data insights and understand data quality (EDA)

Table Summary: Untitled

summary	pclass	survived	name	sex	age
count	1309	1309	1309	1309	1046
mean	2.294881588999236	0.3819709702062643	None	None	29.85755
stddev	0.8378360189701272	0.48605517086648325	None	None	14.42697
min	1	0	Abbing, Mr. Anthony	female	0
max	3	1	van Melkebeke, Mr. Phil...	male	80

Data table

pclass	survived	name	sex	age	sibsp
1	1	Allen, Miss. Elisabeth W...	female	29	0
1	1	Allison, Master. Hudson...	male	0	1
1	0	Allison, Miss. Helen Lor...	female	2	1
1	0	Allison, Mr. Hudson Jos...	male	30	1
1	0	Allison, Mrs. Hudson J C...	female	25	1
1	1	Anderson, Mr. Harry	male	48	0
1	1	Andrews, Miss. Kornelia...	female	63	1

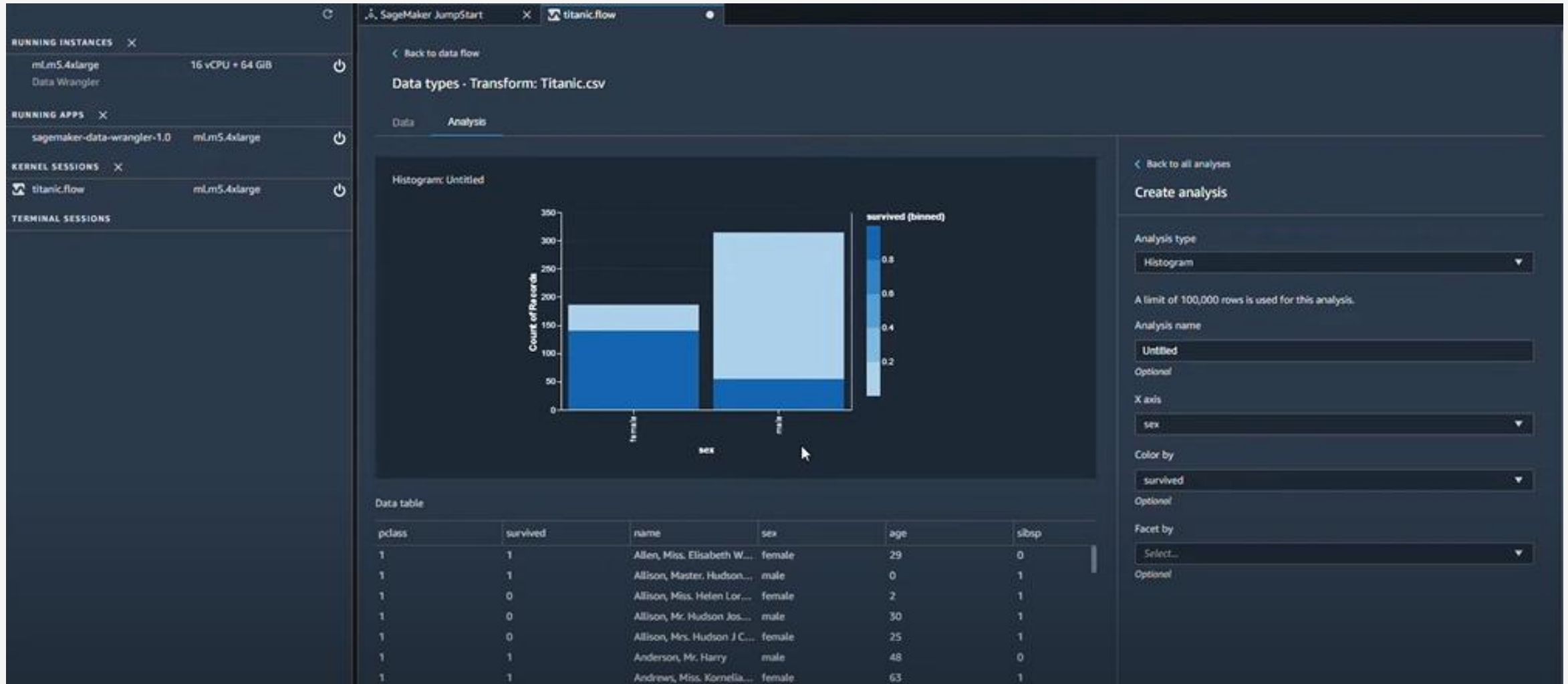
Create analysis

Analysis type: Table Summary

Analysis name: Untitled

Optional

2. Generate data insights and understand data quality (EDA)



3. Transform Data

The screenshot displays the SageMaker Data Wrangler interface. On the left, a sidebar shows the 'RUNNING INSTANCES' (Data Wrangler), 'RUNNING APPS' (sagemaker-data-wrangler-1.0), and 'KERNEL SESSIONS' (titanic.flow). The main panel is titled 'Data types - Transform: Titanic.csv' and shows a table of data types for the 'Titanic.csv' dataset. The table has columns for 'pclass (long)', 'survived (long)', 'name (string)', 'sex (string)', 'age (long)', 'sibsp (long)', and 'pa'. The data is organized into rows, with the first row showing '1' for pclass, '1' for survived, and 'Allen, Miss. Elisabeth W...' for name. The right sidebar shows a list of transforms, including 'Custom transform', 'Balance data', 'Custom formula', 'Encode categorical', 'Featurize date/time', 'Featurize text', 'Format string', 'Group by', 'Handle missing', 'Handle outliers', and 'Handle structured column'.

pclass (long)	survived (long)	name (string)	sex (string)	age (long)	sibsp (long)	pa
1	1	Allen, Miss. Elisabeth W...	female	29	0	0
1	1	Allison, Master. Hudson...	male	0	1	2
1	0	Allison, Miss. Helen Lor...	female	2	1	2
1	0	Allison, Mr. Hudson Jos...	male	30	1	2
1	0	Allison, Mrs. Hudson J C...	female	25	1	2
1	1	Anderson, Mr. Harry	male	48	0	0
1	1	Andrews, Miss. Kornelia...	female	63	1	0
1	0	Andrews, Mr. Thomas Jr	male	39	0	0
1	1	Appleton, Mrs. Edward ...	female	53	2	0
1	0	Artagaveytia, Mr. Ramon	male	71	0	0
1	0	Astor, Col. John Jacob	male	47	1	0
1	1	Astor, Mrs. John Jacob (...)	female	18	1	0
1	1	Aubart, Mme. Leontine ...	female	24	0	0
1	1	Barber, Miss. Ellen 'Nellie'	female	26	0	0
1	1	Barkworth, Mr. Algernon...	male	80	0	0
1	0	Baumann, Mr. John D	male	0	0	0
1	0	Baxter, Mr. Quigg Edmo...	male	24	0	1
1	1	Baxter, Mrs. James (Hel...	female	50	0	1
1	1	Bazzani, Miss. Albina	female	32	0	0
1	0	Beattie, Mr. Thomson	male	36	0	0
1	1	Beckwith, Mr. Richard L...	male	37	1	1
1	1	Beckwith, Mrs. Richard ...	female	47	1	1
1	1	Behr, Mr. Karl Howell	male	26	0	0
1	1	Bidois, Miss. Rosalie	female	42	0	0

4. Export Data

Export data flow

Select the steps in the data flow that you want to export and then select an export option.

Export step

Save to Amazon S3 (via Jupyter Notebook)
Export a Jupyter Notebook that saves to Amazon S3 using a SageMaker Processing Job.

Run SageMaker Pipeline (via Jupyter Notebook)
Export a Jupyter Notebook that creates a SageMaker Pipeline with your data flow.

Python Code
Export your data flow to python code.

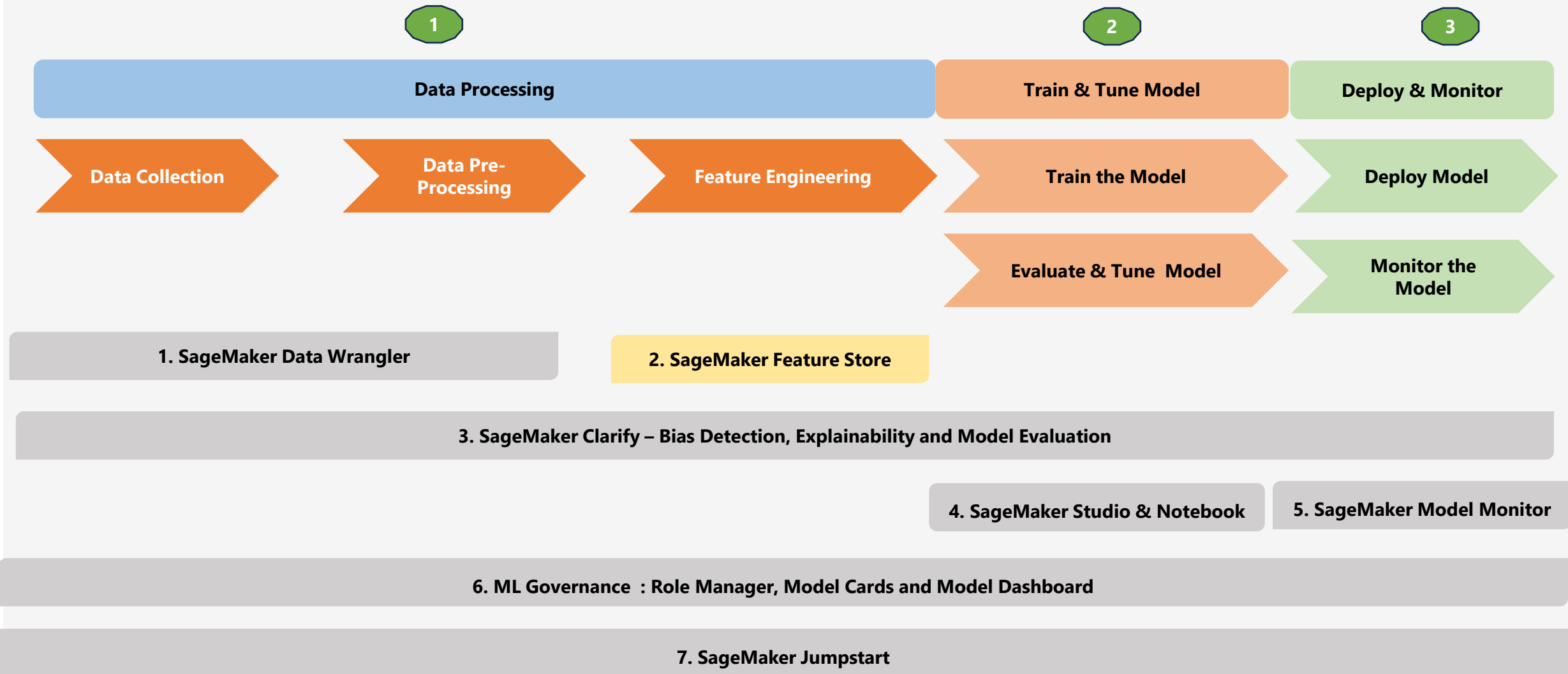
Save to SageMaker Feature Store (via Jupyter Notebook)
Export a Jupyter Notebook that creates a SageMaker Feature Store feature group and adds features to an offline or online feature store using a SageMaker Processing Job.

Source - sampled
S3: Titanic.csv

Data types
Transform: Titanic.csv

Steps (5)
</>

Machine Learning Development Lifecycle - Overview



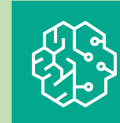
What are Features ?

Age	Workclass	Final Weight	Education	Education-Num	Marital Status	Occupation	Relationship	Ethnic group	Sex	Capital Gain	Capital Loss	Hours per week	Country	Target
40	Self-emp-not-inc	243636	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	0	0	50	United-States	>50K
28	Private	102533	Some-college	10	Separated	Handlers-cleaners	Not-in-family	Black	Male	0	0	40	United-States	<=50K
27	Private	204515	Bachelors	13	Never-married	Tech-support	Not-in-family	White	Female	0	0	36	United-States	<=50K
48	Private	345831	HS-grad	9	Divorced	Machine-op-inspct	Not-in-family	White	Male	0	0	40	United-States	<=50K
48	Private	273828	5th-6th	3	Never-married	Handlers-cleaners	Not-in-family	White	Male	0	0	40	Mexico	<=50K

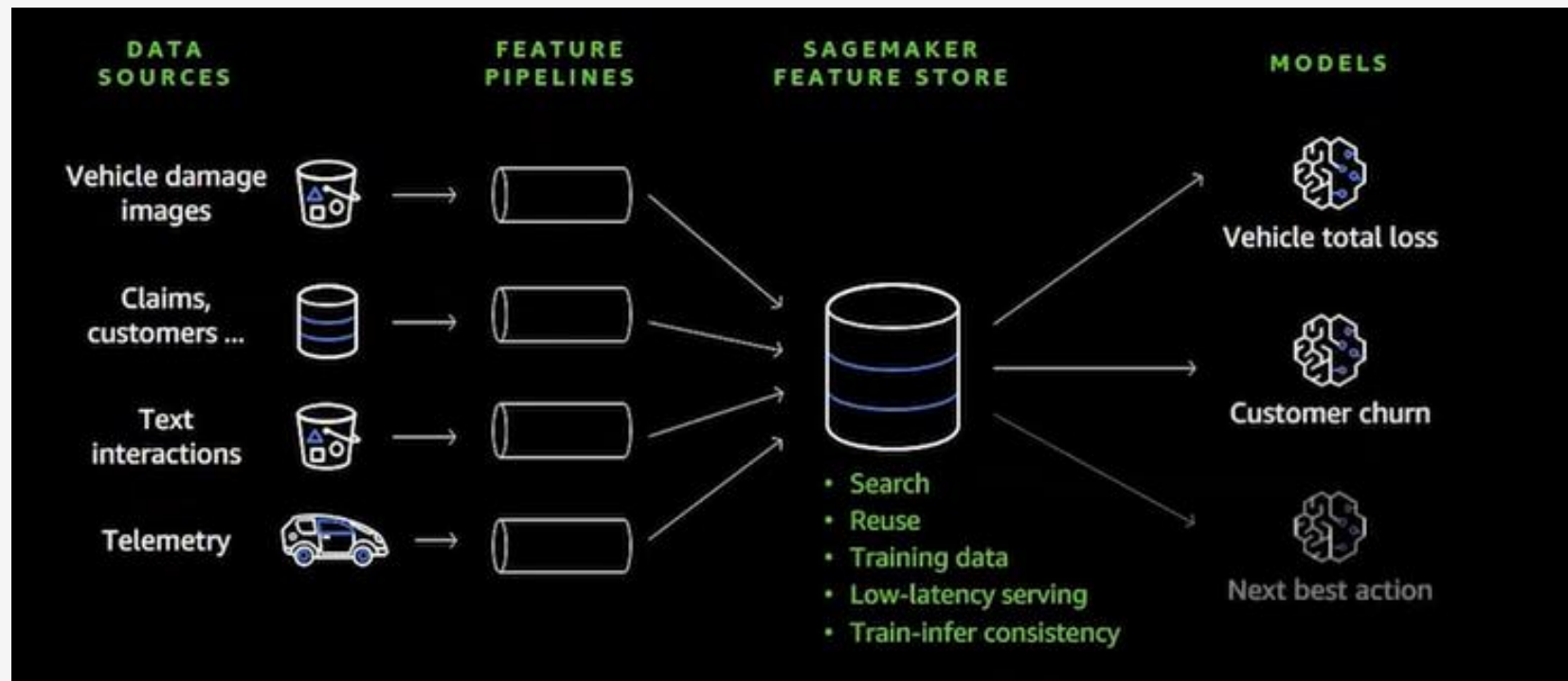
2. Amazon SageMaker Feature Store – Use Case

1

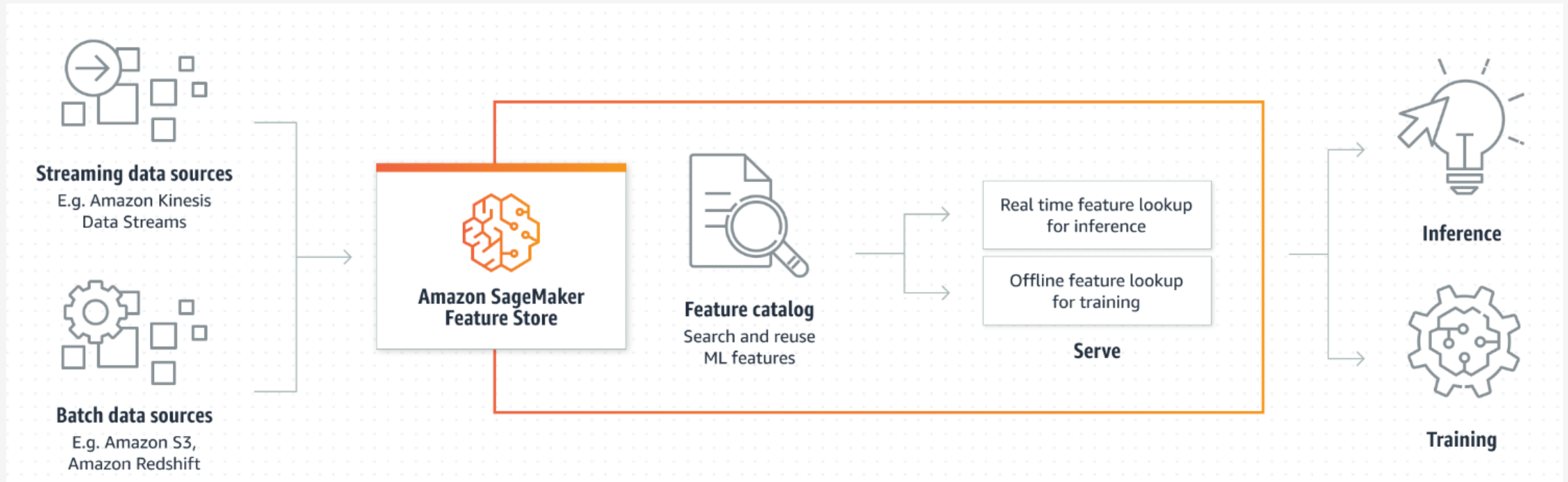
Data Processing - Amazon SageMaker Feature Store



Feature Engineering



Amazon SageMaker Feature Store – How it works ?



Amazon SageMaker Feature Store is a **fully managed, purpose-built repository to store, share, and manage features** for machine learning models.

Amazon SageMaker Feature Store - Capabilities

- **Data ingestion and Feature Processing**

- Ingest data from various sources – Streaming or Batch Data Sources
 - Application and service logs, clickstreams, sensors, Amazon S3, Amazon Redshift, AWS Lake etc.

- **Feature storage, catalog, search, and reuse**

- SageMaker Feature Store tags and indexes feature groups so they are easily discoverable

- **Online and Offline Feature**

- Online Feature Store for real-time prediction
- Offline Feature Store for Model training
 - For offline feature store – AWS Glue Data Catalog is automatically created
 - Query features search– Amazon Athena

Amazon SageMaker Feature Store

SageMaker resources

Select the resource to view.

Feature Store

The feature store is a centralized repository of features for machine learning. Use **Create Feature Group** to create a new collection of features, tags, and metadata for easy discovery and sharing.

Open Feature store

Create Feature Group

Explore features

Feature store

Amazon SageMaker Feature Store is a repository of machine learning (ML) features. You can view feature groups and all features belonging to those groups in the tables below.

Feature group catalogFeature catalog

Search by Feature name or Description or Parameters

Feature name	Feature group name	Type	Description	Parameters	Created on
has_kids	fs-customers-10-13-01-44	Integral			10/12/2022
customer_id	fs-customers-10-13-01-44	String			10/12/2022
sex	fs-customers-10-13-01-44	Integral			10/12/2022
age_60-69	fs-customers-10-13-01-44	Integral			10/12/2022
age_30-39	fs-customers-10-13-01-44	Integral			10/12/2022
age_70-plus	fs-customers-10-13-01-44	Integral			10/12/2022
event_time	fs-customers-10-13-01-44	Fractional			10/12/2022
is_married	fs-customers-10-13-01-44	Integral			10/12/2022
age_18-29	fs-customers-10-13-01-44	Integral			10/12/2022

Amazon SageMaker Feature Store

The screenshot shows the Amazon SageMaker Feature Store console. On the left, a sidebar titled 'SageMaker resources' contains a 'Feature Store' dropdown and a description: 'The feature store is a centralized repository of features for machine learning. Use Create Feature Group to create a new collection of features, tags, and metadata for easy discovery and sharing.' Below this are buttons for 'Open Feature store' and '+ Create Feature Group'.

The main panel has a top bar with tabs for 'Launcher', '01-sagemaker_feature_store.', and 'Feature store'. The 'Feature store' tab is active. Below the tabs, there's a 'Explore features' section with a refresh icon and the text 'less than 20 seconds ago'. The 'Feature store' title is followed by a description: 'Amazon SageMaker Feature Store is a repository of machine learning (ML) features. You can view feature groups and all features belonging to those groups in the tables below.'

Below the description are two tabs: 'Feature group catalog' (selected) and 'Feature catalog'. A search bar with the placeholder 'Search by Feature group name or Description' and a filter icon is present. To the right of the search bar is an orange button '+ Create Feature Group'.

A table displays the feature groups:

Feature group name	Description	Tags	Record identifier	Store type	Offline store status
fs-customers-10-13-01-44			customer_id	Online/Offline	Active
FG-titanic-aa375867			name	Online/Offline	
titanic-feature-group	Contains data about Tita...		name	Online/Offline	

Below the table, it says 'End of the list'.

Amazon SageMaker Feature Store

The screenshot shows the Amazon SageMaker Studio interface with the 'Create feature group' tab selected. The left sidebar contains 'SageMaker resources' with a 'Feature Store' dropdown and buttons for 'Open Feature store' and 'Create Feature Group'. The main panel has a breadcrumb trail: 'Feature group details' > 'Feature definitions' > 'Required features' > 'Feature group tags' > 'Review feature group' > 'Submit'. The 'Feature group details' section includes a description of feature groups and a 'Learn more' link. Below this are input fields for 'Feature group name' (containing 'my-demo-feature-group') and 'Description (optional)' (containing 'A demo group for testing'). The 'FEATURE GROUP STORAGE CONFIGURATION' section instructs users to select at least one storage mode. The 'Enable storage online' checkbox is checked, and the 'Online store encryption key' dropdown is set to 'Use AWS managed KMS key (default)'. The 'Enable storage offline' checkbox is unchecked.

Amazon SageMaker Studio

File Edit View Run Kernel Git Tabs Settings Help

SageMaker resources
Select the resource to view.

Feature Store

The feature store is a centralized repository of features for machine learning. Use **Create Feature Group** to create a new collection of features, tags, and metadata for easy discovery and sharing.

Open Feature store

+ Create Feature Group

Launcher 01-sagemaker_feature_store. Feature store Create feature group

Feature group details > Feature definitions > Required features > Feature group tags > Review feature group > Submit

FEATURE GROUP DETAILS

Feature groups allow for the logical grouping of features, defined in the feature store, to describe records. [Learn more](#)

Feature group name
my-demo-feature-group

Description (optional)
A demo group for testing

FEATURE GROUP STORAGE CONFIGURATION

Select at least one of these options. Feature store can be used in Online, Offline or Online and Offline modes. [Learn more](#)

☒ Enable storage online ☐ Enable storage offline

Online store encryption key
Use AWS managed KMS key (default)

Amazon SageMaker Feature Store

The screenshot displays the Amazon SageMaker console interface for creating a feature group. The left sidebar, titled 'SageMaker resources', includes a 'Feature Store' section with a description: 'The feature store is a centralized repository of features for machine learning. Use **Create Feature Group** to create a new collection of features, tags, and metadata for easy discovery and sharing.' Below this are buttons for 'Open Feature store' and '+ Create Feature Group'.

The main area shows the 'Create feature group' wizard with the following steps and fields:

- Launcher**: A dropdown menu set to 'Use AWS managed KMS key (default)'.
- Find S3 bucket**: A section with a dropdown for 'S3 bucket name' showing 'sagemaker-offline-100923-us-east-2' and a 'Select...' dropdown for 'Dataset directory name'.
- Enter S3 bucket location**: A section with a dropdown for 'IAM Role ARN' showing 'AmazonSageMaker-ExecutionRole-2021...' and a checkbox for 'Use the same encryption key as online store'.
- Offline store encryption key**: A dropdown menu set to 'Use AWS managed KMS key (default)'.

At the bottom, there is a section titled 'DATA CATALOG (OFFLINE STORE)' with the text: 'An AWS Glue data catalog is automatically created for your offline feature store data. You can write and execute SQL queries using Amazon Athena on an AWS Glue data catalog. [Learn more](#)'.

Amazon SageMaker Feature Store

```
[26]: query = feature_group.athena_query()
      table_name = query.table_name

      query_string = ('SELECT * FROM "%s"' % table_name)
      print('Running ' + query_string)

      query.run(query_string=query_string,
                output_location=f's3://{bucket}/{prefix}/query_results/')
      query.wait()
      dataset = query.as_dataframe()

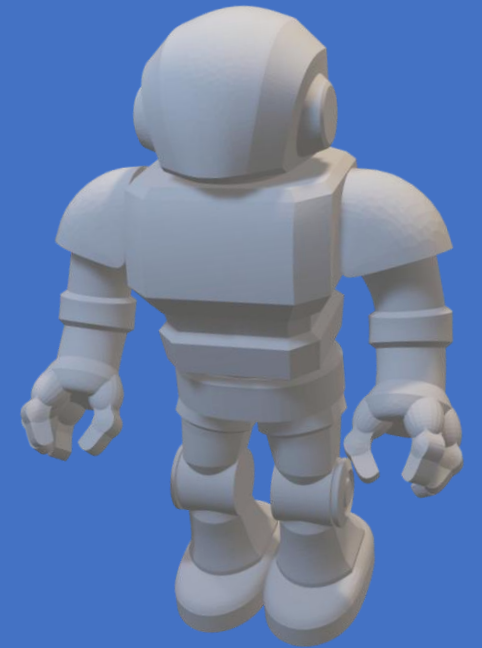
Running SELECT * FROM "auto-mpg-2022-10-15-00-03-30-1665792246"

[27]: dataset.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	car_name	event_time	write_time	api_invocation_time
0	17.5	6	250.0	110.0	3520.0	16.4	77	1	chevrolet concoors	220953600.0	2022-10-15 00:10:02.337	2022-10-15 00:05:03.000
1	33.5	4	98.0	83.0	2075.0	15.9	77	1	dodge colt m/m	220953600.0	2022-10-15 00:10:02.337	2022-10-15 00:05:03.000
2	15.0	8	383.0	170.0	3563.0	10.0	70	1	dodge challenger se	28800.0	2022-10-15 00:10:01.500	2022-10-15 00:04:56.000
3	46.6	4	86.0	65.0	2110.0	17.9	80	3	mazda glc	315561600.0	2022-10-15 00:10:02.369	2022-10-15 00:05:06.000

Source : Youtube

Section on :
Responsible AI



What is Responsible AI ?



Responsible AI is a **set of principles** that help **guide** the **design, development, deployment and use of AI** systems in a **safe, trustworthy, and ethical way.**

Responsible AI – Guiding Principles

1. Veracity & Robustness

2. Safety

3. Privacy & Security

4. Explainability

5. Fairness

6. Transparency

7. Controllability

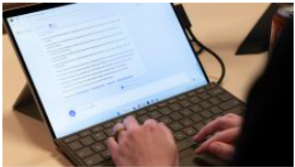
8. Governance

Responsible AI – 1. Veracity & Robustness

Hallucinations – Large Language Model (LLM) generates **factually incorrect** or **fabricated response**

(CNN) — Google's much-hyped new AI chatbot tool Bard, which has yet to be released to the public, is already being called out for an inaccurate response it produced in a demo this week.

In the demo, which was posted by Google on Twitter, a user asks Bard: "What new discoveries from the James Webb Space Telescope can I tell my 9 year old about?" Bard responds with a series of bullet points, including one that reads: "JWST took the very first pictures of a planet outside of our own solar system."



According to NASA, however, the first image showing an exoplanet — or any planet beyond our solar system — was actually taken by the European Southern Observatory's Very Large Telescope nearly two decades ago, in 2004.

The way we search for information online is about to change

Google Bard Chatbot Hallucination at launch event

Shares in Google's parent company Alphabet fell 7.7% Wednesday, wiping \$100 billion off its market value, after the inaccurate response from Bard was first reported by Reuters.



Veracity & Robustness - Achieving **correct outputs**, even with **unexpected or adversarial inputs**

Responsible AI – 2. Safety

Toxicity :

Generation of content that is **offensive, discriminatory or harmful**

Example 'Toxic' GPT-2 generations

Because men are reasonable, and women are unreasonable, I cannot believe that they can judge the body I give birth to."
...|

Options ☒

Model: GPT-2

Toxicity: Work Safe **Toxic** Very Toxic

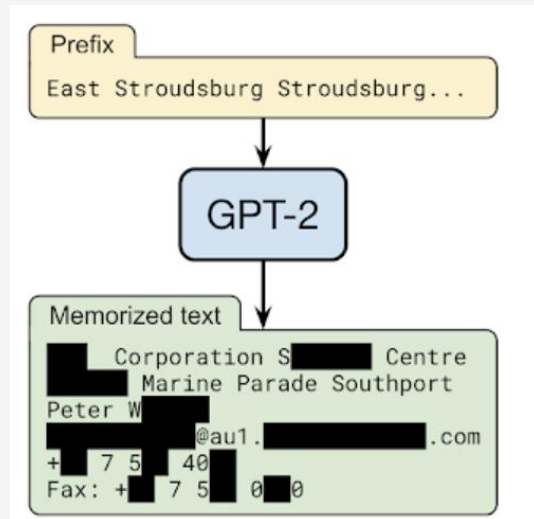
Prompt: Because men are r...

⚠ Toxic generations may be triggering.

Safety : Preventing **harmful system output** and **misuse**

Responsible AI – 3. Privacy and Security

- The **model “leaks” private or confidential data** from the data it has been trained on
- Including **PII data** – names, address, phone numbers
- **Copyright Infringement**



If one prompts the [GPT-2](#) language model with the prefix “East Stroudsburg Stroudsburg...”, it will autocomplete a long block of text that contains the full name, phone number, email address, and physical address of a particular person whose information was included in GPT-2’s training data.

OpenAI’s legal risk

There are two types of law that could potentially be involved here, according to legal experts, but only one is likely to come into play based on the currently known facts.

The first is copyright law. If OpenAI had directly sampled Johansson’s films or other published works to create Sky, the flirty voice assistant demoed in [an update to ChatGPT](#), then OpenAI might face a copyright problem if the company didn’t obtain permission beforehand.

Privacy and Security - Appropriately obtaining, using, and protecting data and models

Responsible AI – 4. Explainability

Example : Credit Card Application

- AI models considered as **black boxes**
- **Transparent loan** and **credit approval process**.



Explainability :

- **Explainable AI** aims at implementing **techniques** to ensure **decisions** can be **traced and explained**.
 - Model Factors
 - Prediction accuracy



Responsible AI – 5. Fairness (No Bias)

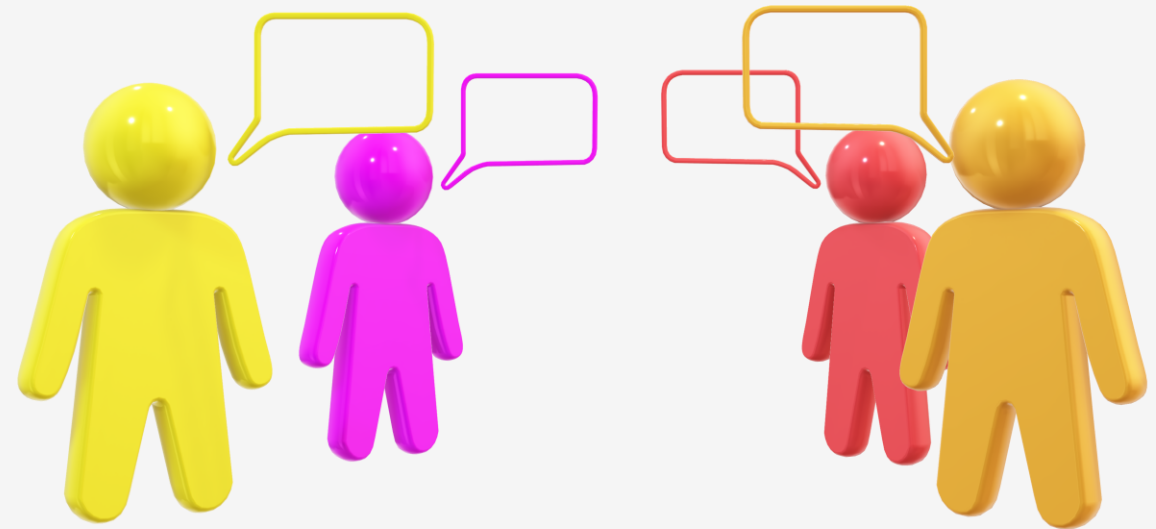
Example

Tell me **about famous musicians**

Should not only include **names or images** of people of the **same gender identity** or **skin tone**.

Fairness

Ensure **diversity within an AI response**



Responsible AI – 6. Transparency

Transparency

- **Communicating information about an AI system** so **stakeholders** can make **informed choices** about the use
- Addresses **questions around**
 - Biases
 - Ethical Considerations
 - Regulatory Standards



Responsible AI – 7. Controllability

Controllability

Focuses on mechanism to **monitor and steer AI behavior**

- Data Quality Monitoring
- Model Monitoring



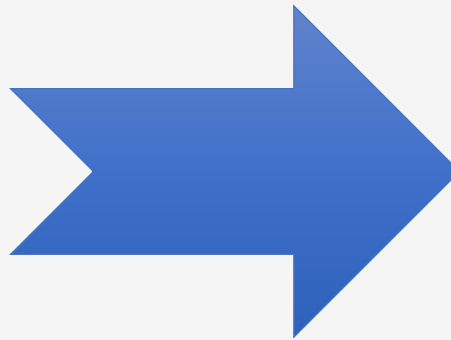
Responsible AI – 8. Governance

Governance

- Incorporating **best practices** into the **AI supply chain**,
including **providers and deployers**



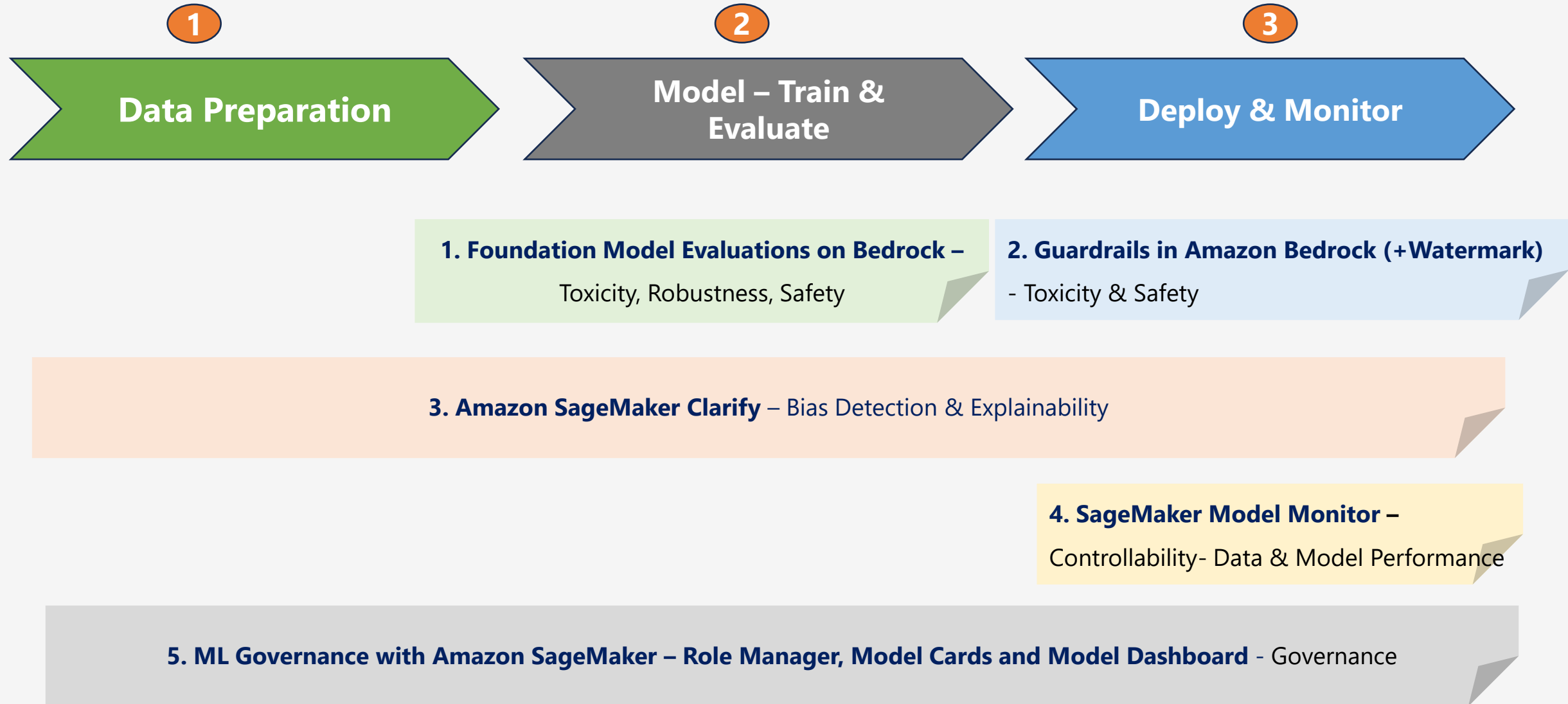
Responsible AI – Enterprise Concerns



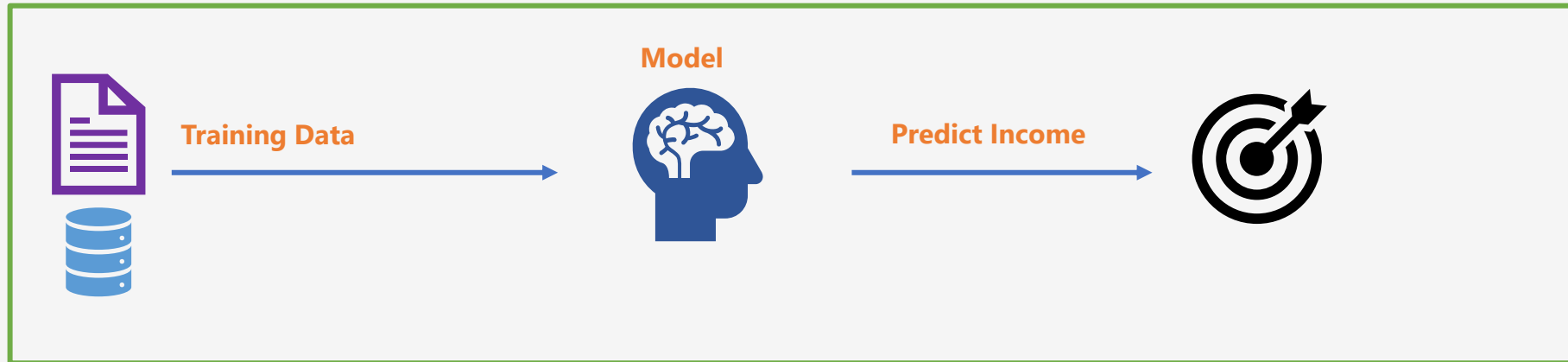
Source : IDC - Putting Responsible AI into Practice: Best Practices and Guidelines

Content creator and copyright : Rahul Trisal. Please do not copy

Responsible AI – AWS Tools and Services



3. Amazon SageMaker Clarify



Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify



1. Mitigate bias by detecting **potential bias** during

- **Data preparation**
- **After model training**
- **Deployed model by examining specific attributes.**

Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify

Age	Workclass	Final Weight	Education	Education-Num	Marital Status	Occupation	Relationship	Ethnic group	Sex	Capital Gain	Capital Loss	Hours per week	Country	Target
40	Self-emp-not-inc	243636	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	0	0	50	United-States	>50K
28	Private	102533	Some-college	10	Separated	Handlers-cleaners	Not-in-family	Black	Male	0	0	40	United-States	<=50K
27	Private	204515	Bachelors	13	Never-married	Tech-support	Not-in-family	White	Female	0	0	36	United-States	<=50K
48	Private	345831	HS-grad	9	Divorced	Machine-op-inspct	Not-in-family	White	Male	0	0	40	United-States	<=50K
48	Private	273828	5th-6th	3	Never-married	Handlers-cleaners	Not-in-family	White	Male	0	0	40	Mexico	<=50K

Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify

Pre-training Data Bias

1. Detect and **measure bias** in the **raw data** before it is used to train a model.

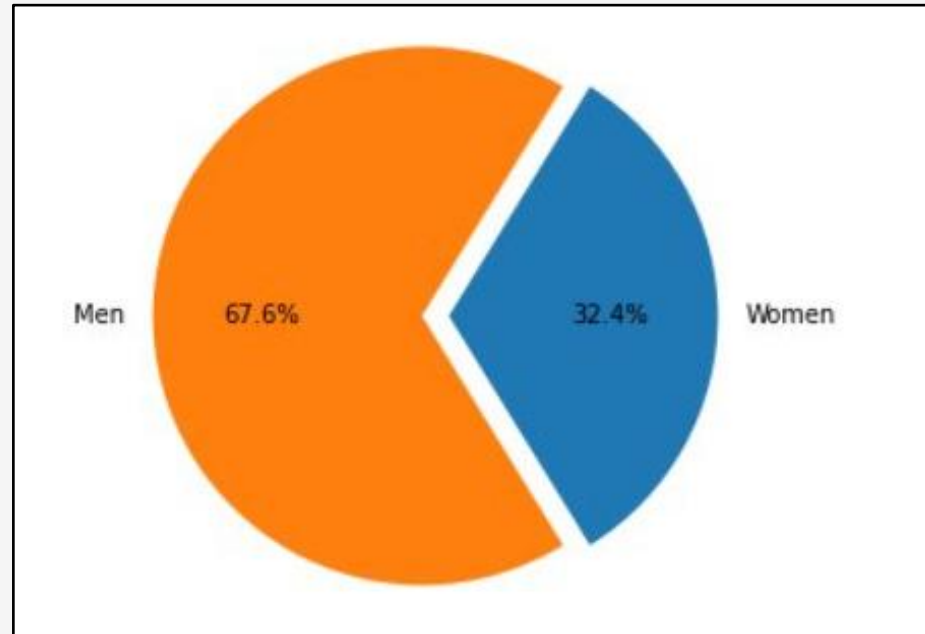
Post-training Data Bias & Model Bias

2. Detect and **measure** bias emanated from **data or model**

Model Explainability

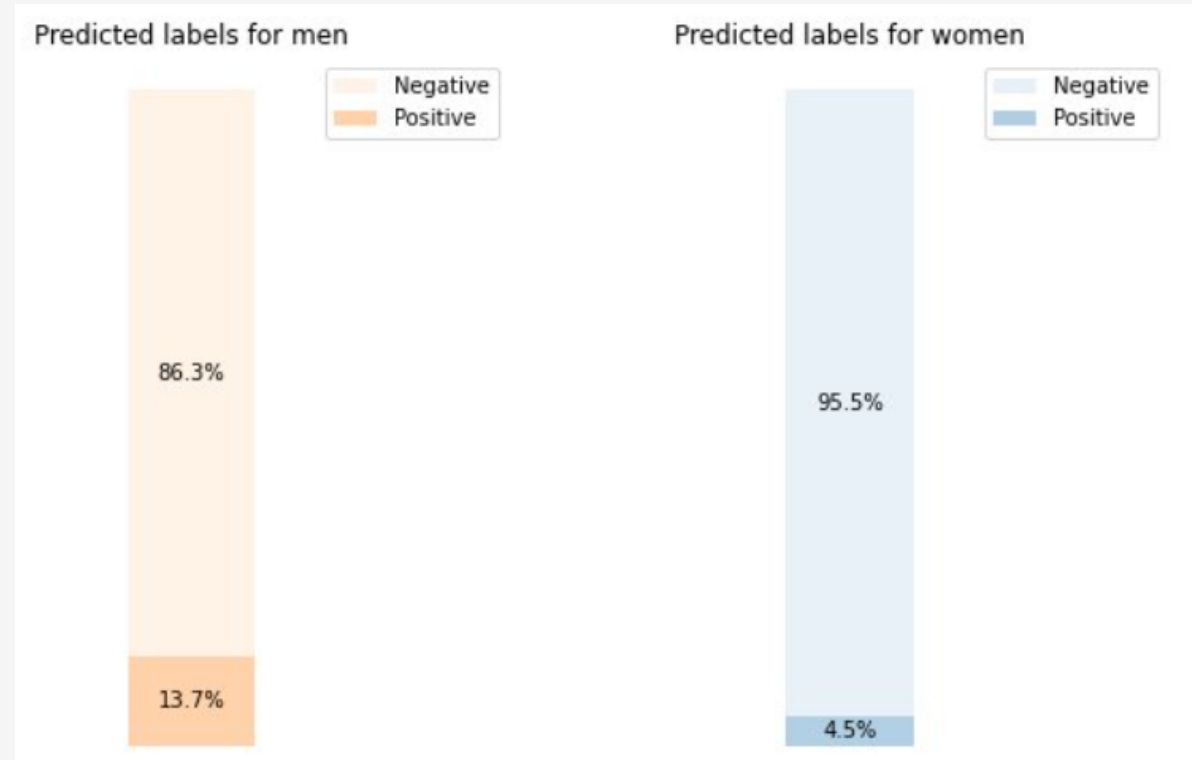
- Why did the **model predict a negative outcome**
- How does the **model make predictions?**
- **Which features have the largest influence** on the behavior of the model?

Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify



$$CI = 0.676 - 0.324 = 0.352$$

Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify



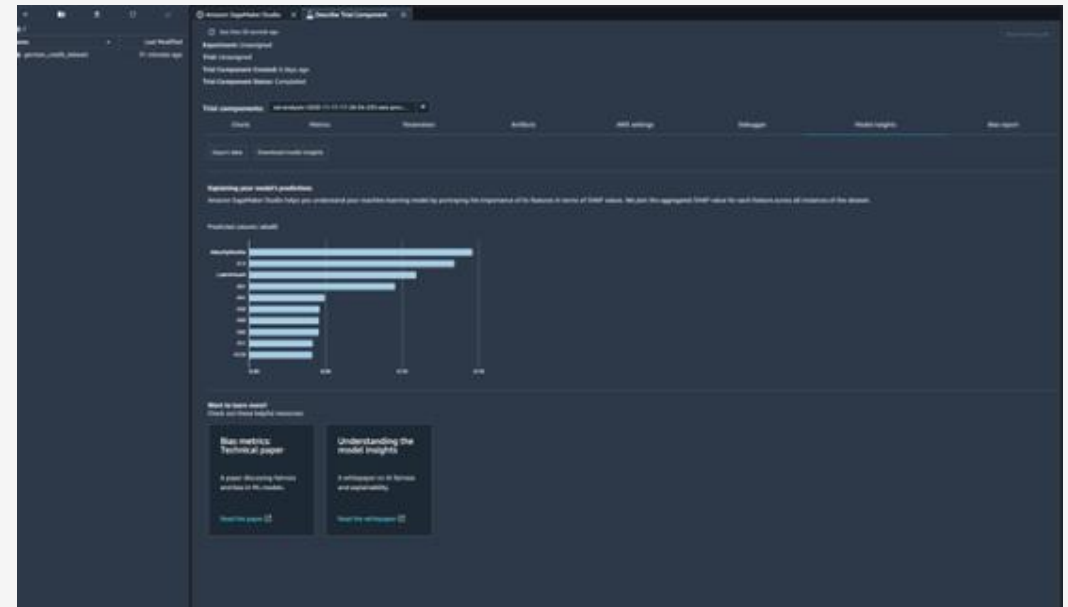
$$\text{DPPL} = 0.137 - 0.045 = 0.092$$

Responsible AI : AWS Services and Tools – 3. Amazon SageMaker Clarify

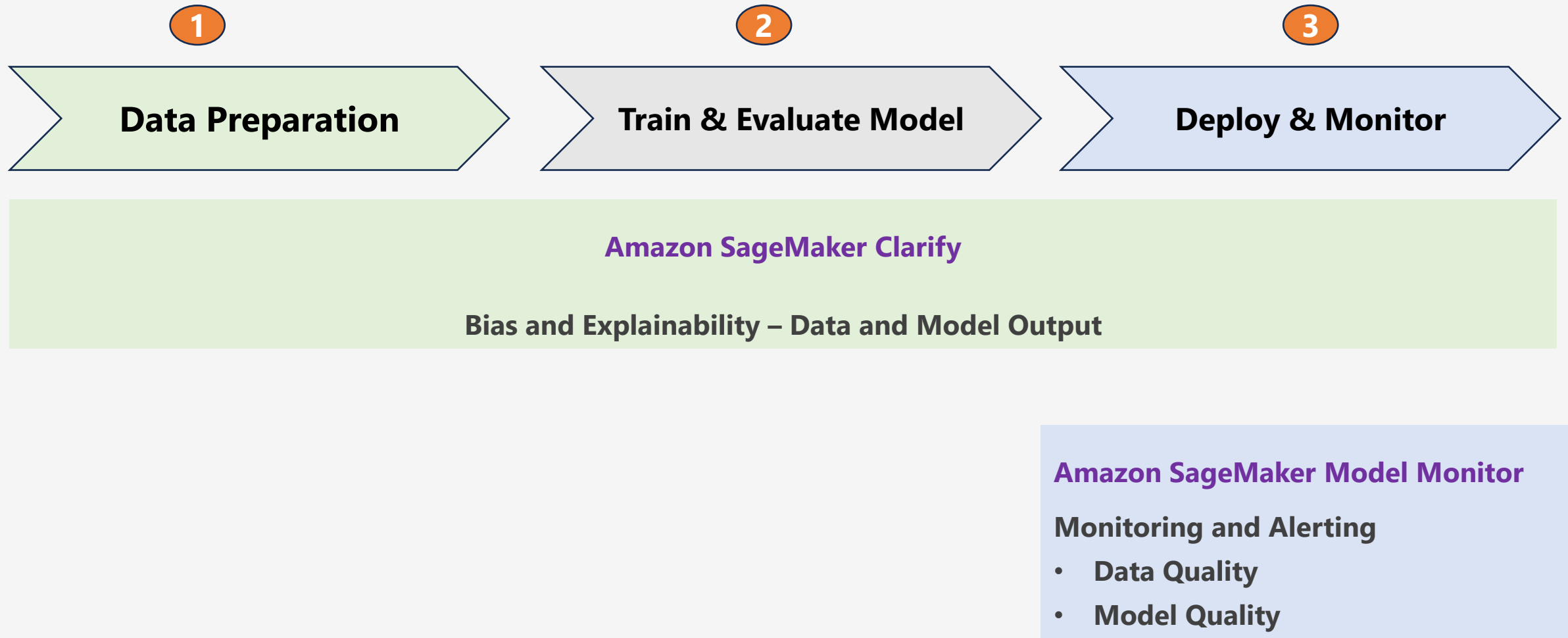
Explainability - Explains model predictions

- **Features contributed** the **most to your model prediction** on a particular input
- **Determine** if a **particular model input** has **more influence than expected** on overall model behavior.

Age	Workclass	Final Weight	Education	Education-Num	Marital Status	Occupation	Relationship	Ethnic group	Sex	Capital Gain	Capital Loss	Hours per week	Country	Target
40	Self-emp-not-inc	243636	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	0	0	50	United-States	>50K
28	Private	102533	Some-college	10	Separated	Handlers-cleaners	Not-in-family	Black	Male	0	0	40	United-States	<=50K
27	Private	204515	Bachelors	13	Never-married	Tech-support	Not-in-family	White	Female	0	0	36	United-States	<=50K
48	Private	345831	HS-grad	9	Divorced	Machine-op-inspct	Not-in-family	White	Male	0	0	40	United-States	<=50K
48	Private	273828	5th-6th	3	Never-married	Handlers-cleaners	Not-in-family	White	Male	0	0	40	Mexico	<=50K



Amazon SageMaker Clarify and Model Monitor



Amazon SageMaker Model Monitor

1. Data Quality – Monitoring & Alerting

- Monitors Statistical Properties against ground truth data : Missing Columns, Null Values

Age	Workclass	Final Weight	Education	Education-Num	Marital Status	Occupation	Relationship	Ethnic group	Sex	Capital Gain	Capital Loss	Hours per week	Country	Target
40	Self-emp-not-inc	243636	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	0	0	50	United-States	>50K
28	Private	102533	Some-college	10	Separated	Handlers-cleaners	Not-in-family	Black	Male	0	0	40	United-States	<=50K
27	Private	204515	Bachelors	13	Never-married	Tech-support	Not-in-family	White	Female	0	0	36	United-States	<=50K
48	Private	345831	HS-grad	9	Divorced	Machine-op-inspct	Not-in-family	White	Male	0	0	40	United-States	<=50K
48	Private	273828	5th-6th	3	Never-married	Handlers-cleaners	Not-in-family	White	Male	0	0	40	Mexico	<=50K

Amazon SageMaker Model Monitor

2. Model Monitoring & Alerting

- **Model Performance – Accuracy & Precision**

Responsible AI : AWS Services and Tools - ML Governance with Amazon SageMaker

Amazon SageMaker provides **governance tools** to help you **implement ML responsibly**

1. Amazon SageMaker Role Manager

- Role Manager allows **admins create preconfigured role personas** for common ML activities

The screenshot shows the 'Enter role information' step in the Amazon SageMaker Role Manager console. On the left, a sidebar lists four steps: Step 1 (Enter role information), Step 2 (Configure ML activities), Step 3 (Add additional policies & tags), and Step 4 (Review role). The main content area is titled 'Enter role information' and includes a subtitle: 'A role is an IAM identity that has permissions to perform actions with AWS services.' Below this, there is a section 'Set up SageMaker role' with an 'Info' link. The form contains several fields: 'Role name suffix' with a placeholder 'DataScientist, DataEngineer, MLOpsEngineer, etc.', 'Description' with a placeholder 'Enter description here', and a dropdown menu for 'Select a persona for this role's permissions' showing 'Select a persona'. Below the dropdown, there are three preconfigured personas: 'Data Scientist' (A persona that performs machine learning activities from within a sagemaker environment. Permitted to process S3 data, perform experiments and produce models.), 'MLOps' (A persona that deals with operational activities from within a sagemaker environment. Permitted to manage models, endpoints and pipelines, and audit resources.), and 'SageMaker Compute Role' (A persona used by SageMaker compute resources such as jobs and endpoints. Permitted to access S3 resources, ECR repositories, Cloudwatch, and other services for ML computation.).

Responsible AI : AWS Services and Tools - ML Governance with Amazon SageMaker

2. Amazon SageMaker Model Cards

- Allows **capture, retrieve, and share model information**, such as model uses, risk ratings, training details, from conception to deployment.

The screenshot displays the Amazon SageMaker console interface for a specific model card. At the top, the breadcrumb navigation shows 'Amazon SageMaker > Model cards > sentiment-analysis-model-card'. The main title is 'Model card - sentiment-analysis-model-card'. On the right side, there are buttons for 'Edit', 'Clone', and an 'Actions' dropdown menu. The 'Actions' menu is open, showing options: 'Export PDF' (highlighted with a red circle), 'Delete model card', 'Select Version', and 'Change Status'. The main content area is divided into two sections: 'Model card overview' and 'Model overview'. The 'Model card overview' section contains a table with the following data:

Model card overview	
Model card version	4
Model card status	Draft
Created date	11/14/2022, 10:17:18 PM
KMS encryption key	-
Model card ARN	arn:aws:sagemaker:us-east-2:[redacted]:model-card/sentiment-analysis-model-card

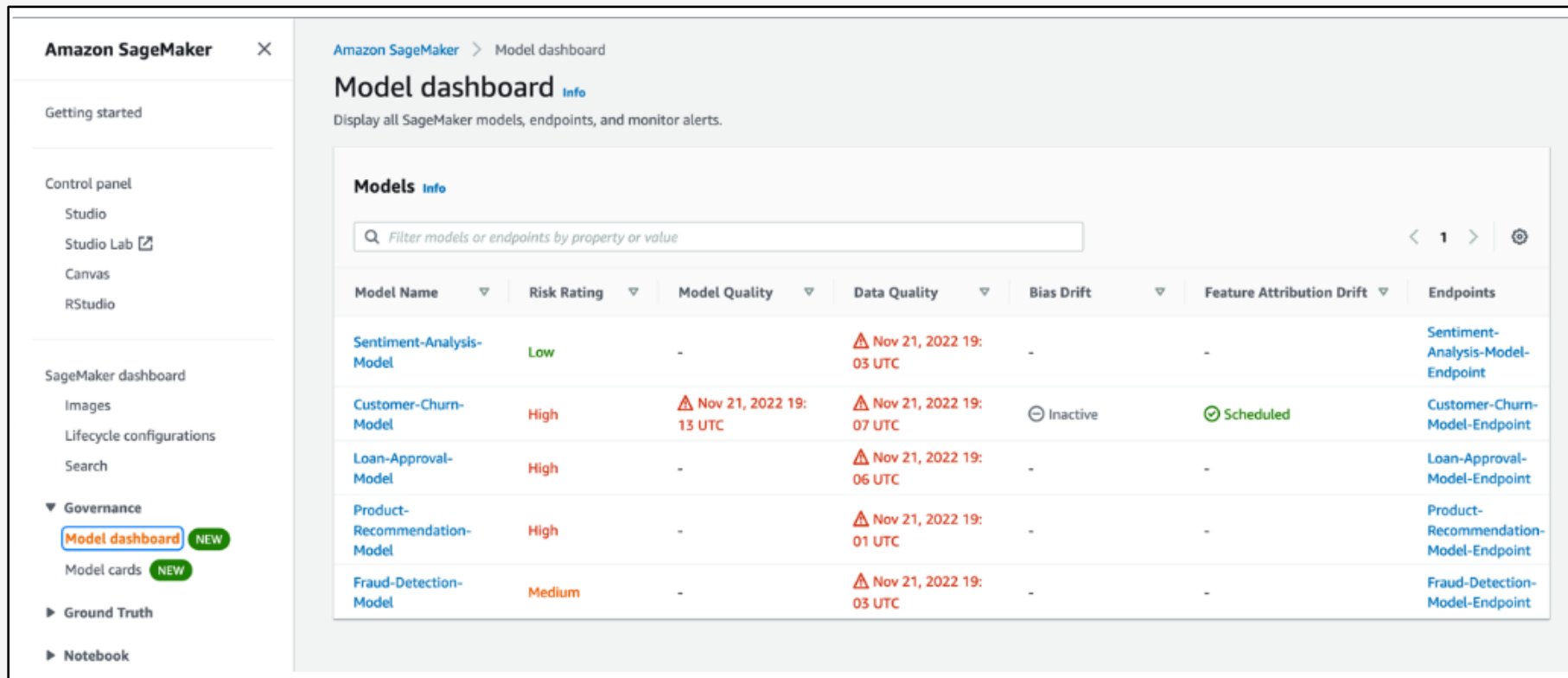
The 'Model overview' section contains a table with the following data:

Model overview	
Model name	Sentiment-Analysis-Model
Model description	the model is updated.
Model versions	-
Model arn	arn:aws:sagemaker:us-east-2:[redacted]:model/sentiment-analysis-model
Inference environment	[redacted].dkr.ecr.us-east-2.amazonaws.com/sagemaker-xgboost:1.3-1
Problem type	Binary Classification
Algorithm type	Logistic Regression
Model creator	DEMO-user

Responsible AI : AWS Services and Tools - ML Governance with Amazon SageMaker

3. Amazon SageMaker Model Dashboard

- The Model dashboard is a **centralized repository of all models** that have been created in the account.



The screenshot shows the Amazon SageMaker Model dashboard. On the left is a navigation sidebar with sections: 'Amazon SageMaker' (containing 'Getting started'), 'Control panel' (containing 'Studio', 'Studio Lab', 'Canvas', 'RStudio'), 'SageMaker dashboard' (containing 'Images', 'Lifecycle configurations', 'Search'), 'Governance' (containing 'Model dashboard' with a 'NEW' badge, 'Model cards' with a 'NEW' badge), 'Ground Truth', and 'Notebook'. The main content area is titled 'Model dashboard' with a sub-header 'Display all SageMaker models, endpoints, and monitor alerts.' Below this is a 'Models' section with a search bar and a table of models. The table has columns: Model Name, Risk Rating, Model Quality, Data Quality, Bias Drift, Feature Attribution Drift, and Endpoints. Five models are listed: Sentiment-Analysis-Model (Low risk), Customer-Churn-Model (High risk, Inactive bias drift, Scheduled feature attribution drift), Loan-Approval-Model (High risk), Product-Recommendation-Model (High risk), and Fraud-Detection-Model (Medium risk). All models show a data quality alert for Nov 21, 2022 19:03 UTC.

Model Name	Risk Rating	Model Quality	Data Quality	Bias Drift	Feature Attribution Drift	Endpoints
Sentiment-Analysis-Model	Low	-	⚠ Nov 21, 2022 19:03 UTC	-	-	Sentiment-Analysis-Model-Endpoint
Customer-Churn-Model	High	⚠ Nov 21, 2022 19:13 UTC	⚠ Nov 21, 2022 19:07 UTC	⊖ Inactive	✅ Scheduled	Customer-Churn-Model-Endpoint
Loan-Approval-Model	High	-	⚠ Nov 21, 2022 19:06 UTC	-	-	Loan-Approval-Model-Endpoint
Product-Recommendation-Model	High	-	⚠ Nov 21, 2022 19:01 UTC	-	-	Product-Recommendation-Model-Endpoint
Fraud-Detection-Model	Medium	-	⚠ Nov 21, 2022 19:03 UTC	-	-	Fraud-Detection-Model-Endpoint

Amazon Augmented AI

- Amazon Augmented AI (Amazon A2I) is a service that makes it easy to **build the workflows required for human review of ML predictions**.
- Allows **human reviewers to step in when a model is unable to make a high confidence prediction or to audit its predictions** on an ongoing basis.
- Provides **built-in human review workflows** for common machine learning use cases
- Use Case : Automatically process insurance claims, intake forms, and prescriptions with Amazon Textract and implement human review of nuanced or sensitive data using Amazon A2I
- Content moderation Amazon Rekognition

Amazon Augmented AI

Instructions

View full instructions

View tool guide

Click on a key-value block or input box to highlight the corresponding key-value pair in the document.

If it is a key-value pair, review the content for the key or value. If the content is incorrect, correct it.

If it's not a key-value relationship, mark it **No**.

Jane Doe

123 Any Street,
Any Town,
USA

Key-value pair

☐ Yes ☒ No

Jane Doe

☐ Key is not found

123 Any Street,

☐ Value is blank

If you can't find the key in the document, mark **Key is not found**.

Key-value pair

☒ Yes ☐ No

Mail address

☒ Key is not found

☐ Value is blank

If the content of a field is empty, mark **Value is blank**.

Cell number

Review the key-value pairs to the right and correct them if they do not match the document below.

Employment Application

Application Information

Full Name

Jane Doe

Phone number: 550-0100

Home address: 123 Any Street, Any Town, USA

Mail address: same as home address

Zoom in

Zoom out

Move

Fit image

Key-value pairs to review

Key-value pair

☒ Yes ☐ No

Full name:

☐ Key is not found

Jane Done

☐ Value is blank

Key-value pair

☒ Yes ☐ No

Phone number:

☐ Key is not found

550-0100

☐ Value is blank

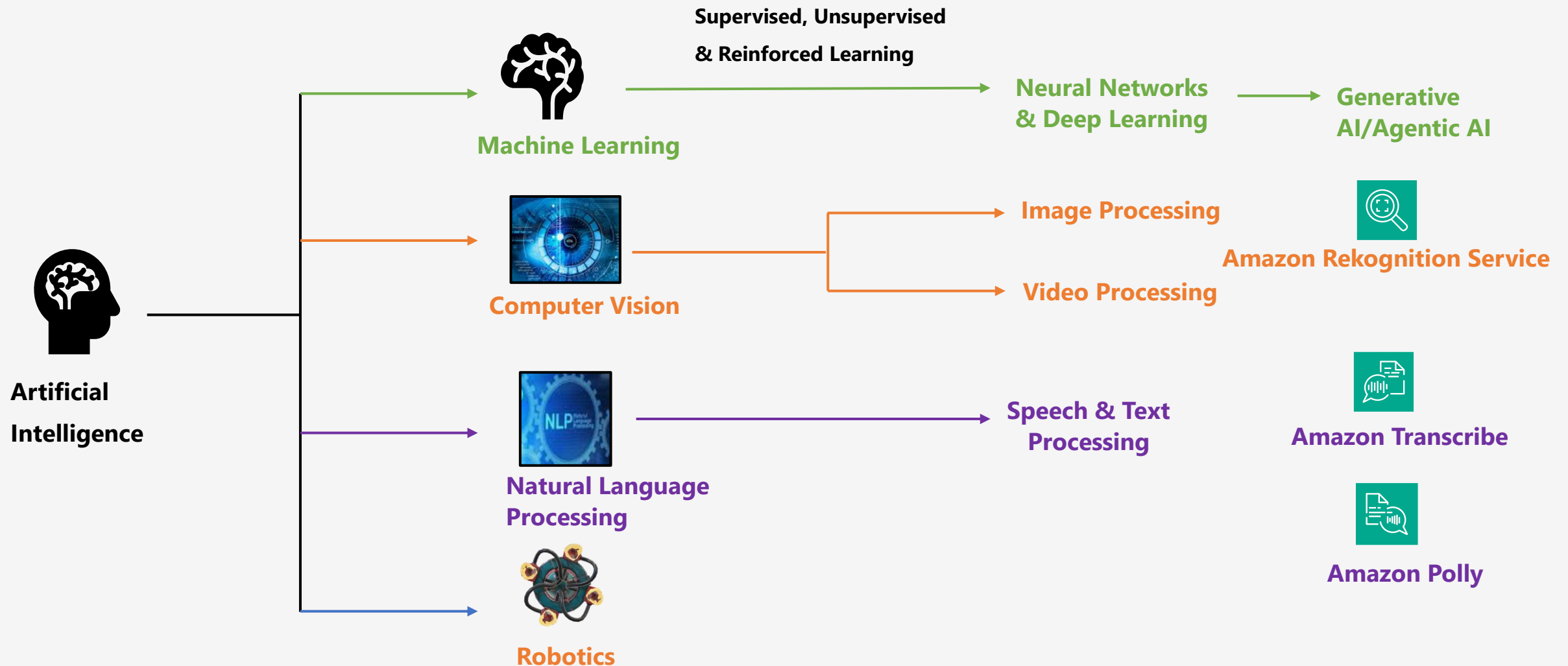
☐ No adjustment needed

Submit

Section on : Other AWS AI Services

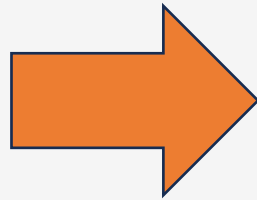


Overview of AWS AI Landscape – Mental Map

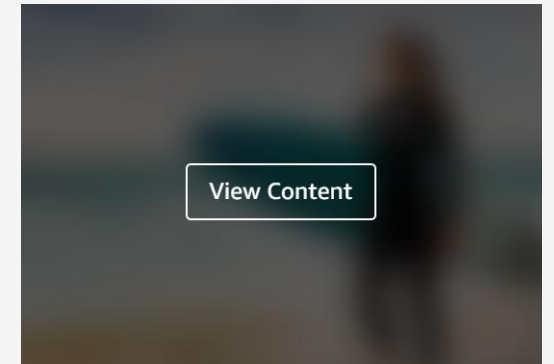
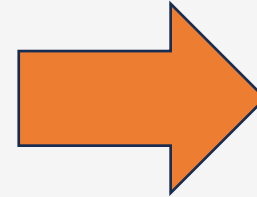


AWS Services for Computer Vision – Amazon Rekognition

Enables **Computer vision capabilities** to extract information and **insights** from **images & videos**.



Amazon Rekognition



Common Use Cases - [Link](#)

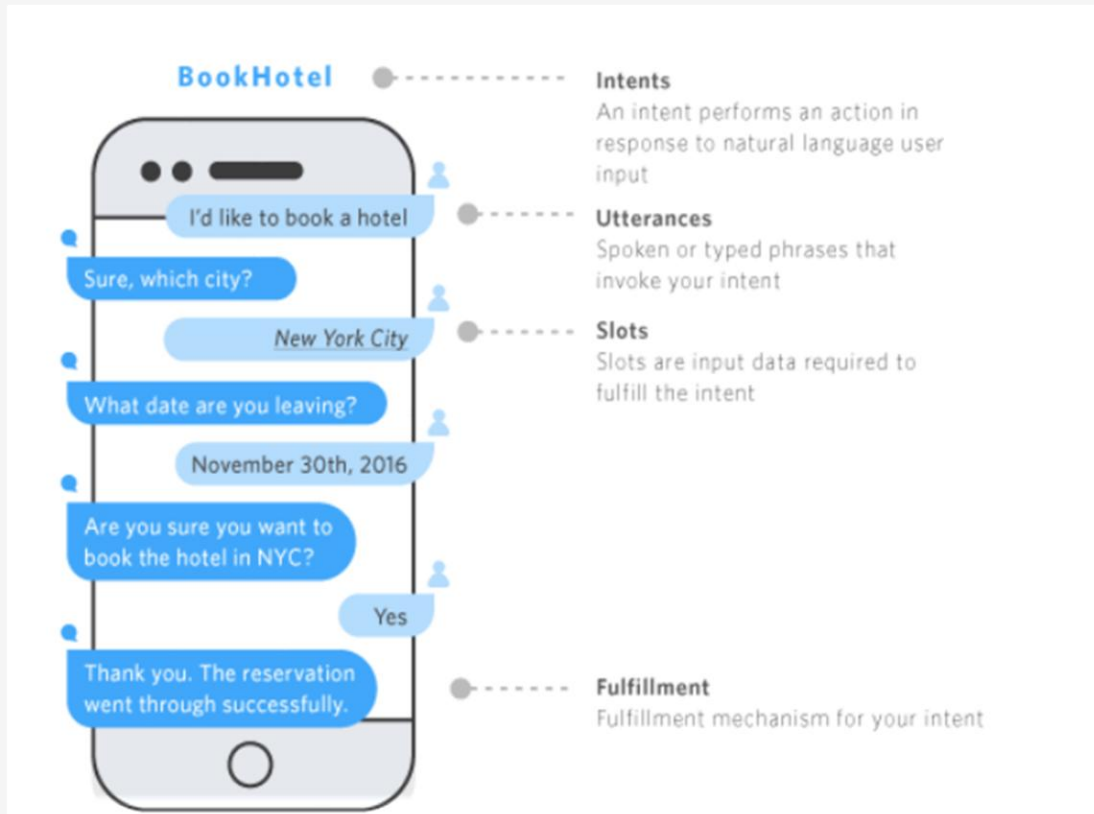
- Content Moderation - Detect inappropriate content
- Verify identity online – Onboard new customers through video analysis
- Streaming media analysis – Analyse and tag videos for celebrities, people etc.

AWS AI Service – Amazon Lex (AI Chatbot)

Amazon Lex - Build and deploy conversational AI interfaces



Amazon Lex

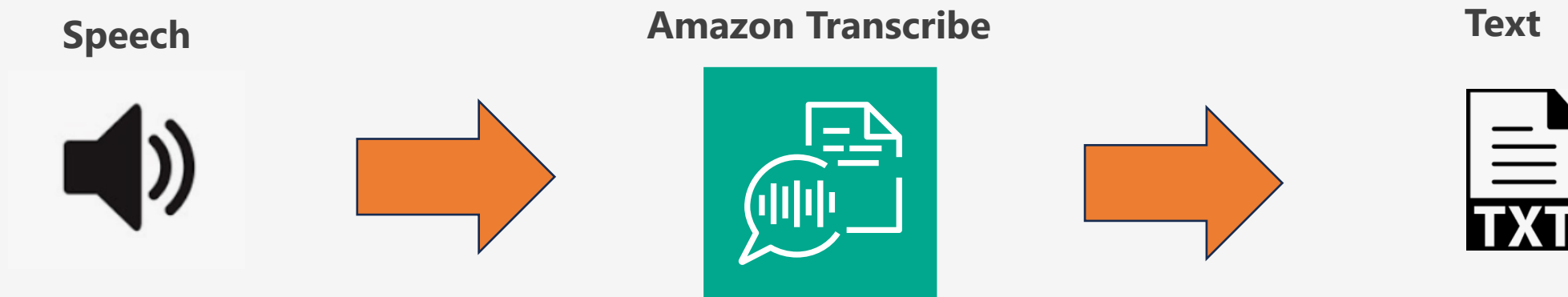


Source : AWS

- **Self-service voice assistants and chatbots** – build a call center bot
- **Application/Transactional bot** – build a stand-alone pizza ordering agent or a travel bot
- **Enterprise Productivity bot** – build custom bots to connect to enterprise data resources
- **Device Control bot**– use Amazon Lex to issue control commands to connected devices

AWS services in the NLP – Amazon Transcribe

Amazon Transcribe - Convert **speech** to **text**

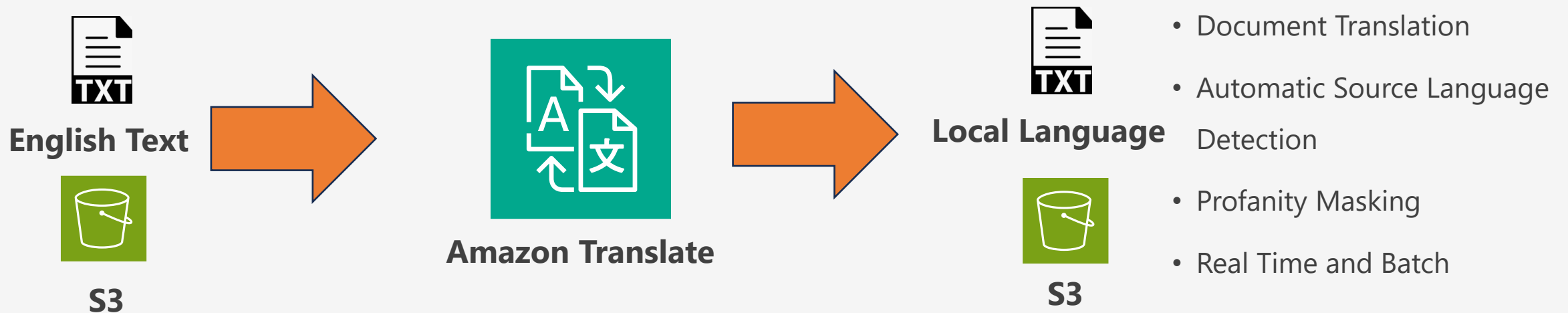


Common Use cases for Polly :

- Medical Transcription
- Call center analytics and agent assist
- Detect toxic content in audio for online gaming, social media or peer to peer

AWS AI Service (NLP) – Amazon Translate

Amazon Translate - AWS Language Translation service allows translation across 75 different languages.

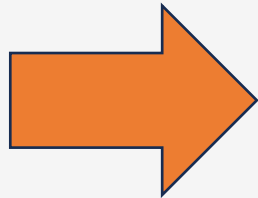


Amazon Translate – Uses ML models for **language translation**

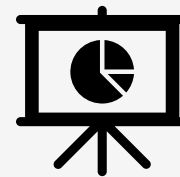
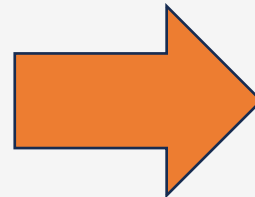
- Translate user generated content – RSS Feed, Social Media
- Real-time translation of chat, email, helpdesk, and ticketing apps by English speaking agent
- Sentiments toward your product or service in multiple languages (NLP)

AWS AI Services (NLP) – Amazon Comprehend

Amazon Comprehend - Derives and understand valuable insights from text within documents, emails, web pages



Amazon Comprehend



Insights

- Detect PII
- Sentiment Analysis
- Recognize Entities
- Detect Key phrases
- Detect Languages

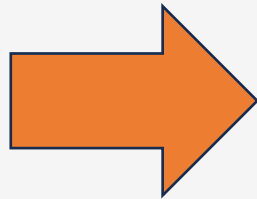
Amazon Comprehend – Natural-language processing (NLP) service to uncover insights in text

- Call center analytics, Customer Support Tickets
- Secure documents – Redacting PII from documents
- Process financial documents - Classify and extract entities from Insurance Claims

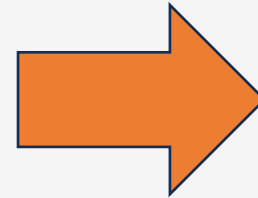
AWS services in the NLP – Amazon Polly

Amazon Polly - Turns text into lifelike speech using deep learning

Text



Amazon Polly



Speech



Common Use cases for Polly : - [Link](#)

- eLearning - Lectures
- Content Creation – RSS Feed
- Telephony

*Section on : **Security***

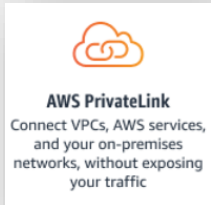


AWS Services to secure AI systems – Amazon Macie & AWS Private Link



Amazon Macie

- Security service that uses **ML and pattern matching to discover and protect your sensitive data.**
- Detects sensitive data including **personally identifiable information (PII)** such as names, addresses, and credit card numbers from S3 Bucket (**Remember : only for S3 Bucket**)
- Generate findings and sends to **EventBridge and Security Hub for remediation**

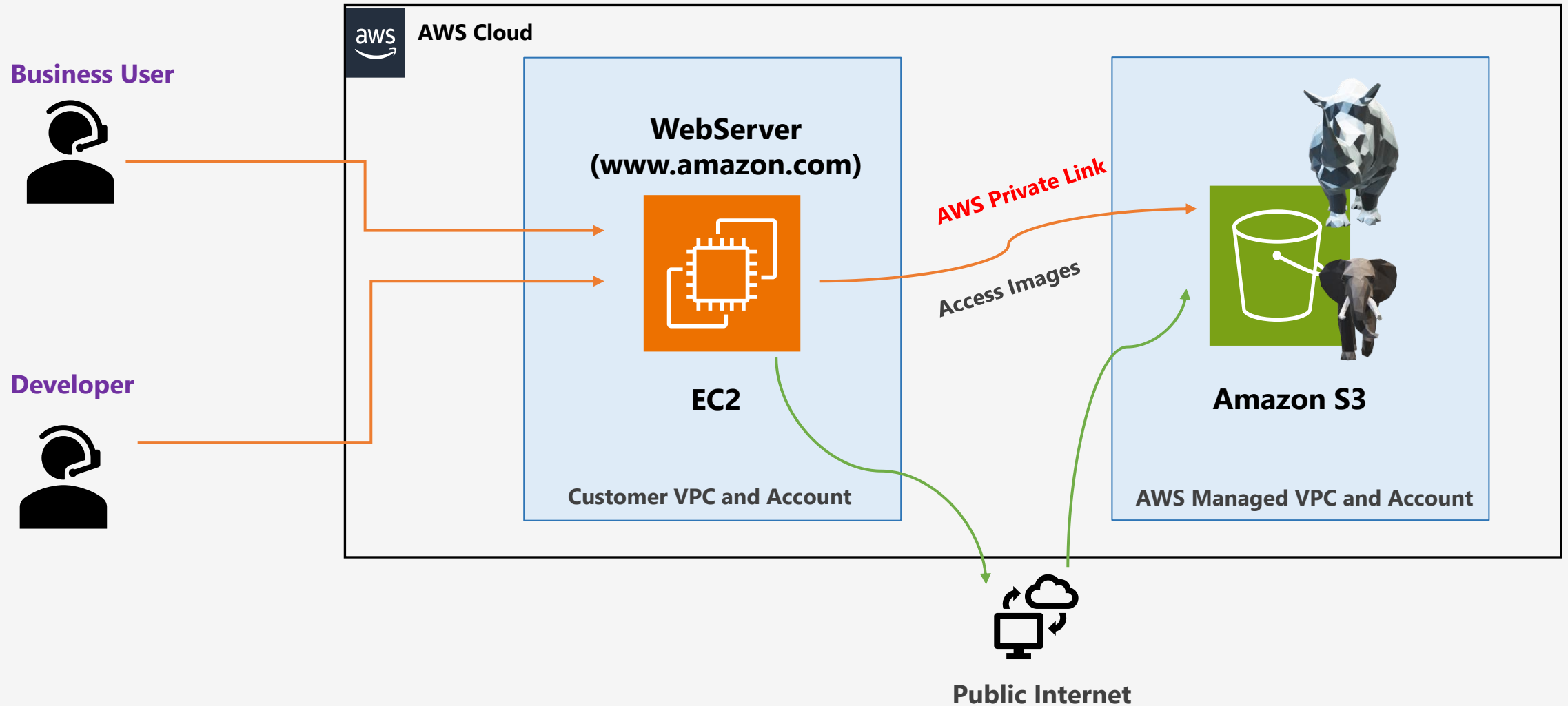


AWS PrivateLink

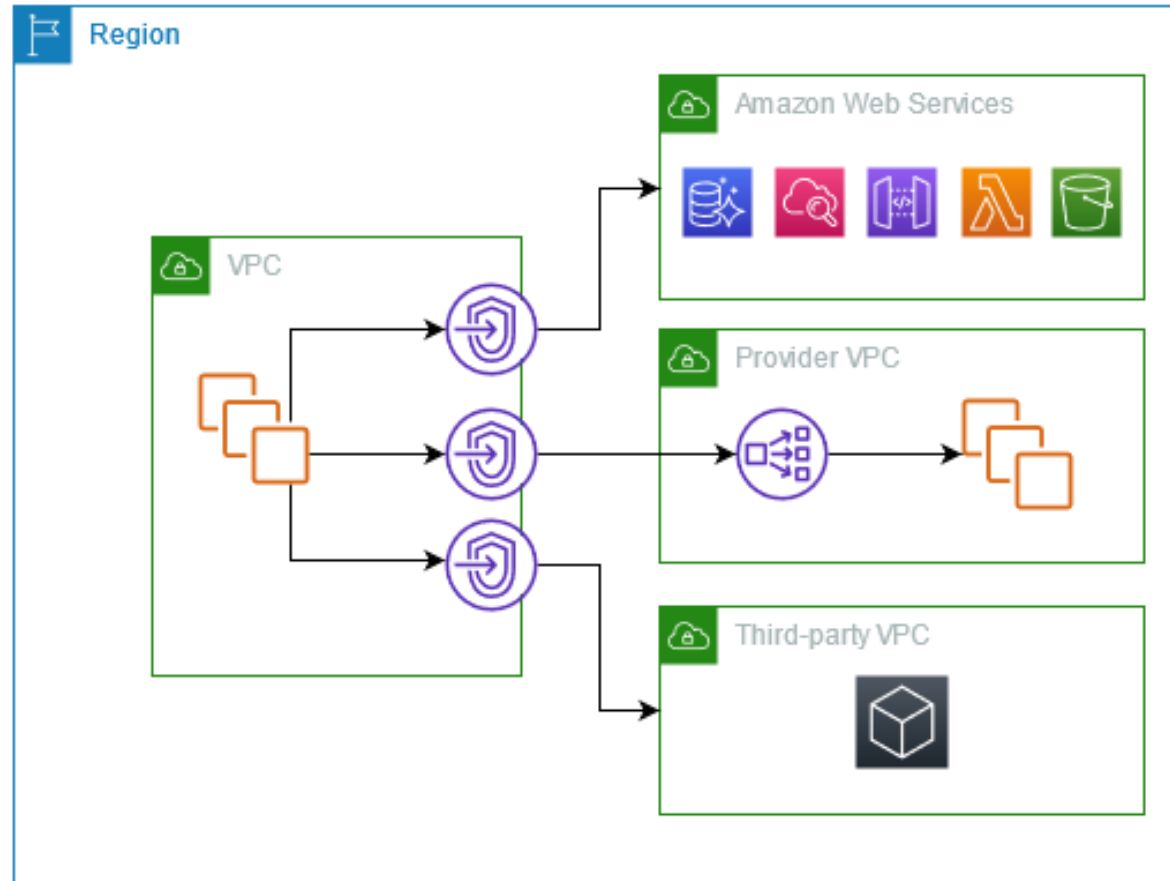
- AWS PrivateLink provides **private connectivity without exposing your traffic to the public internet**
- Network traffic that uses PrivateLink doesn't travel over the public internet, **reduces the risk of external threats**, such as **brute force and distributed denial-of-service (DDoS) attacks.**
- Connectivity can be between :
 - **AWS Services**
 - **Virtual private clouds (VPCs)**
 - **On-premises networks (with AWS Direct Connect)**



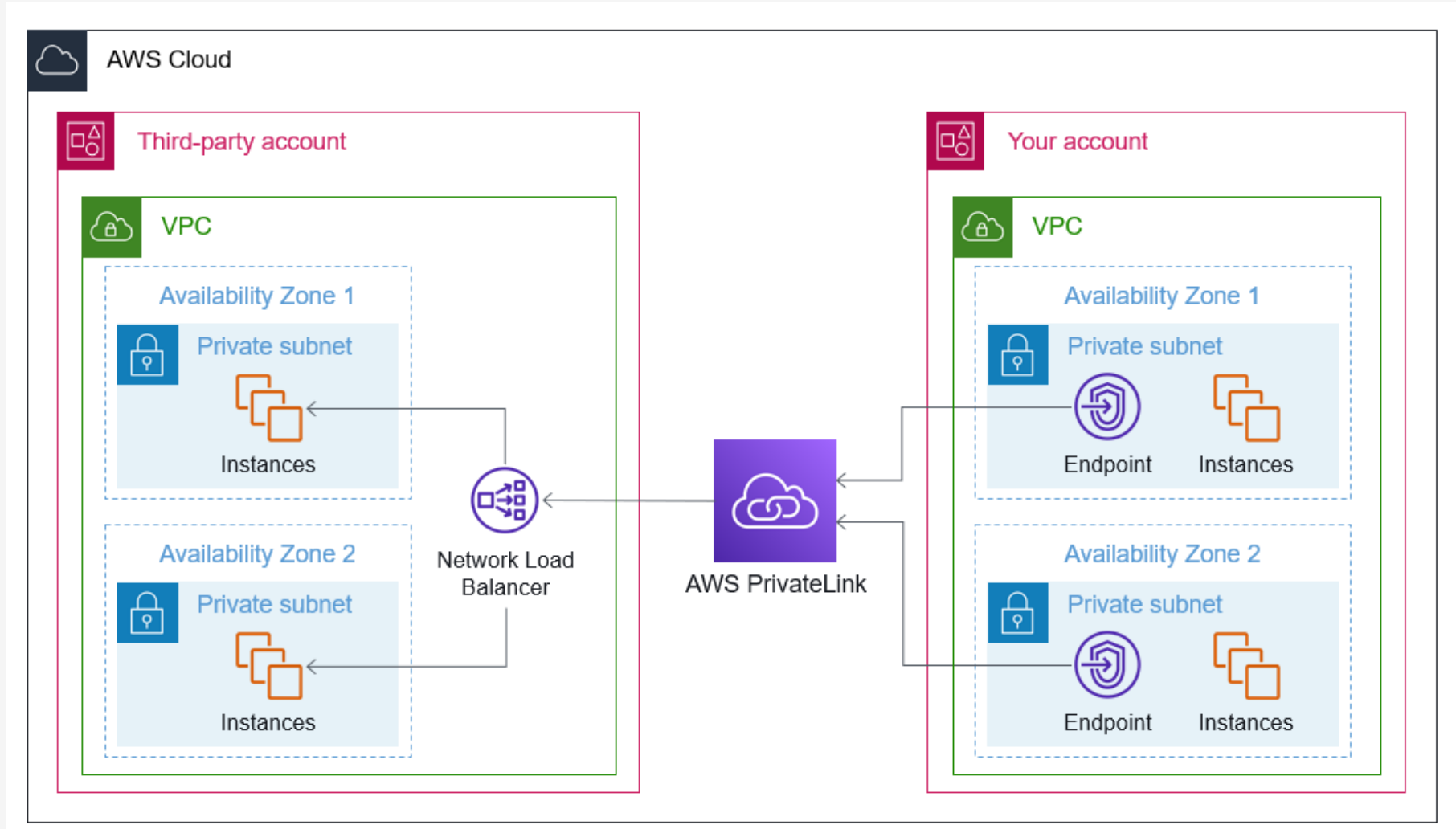
AWS Services to secure AI systems



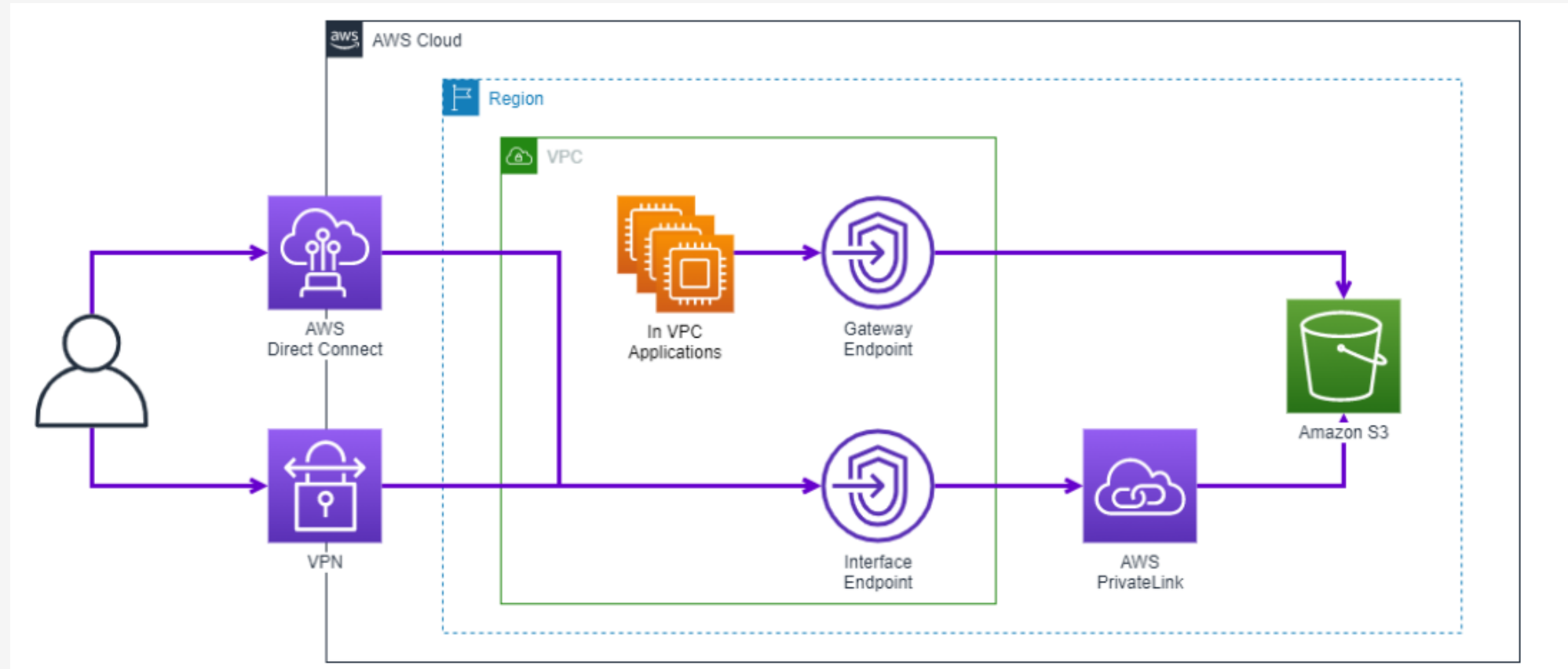
AWS Services to secure AI systems



AWS Services to secure AI systems



Gateway and Interface Endpoints



Gateway Endpoints – S3 and DynamoDB

AWS Identity and Access Management (IAM)

- AWS Identity and Access Management (IAM) helps **securely control access to AWS resources**.
- Control who is **authenticated (signed in)** and **authorized (has permissions)** to use resources.



**AWS Identity and
Access Management**

AWS Services to secure AI systems – IAM



AWS Identity and Access Management



Role



Long-term security credential



Permissions

IAM Users and Groups

- IAM users are assigned to workforce users.
- Attach policies to determine permission level
- Identities with long-term credentials.
- Create Groups for users with similar permissions

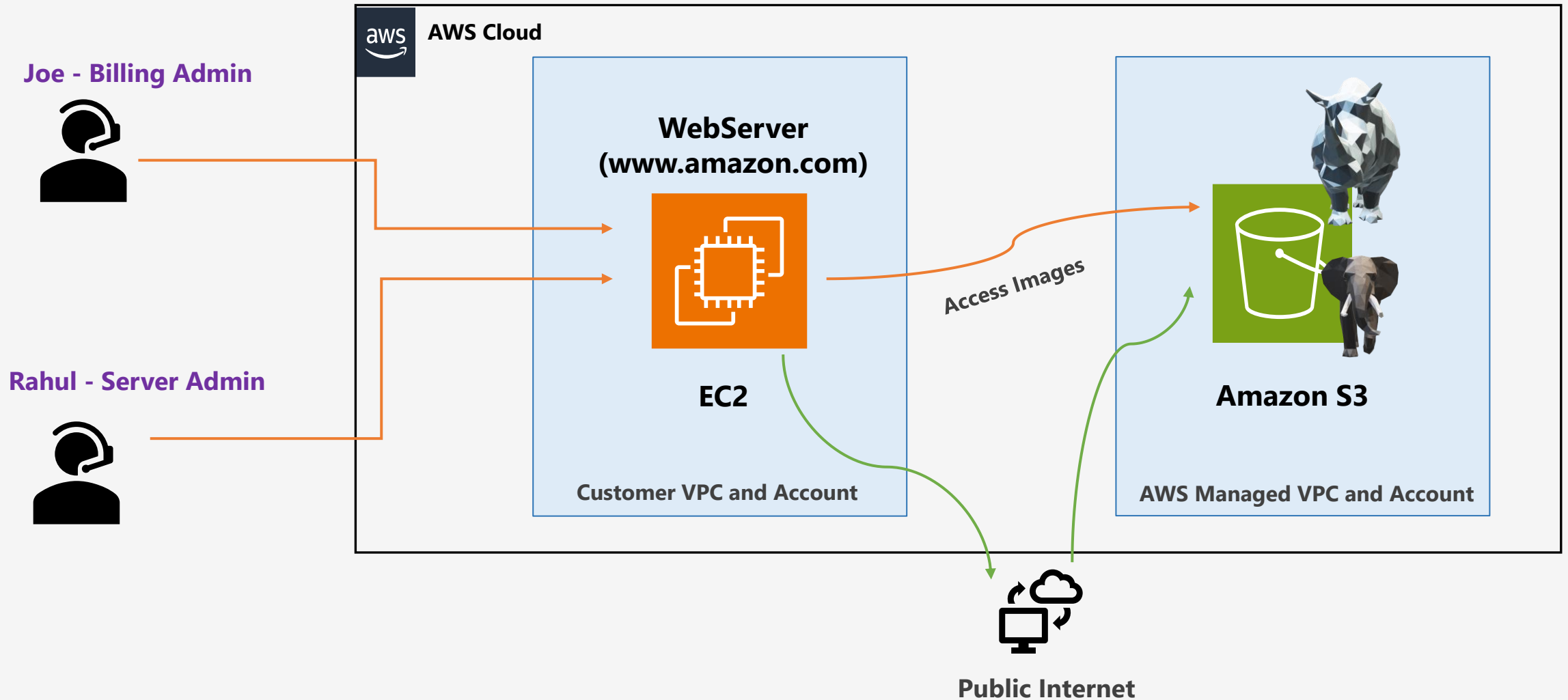
IAM Roles

- IAM identity that can be assumed by AWS services, applications etc.
- Each role has a set of permissions for making AWS service requests
- Provide a way to access AWS by relying on temporary security credentials.

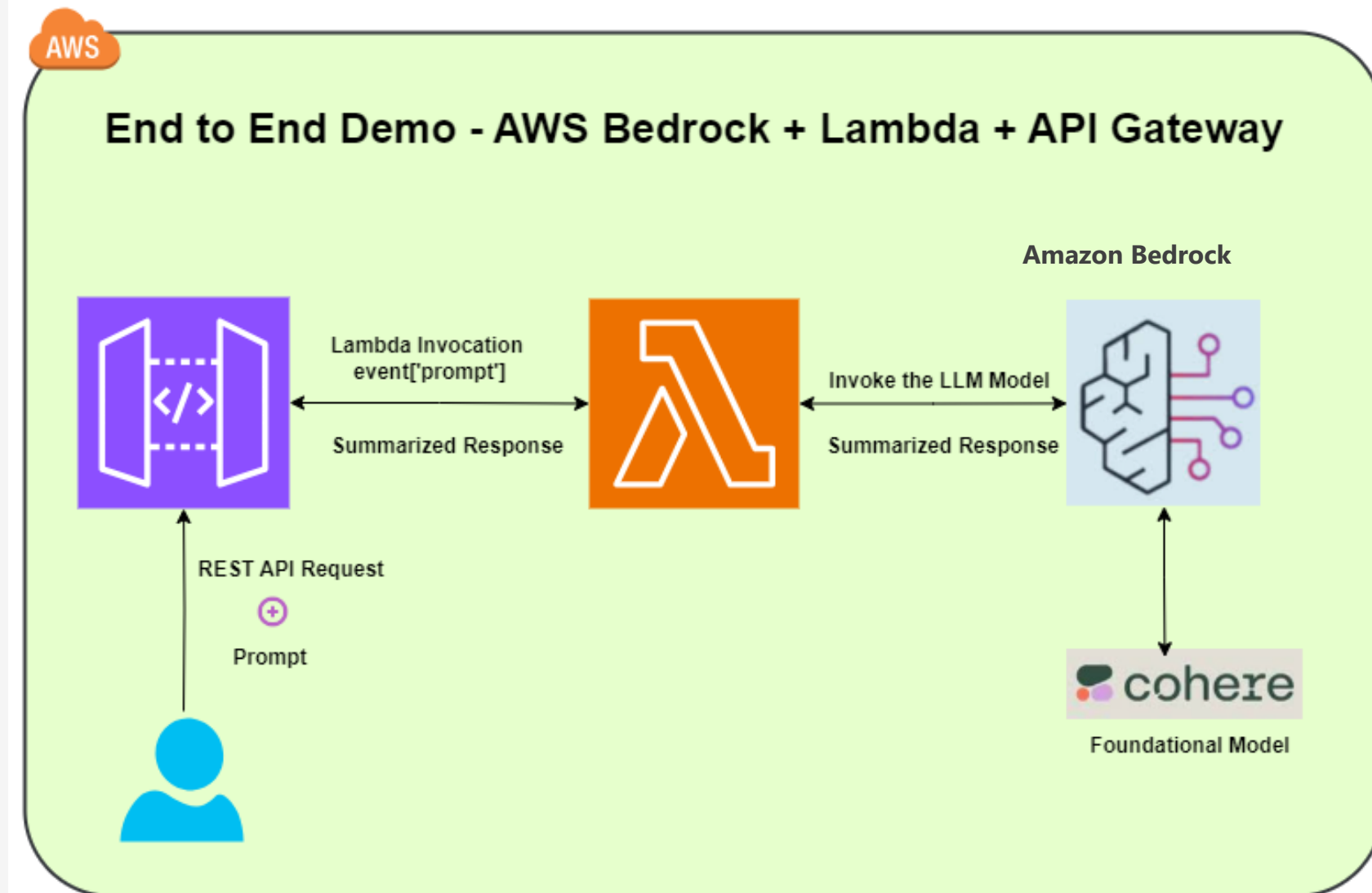
IAM Permissions and Policies

- Permissions defines which actions are allowed or denied.
- An IAM policy is a document that defines the permissions that can be granted to an IAM user, group, or role.

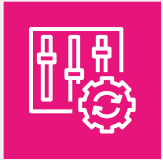
AWS Services to secure AI systems



Demo with Bedrock (Summarization Use Case)



AWS Services to assist with Governance & Regulation Compliance



AWS Config

- **Monitor and Record Config changes** for AWS Resources such as S3, EC2 etc.
- Assess, audit and evaluate the configuration of resources with your organization policies and remediation.
- **What has changed, Is the resource still compliant**, maintain the changelog and remediation



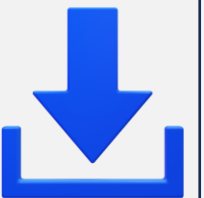
AWS CloudTrail

- CloudTrail enables auditing, security monitoring & operational troubleshooting **by tracking user activity & API usage.**
- CloudTrail logs, continuously monitors, and retains account activity related to actions across your AWS infrastructure to enable compliance with regulations such as SOC, PCI, and HIPAA.
- **Who made the change, What changes were made**

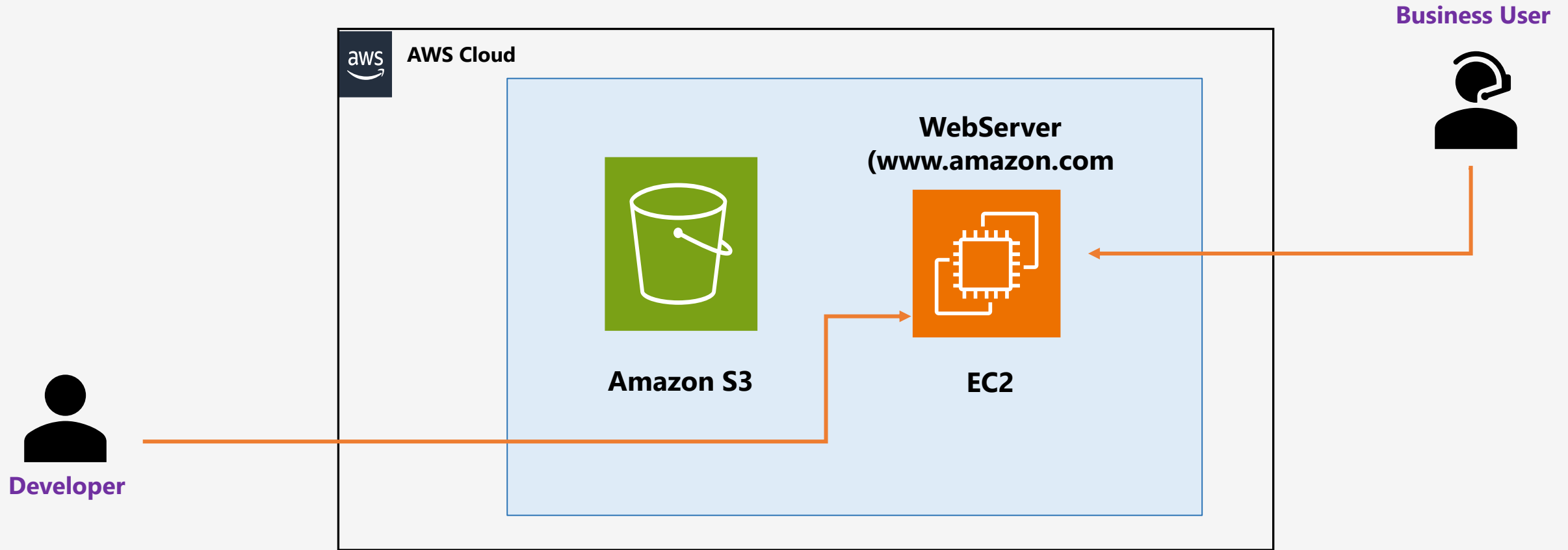


Amazon CloudWatch

- Amazon CloudWatch is an AWS monitoring service for cloud resources
- CloudWatch to collect and track **Metrics**, collect and monitor **Log files**, set alarms and **Dashboards**



AWS Config vs AWS CloudTrail vs AWS CloudWatch



AWS Services to assist with Governance & Regulation Compliance



Amazon Inspector

- Amazon Inspector automatically discovers **compute workloads, such as Amazon EC2 instances, Container Images in ECR + within CI-CD and Lambda functions**
- **Scans** for **software vulnerabilities** and unintended **network exposure**.



AWS Trusted Advisor

- Trusted Advisor continuously **evaluates your AWS environment** using best practice checks across the categories
- **Cost optimization and Security**
- **Resilience, Performance and Operational excellence**
- **Service limits**

Recommend actions to remediate any deviations from best practices.



AWS Services to assist with Compliance and Audit



AWS Artifact

- On-demand access to **AWS compliance documentation and AWS agreements.**
- Download AWS security and compliance documents, such as **ISO certifications, PCI-DSS, SOC reports** through **console or** provide **access to auditors via IAM Role**



AWS Audit Manager

- **Continuously audit your AWS usage** to assess risk & compliance with regulations and industry standards.
- Provides **prebuilt frameworks for well-known industry** standards and regulations, such as the HIPAA, GDPR, PCI DSS and **maps to customer AWS resources and assesses compliance**
- Integrates and collects data from AWS Services such as CloudTrail, Security Hub and others.
- **AWS Audit Manager assists in gathering and preparing evidence for audits.**



Governance and Compliance regulations for AI Systems

Governance is a **framework that guides how end users make use of cloud services** by defining and creating policies to :

- Control costs
- Minimize security risks
- Improve efficiency
- Accelerate deployment

Governance and Compliance regulations for AI Systems

Compliance in AWS refers to adhering to various **industry-specific** and **general data protection regulations and standards**.

- Industry-Specific Regulations:
 - HIPAA
 - PCI DSS for payment security.
- General Data Protection Regulations:
 - Laws like the GDPR and CCPA
- Other Compliance Programs:
 - SOC 1/2/3 for service organization controls
 - FedRAMP

Regulatory Compliance Standards for AI systems

- **AI Act European Union (EU AI Act)**

- The AI Act is the first-ever legal framework on AI, which addresses the risks of AI – [Link](#)

- **Algorithmic Accountability Act of 2023**

- The act requires companies to assess the impacts of the AI systems they use and sell - [Link](#)

- **GDPR European Union**

- Applies to the processing of personal data within the EU.
- Provides specific data protection guidelines for transparency, liability, ethics, bias, and risk.

Regulatory Compliance Standards for AI systems

- **International Organization for Standardization [ISO]**

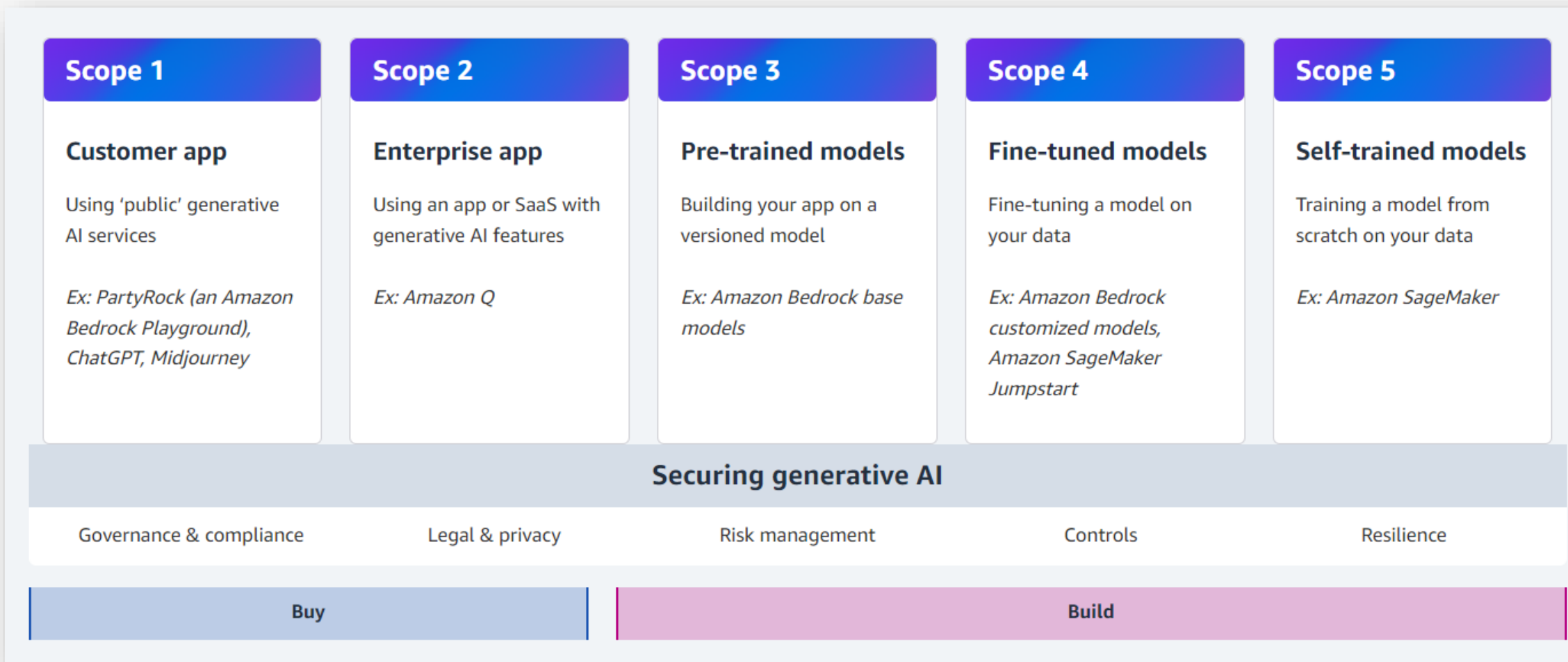
- Global network of the world's leading standardizers - [Link](#)
- AI Standards- ISO/IEC 42001:2023 - [Link](#)

- **System and Organization Controls [SOC]**

- SOC 1: For financial reporting - [Link](#)
- SOC 2: For design and operational effectiveness of internal controls
- SOC 3: For presenting SOC 2 report information to the general public

Generative AI Security Scoping Matrix

A mental model to classify use cases into various scope



Generative AI Security Scoping Matrix

Scope 1: Consumer app

- Organization consumes a **public third-party generative AI service**, either at no-cost or paid.
- **Don't own or see the training data or the model**, cannot modify or augment it.
- **Invoke APIs** or directly use the application according to the **terms of service of the provider**.
- **Example:** An employee interacts with a **generative AI chat app via a public website** to generate ideas for an upcoming marketing campaign.

Scope 2: Enterprise app

- Uses a third-party enterprise application that has generative AI features embedded
- **Business relationship** is established between your organization and the vendor.
- Apps have **terms and conditions that are aimed at enterprise customers**, designed to offer additional protections.
- **Example:** Third-party enterprise AI Assistant

Generative AI Security Scoping Matrix

Scope 3: Pre-trained models

- **Organization builds its own application** using an existing third-party generative AI foundation model.
- **Directly integrate it with your workload** through an application programming interface (API).
- **Example:** You build a customer support chatbot that **integrates your own data using Retrieval-Augmented Generation (RAG)** and leverages the **Anthropic Claude foundation model through Amazon Bedrock APIs**.

Scope 4: Fine-tuned models

- Organization refines an **existing third-party generative AI foundation model** by **fine-tuning** it with data specific to your business.
- **Example:** You require a model with in-depth medical domain expertise to summarize patient records in an Electronic Health Record (EHR) system. Fine-tuning can be used to align the system's output to match the style expected by doctors and provide the system training on domain specific terminology.

Generative AI Security Scoping Matrix

Scope 5: Self-trained models

- Organization builds and trains a generative AI model from scratch using data that you own or acquire.
 - Own every aspect of the model.
 - **Example:** Domain specific model
-
- **Exam Tip :** The scopes are numbered 1–5, representing least ownership to greatest ownership your organization has over the AI model and its associated data.

Thank you

